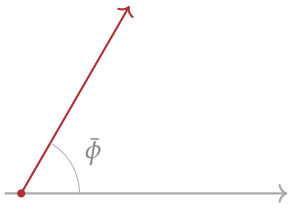
THE COSMIC MASS

VOLUME 1

ONE NUMBER FOR MASS AND CHARGE



MOSTAFA DERRAZ

Morocco 2026

First draft

[First draft, November 9, 2026]

Mostafa Derraz: *The Cosmic Mass, Volume 1. One Number for Mass and Charge*, © 2026

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SOURCES
Chapter 3 presents the essentials of the General Sinus (Derraz, 2026a). The mathematics of Chapter 4 is also published as a stand-alone paper, with complete proofs and a verification script (Derraz, 2026b). Parts of Chapter 1 and Appendix C adapt the teaching approach of Mokhithi and Shock (2020), released under the Creative Commons Attribution 4.0 license.

TYPESETTING
Set in Palatino with LaTeX, using the IMISE classicthesis template, which is based on *A Classic Thesis Style* by André Miede and Ivo Pletikosić.

VERSION
First draft

Egli è scritto in lingua matematica, e i caratteri son triangoli, cerchi, ed altre figure geometriche.

It is written in the language of mathematics, and its characters are triangles, circles, and other geometric figures.

Galileo Galilei, *Il Saggiatore* (The Assayer), Rome, 1623, speaking of the great book of nature. English rendering for this book.

PREFACE

Every particle carries two numbers. One tells how much matter it holds: its mass. The other tells how strongly it takes part in electricity: its charge. Physics has always written these two numbers apart, each in its own law, each with its own constant. This book asks a simple question with a long history behind it: could one number hold them both?

The question did not arrive all at once. It grew over many years, out of three pieces of work that only make full sense when they are read together.

The first piece is the *General Sinus*. It begins with a small doubt about something everyone learns at school. The sine and the cosine are defined on a right triangle. But why a right triangle? Most triangles in the world are not right-angled. The General Sinus gives the sine and the cosine a second variable, so that they can measure any triangle, and it gives back the familiar functions when the angle happens to be right (Derraz, 2026a).

The second piece is the *General Numbers*. The General Sinus does more than add a variable to two old functions. It frees trigonometry from a single point of view, the vertical projection, and lets us look at the same point from any projection angle we choose. That freedom of perspective is what unlocks the complex plane. For about two centuries the complex plane has been built on one fixed frame, with its second axis standing at a right angle to the first. Once the projection is free, the frame is free too: the second axis may lean at any angle, and its unit may take a length of its own. This is the General Number Plane. It extends the complex plane and its frame, and it keeps the complex numbers inside it, as the one frame in which the second axis stands upright. The mathematics of the General Numbers is also published on its own, as a paper with complete proofs (Derraz, 2026b).

The third piece is the *Cosmic Mass*. Inside the General Number Plane there is room for a number whose first part is a mass and whose second part is a charge. That number is the cosmic mass. With it, a single law of interaction can be written for two particles, and that law contains the gravity of Newton, a term shaped like the electrostatics of Coulomb, and a third term that neither of them contains. Read by its real part, the law gives back the laws of Newton and Coulomb exactly, and the third term adds no force between the two particles.

The heart of the book is Chapter 4, on General Numbers, and Chapter 6, on the Interaction Law. The chapters before them are there to carry you to them and to make them easy to read: a first chapter that shows you the destination at once, a second on how a new theory is

[First draft, November 9, 2026]

judged and on what this volume shows, a chapter on the General Sinus, and a prologue on why the language of physics must sometimes grow. The chapters after them put the cosmic mass to work. Chapter 7 draws every particle, ion and nucleus in one frame; Chapter 8 reads the mass of a nucleus off its neighbours, and the energy of a fusion before anyone weighs it; and Chapter 9 writes, for slow bodies, one law of fields and motion that holds Newton's gravity, Maxwell's equations and the force of Lorentz. Chapter 10, in Part iv, opens the question that the second volume takes up: what the third term of the law could mean, and what measurement already says about it. Chapter 11 draws the whole frame: antimatter, the half of negative mass, and the black holes. Short scenes from the long history of numbers, angles, matter and force run through them all.

This is the first of two volumes. It shows that the language of the cosmic mass can speak the old laws of physics, all of them in one form. The second volume asks what new results the same language can predict, beginning with the Coupling Force.

A word on honesty. This book presents a proposal. Some of its statements are definitions that we choose, some are results that follow from those choices, and some are ideas that still wait for the verdict of measurement. I have tried to label each one clearly, so that you can always tell which is which, and so that you can judge the proposal for yourself. That is what the reader of a new idea deserves.

[A personal note from the author on how the journey began.]

ACKNOWLEDGMENTS

I gratefully acknowledge the support of Morocco Aerospace Industries S.A.S. [Further acknowledgments.]

Mostafa Derraz
[Place], [Month] 2026

HOW TO READ THIS BOOK

WHAT YOU NEED

If you remember a little algebra from school (how to multiply out brackets) and the basic idea of sine and cosine, you have everything you need. Each new idea is introduced slowly, first with a picture, then in words, and only then with symbols.

TWO WAYS THROUGH THE BOOK

THE SHORT ROAD. Read Chapters 1 and 2, then Chapter 4, then Chapter 6, and finish with the Coda. This takes you straight to the heart of the theory. A short reminder of the General Sinus is given where Chapter 4 needs it, and Appendix A lists its formulas. To see where the theory leads, read Chapter 9 before the Coda.

THE FULL ROAD. Read everything in order. Chapter 3 shows where the General Sinus comes from, and Chapter 5 explains why physics sometimes needs its language to grow. Chapter 7 draws all the particles, ions and nuclei in one frame, and Part iii follows the cosmic mass into the nuclei (Chapter 8) and into bodies in motion and their fields (Chapter 9). Part iv ends the road with an open discussion of the Coupling Force (Chapter 10): what holds, what is excluded, and what is still open; and with the whole frame (Chapter 11): antimatter, the half of negative mass, and the black holes.

WHAT THE BOXES MEAN

> **IN PLAIN WORDS**
>
> Boxes like this one explain an idea in everyday language. You can read them before or after the formal statement they accompany. They never introduce new results.

> **WORKED EXAMPLE**
>
> Boxes like this one do a short calculation, step by step, using the formulas exactly as they are stated in the text.

A line in small capitals, such as

opens a scene from history: a person, a place, and a moment when an idea changed.

THE FORMAL STATEMENTS

The theory itself is stated in numbered *axioms*, *definitions*, *theorems* and *remarks*, and, in Chapters 8 and 9, *laws*: statements that are not proved but tested against measurement. Chapter 10 adds *views*: readings put forward for discussion, neither proved nor yet tested, each with its status.

SYMBOLS

A table of all the symbols, units and constants used in the book is given in Appendix D.

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Part I

BEYOND THE RIGHT ANGLE

1

ONE NUMBER FOR TWO LAWS

Two old laws that look like twins, a particle that carries the numbers of both, and the idea of this book.

Pick up a pebble and let it go. It falls. Now rub a balloon on your sleeve for a few seconds and hold it close to your hair. Your hair lifts to meet it.

You have just felt the two oldest pulls in physics. The first comes from *mass*. Every piece of matter pulls every other piece of matter towards it, and the Earth is simply a very large piece, so the pebble falls. The second comes from *electric charge*. Rubbing moved a tiny amount of charge between your sleeve and the balloon, and charges pull on one another, or push one another away, far more fiercely than masses do. A few seconds of rubbing are enough to lift a lock of hair against the pull of the entire planet.

For more than three centuries, physics has described these two pulls with two separate laws. What is strange, and what this book is about, is how much those two laws look alike.

1.1 TWO LAWS WITH THE SAME SHAPE

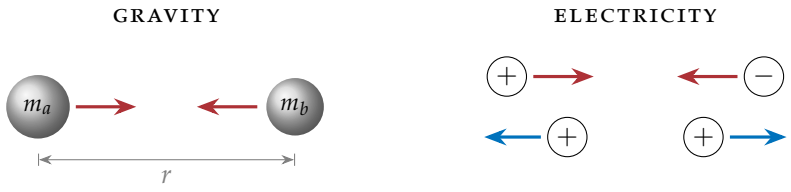
CAMBRIDGE, 1687

The first law was written down by Isaac Newton in the *Principia* (Newton, 1687, 1999). Newton did not write it as an equation with a constant in front. He wrote proportions: the pull grows with the quantity of matter in each body and falls off with the square of the distance between them. The constant came later, once the world had agreed on how to measure a mass. In today's notation, and in the exact form we will use throughout this book, the gravitational force between two bodies A and B, with masses m_a and m_b, at a distance r from each other, is

$$\vec{F}_N = G \cdot \frac{m_a m_b}{r^2} \cdot \vec{e}_r.$$

It is worth reading this slowly, one piece at a time.

- $m_a m_b$ is the product of the two masses. Double one of them and the force doubles.
- r^2 is the square of the distance. Move twice as far away and the force becomes four times weaker.



masses always attract

$$\vec{F}_N = G \cdot \frac{m_a m_b}{r^2} \cdot \vec{e}_r$$

opposite charges attract, like charges repel

$$\vec{F}_C = K \cdot \frac{q_a q_b}{r^2} \cdot \vec{e}_r$$

Figure 1.1: Two laws with the same shape. Newton's law of gravitation (left) and Coulomb's law of electrostatics (right): a constant, times a product of two amounts, divided by the square of the distance.

- G is the gravitational constant. It is a fixed number that nature has chosen and that we can only measure.
- $\vec{e}_r$ is an arrow of length one that points along the line joining the two bodies. It tells us in which direction the force acts.

PARIS, 1785

The second law came a century later. The French engineer Charles-Augustin de Coulomb hung a light rod from a thin wire, fixed a small charged ball to it, and measured how far the wire twisted when another charged ball came near (de Coulomb, 1788). He faced a problem that sounds impossible: there was no instrument that could measure an amount of charge. So he invented a way around it. Touch a charged ball with an identical uncharged ball, and by symmetry the charge is shared equally between them. Touch again, and you have a quarter. Coulomb could not say how much charge there was, but he could make known fractions of it, and that was enough to find the law. In the same notation,

$$\vec{F}_C = K \cdot \frac{q_a q_b}{r^2} \cdot \vec{e}_r,$$

where q_a and q_b are the two charges, measured in coulombs [C], and K is the Coulomb constant.

Now put the two laws side by side, as in Figure 1.1. They have the same shape: a constant, times a product of two amounts, divided by the square of the distance, acting along the line between the bodies. The only differences are *which* amount (mass or charge) and *which* constant (G or K).

WHY THE SQUARE OF THE DISTANCE? There is a beautiful reason, and it is the same for both laws. Imagine something that streams out from a point in every direction without being used up, like light from a small lamp. At a distance r, it is spread over the surface of a sphere of area $4\pi r^2$. At twice the distance, the same total is spread

over four times the area, so each small patch receives a quarter as much. Anything that spreads out this way weakens as $1/r^2$. Gravity and electricity both behave like this, which is why both laws carry the same r^2 underneath.

TWO IMPORTANT DIFFERENCES. Mass comes in only one kind, so masses always attract. Charge comes in two kinds, positive and negative: opposite charges attract and like charges repel. This is why ordinary matter hides its charge so well (positive and negative cancel out) while it can never hide its mass. The second difference is strength. Between two protons, the electric push is about 10^{36} times stronger than their gravitational pull. That is a one followed by thirty-six zeros, which is why a rubbed balloon can beat the whole Earth.

LAWS NEED UNITS. Neither law could be written with its constant until people agreed on how to measure a mass and a charge. The kilogram was first defined in 1799 through a cylinder of platinum kept in Paris, and from 1889 until 2019 the kilogram simply *was* a cylinder of platinum and iridium kept near Paris. The coulomb was named in 1881 as the charge carried by a current of one ampere in one second. In 2019 both units were redefined through constants of nature: the kilogram through Planck's constant, and the coulomb by fixing the charge of a single electron exactly (Bureau International des Poids et Mesures, 2019). The value of G was first reached, in effect, by Henry Cavendish in 1798, when he weighed the Earth with a twisting wire in a shed (Cavendish, 1798).

1.2 ONE PARTICLE, TWO NUMBERS

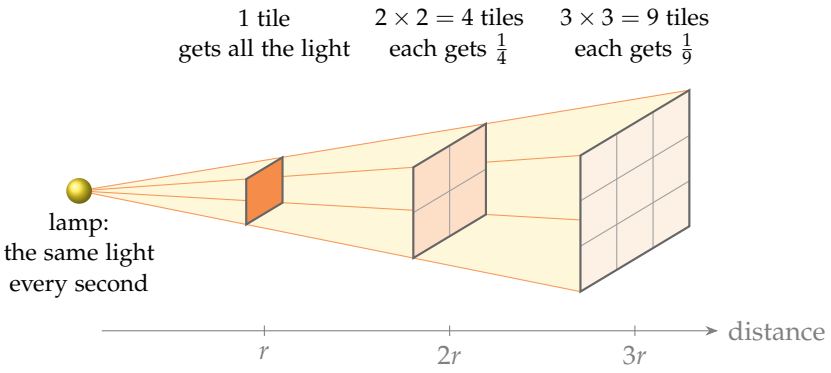
Now look at a single particle. An electron has a mass of about 9.109×10^{-31} kg and a charge of about -1.602×10^{-19} C. A proton has a mass of about 1.673×10^{-27} kg and a charge of $+1.602 \times 10^{-19}$ C. A neutron has almost the same mass as a proton and no charge at all.

Nature does not hand us a mass in one place and a charge in another. It hands us one particle, and the particle carries both at once. Physics, however, looks at the particle through two separate windows. Through the window of gravity it sees only the mass. Through the window of electricity it sees only the charge.

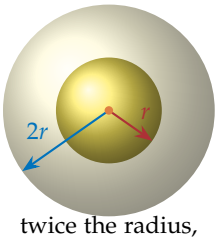
This is where the journey behind this book began. In nature a particle possesses both identifiers, its mass and its charge, while physics still approaches it from separate angles, either by its mass or by its charge. Why not by both?

So the question of this book can be stated in one sentence:

> *Can we write one number for a particle that holds both its mass and its charge, and one law of interaction built from that number?*

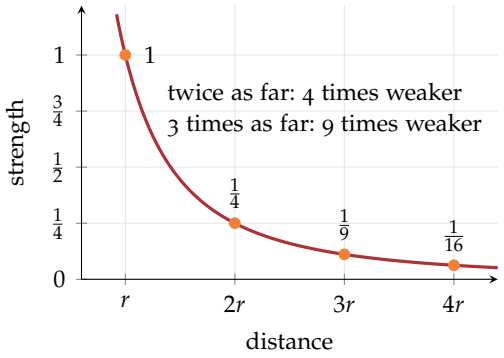


(a) the same light, shared by more and more tiles



twice the radius,
four times the area:
$4\pi(2r)^2 = 4 \times 4\pi r^2$

(b) the same light crosses both spheres



(c) the strength falls as $1/r^2$

Figure 1.2: Why the square of the distance. (a) A small lamp sends out the same amount of light every second. At the distance r, a bundle of its rays covers one tile. At $2r$ the same rays cover a square twice as wide and twice as tall, so $2 \times 2 = 4$ tiles share the light and each receives a quarter of it. At $3r$, $3 \times 3 = 9$ tiles share it, and each receives a ninth. The fading colour shows the light growing thinner. (b) The same idea for the whole lamp: all its light crosses the sphere of radius r and then the sphere of radius $2r$, whose area is four times larger. (c) The rule drawn as a curve: the strength falls from 1 at r to $\frac{1}{4}$ at $2r$, $\frac{1}{9}$ at $3r$ and $\frac{1}{16}$ at $4r$. Gravity spreads out from a mass, and the electric pull spreads out from a charge, in the same way, which is why both laws divide by r^2.

1.3 THE IDEA OF THIS BOOK IN ONE LINE

Here is the answer this book proposes. We show it now, at the very start, so that you always know where we are going.

For a particle with mass m and charge q, we define its *cosmic mass* as

$$\mu = m + q \cdot \mathcal{G}_{\bar{\phi}}.$$

It looks like an ordinary sum, but it is not one. The symbol $\mathcal{G}_{\bar{\phi}}$ is a new kind of unit, which we call the *Composer Unit*. It does not live on the number line. It points along a second axis of a new number plane, an axis that does not have to stand at a right angle to the first. The mass m is measured along the first axis. The charge q is counted in steps of $\mathcal{G}_{\bar{\phi}}$ along the second axis. The Composer Unit also carries the physical unit [kg/C], kilograms per coulomb, so that $q \cdot \mathcal{G}_{\bar{\phi}}$, just like m, is measured in kilograms. One number, two parts, one unit.

With one number per particle, one law can be written. For two particles with cosmic masses μ_a and μ_b at a distance r, the *Cosmic Interaction Law* reads

$$\vec{F}_G = G \cdot \frac{\mu_a \mu_b}{r^2} \cdot \vec{e}_r.$$

It has exactly the shape of Newton's law, with cosmic masses where the masses used to be. Now multiply out the product $\mu_a \mu_b$, the way you multiply out brackets at school, and the single law opens up into three terms:

$$\begin{aligned} \vec{F}_G = {} & G \cdot \frac{m_a m_b}{r^2} \cdot \vec{e}_r \\ & + \mathcal{G}_{\bar{\phi}}^2 \cdot G \cdot \frac{q_a q_b}{r^2} \cdot \vec{e}_r \\ & + \mathcal{G}_{\bar{\phi}} \cdot G \cdot \frac{m_a q_b + q_a m_b}{r^2} \cdot \vec{e}_r. \end{aligned}$$

The first term is Newton's gravity. The second term pairs charge with charge, like Coulomb's law, and it is scaled by $\mathcal{G}_{\bar{\phi}}^2$. The third term is new. It pairs the mass of each particle with the charge of the other. We call it the *Coupling Force*, and it introduces a possible new fundamental interaction. Chapter 6 shows how the law is read: by the real part of the product. Read that way, the first two terms are Newton's law and Coulomb's law, each with its full constant, and the Coupling Force has no real part, so it adds nothing to the force along the line between the two particles.

> **IN PLAIN WORDS**
>
> What to take away from this section:
>
> - one number per particle, the cosmic mass μ, with a mass part and a charge part;

- one law, shaped like Newton's law, written with cosmic masses;
- three terms inside that law: gravity, a Coulomb-like term, and a new coupling term;
- one way to read that law, by its real part: it gives Newton's law and Coulomb's law exactly, and the coupling term adds no force between the two particles.

Do not worry yet about what $\mathcal{G}_{\bar{\phi}}$ really is. Chapter 4 builds it from the ground up.

Two words of fairness before we go on. First, what you have just read is a proposal, not an established result of physics. Every step that leads to it will be stated clearly, so that you can follow it and judge it for yourself. Second, the symbol $\mathcal{G}_{\bar{\phi}}$ is doing a great deal of work in those two lines, and it deserves a proper introduction. It will get one.

Chapter 2 comes first. It says how a new theory is judged, by the old laws it must give back and by the new results it must predict, and it gathers what this volume shows. To build the law itself, we need to travel in four steps:

1. Chapter 3 frees the triangle from its right angle. This gives us the *General Sinus*, a sine and a cosine with two variables.
2. Chapter 4 frees the number plane from its right angle. This gives us the *General Numbers* and the Composer Unit $\mathcal{G}_{\bar{\phi}}$.
3. Chapter 5 explains why the language of physics must sometimes grow, and what this book does and does not try to do.
4. Chapter 6 puts mass and charge into the new number plane and derives the Cosmic Interaction Law.

But first, a short walk through the history of numbers. It will show us that what we are about to do has been done before, several times, and always in the same way.

1.4 HOW NUMBERS GROW

Have you ever seen a three? Not three goats, and not the symbol 3 printed on a page, but three itself? Nobody has. Numbers live in a world of ideas, and yet they fit the real world astonishingly well. How did we come to have them?

BORDER CAVE, SOUTHERN AFRICA, ABOUT 43,000 YEARS AGO

Before there were numbers, there were notches. A small bone found at Border Cave, in the Lebombo Mountains on today's border between eSwatini and South Africa, carries twenty-nine notches cut by a human

hand (d'Errico et al., 2012). The Ishango bone, found near Lake Edward in today's Democratic Republic of the Congo and some twenty to twenty-five thousand years old, carries marks arranged in groups (Brooks and Smith, 1987). We do not know what they counted. We do know the idea behind them: someone decided that a mark could stand for a thing, one mark for one goat, one mark for one day. Everything that follows in this book is that same decision, taken again and again at higher and higher stakes.

Later, something even more important happened, and nobody wrote down when. People noticed that three fish and three days share something, and that this something can be named on its own. That shared thing is the number three. Once numbers came loose from the things they counted, they could be carried anywhere, combined, and argued about.

From there, numbers grew. And they grew in a very particular way: each time, a simple question was asked that the numbers of the time could not answer.

- *What is* $3 - 5$*?* With only counting numbers, there is no answer. The Chinese calculators of the Han period used red and black counting rods for amounts owned and amounts owed (Shen et al., 1999). In India in 628, Brahmagupta wrote down rules for fortunes and debts, and treated zero as a number in its own right (Colebrooke, 1817; Plofker, 2009). The answer to the question was a new kind of number: the negative numbers.

- *What is* $1 \div 3$*?* The answer needed fractions, which Egyptian and Babylonian scribes had used for thousands of years.

- *How long is the diagonal of a square whose side is* 1*?* Its square is 2, and the Greeks discovered that no fraction whatsoever has a square equal to 2. Legend says the discovery cost a man named Hippasus his life. The legend is almost certainly a later decoration, but the shock was real. The answer was a new kind of number: the irrational numbers, such as $\sqrt{2}$.

- *Which number, multiplied by itself, gives* -1*?* Every number on the line gives a positive result when squared. The answer was the strangest new number of all.

Table 1.1 gathers these steps. Read from top to bottom, it is a ladder, and each rung was built because someone refused to accept that a reasonable question had no answer.

MILAN, 1545, AND BOLOGNA, 1572

The last question arrived in the middle of a quarrel between Italian mathematicians over how to solve equations. In 1545 Gerolamo Cardano set a problem of schoolboy simplicity: divide ten into two parts

THE QUESTION		THE NEW NUMBERS
How many?	$\mathbb{N}$	counting numbers $1, 2, 3, \ldots$
What is $3 - 5$?	$\mathbb{Z}$	negative numbers and zero
What is $1 \div 3$?	$\mathbb{Q}$	fractions
Which number squares to 2?	$\mathbb{R}$	irrational numbers such as $\sqrt{2}$
Which number squares to -1?	$\mathbb{C}$	complex numbers

Table 1.1: The ladder of numbers. Each new kind of number answered a question that the old numbers could not: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

whose product is forty (Cardano, 1545). Calling the parts x and $10 - x$, the algebra gives

$$x\,(10 - x) = 40 \quad \Longrightarrow \quad x^2 - 10x + 40 = 0 \quad \Longrightarrow \quad x = 5 \pm \sqrt{-15}.$$

There is no such pair of ordinary numbers; the largest product you can get by splitting ten is twenty-five. Yet Cardano noticed that if he multiplied the two impossible answers together, obeying the usual rules, he got exactly forty. He called the exercise as subtle as it was useless, and moved on.

Twenty-seven years later, Rafael Bombelli, an engineer who drained marshes for a living, met a case he could not walk away from (Bombelli, 1572). The equation $x^3 = 15x + 4$ has the perfectly ordinary solution $x = 4$, as you can check. But the standard formula of the day produced

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}.$$

Bombelli had what he called a wild thought. What if these strange cube roots were numbers of the form "two plus something impossible" and "two minus the same thing"? Writing i for the unit whose square is -1 (so that $\sqrt{-121} = 11i$), he found

$$(2 + i)^3 = 2 + 11i, \qquad (2 - i)^3 = 2 - 11i,$$

so the two cube roots are $2 + i$ and $2 - i$, and their sum is exactly 4. The impossible numbers were not a sign that the question was wrong. They were the only road to the right answer.

Still, they were not welcome. René Descartes called them *imaginary* in 1637, and he did not mean it kindly (Descartes, 1637). They took another two centuries to become respectable.

THE FREEDOM TO DEFINE. How can anyone simply decide that a number whose square is -1 exists? Two teachers at the University of Cape Town explain it to their students in a way worth keeping: in mathematics we are free to define new objects, as long as they do

not contradict the rules we already have, and then we discover what our definitions commit us to (Mokhithi and Shock, 2020). Nobody discovered i lying on the ground. Somebody defined it. What they could not choose were its consequences, and the consequences turned out to be wonderful.

1.5 THE QUARTER TURN

So what *is* i? The clearest answer is a picture, and it is the picture this whole book grows from (Mokhithi and Shock, 2020; Needham, 1997).

Think of the number line. Adding a number moves you to the right or to the left. Multiplying by 2 stretches you away from zero. Multiplying by -1 flips you to the other side of zero. Now try a puzzle: which single step, done twice, takes you from 1 to -1? Stretching never crosses zero. Flipping twice brings you back to where you started. On the line, there is no such step.

In a plane, there is. Turn a quarter of a full turn, counterclockwise, around zero. The number 1 moves to a point one unit *above* zero. Turn once more and you arrive at -1. That halfway point is i (Figure 1.4). Multiplying by i is a quarter turn, and the famous rule $i^2 = -1$ says nothing more mysterious than this: two quarter turns make a half turn.

Keep turning, and the powers of i go round in a cycle of four:

$$i^1 = i, \qquad i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1.$$

Each power adds one more quarter turn: 90°, 180°, 270°, and finally 360°, back where we started. Remember this cycle. We will meet it again in Chapter 4, wearing new clothes.

COPENHAGEN, 1797

This picture was first drawn by a Norwegian surveyor working in Denmark, Caspar Wessel. He spent his days turning bearings and distances into maps, which meant adding and combining directions. In a paper presented in 1797, whose title means *On the Analytical Representation of Direction*, he showed that if you describe a direction by a length and an angle, then the rule "multiply the lengths and add the angles" behaves exactly like multiplication (Wessel, 1799). His paper went almost unnoticed for a hundred years. Jean-Robert Argand found the same picture in 1806 (Argand, 1806), and Carl Friedrich Gauss, who had kept it to himself for decades, published it in 1831 with all his authority and gave these numbers the name they still carry: not imaginary, but *complex* (Gauss, 1831). In 1837 William Rowan Hamilton removed the last mystery by showing that a complex number is simply an ordered pair of real numbers with a clear rule for multiplying pairs (Hamilton, 1837).

So a complex number $a + bi$ is a point of the plane: a steps along the real axis and b steps along the imaginary axis. Multiplying two

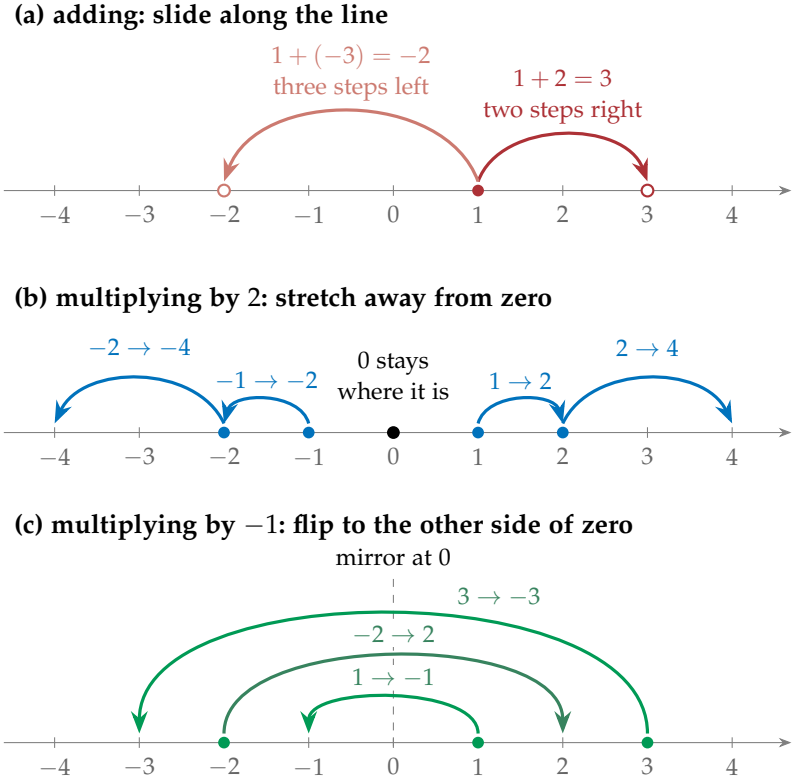


Figure 1.3: Three moves on the number line. (a) Adding slides you along the line: adding 2 takes 1 to 3, two steps to the right, and adding -3 takes 1 to -2, three steps to the left. (b) Multiplying by 2 stretches the whole line away from zero: 1 goes to 2 and 2 goes to 4, -1 goes to -2 and -2 goes to -4. Every number ends up twice as far from zero, on the same side, and zero itself does not move. (c) Multiplying by -1 flips the line around zero, as in a mirror: 1 goes to -1, -2 goes to 2 and 3 goes to -3. Every number keeps its distance from zero and changes side. Flip twice, and you are back where you started.

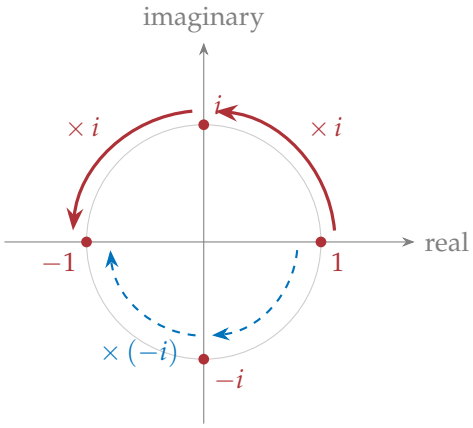


Figure 1.4: The quarter turn. Multiplying by i turns a number by a quarter of a full turn around zero; two quarter turns carry 1 to -1. Turning the other way, by $-i$, also reaches -1 in two steps.

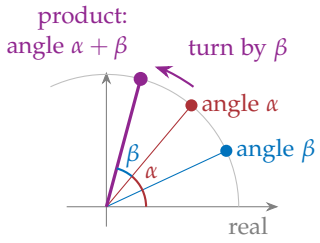
complex numbers multiplies their lengths and adds their angles. For numbers of length one, written with the angles α and β, this rule reads

$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta) = \cos(\alpha+\beta) + i\sin(\alpha+\beta),$$

and it is nothing other than the addition formulas for sine and cosine that astronomers had used for sixteen centuries to follow the Moon. Leonhard Euler packed the whole story into one line, $e^{i\theta} = \cos\theta + i\sin\theta$ (Euler, 1748). And in 1893 the engineer Charles Proteus Steinmetz showed that alternating currents can be computed with complex numbers, because the phase of a current is an angle, and multiplying by a complex number adds an angle (Steinmetz, 1893). Every power grid on Earth is designed that way today.

TWO READINGS, ONE NUMBER. Connect any device to the mains, a lamp, a motor or a phone charger, and measure it with two instruments. A voltmeter tells you how hard the electricity is being pushed, in volts. An ammeter tells you how much of it flows, in amperes. With alternating current both readings swing back and forth fifty times a second, and they do not always swing together: in a coil the push reaches its peak a quarter of a swing before the flow does, and in a capacitor a quarter of a swing after it. So the relation between the two readings holds two facts, not one. The first is a size: how many volts it takes to drive one ampere through the device. The second is a delay: how far the swing of the voltage runs ahead of, or behind, the swing of the current.

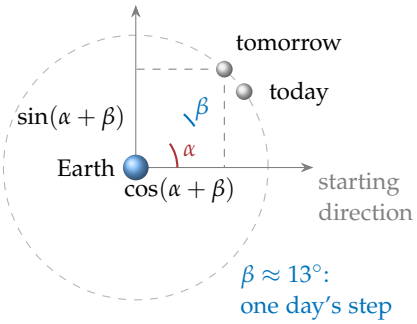
Steinmetz kept both facts in a single complex number, the *impedance* Z. Its length is the size, measured in ohms, which are volts per ampere. Its angle is the delay. Write the voltage and the current as turning



$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)$$
$$= \cos(\alpha+\beta) + i\sin(\alpha+\beta)$$

for any lengths: the lengths multiply, the angles add

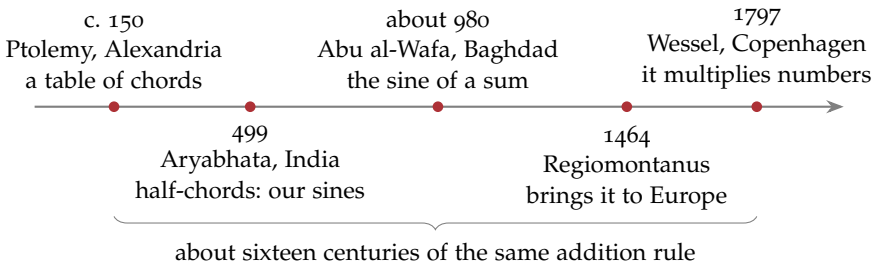
(a) multiplying: the angles add



$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$
$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

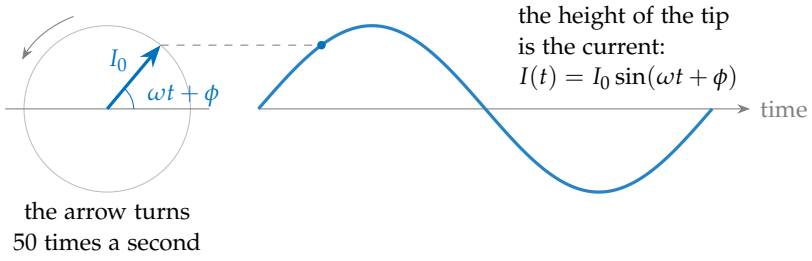
tables give α and β, the rule gives $\alpha + \beta$

(b) following the Moon

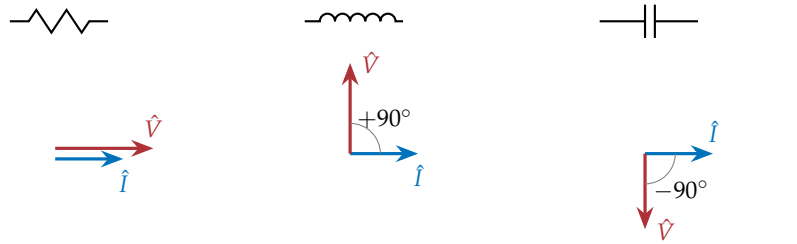


(c) from the chords of Ptolemy to the numbers of Wessel

Figure 1.5: Multiplying turns, and an old rule of the sky. (a) On the circle of radius one, the number at angle α times the number at angle β lands at angle $\alpha + \beta$: multiplying by the second number turns the first one by β. For numbers of other lengths, the lengths multiply as well. (b) Astronomers met the same rule when they followed the Moon. The Moon moves about 13° a day against the stars, so tomorrow's angle is today's angle α plus one day's step β. To turn the new angle into a position, how far across and how far up, they needed the sine and the cosine of a sum, and the addition formulas give these from the values for α and for β, which they read from their tables. (c) The same rule, first written with chords and later with sines, served astronomers from Ptolemy to Wessel, about sixteen centuries, until Wessel's picture showed that it is also the rule for multiplying numbers.



(a) an alternating current is a turning arrow, and its phase is an angle



resistor: $Z = R$
no turn: $\hat{V}$ in step with $\hat{I}$

coil: $Z = i\omega L$
a quarter turn ahead

capacitor: $Z = 1/(i\omega C)$
a quarter turn behind

(b) $\hat{V} = Z\,\hat{I}$: multiplying by Z turns the arrow

Figure 1.6: Alternating current with complex numbers. (a) An alternating current is the height of an arrow of length I_0 that turns at a steady rate, 50 times a second on the electricity grids of Morocco and Europe. The angle of the arrow is the phase of the current. (b) In a circuit, the voltage across each part is the current multiplied by a complex number Z, called the impedance. A resistor multiplies by a real number and leaves the angle as it is. A coil multiplies by $i\omega L$, which turns the voltage a quarter turn ahead of the current. A capacitor multiplies by $1/(i\omega C)$, which turns it a quarter turn behind. Multiplying by a complex number adds its angle: this is why Steinmetz could compute alternating currents with complex numbers.

arrows, that is, as complex numbers $\hat{V}$ and $\hat{I}$, and the whole relation becomes one division:

$$Z = \frac{\hat{V}}{\hat{I}} = \frac{V_0}{I_0}\, e^{i(\phi_V - \phi_I)} = R + iX.$$

The last form tells the same story in another way. Its real part R is the *resistance*: the share of the electrical energy that the device turns into heat and light, as a toaster or a lamp does. Its imaginary part X is the *reactance*: the energy that coils and capacitors store during part of each swing and then give back. Two different physical behaviours, measured in the same unit, travel together in one number, and the rules of complex arithmetic do all the bookkeeping. Multiply a current by Z, and out comes the voltage, with its size and its delay already right.

The impedance as one complex number

- Voltage and current as turning arrows: $\hat{V} = V_0\, e^{i\phi_V}$ and $\hat{I} = I_0\, e^{i\phi_I}$.
- The impedance is their ratio, $Z = \hat{V}/\hat{I} = |Z|\, e^{i\theta}$, with the size $|Z| = V_0/I_0$ in ohms and the angle $\theta = \phi_V - \phi_I$, the delay between the two swings.
- The same number in two parts: $Z = R + iX$, with the resistance $R = |Z|\cos\theta$ and the reactance $X = |Z|\sin\theta$. Back again: $|Z| = \sqrt{R^2 + X^2}$ and $\tan\theta = X/R$.
- A resistor, a coil and a capacitor in a row give $Z = R + iX$ with $X = \omega L - \dfrac{1}{\omega C}$.

WHAT ELSE BELONGS IN ONE NUMBER? The impedance invites a bolder question. If two facts that seemed to need two separate lines, a size and a delay, can live together in one number, are there other notions of physics that belong together in the same way? The first candidates are the ones we met at the start of this chapter. Every particle carries a mass and a charge. They are measured with different instruments and in different units, kilograms and coulombs, and they enter two different laws that nevertheless have exactly the same shape (Section 1.1). Could they, too, be written as one number, as the cosmic mass $\mu = m + q \cdot \mathcal{G}_{\tilde{\phi}}$ of Section 1.3 proposes?

And if they can, which kind of number should hold them? History advises us not to take the answer for granted. Look again at the questions of Table 1.1. With the counting numbers alone, $3 - 5$ has no answer: someone had to make the negative numbers. The division $1 \div 3$ needed the fractions. A number whose square is 2 needed the irrational numbers. A number whose square is -1 needed the com-

plex numbers. Each time the question first looked impossible, and each time the way out was not to drop the question but to enlarge the numbers, keeping all the old ones inside the new. So the question for mass and charge has two halves. Can they share one number? And are the complex numbers, with their two axes fixed at a right angle, the right home for them, or is it time, once again, for a new kind of number?

What we have so far

- Gravity and electricity obey two laws with the same shape, and every particle carries the numbers of both: a mass and a charge.
- This book proposes one number for both, the cosmic mass $\mu = m + q \cdot \mathcal{G}_{\bar{\phi}}$, and one law built from it, which opens into three terms.
- Numbers grow when a reasonable question has no answer. The last great step gave us i, the quarter turn.
- The complex plane is built on a right angle. The rest of the book asks what happens when that right angle is set free.

2

OLD LAWS, NEW RESULTS

A new theory owes two debts. It must say again, exactly, what the old laws say wherever they have been tested, and it must then say something they do not, which a measurement could prove wrong. This chapter states the two debts, shows how much of the first this volume pays, and names the candidate for the second.

LONDON, 6 NOVEMBER 1919

On 6 November 1919 the rooms of the Royal Society in London were full. The Society and the Royal Astronomical Society had called a joint meeting to hear the results of two expeditions sent to watch the total eclipse of the Sun of 29 May, one to Sobral, in the north of Brazil, the other to the island of Príncipe, in the Gulf of Guinea. Each team had photographed the stars around the darkened Sun and compared their places with photographs of the same stars taken when the Sun was far from them (Dyson et al., 1920).

Starlight that grazes the Sun should bend. Newton's law of gravity, applied to light as if it were a falling body, gave a bend of 0.87 seconds of arc at the edge of the Sun. Einstein's new theory of gravity gave twice as much, 1.75 seconds (Einstein, 1916; Dyson et al., 1920). Frank Dyson, the Astronomer Royal, gave the result of the Sobral plates: 1.98 seconds, with a probable error of 0.12. Arthur Eddington, back from Príncipe, gave his: 1.61 seconds, with a probable error of 0.30. A third set of plates, from a second telescope at Sobral, gave a smaller value; the team judged its star images faulty and set it aside, a decision that was argued over for years. Later measurements with radio telescopes have confirmed Einstein's value to better than one part in ten thousand (Will, 2014).

J. J. Thomson, who presided, called it "the most important result obtained in connection with the theory of gravitation since Newton's day" (Royal Society and Royal Astronomical Society, 1919). The philosopher Alfred North Whitehead was in the room. "The whole atmosphere of tense interest was exactly that of the Greek drama," he wrote six years later, and he remembered, in the background, "the picture of Newton to remind us that the greatest of scientific generalisations was now, after more than two centuries, to receive its first modification" (Whitehead, 1925).

Two things made that meeting decisive, and their order matters. First, Einstein's theory gave back Newton's. For slow bodies in weak gravity its equations become Newton's law, so it agreed with two centuries of measurements of falling bodies, tides, planets and comets

(Einstein, 1916). It also settled an old puzzle: the orbit of Mercury turns by 43 seconds of arc per century more than Newton's law allows, even with the pull of all the other planets, and in 1915 Einstein's theory gave exactly that (Einstein, 1915). But the turning of Mercury was known before the theory, which explained it after the fact. Second, the theory said something new, before anyone had measured it: light grazing the Sun bends by 1.75 seconds, not by 0.87. That number could have come out wrong. It did not.

This chapter is about those two duties of a new theory, and about where this volume stands with each. Section 2.1 states them, with five theories that met both. Section 2.2 sets out the rules under which this volume tests itself. Section 2.3 gathers what the cosmic mass language has been found to say, the laws of Newton, Coulomb, Maxwell and Lorentz, Heaviside's gravity, the balance of nuclear fusion, and a mass first written down by George Johnstone Stoney, and asks what it means for physics to have all of them in one language. Section 2.4 turns to the Coupling Force, the candidate for new results, which is the subject of Volume 2.

2.1 THE TWO DUTIES OF A NEW THEORY

The first duty is to agree. A new theory must give back every old law wherever the old law has been tested, with the same measured constants. Niels Bohr made this a rule for the quantum theory of the atom, which had to give back the classical laws of radiation for slow vibrations and large orbits; in 1920 he gave it a name, the principle of correspondence (Bohr, 1920). The great theories since have kept it. Relativity gives back Newton's mechanics for slow bodies and Newton's gravity for weak fields; quantum mechanics gives back classical mechanics for large bodies. The old law is not thrown away. It becomes a special case, true inside a domain whose borders the new theory draws.

The second duty is to risk being wrong. Karl Popper put it in one sentence: "it must be possible for an empirical scientific system to be refuted by experience" (Popper, 1959, section 6). A theory that agrees with everything known but forbids nothing has told us nothing. Imre Lakatos added a measure of progress for a whole line of research: it advances when its new versions predict new facts, and some of those facts are then found; it degenerates when it only adjusts itself, after the event, to facts that others found (Lakatos, 1970).

THEORY	OLD LAWS KEPT	NEW RESULT	FOUND
Newton's gravity, in Le Verrier's hands, 1846	the motions of the known planets	an unseen planet beyond Uranus, and where to look for it	Neptune, by Galle and d'Arrest, 23 September 1846, within one degree of the place (Le Verrier, 1846)
Maxwell's equations, 1865	the laws of Coulomb, Ampère and Faraday	electric waves that travel at the speed of light	made and detected by Hertz, 1887 to 1888 (Maxwell, 1865; Hertz, 1893)
Mendeleev's periodic law, 1871	the known elements and their chemistry	an element below aluminium, with its weight and properties	gallium, by Lecoq de Boisbaudran, 1875 (Mendeleev, 1871; Lecoq de Boisbaudran, 1875)
Einstein's gravity, 1915	Newton's law for slow bodies and weak fields	starlight bent by 1.75 seconds of arc at the edge of the Sun	the eclipse of 1919 (Einstein, 1916; Dyson et al., 1920)
Dirac's electron, 1928 to 1931	quantum mechanics and special relativity	a particle with the mass of the electron and the opposite charge	the positron, by Anderson, 1932 (Dirac, 1928, 1931; Anderson, 1933)

Table 2.1: Five theories that met both duties. Each kept the old laws whole, and each predicted a result that nobody had seen and that a measurement could have refuted.

Put together, the two duties read:

> *A new theory must give back every old law wherever that law has been tested, with the same constants; and it must predict at least one new result, a number with its error, stated before it is measured, that a measurement could prove wrong.*

Table 2.1 shows five theories that met both duties. In each of them the old laws were kept whole, and the new result was not put in on purpose.

Every row has the same shape. The new theory first contained the old one, and only then did its new prediction mean something, because it came from a theory that was already right wherever it had been checked. A theory that only agrees is a new way of writing

known physics: a language. A theory that also predicts, and survives the test, is new physics.

> **In plain words**
>
> A new theory has two jobs. First, it must get the old answers right: wherever the old laws were measured and found true, the new theory must give the same numbers. Second, it must say something new that could turn out to be false, and say it before anyone measures it. Einstein's theory did both. It gave Newton's answers for the planets, and it said that starlight would bend twice as much as Newton's law said. The eclipse of 1919 could have proved it wrong, and it did not.

2.2 THE RULES THIS VOLUME KEEPS

This volume holds itself to the first duty, under rules that it fixes in advance.

1. *Two inputs.* Each body enters with two measured numbers, its mass in kilograms and its charge in coulombs, and with nothing else (Table 5.1).
2. *Measured constants only.* The constants are those of physics: G, K, c, e and $\hbar$ (Appendix D). The size of the Composer Unit, $|\mathcal{G}_{\bar{\phi}}| = \sqrt{K/G}$, is not fitted to anything; G and K fix it (Section 6.3.1).
3. *One reading.* Every measured force, energy and momentum is read as the real part of a General Number. The rule is stated once, in Chapter 6 (Definition 6.17), and used without exception afterwards.
4. *One angle.* The angle of the Composer axis is held at 79.86° from Chapter 6 to the end of the book (Section 6.6.3). It is a choice of coherence, not a fit to data.
5. *Every number checked.* Each result is set beside a measurement, or beside the value of the standard laws, with its source, and the difference is reported.

Under these rules, each agreement is a test that could have failed. The last rule also says what an agreement is worth. When the language gives back a known law, it has passed a test of alignment, and it has discovered nothing yet. Chapter 9 says so of its own results: each is a check on the reading rather than a discovery.

2.3 WHAT THE COSMIC MASS LANGUAGE ALREADY SAYS

Chapters 6 to 9 test the language against the laws of physics, one after another. Table 2.2 gathers what they find, and Figure 2.1 draws it as a

OLD LAW	IN COSMIC MASSES	WHERE, HOW WELL
Bodies at rest, and matter		
the gravity of Newton (1687) and the law of Coulomb (1788)	the real part of one product of two cosmic masses	Proposition 6.16: both exactly, each with its full constant, at every angle
the rule of Du Fay (1733): like charges push apart	the minus sign of the Rule of the Square	Proposition 4.22: in every true plane
the hydrogen atom of Bohr (1913)	Coulomb's law at its full strength	Section 6.5.3: 13.598 eV, as measured (Sansonetti and Martin, 2005)
the neutral neutron	the Coupling term has no real part	Section 6.5.3: zero; measured, $(-0.2 \pm 0.8) \times 10^{-21}\, e$ (Baumann et al., 1988)
the masses of Stoney (1881) and Planck (1899)	the charge part of one elementary charge, $m_S = e\,\lvert\mathcal{G}_{\tilde{\phi}}\rvert$	Definition 7.2, Remark 7.3: $m_S = \sqrt{\alpha_{\text{fs}}}\, m_P$
the charge-to-mass ratios of Thomson (1897) and Aston (1919)	one angle for each body, on one frame	Chapter 7: 306 bodies, no measured value moved
charge and mass in fusion; the mass relations of Garvey and Kelson (1966)	two coordinates of one conservation law; six neighbours whose charge parts cancel exactly	Chapter 8: the masses of 2,481 nuclei to 0.25 MeV, and to 0.11 MeV for $A \geq 40$
Bodies in motion, and fields		
the equations of Maxwell (1865)	the Composer coordinate of the field equations, with $\mu_0 = 4\pi K/c^2$	Theorem 9.5: exactly
Newton's mechanics, and the gravity of Heaviside (1893)	the real coordinate of the same equations, and of the law of motion	Law 9.3: as Newton and Heaviside
the force of Lorentz	the real part of $\mu\,(\boldsymbol{\mathcal{E}} + \mathbf{v} \times \boldsymbol{\mathcal{B}})$	Proposition 9.7: Thomson's tube, the cyclotron, the drift in crossed fields
waves at the speed of light	both coordinates of the fields	Theorem 9.6: gravity as fast as light within 3 parts in 10^{15} (Abbott et al., 2017)
the conservation of energy and of momentum	one balance for both parts; one action for everything	Theorems 9.9 and 9.13
the Moon, and hydrogen in motion	one program for every body	Table 9.2: 27.28 days against 27.32; 13.598 eV

Table 2.2: The old laws in the cosmic mass language: where each is found in this volume, and how closely it agrees.

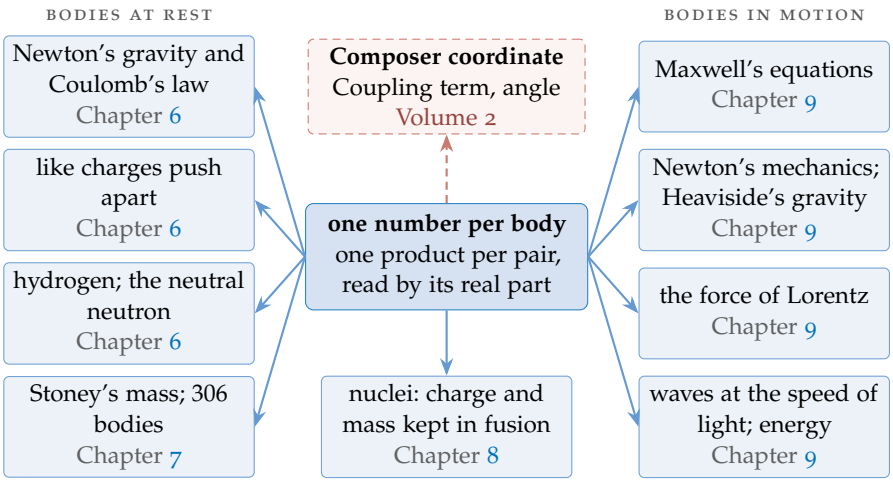


Figure 2.1: The old laws inside one law. Read by its real part, the product of two cosmic masses gives back the laws of bodies at rest (left), of bodies in motion (right) and of nuclei (below), each in the chapter shown. What the real part leaves, the Composer coordinate, holds the Coupling term and the angle of the Composer axis: the questions of Volume 2 (above).

map: one number for each body, one product for each pair, one way of reading it, and inside them the old laws. This section walks through the map in four steps, from two bodies at rest to moving bodies and their fields, and then asks what the result is worth.

2.3.1 *One product, every sign*

Physics has always written gravity and electricity as two laws with two constants, and it adds a rule of signs by hand. Masses always pull. Charges pull when their signs differ and push when they agree, as Du Fay found in 1733 (Du Fay, 1733). Nothing in the two laws says why: the rule lives in the signs given to the constants and to the charges, and it is learnt as a fact of nature.

In the cosmic mass language one law serves both, and no rule of signs is added. For two bodies at rest, the force is the real part of the product of their cosmic masses, multiplied by G and divided by the square of the distance, and it is counted positive for a pull (Proposition 6.16):

$$F = \frac{G\,\Re(\mu_a \mu_b)}{r^2} = \frac{G\,m_a m_b - K\,q_a q_b}{r^2}.$$

The minus sign between the two terms is not a convention. It is the real part of the square of the Composer Unit, $\Re(\mathcal{G}_{\bar{\phi}}^2) = -|\mathcal{G}_{\bar{\phi}}|^2$, the same at every angle of a true plane (Proposition 4.22): a unit that leans out of the line has a square that points backward. The pull between masses and the push between like charges come from the geometry of one plane.

Worked example: one product, three answers

- *Two electrons, one metre apart.* The charge part of each is $e\,|\mathcal{G}_{\bar{\phi}}| = m_S$, the Stoney mass, so $\Re(\mu_e\mu_e) = m_e^2 - m_S^2 \approx 8.3 \times 10^{-61} - 3.457 \times 10^{-18} \approx -3.457 \times 10^{-18}$ kg^2. Multiplied by G, the force is -2.307×10^{-28} N: a push, Coulomb's force between two electrons. The share of gravity in it is 2.4×10^{-43}.
- *An electron and a proton, one metre apart.* Now $q_a q_b = -e^2$, and $\Re(\mu_e\mu_p) = m_e m_p + m_S^2 \approx 3.457 \times 10^{-18}$ kg^2: a pull of 2.307×10^{-28} N.
- *An apple of* 0.1 *kg at the surface of the Earth.* Both bodies are neutral, so the real part is the product of the masses, and the force is $G \times 0.1$ kg $\times\, 5.972 \times 10^{24}$ kg$/(6.371 \times 10^6$ m$)^2 \approx$ 0.98 N: a pull, the weight of the apple.

One expression gave a push, a pull and a weight. Each time the sign came from the plane, and the size from the two parts of the cosmic masses.

The same expression shows why electricity is so much stronger than gravity between particles. Between two electrons the push and the pull stand in the ratio $(m_S/m_e)^2 \approx 4.2 \times 10^{42}$, because the charge part of an electron is about 2×10^{21} times its mass: its cosmic mass lies almost along the Composer axis. Physics has long quoted this ratio as a bare number. Here it is the same number, with a picture: the lean of one arrow.

2.3.2 *One number for every body*

The Stoney mass of the example above shows the kind of coherence that the language brings. In 1874 Stoney proposed units of nature in which nothing would be arbitrary, built from three constants: the speed of light, the constant of gravity, and the portion of electricity that every broken chemical bond carries through an electrolyte, which he later named the electron (Stoney, 1881, 1894). His unit of mass reappears here with a new meaning: it is the charge part of one elementary charge.

Worked example: the Stoney mass

- *The size of the Composer Unit.* $|\mathcal{G}_{\tilde{\phi}}| = \sqrt{K/G} \approx 1.16043 \times 10^{10}$ kg per coulomb (Section 6.3.1).
- *The charge part of one elementary charge.* $m_S = e\,|\mathcal{G}_{\tilde{\phi}}| \approx 1.602177 \times 10^{-19}\ \mathrm{C} \times 1.16043 \times 10^{10}\ \mathrm{kg/C} \approx 1.85921 \times 10^{-9}$ kg (Definition 7.2).
- *A balanced pair.* $G\,m_S^2 \approx 6.67430 \times 10^{-11} \times (1.85921 \times 10^{-9})^2 \approx 2.30708 \times 10^{-28}\ \mathrm{N\,m^2}$, and $Ke^2 \approx 8.98755 \times 10^{9} \times (1.602177 \times 10^{-19})^2 \approx 2.30708 \times 10^{-28}\ \mathrm{N\,m^2}$. Two bodies of mass m_S, each with one elementary charge, pull by gravity exactly as hard as they push electrically.
- *The Planck mass.* $m_P = \sqrt{\hbar c/G} \approx 2.17643 \times 10^{-8}$ kg, so $m_P/m_S \approx 11.706$ and $(m_S/m_P)^2 \approx 1/137.036$, the fine-structure constant (Remark 7.3).

None of these numbers is new, and honesty requires saying so. The Stoney mass follows at once from the size given to the Composer Unit, and its link with the fine-structure constant is an identity among the constants, true in any language. What is new is its place. In the cosmic mass language, Stoney's unit is the step by which every charged body climbs the Composer axis: an electron or a proton carries one such step, and a nucleus with z protons carries z of them.

Chapter 7 builds on these steps. With the angle fixed, every body becomes one triangle, with one magnitude and two angles, and all 306 bodies of its catalogue stand on one frame: every neutral body on one line, every positive body near a second line and every negative body near a third, each tipped by its mass per charge. Thomson in 1897 and Aston in 1919 measured the ratio of charge to mass of electrons and ions (Thomson, 1897; Aston, 1919). In the cosmic mass language that ratio is an angle, and the whole of ordinary matter, from the electron to the heaviest nucleus, is one drawing. No measured mass or charge is moved to make it.

2.3.3 *One law for nuclei*

When two nuclei fuse, two old rules meet. Charge is kept exactly. Mass is not: the product is lighter than the two nuclei that made it, and the missing mass leaves as energy, $Q = \Delta m\,c^2$, as Einstein showed in 1905 (Einstein, 1905) and as Eddington proposed in 1920 for the energy of the stars (Eddington, 1920). Physics states these as two rules. In the cosmic mass language they are one statement. The cosmic masses add part by part, so the charge parts of the two nuclei add exactly to the charge part of the product, and the loss of cosmic

mass, $\Delta\mu = \Delta m$, is a real number: a neutral body, with no charge part at all (Proposition 8.2). The charge part is never lost; only the real part is, and it leaves as the energy $Q = \Delta m\, c^2$.

Chapter 8 takes one step more. Six neighbouring nuclei, added with alternating signs, close on themselves: their charge parts cancel exactly around the loop, at every angle, and their binding nearly cancels (Theorem 8.4). From that one fact the mass of a nucleus can be read off its neighbours, for 2,481 measured nuclei to 0.25 MeV, and to 0.11 MeV for those with $A \geq 40$ (Law 8.7), and the energy of a fusion can be known before it is weighed.

2.3.4 *One law for fields and motion*

Chapter 9 carries the same number into motion. Two fields whose values are General Numbers obey four equations, and one law of motion moves every body (Law 9.3). Opened into its two coordinates, the law is two sets of equations at once (Proposition 9.4). The Composer coordinate is Maxwell's set, with the electric and magnetic fields $\mathbf{E} = -|\mathcal{G}_{\bar{\phi}}|^2\,\mathbf{e}$ and $\mathbf{B} = -|\mathcal{G}_{\bar{\phi}}|^2\,\mathbf{b}$ and the magnetic constant $\mu_0 = 4\pi K/c^2$ (Theorem 9.5). The real coordinate is Newton's gravity, with the magnetic part that Heaviside wrote down in 1893 (Heaviside, 1893). Opened in the same way, the law of motion is Newton's law of motion with gravity and the force of Lorentz in one line (Proposition 9.7):

$$m\,\frac{d\mathbf{v}}{dt} = m\,(\mathbf{g} + \mathbf{v} \times \mathbf{B}_g) + q\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Both parts of the field travel at the speed of light (Theorem 9.6); one equation keeps the source, mass and charge together (Theorem 9.8); and all of it follows from a single action (Theorem 9.13). Run on Thomson's tube, the cyclotron, the drift in crossed fields, the hydrogen atom and the Moon, the law gives what Newton, Maxwell and Lorentz give (Table 9.2).

Here the language does something that Maxwell could not. In 1864 he saw that a field theory of gravity would give the field of gravitation a negative energy, and he could not place the sign (Chapter 9). In the cosmic mass language one formula gives the energy of both parts (Theorem 9.9), and opened into its coordinates it reads

$$u = -\frac{g^2 + c^2 B_g^2}{8\pi G} + \frac{E^2 + c^2 B^2}{8\pi K}$$

(Proposition 9.10): negative for gravity, positive for the Maxwell part. The minus sign is the one that pushes like charges apart, the real part of the square of the Composer Unit. One fact of geometry gives both the sign of the force between charges and the sign of the energy of gravity. What that sign costs for the waves of gravity, Chapter 9 explains.

GIVEN	GAINED
one number for each body, $\mu = m + q\,\mathcal{G}_{\bar{\phi}}$, with its two measured parts	the opposite signs of gravity and electricity, from the plane itself
the shape of Newton's law, written for that number	Stoney's mass as the charge part of one elementary charge, and a picture of why electricity outweighs gravity between particles
the size of the Composer Unit, $\lvert\mathcal{G}_{\bar{\phi}}\rvert = \sqrt{K/G}$, chosen so that the electric term has Coulomb's strength	one frame for 306 bodies, with the charge-to-mass ratio as an angle
one rule for reading a product: its real part	the conservation of charge and the energy of fusion as two coordinates of one statement, and the exact closure of charge around six neighbouring nuclei
for moving bodies, four field equations of Maxwell's form, written for General Numbers, and one law of motion	Heaviside's gravity beside Maxwell's equations, with Maxwell's constants drawn from G and $\lvert\mathcal{G}_{\bar{\phi}}\rvert$, and waves at the speed of light in both
	Newton's gravity and the force of Lorentz in one law of motion; the opposite signs of the two energies; mass and charge kept by one equation; all of it from one action

Table 2.3: What the cosmic mass language was given, and what it gave back. The items on the right were not put in one by one: they come out of the few on the left.

2.3.5 *Why unification matters*

Unification is one of the oldest ways in which physics has moved forward, and its history shows what kind of gain it is.

In Book III of the *Principia*, Newton compared the fall of the Moon towards the Earth with the fall of a stone at its surface, and found one force behind both (Newton, 1687). With one law for the heavens and the Earth, tides, comets and the shape of the planets became problems that could be calculated, and in 1846 the same law found a planet that nobody had seen (Table 2.1). In 1865 Maxwell joined electricity and magnetism, and found light inside them (Maxwell, 1865); some twenty years later Hertz made the waves that his equations described (Hertz, 1893), and radio followed. In 1908 Hermann Minkowski told a meeting in Cologne that space by itself and time by itself would

sink into shadows, and that only a union of the two would keep its independence (Minkowski, 1909). On that union Einstein built his theory of gravity seven years later.

Not every unification brought a new prediction at once, and some of the most fruitful brought none. In 1788 Lagrange rewrote Newton's mechanics without a single new law, in a form built on energies instead of forces (Lagrange, 1788), and in 1834 Hamilton gave it another form, built on one function, the action (Hamilton, 1834). For the planets they predicted nothing that Newton had not. Nearly a century later Schrödinger built wave mechanics on Hamilton's analogy between mechanics and optics (Schrödinger, 1926), and Feynman built a whole form of quantum mechanics on the action (Feynman, 1948). In 1893 Steinmetz changed no law of electricity: he wrote alternating currents with complex numbers, as in the impedance of Chapter 1, and the arithmetic of circuits became routine (Steinmetz, 1893).

Three lessons follow. A language that holds known laws in one form is shorter: fewer starting points carry more results. It explains: what looked like coincidences of form become consequences of one structure. And it is often the door through which the next theory enters, sometimes long after it was built.

2.3.6 *What the cosmic mass language gives*

Measured against these examples, the result of this volume is of the second kind: a new form for old laws, with the values that such forms have shown before. A fair way to weigh it is to set what the language was given beside what it gave back (Table 2.3).

The second column is longer than the first, and its entries were not put in one by one: they come out of the few things in the first. That is the sense in which the cosmic mass language does not only keep the old laws but finds them again. For physics, five values follow.

1. *Economy.* Where physics uses separate laws for gravity, for electricity, for the sign of charges, for the fields and for motion, and two laws of conservation, the language uses one number for each body, one product, one rule of reading and one action.

2. *Explanation.* The shared shape of the laws of Newton and Coulomb is no longer a coincidence: they are the two parts of one product. The rule of signs, the sign of the energy of gravity and the strength of electricity over gravity become facts of one geometry.

3. *One picture of matter.* From the electron to the heaviest nucleus, every body has one card on one frame.

4. *One calculation.* One line of force serves every body, as the program of Chapter 9 shows: the kind of value that the complex impedance had for circuits.

5. *An address for the new.* Every law that has been measured lives in the real part. Anything new that the language can say must therefore live in the Composer coordinate, with the Coupling term and the angle. Faraday's hope of a link between gravity and electricity, which had no place to stand, now has an exact address.

None of this is yet a new measurement. It is the value of a language, the kind of value that the forms of Lagrange, Hamilton and Steinmetz had before their full use was seen. Whether the cosmic mass language also carries new physics is the question of the next section, and of Volume 2.

> IN PLAIN WORDS
>
> Every old law this volume has tried comes out of the same few things: one number for each body, one product for each pair of bodies, and one rule for reading the product. Newton's gravity and Coulomb's law come out of the same product, with their opposite signs; so do Maxwell's equations, the force of Lorentz and the balance of mass and charge when nuclei fuse. Even an old unit of mass, Stoney's, turns up in a new place: it is the charge part of one electron. Physics has gained from such unions before, by writing less, by explaining more, and by opening doors that were found only later. That is what the cosmic mass language offers now; whether it also finds something new is the next question.

2.4 THE COUPLING FORCE: A CANDIDATE FOR NEW RESULTS

If the language is to pay its second debt, the natural place to look is the term that neither Newton's law nor Coulomb's contains. Multiplied out, the product of two cosmic masses has three terms: mass with mass, charge with charge, and the cross term $m_a q_b + q_a m_b$, which carries the Composer Unit once (Theorem 6.6). The book calls it the Coupling Force (Remark 6.7). It vanishes between neutral bodies, and between a particle and its antiparticle, which have equal masses and opposite charges.

The term is not small. Between two equal bodies it is exactly twice the geometric mean of the other two terms, so it sits halfway between gravity and electricity on a scale of powers of ten.

Worked example: the size of the Coupling term

- *Two protons one metre apart.* The gravity term is about 1.9×10^{-64} N, the electric term 2.3×10^{-28} N, and the Coupling term 4.2×10^{-46} N $= 2\sqrt{1.9 \times 10^{-64} \times 2.3 \times 10^{-28}}$ N (Figure 6.9): $2\,m_S/m_p \approx 2.2 \times 10^{18}$ times gravity, and $2\,m_p/m_S \approx 1.8 \times 10^{-18}$ times the electric push.
- *The Earth and one electron at the ground.* The gravity term is the weight of the electron, $m_e\,g \approx 8.9 \times 10^{-30}$ N. The Coupling term pairs the mass of the Earth with the charge of the electron: $g\,m_S \approx 9.8$ m/s$^2 \times 1.859 \times 10^{-9}$ kg $\approx 1.8 \times 10^{-8}$ N, or $m_S/m_e \approx 2 \times 10^{21}$ times the weight.

Measurement already says what the term is not. If any sizeable share of it acted along the line between two bodies, even the share $\cos\bar{\phi} \approx 0.176$ that the reading by the shadow keeps (Section 6.5), the neutron would carry an apparent charge about two hundred standard errors beyond what is measured (Section 6.5.3), and every free electron near the ground would feel a force more than 10^{20} times its weight, which any electron beam in a laboratory would show. Read by its real part, the law gives the term no part along the line, which is what these measurements require. Whatever the Coupling term means, it is not a force between the mass of one body and the charge of another of anything like its own size.

Where, then, could it show itself? Read by its real part, the Coupling term lives in the Composer coordinate of the product, together with the angle of the Composer axis (Definition 6.17). The same holds for bodies in motion: the law of motion of Chapter 9 reads only the real part of $\mu\,(\boldsymbol{\mathcal{E}} + \mathbf{v} \times \boldsymbol{\mathcal{B}})$, and its Composer coordinate, which pairs the mass of the body with the charge part of the field and its charge with the gravity part, is left unread (Law 9.3).

Volume 2 begins from this. Its question is whether any measurable quantity reads the Composer coordinate of a product. For slow bodies, Chapter 9 shows that forces, energies and momenta do not. What remains are the places this volume does not reach: fast bodies and strong fields, where its law gives way to general relativity, and the quantum world, where what is measured is not always a force. A prediction found there must meet the standard of Section 2.1:

1. a number, with its error, drawn from the law with no new constant fitted to the effect it predicts;
2. the conditions under which it holds, and an experiment able to see it;
3. agreement with every measurement that already limits the term, among them those of the neutron and of hydrogen;

4. the result that would refute it, stated before the measurement is made.

No such prediction is claimed in this volume. The Coupling term is a candidate, not a result.

The second open question meets the first in the same place. No force fixes the angle of the Composer axis, because the real part of a product does not contain it (Section 6.6.3); the angle lives, with the Coupling term, in the Composer coordinate. A measurement that reads that coordinate, if one exists, would be the first that could fix the angle, and with it the choice of 79.86°.

One set of predictions in this volume is already at risk, and it should be named. Chapter 8 registers the masses of 61 nuclei that have not been measured, each with its stated error, after a first list of 67 in which 24 masses turned out to be measured already, and 20 of those fell within one stated error (Section 8.7). These predictions test the Closure Law, and with it the claim that mass and charge are kept as the two coordinates of one number. They do not test the Coupling term, or the angle.

> **In plain words**
>
> The Coupling term pairs the mass of one body with the charge of the other. As a number it is large: between the Earth and an electron, far larger than the electron's weight. As a force it cannot exist, or laboratories would have seen it long ago, and the real reading gives it no force at all. It lives instead in the second part of the product, together with the angle. Volume 2 asks whether anything we can measure reads that second part. If something does, the Coupling term could give the first new prediction of the theory; if nothing does, the cosmic mass remains a language for the physics we already know.

TWO VOLUMES

The Coda of this book lists what every new kind of number has needed: a question that the old numbers could not answer, a definition that contradicts nothing, consequences that nobody put in on purpose, and in the end a fact about the world that could not be said without them. This volume offers a candidate for each of the first three. Its question is the one of Chapter 1: can one number hold the mass and the charge of a particle? Its definition is the General Number, which contains the complex numbers as one of its planes. Its consequences include the minus sign that pushes like charges apart, which comes from the plane itself, and the laws gathered in Table 2.2, which the language gives back in one form.

The fourth need is the work of Volume 2. There the Coupling term and the angle of the Composer axis will be asked for numbers that can be measured, and those numbers will be set against the world.

The rest of this volume builds the language and tests it. Chapters 3 and 4 build the numbers, from a triangle that leans to a plane whose second axis leans. Chapter 5 explains why the language of physics must sometimes grow. Chapter 6 writes the law for two bodies and the rule for reading it, Chapter 7 draws every particle, ion and nucleus on one frame, Chapter 8 weighs the nuclei, and Chapter 9 writes one law for fields and moving bodies. The next chapter begins with the oldest piece of that road: a triangle that does not stand upright.

What we have so far

- A new theory must first give back the old laws, exactly, wherever they have been tested, and then predict something new that a measurement could prove wrong.
- This volume pays the first debt: one number for each body, read by its real part, gives back the laws of Newton, Coulomb, Maxwell and Lorentz, Heaviside's gravity and the balance of nuclear fusion, and it gives Stoney's unit of mass a new place.
- Such a union has value before any new prediction: fewer starting points, coincidences explained, one picture of matter, one calculation for every body, and an exact address for what is new.
- None of this depends on the angle of the Composer axis, and none of it is a new prediction.
- The candidate for the second debt is the Coupling term, which lives with the angle in the Composer coordinate of a product. Finding what, if anything, measures that coordinate is the task of Volume 2.

3

THE OBLIQUE TRIANGLE

The sine and the cosine were born in a right triangle. In this chapter we let the triangle lean, and we watch what the two old functions become.

EGYPT AND BABYLON, BEFORE 1500 BCE

Drive a stick upright into flat ground on a sunny day. The sun writes on the soil beside it. The shadow is long at dawn, short at noon and long again at dusk, and on a given day at a given hour it has the same length this year as it had last year. A stick used in this way is called a *gnomon*, and it is one of the oldest scientific instruments we know.

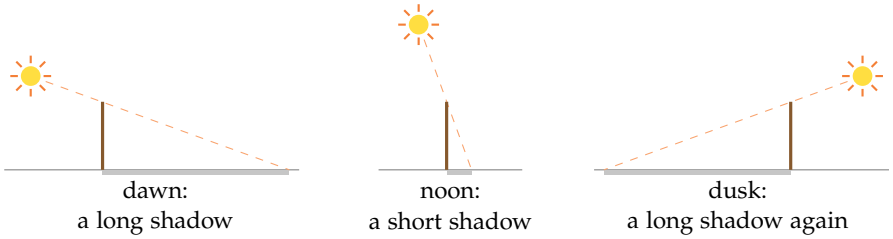
Notice what the gnomon really measures. It does not give you an angle in degrees. It gives you a *ratio*: so much shadow for so much stick. Two sticks of different heights, standing in the same sun, cast shadows in the same proportion, and it is the proportion, not either length, that tells you where the sun is. The Egyptian scribes who wrote the Rhind papyrus used the same idea to describe the slope of a pyramid, as so many palms of run for each cubit of rise. The angle entered history in disguise, as a fraction.

The Greeks gave that fraction its deepest reason. Two triangles with the same angles have sides in the same proportions, however large or small they are. This is the idea of *similar* triangles, and it is what makes a table of trigonometry possible at all: if the ratios depended on the size of the triangle, every triangle would be its own private case and nothing could be written down once and for all.

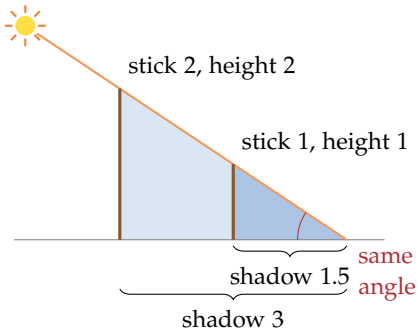
RHODES, ALEXANDRIA, INDIA AND BAGHDAD, 150 BCE TO 1000 CE

Astronomers needed numbers for angles, to follow the Sun, the Moon and the planets. Hipparchus and later Ptolemy built tables of *chords*: for each arc of a circle, the length of the straight line joining its two ends. Indian astronomers made a small change that turned out to be a large one. They cut the chord in half. The half-chord sits inside a right triangle whose longest side is the radius of the circle, and it is exactly what we now call the sine (Plofker, 2009; Van Brummelen, 2009).

The name has a curious story. The Sanskrit word for the half-chord, *jya*, was written in Arabic letters as *jyb*, a word with no meaning in Arabic. A later reader, seeing the bare consonants, read it as *jaib*, a fold or a pocket, and by extension a bay. Latin translators then rendered that word faithfully as *sinus*. Historians add some reservations to this

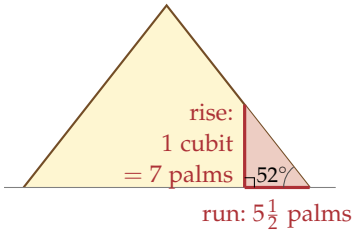


(a) the gnomon: the same hour gives the same shadow, year after year



$$\frac{\text{shadow}}{\text{stick}} = \frac{1.5}{1} = \frac{3}{2} = 1.5 \text{ for both:}$$
the ratio tells where the sun is

(b) two sticks, one sun: the same ratio



the seked: $5\frac{1}{2}$ palms of run
for each cubit of rise;
no angle was written down

(c) a pyramid's slope as a fraction

Figure 3.1: The angle in disguise. (a) A gnomon, an upright stick, casts a long shadow at dawn, a short one at noon and a long one again at dusk; at the same hour of the same day of the year, its shadow has the same length. (b) Two sticks in the same sun. The taller stick is twice as tall, and its shadow is twice as long, so the ratio of shadow to stick, here 1.5, is the same for both. It is the ratio, not the size, that tells where the sun is. The Greeks gave the reason: triangles with the same angles have their sides in the same proportions. (c) Egyptian scribes wrote the slope of a pyramid as a fraction, the *seked*: so many palms of horizontal run for each cubit of rise, a cubit being seven palms. A seked of $5\frac{1}{2}$ palms, about the slope of the Great Pyramid, belongs to an angle of about 52°, but the angle itself was never written down: only the fraction.

traditional account, but its lesson is sound: the idea crossed three languages intact, while its name slipped along the way (Van Brummelen, 2009; Maor, 1998). The General Sinus of this chapter keeps the old Latin word on purpose.

By 1600 the functions were defined, as they still are at school, on the sides of a right triangle. Everything you learned about sine and cosine rests on that one silent condition: somewhere in the triangle there is a right angle.

3.1 THE SILENT CONDITION

Let us make the classical definitions precise, in the notation of the General Sinus (Derraz, 2026a). Take an angle α with its vertex at a point O and its first side along the x-axis. Pick any point Z on the second side of the angle. Now drop Z straight down onto the first side, so that it lands at a point we call $\bar{Z}$ (read "Z bar"). The segment $Z\bar{Z}$ is perpendicular to the first side, and we obtain the right triangle $OZ\bar{Z}$ of Figure 3.2(a). Its sides have the familiar names: $[OZ]$ is the hypotenuse (hypo), $[O\bar{Z}]$ is the adjacent side (adja), and $[Z\bar{Z}]$ is the opposite side (oppo).

Definition 3.1 (Relative trigonometric functions)**.** The *relative* trigonometric functions are the three ratios

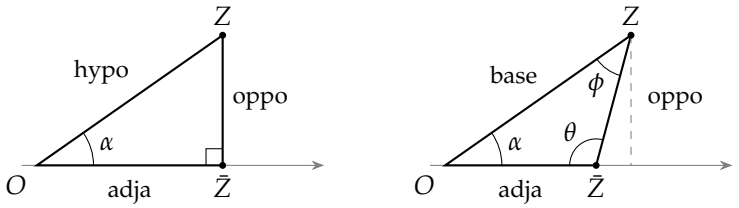
$$\sin(\alpha) = \frac{[Z\bar{Z}]}{[OZ]} = \frac{\text{oppo}}{\text{hypo}},$$
$$\cos(\alpha) = \frac{[O\bar{Z}]}{[OZ]} = \frac{\text{adja}}{\text{hypo}},$$
$$\tan(\alpha) = \frac{[Z\bar{Z}]}{[O\bar{Z}]} = \frac{\text{oppo}}{\text{adja}},$$

together with their reciprocals $\csc(\alpha) = 1/\sin(\alpha)$, $\sec(\alpha) = 1/\cos(\alpha)$ and $\cot(\alpha) = 1/\tan(\alpha)$.

The General Sinus papers call these the *relative* functions, because they are relative to one particular way of projecting: straight down. They are well defined only when the angle at $\bar{Z}$ is a right angle, $\pi/2$. That is an *existential condition*. It quietly restricts the whole classical theory to a single kind of triangle.

3.2 LETTING THE PROJECTION LEAN

Here is the one change that starts everything. Instead of dropping Z straight down, slide it down along a slanted line until it meets the first side. The slanted line makes an angle ϕ with the segment ZO at the point Z. We call ϕ the *projection angle*, and α the *studied angle*. The point where the slanted line lands is again called $\bar{Z}$, and we obtain



(a) straight down (b) at the angle ϕ

Figure 3.2: Two ways to project a point Z onto the first side of the angle α. (a) Straight down, which gives the right triangle of classical trigonometry. (b) Along a slanted line that makes the projection angle ϕ with ZO, which gives an oblique triangle with angles α, ϕ and $\theta = \pi - (\alpha + \phi)$.

the triangle $OZ\bar{Z}$ of Figure 3.2(b). It is no longer a right triangle. Its angles are α at O, ϕ at Z, and $\theta = \pi - (\alpha + \phi)$ at $\bar{Z}$.

Why can we build functions on this new triangle? For the same reason as before: similarity.

Proposition 3.2 (Constancy of the projection ratio)**.** *For a fixed angle α and a fixed projection angle ϕ, the ratios $[Z_n\bar{Z}_n]/[OZ_n]$, $[O\bar{Z}_n]/[OZ_n]$ and $[Z_n\bar{Z}_n]/[O\bar{Z}_n]$ are constant for all points Z_n on the terminal side of α. They depend only on α and ϕ, not on the distance $[OZ_n]$.*

Proof. For any two points Z_m and Z_n on the terminal side of α, the triangles $OZ_m\bar{Z}_m$ and $OZ_n\bar{Z}_n$ share the angles α and ϕ. By the angle-angle criterion they are similar, so their corresponding sides are proportional. The ratios therefore do not depend on the choice of the point. □

3.3 THE GENERAL SINUS

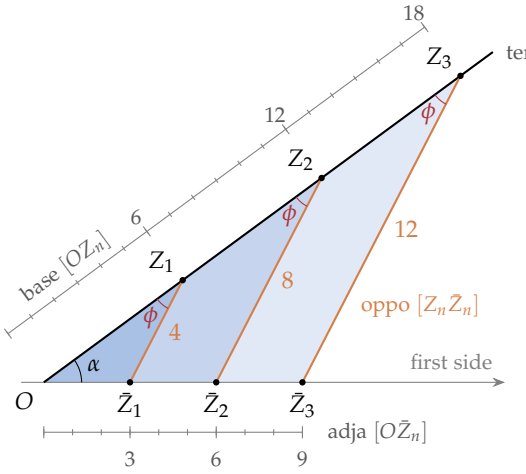
We can now define the three functions of the oblique triangle. They have two variables, the studied angle α and the projection angle ϕ.

Definition 3.3 (General trigonometric functions, geometric form)**.** The *general* trigonometric functions are the three ratios

$$\sin(\alpha,\phi) = \frac{[Z\bar{Z}]}{[OZ]} = \frac{\text{oppo}}{\text{base}},$$
$$\cos(\alpha,\phi) = \frac{[O\bar{Z}]}{[OZ]} = \frac{\text{adja}}{\text{base}},$$
$$\tan(\alpha,\phi) = \frac{[Z\bar{Z}]}{[O\bar{Z}]} = \frac{\text{oppo}}{\text{adja}},$$

together with their reciprocals $\csc(\alpha,\phi)$, $\sec(\alpha,\phi)$ and $\cot(\alpha,\phi)$.

A picture is a good start, but we also want formulas that we can compute with. They come from a rule you may remember from school,



Keep the studied angle α and the projection angle ϕ fixed, and pick three points Z_1, Z_2, Z_3 on the terminal side.

The angle ϕ is the same at every point, so the three slanted lines are parallel.

The gray rulers show equal steps on both sides: the middle triangle is the small one zoomed $\times 2$, the big one is the small one zoomed $\times 3$. The sides grow, the angles stay.

(a) one angle, one slant, three points

		Z_1	Z_2	Z_3
distance from O	$[OZ_n]$	6	12	18
oppo / base	$[Z_n\bar{Z}_n]/[OZ_n]$	$4/6 = 2/3$	$8/12 = 2/3$	$12/18 = 2/3$
adja / base	$[O\bar{Z}_n]/[OZ_n]$	$3/6 = 1/2$	$6/12 = 1/2$	$9/18 = 1/2$
oppo / adja	$[Z_n\bar{Z}_n]/[O\bar{Z}_n]$	$4/3$	$8/6 = 4/3$	$12/9 = 4/3$

read along a row: the distance grows, the ratios stay the same

(b) measure and divide

Figure 3.3: Why the ratios do not depend on the point. (a) The studied angle α and the projection angle ϕ are kept fixed, and three points Z_1, Z_2, Z_3 of the terminal side are projected. Since ϕ is the same at every point, the three slanted lines are parallel, and the triangles $OZ_n\bar{Z}_n$ all have the same angles: they are one triangle drawn at three sizes. (b) Measure and divide. The lengths double and triple, but each ratio of two sides is the same for the three points. Here α is about 36° and ϕ about 26°, chosen so that every length is a whole number. The three constant numbers $2/3$, $1/2$ and $4/3$ depend only on α and ϕ; Definition 3.3 will call them $\sin(\alpha, \phi)$, $\cos(\alpha, \phi)$ and $\tan(\alpha, \phi)$.

the *law of sines*: in any triangle, each side divided by the sine of the angle facing it gives the same number.

Theorem 3.4 (General trigonometric functions, analytic form)**.** *For $\alpha \in \mathbb{R}$ and $\phi \in (0, \pi)$ with $\alpha + \phi \notin \pi\mathbb{Z}$,*

$$\begin{aligned}\sin(\alpha, \phi) &= \frac{\sin(\alpha)}{\sin(\alpha + \phi)},\\ \cos(\alpha, \phi) &= \frac{\sin(\phi)}{\sin(\alpha + \phi)},\\ \tan(\alpha, \phi) &= \frac{\sin(\alpha)}{\sin(\phi)}.\end{aligned}$$

Proof. Apply the law of sines to the triangle $OZ\bar{Z}$ of Figure 3.2(b), with $\alpha = \angle ZO\bar{Z}$, $\phi = \angle OZ\bar{Z}$ and $\theta = \angle Z\bar{Z}O$:

$$\frac{\sin(\alpha)}{[Z\bar{Z}]} = \frac{\sin(\phi)}{[O\bar{Z}]} = \frac{\sin(\theta)}{[OZ]}.$$

Since $\theta = \pi - (\alpha + \phi)$ and $\sin(\pi - x) = \sin(x)$, we have $\sin(\theta) = \sin(\alpha + \phi)$. Dividing the first and third terms gives

$$\frac{\sin(\alpha)}{\sin(\alpha + \phi)} = \frac{[Z\bar{Z}]}{[OZ]} = \sin(\alpha, \phi),$$

which is the first formula. Dividing the second and third terms gives the second, and dividing the first and second terms gives the third. □

The next result is the reason the General Sinus deserves its name. It says that the old functions are not replaced. They are found again, inside the new ones, at one special angle.

Corollary 3.5 (Classical recovery)**.** *When $\alpha + \phi = \pi/2$, the general trigonometric functions reduce to the relative trigonometric functions:*

$$\begin{aligned}\sin\left(\alpha, \tfrac{\pi}{2} - \alpha\right) &= \sin(\alpha),\\ \cos\left(\alpha, \tfrac{\pi}{2} - \alpha\right) &= \cos(\alpha),\\ \tan\left(\alpha, \tfrac{\pi}{2} - \alpha\right) &= \tan(\alpha).\end{aligned}$$

Proof. Substitute $\phi = \pi/2 - \alpha$ in Theorem 3.4. Then $\sin(\alpha + \phi) = \sin(\pi/2) = 1$, so $\sin(\alpha, \phi) = \sin(\alpha)/1 = \sin(\alpha)$. Similarly $\sin(\phi) = \sin(\pi/2 - \alpha) = \cos(\alpha)$, which gives $\cos(\alpha, \phi) = \cos(\alpha)/1 = \cos(\alpha)$ and $\tan(\alpha, \phi) = \sin(\alpha)/\cos(\alpha) = \tan(\alpha)$. □

In plain words

When $\alpha + \phi = \pi/2$, the third angle of the triangle is $\pi - \pi/2 = \pi/2$: the triangle is a right triangle again, with its right angle at $\bar{Z}$. So the new functions give back the old ones exactly when the slanted projection happens to be straight down. The second variable ϕ turns the old "existential condition" into a number that we are free to choose.

Worked example

Take $\alpha = \pi/4$ and $\phi = \pi/3$. Then $\alpha + \phi = 7\pi/12$, which is not a multiple of π, so the functions are well defined. By Theorem 3.4,

$$\begin{aligned}\sin\left(\tfrac{\pi}{4}, \tfrac{\pi}{3}\right) &= \frac{\sin(\pi/4)}{\sin(7\pi/12)} = \frac{\sqrt{2}/2}{(\sqrt{6}+\sqrt{2})/4} \\ &= \sqrt{3} - 1 \approx 0.732.\end{aligned}$$

If $[OZ] = 1$, the opposite side has length $\sin(\pi/4, \pi/3) \approx 0.732$. The relative function gives $\sin(\pi/4) = \sqrt{2}/2 \approx 0.707$, a different value, because it only knows how to project straight down.

The three reciprocal functions are given explicitly by

$$\begin{aligned}\csc(\alpha, \phi) &= \frac{\sin(\alpha + \phi)}{\sin(\alpha)}, \\ \sec(\alpha, \phi) &= \frac{\sin(\alpha + \phi)}{\sin(\phi)}, \\ \cot(\alpha, \phi) &= \frac{\sin(\phi)}{\sin(\alpha)},\end{aligned}$$

and, just as in the classical theory, the tangent is the sine divided by the cosine:

$$\frac{\sin(\alpha, \phi)}{\cos(\alpha, \phi)} = \frac{\sin(\alpha)}{\sin(\phi)} = \tan(\alpha, \phi).$$

Table 3.1 lists the values of $\sin(\alpha, \phi)$ for the standard angles. Look at the three entries in color. They sit where $\alpha + \phi = \pi/2$, and they agree with the classical sines in the last row, exactly as Corollary 3.5 promises.

The general functions have many more properties: the sine and the cosine trade places when the two angles are exchanged, special triangles give special values, and there are rules for adding angles. They are collected in Appendix A. The rest of this book needs only the definitions above and the circle of the next section, where the angle of the projection becomes the angle of a new axis.

$\sin(\alpha,\phi)$	$\alpha=\pi/6$	$\alpha=\pi/4$	$\alpha=\pi/3$	$\alpha=\pi/2$
$\phi=\pi/6$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}-1$	$\frac{\sqrt{3}}{2}$	$\frac{2\sqrt{3}}{3}$
$\phi=\pi/4$	$\frac{\sqrt{6}-\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{3\sqrt{2}-\sqrt{6}}{2}$	$\sqrt{2}$
$\phi=\pi/3$	$\frac{1}{2}$	$\sqrt{3}-1$	1	2
$\phi=\pi/2$	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	*not defined*
$\sin(\alpha)$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Table 3.1: Values of the general sine $\sin(\alpha,\phi)$ for standard angles. The entries in color lie where $\alpha+\phi=\pi/2$ and equal the classical sine of the last row. The corner entry is not defined because there $\sin(\alpha+\phi)=\sin(\pi)=0$.

3.4 THE COMPOSED CIRCLE

In classical trigonometry, the best picture of sine and cosine is the unit circle. A point at angle α on a circle of radius one has coordinates $(\cos\alpha,\sin\alpha)$, measured along two perpendicular axes. The General Sinus has a circle of its own.

Definition 3.6 (Composed trigonometric circle)**.** Let σ be the unitary circle centered at a point O, with radius 1 and the standard axes (Ox,Oy). A second circle σ_c, centered at O with radius $r_c=1.2\,r$ (chosen only for visual clarity), is called the outer composed circle. A ray from O determines a half-line called the composer axis, or *Com-axis*. The angle between the x-axis and the Com-axis, measured counterclockwise, is the *relative projection angle* $\bar{\phi}$. The *composed trigonometric circle* is the configuration made of the inner circle σ, the outer circle σ_c, the x-axis as the initial reference axis, and the Com-axis as the variable projection axis.

The Com-axis is a generalization of the y-axis. When $\bar{\phi}=\pi/2$ it coincides with the y-axis, and the composed circle reduces to the classical unit circle. For every other value, the Com-axis leans.

How is the relative projection angle $\bar{\phi}$ related to the two angles of our triangle? The projection line through Z is parallel to the Com-axis, and the angles of the triangle add up to π. Following them around gives the simple relation

$$\bar{\phi}=\alpha+\phi.$$

With it, the General Sinus becomes a way of reading coordinates.

Proposition 3.7 (Coordinate interpretation)**.** *Place the triangle $OZ\bar{Z}$ on the composed trigonometric circle with $[OZ]=1$. Then:*

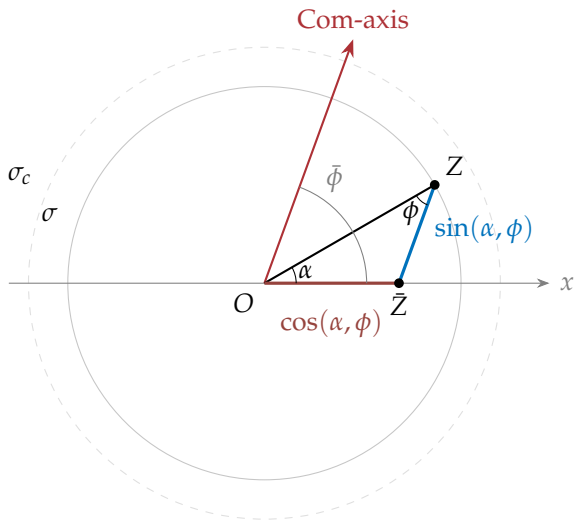


Figure 3.4: The composed trigonometric circle. The point Z lies on the unit circle σ at the studied angle α. Projecting it onto the x-axis along the leaning Com-axis (relative projection angle $\bar{\phi} = \alpha + \phi$) gives its two general coordinates: $\cos(\alpha, \phi)$ along the x-axis and $\sin(\alpha, \phi)$ along the direction of the Com-axis.

- $\cos(\alpha, \phi)$ *is the x-coordinate of Z, projected onto the x-axis along the Com-axis;*
- $\sin(\alpha, \phi)$ *is the Com-coordinate of Z, projected onto the Com-axis.*

> IN PLAIN WORDS
>
> On the classical unit circle, a point is read along two perpendicular axes, and its coordinates are $\cos\alpha$ and $\sin\alpha$. On the composed circle the same point is read along the x-axis and along a *leaning* axis, and its two coordinates are $\cos(\alpha, \phi)$ and $\sin(\alpha, \phi)$ (Figure 3.4). The General Sinus is, at heart, the trigonometry of a slanted frame of reference.

One small warning helps avoid a common mistake. If we keep the *projection angle* ϕ fixed and look at two different studied angles α and β, the two points need two different Com-axes, at the angles $\alpha + \phi$ and $\beta + \phi$. Drawing both angles against a single Com-axis is an error. The other way of looking is just as useful: if we keep the *Com-axis* fixed at the angle $\bar{\phi}$, then each studied angle α comes with its own projection angle $\phi = \bar{\phi} - \alpha$. This second point of view is the one we will use in the next chapter.

3.5 WHERE THE GENERAL SINUS LEADS

Look at what this chapter has given us. We started from a triangle that no longer needs a right angle, and we ended with a point read

in a frame whose second axis leans at the angle $\bar{\phi}$. The two numbers that describe the point, $\cos(\alpha, \phi)$ and $\sin(\alpha, \phi)$, are measured along two axes that are *not* perpendicular.

Now think back to the complex numbers of Chapter 1. A complex number $a + bi$ also describes a point with two numbers, but its two parts are always measured along perpendicular axes: the imaginary unit i stands at a right angle to the real line, by construction. The pair of general coordinates does not fit into that frame. It asks for a number plane whose second axis is allowed to lean, a plane in which numbers can be added and multiplied just as complex numbers are, but whose second unit points in a direction of our choosing.

That plane is the General Number Plane, and its second unit is the Composer Unit $\mathcal{G}_{\bar{\phi}}$. They are the subject of the next chapter, and they are the heart of this book. Two things of this chapter travel on with them. The general cosine and sine become the two parts of a General Number written with a length and two angles (Definition 4.8), and the triangle $OZ\bar{Z}$ becomes the triangle that every General Number draws (Theorem 4.9). In Chapter 6 the same triangle carries the mass and the charge of a particle.

What we have so far

- Classical sine and cosine silently assume a right angle.
- Letting the projection lean at an angle ϕ gives the General Sinus: $\sin(\alpha, \phi) = \sin(\alpha)/\sin(\alpha + \phi)$, $\cos(\alpha, \phi) = \sin(\phi)/\sin(\alpha + \phi)$, $\tan(\alpha, \phi) = \sin(\alpha)/\sin(\phi)$.
- When $\alpha + \phi = \pi/2$, the classical functions come back exactly.
- On the composed circle, $\cos(\alpha, \phi)$ and $\sin(\alpha, \phi)$ are the coordinates of a point read along the x-axis and along a leaning Com-axis at the angle $\bar{\phi} = \alpha + \phi$.
- These coordinates need a number plane whose second axis can lean. Chapter 4 builds it, and the triangle of this chapter becomes the triangle of every General Number.

4

GENERAL NUMBERS

The complex plane stands on a right angle. Here we let its second axis lean, and we build the number system that the rest of the book stands on.

DUBLIN, 16 OCTOBER 1843

William Rowan Hamilton had spent years on a single problem. The complex numbers had turned the plane into arithmetic: every point a number, every rotation a multiplication. Surely, he thought, the same could be done for space, with numbers made of three parts instead of two. Month after month, the rules refused to fit. Then, walking with his wife along the Royal Canal in Dublin, he saw the way through. It works, not with three parts but with four, provided that you give up a rule everybody had believed untouchable: that $a \times b$ is always the same as $b \times a$. He was so pleased that he cut the new rule into the stone of Brougham Bridge with his knife (Kline, 1972).

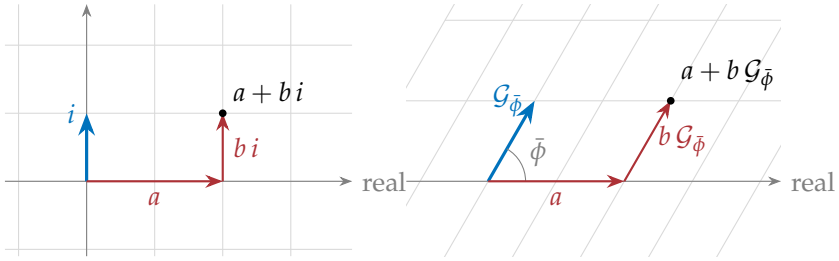
The lesson of Hamilton's walk is that a number system can grow by letting go of one of its habits. Hamilton let go of the order of multiplication, and climbed to four dimensions. In this chapter we stay in the plane, and let go of a different habit: the right angle between the two axes.

4.1 INTRODUCTION

The complex number system, foundational to mathematics and physics, is built upon two orthogonal axes: the real axis and the imaginary axis. The imaginary unit i represents a unit vector that is orthogonal to the real axis and has a modulus of one. However, this orthogonality imposes a geometric rigidity that may not fully describe more general vectorial or algebraic structures.

In this chapter, we introduce a novel extension of the complex number system: the *General Number Plane*, denoted $\mathbb{G}$. This system generalizes the imaginary unit i to a new mathematical object called the *Composer Unit*, denoted $\mathcal{G}_{\bar{\phi}}$. The Composer Unit defines a second axis, the *Composer axis*, which is not necessarily orthogonal to the real axis but is instead rotated by an arbitrary projection angle $\bar{\phi} \in [0, \pi)$.

The Composer Unit $\mathcal{G}_{\bar{\phi}}$ is defined as the product of an amplitude $\mathcal{A}_{\bar{\phi}}$ and a direction vector $\vec{u}_{\bar{\phi}}$. This structure allows the construction of complex-like numbers that live in a skewed, two-dimensional reference frame, wherein the traditional imaginary part is replaced by a generalized *Composer part*.



(a) the complex plane $\mathbb{C}$ (b) a relative plane $\mathbb{G}_{\bar\phi}$, here $\bar\phi = \pi/3$

Figure 4.1: Square and slanted graph paper. (a) In the complex plane a number $a + bi$ is reached by a steps along the real axis and b steps of length one straight up. (b) In a relative plane of the General Number Plane the second family of lines leans at the angle $\bar\phi$, and a number $a + b\,\mathcal{G}_{\bar\phi}$ is reached by a steps along the real axis and b steps of the Composer Unit. The picture uses $a = 2$ and $b = 1$.

Remarkably, when the rotation angle $\bar\phi = \frac{\pi}{2}$, the system naturally reduces to the standard complex plane, and $\mathcal{G}_{\pi/2} = i$. Thus, the General Number Plane extends and contains the complex number system as a special case, offering new degrees of freedom in both geometric interpretation and algebraic manipulation.

This chapter establishes the axiomatic definition of the General Number Plane, explores the properties of the Composer Unit, and proves that the classical complex system is a subset within this broader numerical framework. The mathematics of this chapter is also published as a stand-alone paper (Derraz, 2026b), with complete proofs and a script, `verify_paper.py`, that checks each of its numbered statements.

> IN PLAIN WORDS
>
> Think of the complex plane as a sheet of graph paper with square cells: one family of lines runs along the real axis, the other runs straight up, and every step has length one. The General Number Plane is graph paper whose cells are parallelograms. The second family of lines leans at the angle $\bar\phi$, and the step along it has its own length. Every point of the sheet is still reached by "so many steps along the first axis, then so many steps along the second" (Figure 4.1).

THE ROAD THROUGH THIS CHAPTER. The chapter climbs in eight steps, set out in two rows in Figure 4.2. The first row builds the new numbers: one axiom and what its unit does (Section 4.2), the old numbers found again inside them (Section 4.3), and the triangle that every General Number draws (Section 4.4). The second row gives the new numbers the words that Part ii will use, each defined once, in mathematics alone: the parts of a number (Section 4.5), products and

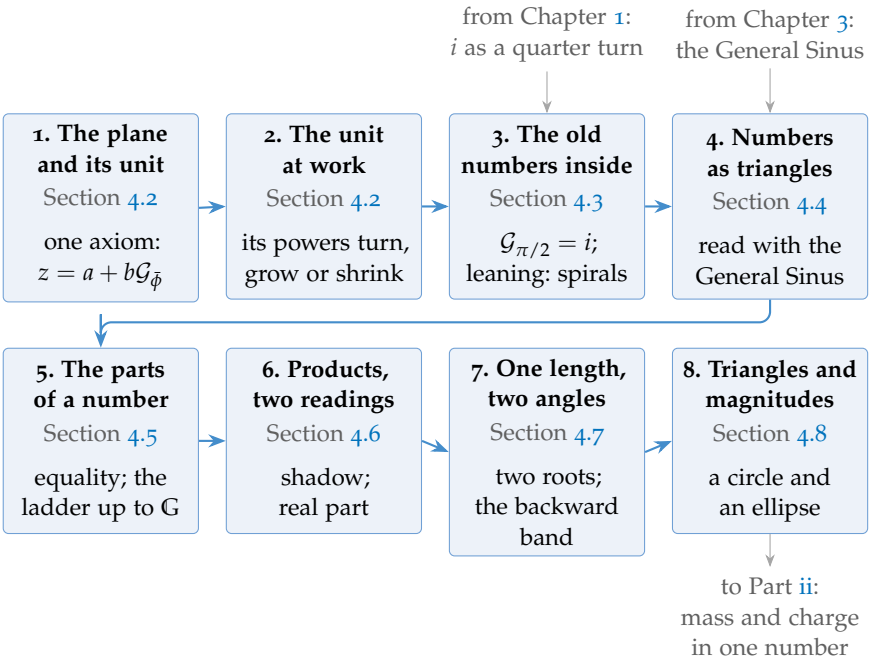


Figure 4.2: The road through the chapter. Each box is one step, with the section where it is taken. Two ideas come in from the earlier chapters, and the last step leads into Part ii.

their two readings, the shadow and the real part (Section 4.6), the two angles that share one length (Section 4.7), and all the numbers of one magnitude (Section 4.8). Two earlier ideas are all you need along the way: the complex numbers as arrows, with i as a quarter turn (Chapter 1), and the general sine and cosine of the oblique triangle (Chapter 3). A ninth section stands beside the road: Section 4.9 completes the algebra of the new numbers, law by law beside the complex numbers, and Part ii does not need it.

4.2 THE GENERAL NUMBERS PLANE AND THE COMPOSER UNIT

This section lays the foundation. It states the axiom on which the whole chapter rests, and then looks at the new unit from three sides: what happens when we raise it to a power, how long it is, and how each choice of the angle $\bar{\phi}$ gives a plane of its own.

4.2.1 *The Axiom*

Axiom 4.1 (General Number Plane and Composer Unit)**.** *Let $\bar{\phi} \in [0, \pi)$ be a projection angle relative to the x-axis (real axis). Then, there exists a two-dimensional number space $\mathbb{G} \subset \mathbb{R}^2$, called the* General Number Plane, *such that each element $z \in \mathbb{G}$ is conventionally defined as:*

$$z = a + b\mathcal{G}_{\bar{\phi}} \qquad \text{where } a, b \in \mathbb{R}.$$

Here, $\mathcal{G}_{\bar{\phi}}$ is the Composer Unit, *also conventionally defined as:*

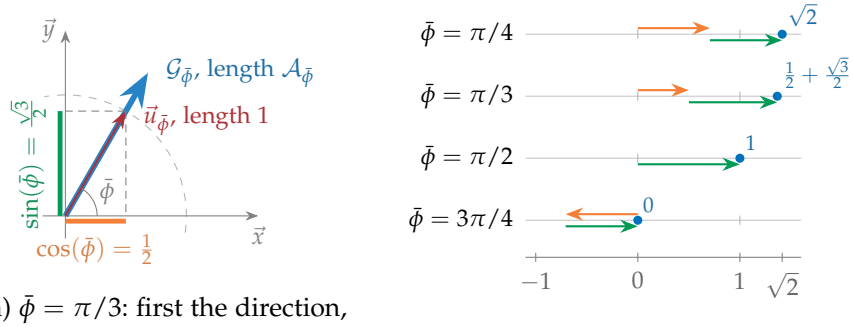
$$\mathcal{G}_{\bar{\phi}} := \mathcal{A}_{\bar{\phi}} \cdot \vec{u}_{\bar{\phi}}$$

with:

$$\mathcal{A}_{\bar{\phi}} = |\cos(\bar{\phi}) + \sin(\bar{\phi})| \qquad \textit{(Amplitude)}$$
$$\vec{u}_{\bar{\phi}} = \cos(\bar{\phi})\vec{x} + \sin(\bar{\phi})\vec{y} \qquad \textit{(Direction)}$$

> IN PLAIN WORDS
>
> A General Number has two parts, just like a complex number. The number a is the *real part*: how many steps you take along the real axis. The number b says how many steps you take along the Composer axis, and those b steps, the term $b\mathcal{G}_{\bar{\phi}}$, make the *Composer part* (Definition 4.11 states these names exactly). The Composer Unit $\mathcal{G}_{\bar{\phi}}$ is one such step. It is an arrow with two properties:
>
> - its *direction* $\vec{u}_{\bar{\phi}}$ is the unit arrow at the angle $\bar{\phi}$ from the real axis, written with the two perpendicular unit arrows $\vec{x}$ and $\vec{y}$ of the page;
> - its *amplitude* $\mathcal{A}_{\bar{\phi}}$ is its length, and that length depends on the angle.
>
> The angle $\bar{\phi}$ is the one we met on the composed circle of Chapter 3: the angle between the real axis and the axis that leans. The Axiom calls it a projection angle, but the General Sinus has a projection angle of its own, ϕ, which returns in Section 4.4; so this book calls $\bar{\phi}$ the *angle of the Composer axis*, and calls the amplitude $\mathcal{A}_{\bar{\phi}}$ the *length of the Composer Unit*.

A PICTURE OF THE AMPLITUDE. The Axiom builds the Composer Unit in two moves. First it takes the unit arrow $\vec{u}_{\bar{\phi}}$, whose shadows on the two perpendicular axes of the page are $\cos(\bar{\phi})$ and $\sin(\bar{\phi})$. Then it stretches or shrinks that arrow by the amplitude $\mathcal{A}_{\bar{\phi}}$, which is the sum of the two shadows, taken without its sign. While the axis leans to the right of the upright, both shadows are positive, and the arrow comes out longer than one. Once it leans past the upright, the shadow on the real axis turns negative and eats into the other one, and the arrow comes out shorter (Figure 4.3).



(a) $\bar{\phi} = \pi/3$: first the direction, then the length

(b) the amplitude: add the two shadows

Figure 4.3: How the Axiom builds the Composer Unit. (a) For $\bar{\phi} = \pi/3$, the unit arrow $\vec{u}_{\bar{\phi}}$ (dark red) has the shadows $\cos(\bar{\phi}) = \frac{1}{2}$ on $\vec{x}$ (orange) and $\sin(\bar{\phi}) = \frac{\sqrt{3}}{2}$ on $\vec{y}$ (green). The Composer Unit $\mathcal{G}_{\bar{\phi}} = \mathcal{A}_{\bar{\phi}} \cdot \vec{u}_{\bar{\phi}}$ (blue) points the same way, and its length is the sum of the two shadows, $\mathcal{A}_{\bar{\phi}} = \frac{1}{2} + \frac{\sqrt{3}}{2}$. (b) The same sum for the four angles of the worked example: first the shadow $\cos(\bar{\phi})$ (orange), then the shadow $\sin(\bar{\phi})$ (green); the blue dot is the amplitude $\mathcal{A}_{\bar{\phi}}$. The bars in $|\cos(\bar{\phi}) + \sin(\bar{\phi})|$ only matter for $\bar{\phi}$ between $3\pi/4$ and π, where the sum is negative.

Worked example: a few Composer Units

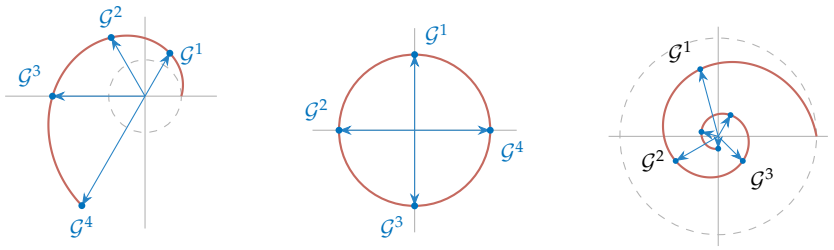
The Axiom gives the Composer Unit for any angle. Here are some of them.

- $\bar{\phi} = \pi/2$: $\mathcal{A}_{\bar{\phi}} = |\cos(\pi/2) + \sin(\pi/2)| = |0 + 1| = 1$ and $\vec{u}_{\bar{\phi}} = \vec{y}$. The Composer Unit is one step straight up: it is the imaginary unit i of the complex plane (Theorem 4.5).
- $\bar{\phi} = \pi/4$: $\mathcal{A}_{\bar{\phi}} = |\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}| = \sqrt{2}$. The Composer Unit points halfway between the axes and is at its longest.
- $\bar{\phi} = \pi/3$: $\mathcal{A}_{\bar{\phi}} = \frac{1}{2} + \frac{\sqrt{3}}{2}$, and $\vec{u}_{\bar{\phi}} = \frac{1}{2}\vec{x} + \frac{\sqrt{3}}{2}\vec{y}$. This is the case drawn in Figure 4.1(b).
- $\bar{\phi} = 3\pi/4$: $\mathcal{A}_{\bar{\phi}} = |-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}| = 0$. Here the amplitude vanishes.

Panel (b) of Figure 4.3 shows these four sums on the number line.

4.2.2 *Power-Rotation Duality of the Composer Unit*

What happens when we multiply the Composer Unit by itself? In the complex plane, Chapter 1 gave us Wessel's rule: to multiply two numbers, multiply their lengths and add their angles. The Composer Unit follows the same rule, and the result is one of the most striking features of the General Number Plane.



(a) $\bar{\phi} = \pi/3$: $\mathcal{A}_{\bar{\phi}} > 1$, the powers spiral out

(b) $\bar{\phi} = \pi/2$: $\mathcal{A}_{\bar{\phi}} = 1$, the powers go round

(c) $\bar{\phi} = 7\pi/12$: $\mathcal{A}_{\bar{\phi}} < 1$, the powers spiral in

Figure 4.4: What the powers of the Composer Unit do (Theorem 4.1). The powers in (a) return in Figure 4.7(a), and those in (b) are the powers of i (Remark 4.6). In (c), $\bar{\phi} = 7\pi/12 = 105°$ and the length $\mathcal{A}_{7\pi/12} = \sqrt{2}/2$ halves every two steps. In each panel the dashed circle has radius 1; each panel has its own scale.

Theorem 4.1 (Powers of the Composer Unit)**.** *Let $\mathcal{G}_{\bar{\phi}}$ be the Composer Unit defined by*

$$\mathcal{G}_{\bar{\phi}} = \mathcal{A}_{\bar{\phi}} \cdot \vec{u}_{\bar{\phi}},$$

where

$$\mathcal{A}_{\bar{\phi}} = \left|\sqrt{2}\sin\left(\bar{\phi} + \frac{\pi}{4}\right)\right| \quad \textit{and} \quad \vec{u}_{\bar{\phi}} = \cos(\bar{\phi})\vec{x} + \sin(\bar{\phi})\vec{y}.$$

Then, for any integer $n \geq 1$, the n-th power of $\mathcal{G}_{\bar{\phi}}$ is given by

$$\mathcal{G}^n_{\bar{\phi}} = \mathcal{A}^n_{\bar{\phi}} \cdot \vec{u}_{n\bar{\phi}},$$

where

$$\vec{u}_{n\bar{\phi}} = \cos(n\bar{\phi})\vec{x} + \sin(n\bar{\phi})\vec{y}.$$

In particular:

- *The amplitude scales exponentially with the power n as $\mathcal{A}^n_{\bar{\phi}}$.*
- *The direction vector rotates by the angle $n\bar{\phi}$.*

> IN PLAIN WORDS
>
> Two things happen at every multiplication by $\mathcal{G}_{\bar{\phi}}$: the arrow turns by the angle $\bar{\phi}$, and its length is multiplied by $\mathcal{A}_{\bar{\phi}}$. After n multiplications, the arrow has turned by $n\bar{\phi}$ and its length is $\mathcal{A}^n_{\bar{\phi}}$. This is why the section is called a *power-rotation duality*: raising to a power is the same as turning, with a change of length at each step. Whether the arrow grows or shrinks depends only on whether $\mathcal{A}_{\bar{\phi}}$ is larger or smaller than one (Figure 4.4).

Notice also that the amplitude is written in two ways, $|\cos(\bar{\phi}) + \sin(\bar{\phi})|$ in the Axiom and $\left|\sqrt{2}\sin(\bar{\phi} + \pi/4)\right|$ here. They are the same number, because

$$\sqrt{2}\sin\left(\bar{\phi} + \tfrac{\pi}{4}\right) = \sqrt{2}\left(\sin\bar{\phi}\cos\tfrac{\pi}{4} + \cos\bar{\phi}\sin\tfrac{\pi}{4}\right) = \sin\bar{\phi} + \cos\bar{\phi}.$$

4.2.3 *The General Number Plane as a Non-Orthonormal Reference*

Definition 4.2 (Non-Unified Reference in the General Number Plane)**.** In the classical Complex Number Plane $\mathbb{C}$, the coordinate system is orthonormal: the Real axis has a unit length of 1, and the Imaginary unit i also has modulus 1, establishing a *unified reference frame* with equal scaling along both axes.

By contrast, the General Number Plane $\mathbb{G}$, defined via the Composer Unit $\mathcal{G}_{\bar{\phi}}$, introduces a *skewed and amplitude-dependent reference system*. While the Real axis retains its standard unit length, the Composer axis, determined by the direction and magnitude of $\mathcal{G}_{\bar{\phi}}$, generally does not have a fixed modulus. Its magnitude depends explicitly on the projection angle $\bar{\phi}$:

$$\left|\mathcal{G}_{\bar{\phi}}\right| = \left|\sqrt{2}\sin\left(\bar{\phi} + \frac{\pi}{4}\right)\right|.$$

Consequently, the General Number Plane is a *non-unified reference system*: the Composer axis simultaneously rotates by $\bar{\phi}$ and scales non-uniformly. This breaks the classical orthonormality, allowing richer geometric interpretations and algebraic structures that extend beyond the standard complex framework.

In plain words

"Orthonormal" is a compact word for two habits of the complex plane: the axes are perpendicular (*ortho*), and both steps have length one (*normal*). The General Number Plane drops both habits at once. Its second axis leans, and its second step has a length that depends on how far it leans. Figure 4.5 shows that length as the angle $\bar{\phi}$ goes from 0 to π: it starts at 1, rises to its largest value $\sqrt{2}$ at $\bar{\phi} = \pi/4$, comes back to 1 at $\bar{\phi} = \pi/2$ (the complex plane), falls to 0 at $\bar{\phi} = 3\pi/4$, and climbs back to 1 as $\bar{\phi}$ approaches π.

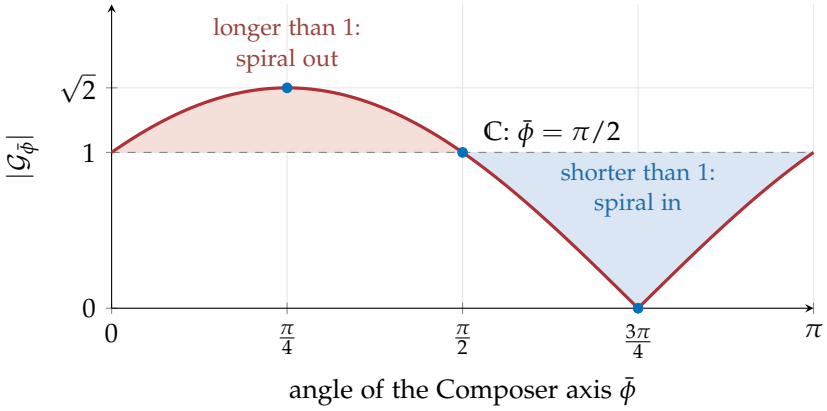


Figure 4.5: The length of the Composer Unit, $|\mathcal{G}_{\bar\phi}| = |\sqrt{2}\sin(\bar\phi + \pi/4)|$, as the angle $\bar\phi$ of the Composer axis goes from 0 to π. Where it is longer than 1 (red), the powers of the Composer Unit spiral out; where it is shorter than 1 (blue), they spiral in (Figure 4.4).

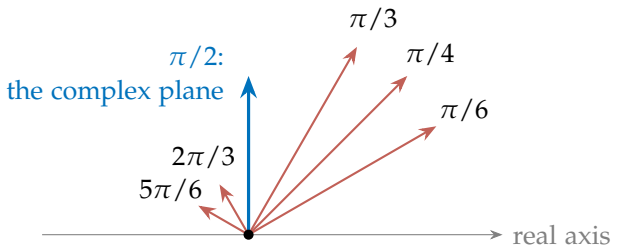


Figure 4.6: A fan of relative planes. Each arrow is the Composer Unit $\mathcal{G}_{\bar\phi}$ of one relative plane, drawn at its angle $\bar\phi$ with its own length $|\mathcal{G}_{\bar\phi}|$. The upright arrow ($\bar\phi = \pi/2$, length 1) is the imaginary unit i. At $\bar\phi = 3\pi/4$ the length is 0.

4.2.4 *Relative Planes*

Definition 4.3 (Relative Plane within the General Number Plane)**.** The General Number Plane $\mathbb{G}$ is a two-dimensional numerical and geometric framework generated by the real axis and the Composer axis associated with the unit $\mathcal{G}_{\bar\phi}$.

To work in a specific geometric realization, one fixes a projection angle $\bar\phi \in [0, \pi)$. The resulting two-dimensional plane, denoted $\mathbb{G}_{\bar\phi}$, is called a *Relative Plane*.

Each Relative Plane represents a particular instance of the General Number Plane relative to the chosen angle $\bar\phi$, providing a concrete reference frame for algebraic and geometric constructions.

Remark 4.4 (The Complex Plane as a Relative Plane). Among all Relative Planes $\mathbb{G}_{\bar\phi}$ within the General Number Plane $\mathbb{G}$, the case $\bar\phi = \pi/2$ recovers the classical complex plane:

$$\mathbb{G}_{\pi/2} = \mathbb{C}, \qquad \mathcal{G}_{\pi/2} = i.$$

This shows that the complex numbers are not separate from the General Number Plane; rather, they correspond to a particular Relative Plane with an orthogonal Composer axis.

Thus, every Relative Plane can be viewed as a generalization of the complex plane, and the classical complex system appears as a special instance within the broader framework.

> IN PLAIN WORDS
>
> Picture a hand fan. Each blade is a Composer axis at its own angle $\bar{\phi}$, with its own step length, and each blade defines its own relative plane $\mathbb{G}_{\bar{\phi}}$ (Figure 4.6). The General Number Plane is the whole fan. The complex plane is one blade among the others, the one that stands exactly upright. A relative plane is also called a *subgeneral plane*; the two names mean the same thing.

4.3 PROPERTIES OF THE COMPOSER UNIT

We now have a whole fan of planes, one for each angle. A new number system earns our trust in two ways: it must give back the old system exactly, and it must show something new. This section checks both at two angles: with the Composer axis upright, where the complex numbers come back, and with the axis leaning, where the powers of Theorem 4.1 trace a spiral.

4.3.1 *Reduction to the Complex Number System*

Theorem 4.5 (Classical Recovery)**.** *Let $\bar{\phi} = \dfrac{\pi}{2}$. Then, the General Number Plane $\mathbb{G}$ reduces to the classical complex plane $\mathbb{C}$, and the Composer Unit becomes the imaginary unit:*

$$\mathcal{G}_{\pi/2} = i$$

Proof. When $\bar{\phi} = \dfrac{\pi}{2}$, we compute:

$$\mathcal{A}_{\pi/2} = \left|\sqrt{2}\sin\left(\frac{3\pi}{4}\right)\right| = \left|\sqrt{2}\cdot\frac{\sqrt{2}}{2}\right| = 1$$

$$\vec{u}_{\pi/2} = \cos\left(\frac{\pi}{2}\right)\vec{x} + \sin\left(\frac{\pi}{2}\right)\vec{y} = \vec{y}$$

Therefore:

$$\mathcal{G}_{\pi/2} = 1\cdot\vec{y} = \vec{y} = i$$

□

Remark 4.6 (The Classical Case $\bar{\phi} = \frac{\pi}{2}$). Let us consider the particular case $\bar{\phi} = \frac{\pi}{2}$, where the Composer Unit $\mathcal{G}_{\bar{\phi}}$ aligns with the classical imaginary unit i. From the previously defined structure:

$$\mathcal{G}_{\bar{\phi}} = \mathcal{A}_{\bar{\phi}} \cdot \vec{u}_{\bar{\phi}}, \quad \text{with } \mathcal{A}_{\bar{\phi}} = \left|\sqrt{2}\sin\left(\bar{\phi} + \frac{\pi}{4}\right)\right|,$$
$$\vec{u}_{\bar{\phi}} = \cos(\bar{\phi})\vec{x} + \sin(\bar{\phi})\vec{y}.$$

For $\bar{\phi} = \frac{\pi}{2}$, we compute:

$$\mathcal{A}_{\pi/2} = \left|\sqrt{2}\sin\left(\frac{3\pi}{4}\right)\right| = 1, \qquad \vec{u}_{\pi/2} = \vec{y}, \qquad \Rightarrow \qquad \mathcal{G}_{\pi/2} = \vec{y}.$$

Thus, successive powers yield:

$$\mathcal{G}^2_{\pi/2} = -\vec{x}, \qquad \mathcal{G}^3_{\pi/2} = -\vec{y}, \qquad \mathcal{G}^4_{\pi/2} = \vec{x}.$$

This result reflects a complete rotation cycle of 90° per power, analogous to the behavior of i in complex analysis:

$$i = \vec{y}, \qquad i^2 = -1 = -\vec{x}, \qquad i^3 = -i = -\vec{y}, \qquad i^4 = 1 = \vec{x}.$$

POWER	RESULT	INTERPRETATION
$\mathcal{G}^1_{\pi/2}$	$\vec{y}$	90° rotation from $\vec{x}$
$\mathcal{G}^2_{\pi/2}$	$-\vec{x}$	180° rotation
$\mathcal{G}^3_{\pi/2}$	$-\vec{y}$	270° rotation
$\mathcal{G}^4_{\pi/2}$	$\vec{x}$	360° rotation (identity)

This example confirms that, in the special orthogonal case, the Composer Unit $\mathcal{G}_{\bar{\phi}}$ reproduces the same rotational algebra as the complex imaginary unit i, but within a geometric and coordinate-based interpretation.

These are the four quarter turns of Chapter 1 in new clothes (Figure 4.4(b)): all of complex arithmetic lives inside the General Number Plane, in the relative plane $\mathbb{G}_{\pi/2}$.

4.3.2 *When the Composer Axis Leans*

Now let the axis lean, and watch what the powers do. We take the angle $\bar{\phi} = \pi/3$, the relative plane of Figure 4.1(b), and follow the powers of the Composer Unit through one full turn.

Remark 4.7 (Extended Non-Orthogonal Case $\bar{\phi} = \frac{\pi}{3}$, Subgeneral Plane). Let us extend the Composer Unit computations for $\bar{\phi} = \frac{\pi}{3}$ up to the 6th power to observe the full rotational cycle of this generalized structure.

Base definitions:

$$\mathcal{A}_{\pi/3} \approx 1.365, \qquad \vec{u}_{\pi/3} = \frac{1}{2}\vec{x} + \frac{\sqrt{3}}{2}\vec{y}$$

POWER	EXPRESSION	APPROX. RESULT	ROTATION ANGLE
$\mathcal{G}^1$	$\mathcal{A} \cdot \vec{u}_{\pi/3}$	$0.6825\vec{x} + 1.182\vec{y}$	$60°$
$\mathcal{G}^2$	$\mathcal{A}^2 \cdot \vec{u}_{2\pi/3}$	$-0.932\vec{x} + 1.614\vec{y}$	$120°$
$\mathcal{G}^3$	$\mathcal{A}^3 \cdot \vec{u}_{\pi}$	$-2.546\vec{x}$	$180°$
$\mathcal{G}^4$	$\mathcal{A}^4 \cdot \vec{u}_{4\pi/3}$	$-1.739\vec{x} - 3.013\vec{y}$	$240°$
$\mathcal{G}^5$	$\mathcal{A}^5 \cdot \vec{u}_{5\pi/3}$	$2.377\vec{x} - 4.121\vec{y}$	$300°$
$\mathcal{G}^6$	$\mathcal{A}^6 \cdot \vec{u}_{2\pi}$	$6.5\vec{x}$	$360°$

We observe a full geometric cycle with $60°$ steps. However, unlike the classical complex system where all magnitudes stay constant, the amplitude here evolves as a power of the base amplitude $\mathcal{A}_{\bar{\phi}}$, leading to expanding spirals or contractions depending on the value of $\bar{\phi}$. This supports the view that $\mathcal{G}_{\bar{\phi}}$ defines a generalized number system with rotational and amplitude dynamics.

The computation of Remark 4.7 works for any angle. Here it is for three more, $\bar{\phi} = \pi/4$, $\bar{\phi} = \pi/5$ and $\bar{\phi} = \pi/6$, with the amplitude and the direction taken from the Axiom.

WORKED EXAMPLE: THE POWERS FOR $\bar{\phi} = \pi/4$, $\pi/5$ AND $\pi/6$

The angle $\bar{\phi} = \pi/4$. Here $\mathcal{A}_{\pi/4} = \left|\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right| = \sqrt{2}$ and $\vec{u}_{\pi/4} = \frac{\sqrt{2}}{2}\vec{x} + \frac{\sqrt{2}}{2}\vec{y}$, so the first power is $\mathcal{G}^1 = \vec{x} + \vec{y}$. Every result is exact: each power turns the arrow by another $45°$ and stretches it by $\sqrt{2}$, and after eight steps it is back on the real axis, sixteen times longer than the unit.

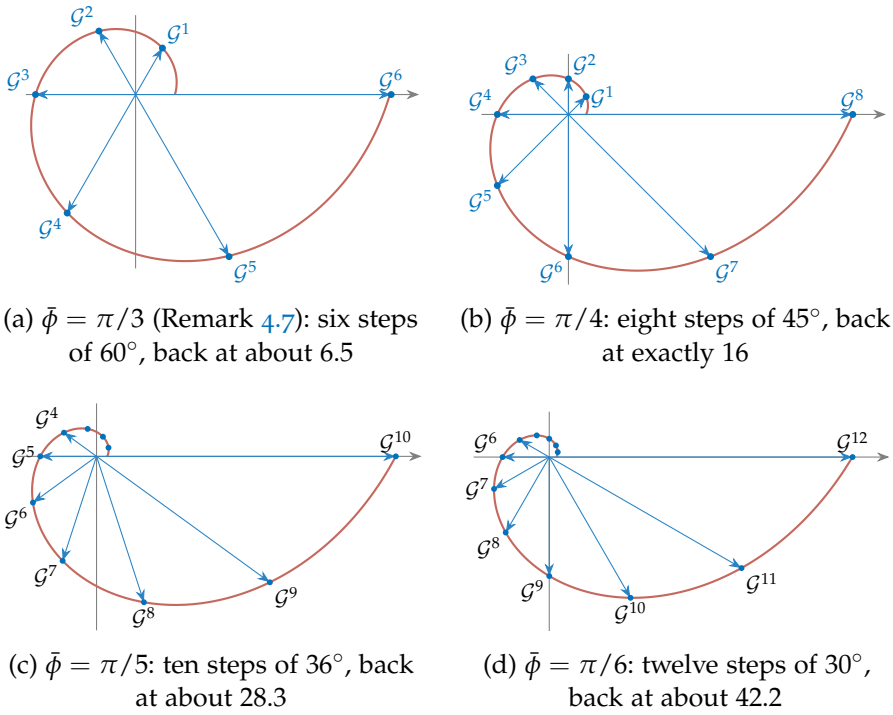
POWER	EXPRESSION	RESULT	ROTATION ANGLE
$\mathcal{G}^1$	$\mathcal{A} \cdot \vec{u}_{\pi/4}$	$\vec{x} + \vec{y}$	$45°$
$\mathcal{G}^2$	$\mathcal{A}^2 \cdot \vec{u}_{\pi/2}$	$2\vec{y}$	$90°$
$\mathcal{G}^3$	$\mathcal{A}^3 \cdot \vec{u}_{3\pi/4}$	$-2\vec{x} + 2\vec{y}$	$135°$
$\mathcal{G}^4$	$\mathcal{A}^4 \cdot \vec{u}_{\pi}$	$-4\vec{x}$	$180°$
$\mathcal{G}^5$	$\mathcal{A}^5 \cdot \vec{u}_{5\pi/4}$	$-4\vec{x} - 4\vec{y}$	$225°$
$\mathcal{G}^6$	$\mathcal{A}^6 \cdot \vec{u}_{3\pi/2}$	$-8\vec{y}$	$270°$
$\mathcal{G}^7$	$\mathcal{A}^7 \cdot \vec{u}_{7\pi/4}$	$8\vec{x} - 8\vec{y}$	$315°$
$\mathcal{G}^8$	$\mathcal{A}^8 \cdot \vec{u}_{2\pi}$	$16\vec{x}$	$360°$

The angle $\bar{\phi} = \pi/5$. Here $\cos(\pi/5) = \frac{1+\sqrt{5}}{4} \approx 0.809$, which is half the golden ratio $(1 + \sqrt{5})/2$ (Appendix A.4), and $\sin(\pi/5) \approx 0.588$, so $\mathcal{A}_{\pi/5} \approx 1.397$ and $\vec{u}_{\pi/5} \approx 0.809\vec{x} + 0.588\vec{y}$. Ten steps of $36°$ bring the arrow back to the real axis. The results are rounded to three decimals. The golden ratio is followed through these powers in Derraz (2026b, Remark 5.7).

POWER	EXPRESSION	APPROX. RESULT	ROTATION ANGLE
$\mathcal{G}^1$	$\mathcal{A} \cdot \vec{u}_{\pi/5}$	$1.130\vec{x} + 0.821\vec{y}$	36°
$\mathcal{G}^2$	$\mathcal{A}^2 \cdot \vec{u}_{2\pi/5}$	$0.603\vec{x} + 1.856\vec{y}$	72°
$\mathcal{G}^3$	$\mathcal{A}^3 \cdot \vec{u}_{3\pi/5}$	$-0.842\vec{x} + 2.592\vec{y}$	108°
$\mathcal{G}^4$	$\mathcal{A}^4 \cdot \vec{u}_{4\pi/5}$	$-3.080\vec{x} + 2.237\vec{y}$	144°
$\mathcal{G}^5$	$\mathcal{A}^5 \cdot \vec{u}_{\pi}$	$-5.317\vec{x}$	180°
$\mathcal{G}^6$	$\mathcal{A}^6 \cdot \vec{u}_{6\pi/5}$	$-6.009\vec{x} - 4.365\vec{y}$	216°
$\mathcal{G}^7$	$\mathcal{A}^7 \cdot \vec{u}_{7\pi/5}$	$-3.206\vec{x} - 9.866\vec{y}$	252°
$\mathcal{G}^8$	$\mathcal{A}^8 \cdot \vec{u}_{8\pi/5}$	$4.478\vec{x} - 13.781\vec{y}$	288°
$\mathcal{G}^9$	$\mathcal{A}^9 \cdot \vec{u}_{9\pi/5}$	$16.375\vec{x} - 11.897\vec{y}$	324°
$\mathcal{G}^{10}$	$\mathcal{A}^{10} \cdot \vec{u}_{2\pi}$	$28.272\vec{x}$	360°

The angle $\bar{\phi} = \pi/6$. Here $\mathcal{A}_{\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2} \approx 1.366$, the same length as $\mathcal{A}_{\pi/3}$ (Remark 4.7 gives it as 1.365; Section 4.7 explains why two angles share one length), and $\vec{u}_{\pi/6} = \frac{\sqrt{3}}{2}\vec{x} + \frac{1}{2}\vec{y}$. The arrow now turns by only 30° at each step, so it needs twelve steps to come back to the real axis, and it keeps growing all the way. The results are rounded to three decimals.

POWER	EXPRESSION	APPROX. RESULT	ROTATION ANGLE
$\mathcal{G}^1$	$\mathcal{A} \cdot \vec{u}_{\pi/6}$	$1.183\vec{x} + 0.683\vec{y}$	30°
$\mathcal{G}^2$	$\mathcal{A}^2 \cdot \vec{u}_{\pi/3}$	$0.933\vec{x} + 1.616\vec{y}$	60°
$\mathcal{G}^3$	$\mathcal{A}^3 \cdot \vec{u}_{\pi/2}$	$2.549\vec{y}$	90°
$\mathcal{G}^4$	$\mathcal{A}^4 \cdot \vec{u}_{2\pi/3}$	$-1.741\vec{x} + 3.016\vec{y}$	120°
$\mathcal{G}^5$	$\mathcal{A}^5 \cdot \vec{u}_{5\pi/6}$	$-4.119\vec{x} + 2.378\vec{y}$	150°
$\mathcal{G}^6$	$\mathcal{A}^6 \cdot \vec{u}_{\pi}$	$-6.498\vec{x}$	180°
$\mathcal{G}^7$	$\mathcal{A}^7 \cdot \vec{u}_{7\pi/6}$	$-7.687\vec{x} - 4.438\vec{y}$	210°
$\mathcal{G}^8$	$\mathcal{A}^8 \cdot \vec{u}_{4\pi/3}$	$-6.062\vec{x} - 10.500\vec{y}$	240°
$\mathcal{G}^9$	$\mathcal{A}^9 \cdot \vec{u}_{3\pi/2}$	$-16.563\vec{y}$	270°
$\mathcal{G}^{10}$	$\mathcal{A}^{10} \cdot \vec{u}_{5\pi/3}$	$11.312\vec{x} - 19.594\vec{y}$	300°
$\mathcal{G}^{11}$	$\mathcal{A}^{11} \cdot \vec{u}_{11\pi/6}$	$26.766\vec{x} - 15.453\vec{y}$	330°
$\mathcal{G}^{12}$	$\mathcal{A}^{12} \cdot \vec{u}_{2\pi}$	$42.219\vec{x}$	360°



(a) $\bar{\phi} = \pi/3$ (Remark 4.7): six steps of 60°, back at about 6.5

(b) $\bar{\phi} = \pi/4$: eight steps of 45°, back at exactly 16

(c) $\bar{\phi} = \pi/5$: ten steps of 36°, back at about 28.3

(d) $\bar{\phi} = \pi/6$: twelve steps of 30°, back at about 42.2

Figure 4.7: The spiral of powers for four angles. Each power of the Composer Unit turns by another $\bar{\phi}$ and grows by the factor $\mathcal{A}_{\pi/3} \approx 1.365$ in (a), $\mathcal{A}_{\pi/4} = \sqrt{2}$ in (b) and $\mathcal{A}_{\pi/5} \approx 1.397$ in (c); in (d) the factor is the same as in (a), but the steps are only 30°. The points are the values of Remark 4.7 and of the worked example; the curves join them. Each panel has its own scale; the first powers in (c) and (d) are too close to 0 to be labelled.

> **In plain words**
>
> In the complex plane, the powers of i walk around a circle: four steps and you are home. In the relative plane at $\bar{\phi} = \pi/3$, the powers of the Composer Unit also come home to the real axis, after six steps of 60° each, but they come home *bigger*. Their path is a spiral that opens outwards (Figure 4.7), and the smaller the angle, the more steps the trip takes and the bigger the powers are when they come home.

The spiral is proved for every angle in Derraz (2026b, Proposition 5.6). The powers show what the Composer Unit does to itself. The next section asks a more practical question: given any General Number, how far is it from zero, and in which direction does it point?

4.4 GENERAL NUMBERS AND GENERAL TRIGONOMETRY FUNCTIONS

We now bring together the two halves of Part i. In Chapter 3, the General Sinus gave us two coordinates, $\cos(\alpha, \phi)$ and $\sin(\alpha, \phi)$, for

reading a point along the real axis and along a leaning axis. In this chapter, the General Number Plane gave us numbers whose two parts are measured along exactly such axes. The two ideas fit together like a key in a lock.

4.4.1 *A Polar Form with Two Angles*

In the complex plane, a number is fixed by its length and one angle. In a relative plane, where the second axis leans, the natural description uses two.

Definition 4.8 (General Scalar Representation with Generalized Trigonometric Functions)**.** In the General Number Plane G, any element

$$z = a + b\mathcal{G}_{\bar{\phi}}$$

can be represented using a generalized polar form, where the real part a and the Composer part $b\mathcal{G}_{\bar{\phi}}$ are expressed in terms of a magnitude M and generalized trigonometric functions that depend on two angles: the projection angle ϕ and a studied angle α:

$$a = M \cdot \cos(\alpha, \phi), \qquad b\mathcal{G}_{\bar{\phi}} = M \cdot \sin(\alpha, \phi).$$

4.4.2 *The Triangle of a General Number*

Walk from the origin O to z in two legs: first a along the real axis, to the corner P, then b steps along the Composer axis, to the tip Q, which is the number z itself. The two legs and the base OQ form a triangle, and the next two theorems read its sides and its angles.

Theorem 4.9 (Triangle Interpretation and Magnitude Formula)**.** *Let $z = a + b\mathcal{G}_{\bar{\phi}}$ be a General Number in the Subgeneral Plane defined by angle $\bar{\phi}$. Then, one can associate z with a triangle where:*

- *The side opposite to angle α represents the* ***Composer part*** *$b\mathcal{G}_{\bar{\phi}}$,*
- *The side opposite to angle ϕ represents the* ***Real part*** *a,*
- *The base of the triangle is the* ***Magnitude*** *M.*

Applying the Law of Cosines to this triangle, the magnitude M satisfies:

$$M^2 = a^2 + b^2\mathcal{G}_{\bar{\phi}}^2 + 2ab\cos(\bar{\phi})\mathcal{G}_{\bar{\phi}}.$$

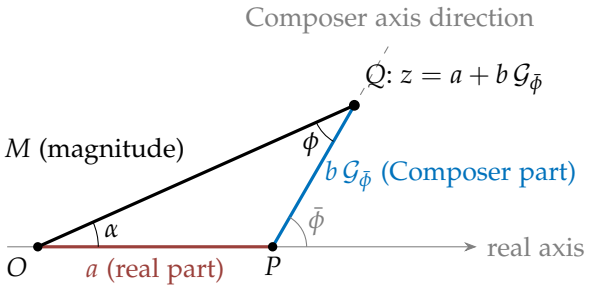


Figure 4.8: The triangle of a General Number (Theorem 4.9): the real part runs from O to P, the Composer part from P to Q, and the magnitude M joins O to Q.

Theorem 4.10 (Phase Angle Expression in the General Number Plane)**.** *Let $z = a + b\mathcal{G}_{\bar{\phi}}$ be a General Number where $a \neq 0$:*

$$\alpha = k\pi + \beta, \qquad \textit{where} \quad \beta = \arctan\left(\frac{\eta \sin(\bar{\phi})}{1 + \eta \cos(\bar{\phi})}\right),$$

with $k \in \mathbb{Z}$, $\eta = \mathcal{G}_{\bar{\phi}} \dfrac{b}{a}$, and $\bar{\phi} \in [0, \pi)$.

Moreover, the complementary angle between the Composer axis and the vector is:

$$\phi = \bar{\phi} - \alpha.$$

> IN PLAIN WORDS
>
> Look at Figure 4.8 and compare it with the oblique triangle of Chapter 3. It is the same triangle. The studied angle α sits at the origin O, the projection angle ϕ sits at the tip Q, and the leaning axis makes the angle $\bar{\phi} = \alpha + \phi$ with the real axis. The General Sinus therefore reads the two parts of a General Number directly: the real part is $M\cos(\alpha, \phi)$ and the Composer part is $M\sin(\alpha, \phi)$. In the two formulas of the theorems, the Composer Unit enters through its length, the amplitude $\mathcal{A}_{\bar{\phi}}$: the side of the triangle opposite α is made of b steps of that length. With this in mind, Theorem 4.9 is the law of cosines of the triangle, and Theorem 4.10 gives the angle at the origin from the ratio η of the two walks.

4.4.3 *Two Worked Examples*

The first example tries the formulas on familiar ground, in the complex plane. The second puts them to work in a plane that leans, and Figure 4.9 draws it to scale.

Worked example 1: the complex plane again

Take $\bar{\phi} = \pi/2$, where $\mathcal{G}_{\pi/2} = i$ has length 1 and $\cos(\pi/2) = 0$. Theorem 4.9 becomes

$$M^2 = a^2 + b^2,$$

the theorem of Pythagoras, which gives the usual modulus of $a + bi$. Theorem 4.10 becomes $\beta = \arctan(\eta)$ with $\eta = b/a$, the usual argument of a complex number. The geometry of the complex plane is found again.

Worked example 2: a number that leans

Take $\bar{\phi} = \pi/3$ and the General Number $z = 1 + 1 \cdot \mathcal{G}_{\pi/3}$, so $a = 1$ and $b = 1$. Use the rounded length of Remark 4.7, $\mathcal{A}_{\pi/3} \approx 1.365$, and $\cos(\pi/3) = 0.5$.

- Magnitude, by Theorem 4.9: $M^2 \approx 1 + 1.365^2 + 2 \cdot 0.5 \cdot 1.365 \approx 1 + 1.863 + 1.365 = 4.228$, so $M \approx 2.056$.
- Phase angle, by Theorem 4.10: $\eta \approx 1.365$, so $\beta = \arctan\big(1.365 \cdot 0.866/(1 + 1.365 \cdot 0.5)\big) \approx \arctan(0.703) \approx 35.1°$, and $\phi = 60° - 35.1° \approx 24.9°$.
- Check with the General Sinus, by Definition 4.8: $M\cos(\alpha, \phi) = M\sin(\phi)/\sin(\bar{\phi}) \approx 2.056 \cdot 0.421/0.866 \approx 1.00$, which is the real part a; and $M\sin(\alpha, \phi) = M\sin(\alpha)/\sin(\bar{\phi}) \approx 2.056 \cdot 0.575/0.866 \approx 1.365$, which is the length of the Composer part.

The three statements of this section agree, as they should: they are three views of one triangle.

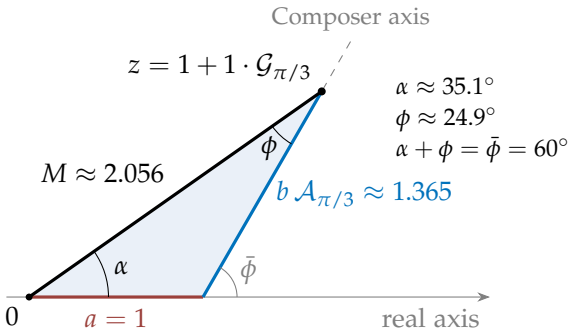
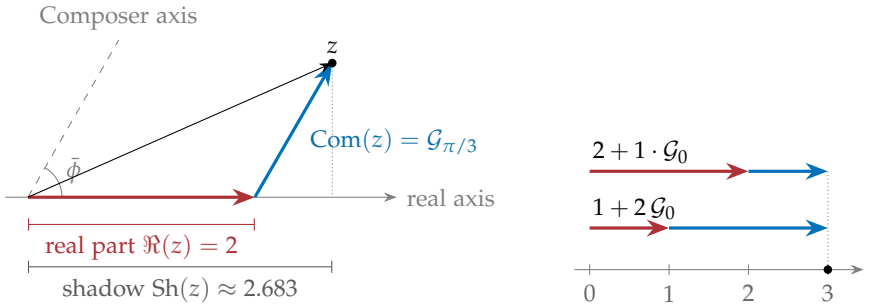


Figure 4.9: Worked example 2 drawn to scale. The number $z = 1 + 1 \cdot \mathcal{G}_{\pi/3}$ is one step along the real axis and one step of the Composer Unit, about 1.365 long. The base is $M \approx 2.056$, and the angles $\alpha \approx 35.1°$ and $\phi \approx 24.9°$ add up to $\bar{\phi} = 60°$.



(a) $z = 2 + \mathcal{G}_{\pi/3}$ and its two parts (b) $\bar{\phi} = 0$: a line, no plane

Figure 4.10: A General Number and its parts. (a) In the relative plane at $\pi/3$, the number $z = 2 + \mathcal{G}_{\pi/3}$ is its real part $\Re(z) = 2$ (dark red) followed by its Composer part $\mathrm{Com}(z) = \mathcal{G}_{\pi/3}$, one step of the Composer Unit, about 1.366 long (blue). Its two readings on the real axis are the real part, 2, and the shadow, about 2.683 (Remark 4.21). (b) At the angle 0 the Composer Unit is $\mathcal{G}_0 = 1$, and the walks $1 + 2\,\mathcal{G}_0$ and $2 + 1 \cdot \mathcal{G}_0$ end at the same point, 3 (Remark 4.13).

4.5 THE PARTS OF A GENERAL NUMBER

With the numbers built, the chapter turns to the words that Part ii will use. Section 4.2.1 met the two parts of a General Number informally; here they get their exact names, together with the rule for when two numbers are equal and the place of the General Numbers among the older numbers.

4.5.1 *Real Part, Composer Part and Equality*

Definition 4.11 (Real Part, Composer Part and Composer Coordinate)**.** Let $z = a + b\mathcal{G}_{\bar{\phi}}$ be a General Number of the relative plane $\mathbb{G}_{\bar{\phi}}$, with $a, b \in \mathbb{R}$.

- The real number a is the *real part* of z, written $\Re(z) = a$.
- The term $b\mathcal{G}_{\bar{\phi}}$ is the *Composer part* of z, written $\mathrm{Com}(z) = b\mathcal{G}_{\bar{\phi}}$: it is b steps of the Composer Unit along the Composer axis, an arrow of length $|b|\,\mathcal{A}_{\bar{\phi}}$.
- The real number b, the number of those steps, is the *Composer coordinate* of z.

Every General Number is the sum of its two parts:

$$z = \Re(z) + \mathrm{Com}(z).$$

> IN PLAIN WORDS
>
> Why three names for two legs? In the complex plane the second leg of $a + bi$ is $|b|$ long, because i has length one, so the number b and the leg bi hardly need two names. Here the step $\mathcal{G}_{\bar{\phi}}$ has a length of its own, and the leg $b\mathcal{G}_{\bar{\phi}}$ is $|b|\,\mathcal{A}_{\bar{\phi}}$ long (Figure 4.10(a)). That is why the Composer coordinate b and the Composer part $b\mathcal{G}_{\bar{\phi}}$ are named apart: it is the non-unified reference of Definition 4.2, in one line.

Definition 4.12 (Equality of General Numbers)**.** Two General Numbers of the same relative plane, $z_1 = a_1 + b_1\mathcal{G}_{\bar{\phi}}$ and $z_2 = a_2 + b_2\mathcal{G}_{\bar{\phi}}$, are equal, $z_1 = z_2$, if and only if

$$a_1 = a_2 \qquad \text{and} \qquad b_1 = b_2,$$

that is, if and only if they have the same real part and the same Composer part.

Remark 4.13 (A Line, No Plane). Definition 4.12 agrees with the picture on the page as long as the Composer Unit leaves the real axis, that is, for $0 < \bar{\phi} < \pi$ with $\mathcal{A}_{\bar{\phi}} \neq 0$. Then the two axes meet at 0 only, and every point of the page is reached by exactly one pair (a, b): the relative plane is a *true plane*. Two angles close it into a line.

- At $\bar{\phi} = 0$, $\mathcal{A}_0 = 1$ and $\vec{u}_0 = \vec{x}$, so $\mathcal{G}_0 = 1$: the Composer axis lies on the real axis, and $1 + 2\,\mathcal{G}_0$ and $2 + 1 \cdot \mathcal{G}_0$ are the same number, 3, although their parts differ.
- At $\bar{\phi} = 3\pi/4$, $\mathcal{A}_{3\pi/4} = 0$, so $\mathcal{G}_{3\pi/4} = 0$, and every General Number is its own real part.

In both cases the relative plane is *a line, no plane*. Every other angle gives a true plane, and every relative plane in this book is a true plane unless the angle 0 is named on purpose.

4.5.2 *The Set of General Numbers and the Ladder*

Definition 4.14 (The Set of General Numbers)**.** The set of all General Numbers, of every relative plane, is written $\mathbb{G}$:

$$\mathbb{G} = \bigcup_{\bar{\phi} \in [0,\pi)} \mathbb{G}_{\bar{\phi}}.$$

General Numbers are added, multiplied and compared within one relative plane, with one Composer Unit.

Proposition 4.15 (The Ladder of Number Sets)**.** *As the counting numbers are a subset of the integers, the complex numbers are a subset of the General Numbers:*

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \subset \mathbb{G}.$$

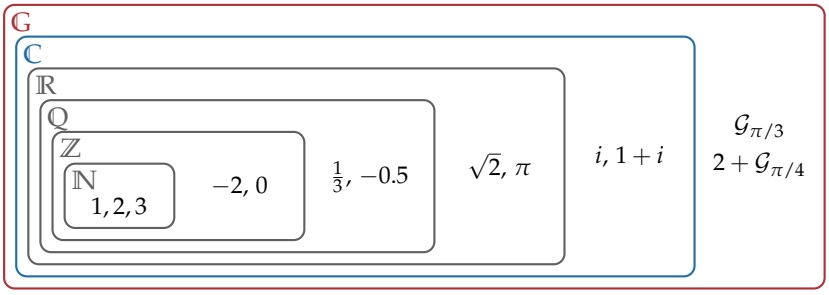


Figure 4.11: The ladder of number sets (Proposition 4.15). Each set holds every set drawn inside it, and a few of its members are written in the ring that it adds: the counting numbers $\mathbb{N}$, the integers $\mathbb{Z}$, the fractions $\mathbb{Q}$, the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ and the General Numbers $\mathbb{G}$. The complex plane is one relative plane, $\mathbb{G}_{\pi/2}$; the outer ring holds the numbers of the leaning planes, such as $\mathcal{G}_{\pi/3}$ and $2 + \mathcal{G}_{\pi/4}$.

Proof. The first four inclusions are the ladder of Chapter 1 (Table 1.1). The last one is Remark 4.4: the complex plane is the relative plane $\mathbb{G}_{\pi/2}$, where $\mathcal{G}_{\pi/2} = i$ (Theorem 4.5). □

Remark 4.16 (Real Numbers and Pure Composer Numbers). A General Number with no Composer part, $\mathrm{Com}(z) = 0$, is a real number. The real numbers lie on the real axis, which all the relative planes share, so $\mathbb{R}$ lies inside every $\mathbb{G}_{\tilde{\phi}}$. A General Number with no real part, $\Re(z) = 0$, is a *pure Composer number*, $z = b\mathcal{G}_{\tilde{\phi}}$; it lies on the Composer axis of its own relative plane. In the complex plane the pure Composer numbers are the imaginary numbers bi.

> **In plain words**
>
> The new rung keeps all the old rungs inside it (Figure 4.11). What it adds is the freedom of the Composer axis to lean, with a step whose length depends on how far it leans. Note what is new and what is not: a true relative plane covers the same points of the page as the complex plane, so what is new is not the points but the frame in which they are named.

4.6 SUMS, PRODUCTS AND TWO READINGS

Numbers are made to be added and multiplied. This section does both within one relative plane, and then reads the result on the real axis in two ways: by its shadow, a drop straight down, and by its real part, a walk back along the Composer axis. The second reading needs one new rule, the Rule of the Square, and it gives every product a real part with one fixed minus sign. The physical chapters read every measured force with it.

4.6.1 *Sum and Product*

Definition 4.17 (Sum and Product)**.** Let $z_1 = a_1 + b_1\mathcal{G}_{\bar\phi}$ and $z_2 = a_2 + b_2\mathcal{G}_{\bar\phi}$ be General Numbers of one relative plane. Their sum is taken part by part,

$$z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2)\,\mathcal{G}_{\bar\phi},$$

and their product $z_1 z_2$ is the product of the two arrows by Wessel's rule: the lengths multiply and the angles add.

Proposition 4.18 (The Three Terms of a Product)**.** *The product of two General Numbers of one relative plane multiplies out like two brackets:*

$$z_1 z_2 = a_1 a_2 + (a_1 b_2 + b_1 a_2)\,\mathcal{G}_{\bar\phi} + b_1 b_2\,\mathcal{G}_{\bar\phi}^2.$$

The three terms are the real term $a_1 a_2$*, along the real axis; the* cross term $(a_1 b_2 + b_1 a_2)\,\mathcal{G}_{\bar\phi}$*, along the Composer axis, at the angle* $\bar\phi$*; and the* square term $b_1 b_2\,\mathcal{G}_{\bar\phi}^2$*, along the arrow* $\mathcal{G}_{\bar\phi}^2$*, of length* $\mathcal{A}_{\bar\phi}^2$ *at the angle* $2\bar\phi$ *(Theorem 4.1).*

Proof. Multiplying by a fixed arrow turns the whole page by its angle and stretches it by its length, and a turn and a stretch carry a sum of arrows to the sum of their images. So each part of z_1 multiplies each part of z_2. The four products are $a_1 a_2$, $a_1 b_2\,\mathcal{G}_{\bar\phi}$, $b_1 a_2\,\mathcal{G}_{\bar\phi}$ and $b_1 b_2\,\mathcal{G}_{\bar\phi}\,\mathcal{G}_{\bar\phi} = b_1 b_2\,\mathcal{G}_{\bar\phi}^2$, and the two middle ones share the factor $\mathcal{G}_{\bar\phi}$. □

In a true plane the sum and the product are arrows of the page, and so, by Remark 4.13, they are again General Numbers of the same relative plane.

4.6.2 *The Shadow on the Real Axis*

Definition 4.19 (The Shadow on the Real Axis)**.** The *shadow* of a General Number z on the real axis, written $\mathrm{Sh}(z)$, is the signed length of its perpendicular projection on that axis: drop a perpendicular from the tip of z to the real axis, and measure from 0 to its foot, positive forward and negative backward. The tip of $z = a + b\mathcal{G}_{\bar\phi}$ lies $a + b\,\mathcal{A}_{\bar\phi}\cos(\bar\phi)$ across and $b\,\mathcal{A}_{\bar\phi}\sin(\bar\phi)$ up, so

$$\mathrm{Sh}(z) = a + b\,\mathcal{A}_{\bar\phi}\cos(\bar\phi).$$

Theorem 4.20 (The Shadows of a Product)**.** *In a relative plane* $\mathbb{G}_{\bar\phi}$*:*

1. *the shadow of a sum is the sum of the shadows, and* $\mathrm{Sh}(c\,z) = c\ \mathrm{Sh}(z)$ *for every real number* c*;*
2. $\mathrm{Sh}(\mathcal{G}_{\bar\phi}^n) = \mathcal{A}_{\bar\phi}^n\cos(n\bar\phi)$ *for every power of the Composer Unit;*
3. *the product of two General Numbers has the shadow*

$$\mathrm{Sh}(z_1 z_2) = a_1 a_2 + (a_1 b_2 + b_1 a_2)\,\mathcal{A}_{\bar\phi}\cos(\bar\phi) + b_1 b_2\,\mathcal{A}_{\bar\phi}^2\cos(2\bar\phi).$$

Each term of a product counts with the shadow of its direction: 1 *for the real term,* $\cos(\bar{\phi})$ *for the cross term and* $\cos(2\bar{\phi})$ *for the square term.*

Proof. (1) Arrows placed head to tail add, and so do the feet of their perpendiculars; stretching an arrow by c stretches its shadow by c. (2) By Theorem 4.1, $\mathcal{G}_{\bar{\phi}}^n$ has the length $\mathcal{A}_{\bar{\phi}}^n$ and leans at $n\bar{\phi}$, so the foot of its perpendicular lies at $\mathcal{A}_{\bar{\phi}}^n \cos(n\bar{\phi})$. (3) Apply (1) and (2) to the three terms of Proposition 4.18. □

Remark 4.21 (Real Part and Shadow). The real part and the shadow both read a General Number on the real axis, in two different ways. The real part $\Re(z) = a$ slides the tip of z back to the real axis along the Composer axis; the shadow $\mathrm{Sh}(z) = a + b\,\mathcal{A}_{\bar{\phi}} \cos(\bar{\phi})$ drops it straight down. The two differ by $b\,\mathcal{A}_{\bar{\phi}} \cos(\bar{\phi})$, so they agree for real numbers and for every number of the complex plane, where the Composer axis stands upright, and nowhere else.

> IN PLAIN WORDS
>
> Picture a lamp high above the real axis. The shadow of z is where its tip falls under that lamp: the component of the arrow along the real axis, in the ordinary sense of geometry, and what a ruler laid along that axis would read. The real part counts steps in the leaning frame instead. For the number $z = 2 + \mathcal{G}_{\pi/3}$ of Figure 4.10(a), the real part is 2, while the shadow is $2 + 1.366 \times 0.5 \approx 2.683$.

4.6.3 *The Real Part of a Product*

The product of Proposition 4.18 has three terms. The first two are already written in the form $a + b\mathcal{G}_{\bar{\phi}}$: the real term is a real number, and the cross term is a number of steps of the Composer Unit. The third is not. It carries the square $\mathcal{G}_{\bar{\phi}}^2$, an arrow that leans at the angle $2\bar{\phi}$, and before a product can be read by its real part we must know the real part of that square. One rule gives it. Here $|\mathcal{G}_{\bar{\phi}}|$ is the length of the Composer Unit, the amplitude $\mathcal{A}_{\bar{\phi}}$ of Definition 4.2.

Proposition 4.22 (The Rule of the Square)**.** *In every true plane,* $0 < \bar{\phi} < \pi$ *with* $\mathcal{G}_{\bar{\phi}} \neq 0$*, the square of the Composer Unit is the General Number*

$$\mathcal{G}_{\bar{\phi}}^2 = 2\,|\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}\; \mathcal{G}_{\bar{\phi}} - |\mathcal{G}_{\bar{\phi}}|^2,$$

with the real part $\Re(\mathcal{G}_{\bar{\phi}}^2) = -|\mathcal{G}_{\bar{\phi}}|^2$ *and the Composer coordinate* $2\,|\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}$*.*

Proof. By Theorem 4.1, $\mathcal{G}_{\bar{\phi}}^2$ is an arrow of length $|\mathcal{G}_{\bar{\phi}}|^2$ at the angle $2\bar{\phi}$, so its tip lies $|\mathcal{G}_{\bar{\phi}}|^2 \cos 2\bar{\phi}$ across and $|\mathcal{G}_{\bar{\phi}}|^2 \sin 2\bar{\phi}$ up. In a true plane it is reached by exactly one walk, a steps along the real axis and then b

steps of the Composer Unit (Remark 4.13). Only the leaning steps go up, each by $|\mathcal{G}_{\bar\phi}| \sin\bar\phi$, which is not zero; so

$$b = \frac{|\mathcal{G}_{\bar\phi}|^2 \sin 2\bar\phi}{|\mathcal{G}_{\bar\phi}| \sin\bar\phi} = 2\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi.$$

Those steps also carry the walk across by $b\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi = 2\,|\mathcal{G}_{\bar\phi}|^2 \cos^2\bar\phi$, and the real part supplies the rest:

$$a = |\mathcal{G}_{\bar\phi}|^2 \cos 2\bar\phi - 2\,|\mathcal{G}_{\bar\phi}|^2 \cos^2\bar\phi = -|\mathcal{G}_{\bar\phi}|^2,$$

since $\cos 2\bar\phi = 2\cos^2\bar\phi - 1$. □

The angle has dropped out of the real part: the terms $2\cos^2\bar\phi$ cancel exactly. It survives only in the Composer coordinate. At the right angle that coordinate is zero and the rule becomes $i^2 = -1$ (Theorem 4.5); at every other angle of a true plane the square of the Composer Unit is still minus its squared length, plus a step along its own axis.

Worked example: the square at 79.86°

Take the length of the Composer Unit as 1 and the angle 79.86°, the angle of the physical chapters, with $\cos\bar\phi \approx 0.17605$ and $\sin\bar\phi \approx 0.98438$ (Figure 4.12). The square leans at 159.72°, so its tip lies at $\cos 159.72° \approx -0.93801$ across and $\sin 159.72° \approx 0.34660$ up.

- *The steps along the Composer axis.* Each rises 0.98438, and the walk must rise 0.34660: $b = 0.34660/0.98438 \approx 0.35210$, which is $2\cos\bar\phi$.
- *The real part.* Those steps also move the walk to the right by $0.35210 \times 0.17605 \approx 0.06199$. The tip is at -0.93801, so the real axis must supply $a = -0.93801 - 0.06199 = -1.00000$.

So $\mathcal{G}^2_{79.86°} = -1 + 0.35210\,\mathcal{G}_{79.86°}$ when the length is 1. The shadow, −0.938, and the real part, −1, differ by the sideways move of the leaning steps, 0.062.

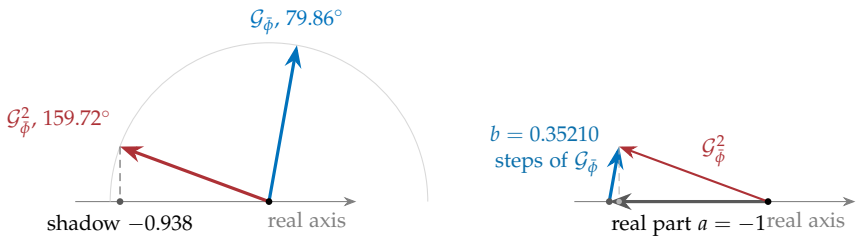
With the Rule of the Square, every product can be written in the form $a + b\mathcal{G}_{\bar\phi}$, and its real part can be read off.

Proposition 4.23 (The Real Part of a Product)**.** *Let $z_1 = a_1 + b_1\mathcal{G}_{\bar\phi}$ and $z_2 = a_2 + b_2\mathcal{G}_{\bar\phi}$ be General Numbers of one true relative plane.*

1. *Their product is*

$$z_1 z_2 = \big(a_1a_2 - |\mathcal{G}_{\bar\phi}|^2 b_1b_2\big) + \big(a_1b_2 + b_1a_2 + 2\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi\; b_1b_2\big)\mathcal{G}_{\bar\phi}.$$

2. No angle. *Its real part, $\Re(z_1z_2) = a_1a_2 - |\mathcal{G}_{\bar\phi}|^2 b_1b_2$, does not depend on $\bar\phi$, and it has one fixed minus sign.*



(a) drop straight down (b) walk back along the axis

Figure 4.12: Two readings of the square of the Composer Unit at the angle 79.86°, with its length taken as 1 (Definition 4.19 and Proposition 4.22). The Composer Unit leans at 79.86°, and its square, of the same length, at twice the angle, 159.72°. (a) The shadow drops the tip straight down and reads −0.938. (b) The real part walks back along the Composer axis: $a = -1$ along the real axis, then $b = 0.35210$ steps of the Composer Unit up to the tip. The walk ends a little beyond the shadow, because its leaning steps carry it back to the right by 0.062.

3. Nothing is missed. *If* $\Re(wX) = 0$ *for every General Number* w*, then* $X = 0$.

Proof. (1) Multiply out the two brackets (Proposition 4.18), $z_1 z_2 = a_1 a_2 + (a_1 b_2 + b_1 a_2)\mathcal{G}_{\bar\phi} + b_1 b_2 \mathcal{G}^2_{\bar\phi}$, and replace $\mathcal{G}^2_{\bar\phi}$ by the Rule of the Square. (2) Read the real part off (1). (3) Write $X = x + y\,\mathcal{G}_{\bar\phi}$. Taking $w = 1$ gives $x = 0$, and taking $w = \mathcal{G}_{\bar\phi}$ gives $\Re(\mathcal{G}_{\bar\phi} X) = -|\mathcal{G}_{\bar\phi}|^2 y = 0$, so $y = 0$. □

WORKED EXAMPLE: THE PRODUCT CHECKED AGAINST THE PAGE

Take a Composer Unit of length 1 at 79.86°, where $\cos\bar\phi = 0.17605$, and multiply $(2 + 3\mathcal{G}_{\bar\phi})(1 + \mathcal{G}_{\bar\phi})$. Proposition 4.23 gives $-1 + 6.05632\,\mathcal{G}_{\bar\phi}$. On the page, $\mathcal{G}_{\bar\phi}$ is the arrow $0.17605 + 0.98438\,i$, and multiplying the two arrows directly, lengths times lengths and angles plus angles, gives $0.06624 + 5.96173\,i$. The formula's result, $-1 + 6.05632\,(0.17605 + 0.98438\,i)$, is the same arrow. The rule also reproduces the square: $\mathcal{G}^2_{\bar\phi} = -1 + 0.35211\,\mathcal{G}_{\bar\phi}$, which lies $-1 + 0.35211 \times 0.17605 = -0.93801$ across and $0.35211 \times 0.98438 = 0.34661$ up, the cosine and the sine of 159.72°.

Remark 4.24 (The Rules of the Arithmetic). A true relative plane holds the same points as the complex plane, and Definition 4.17 multiplies them by the same rule: lengths multiply and angles add. So the General Numbers of one relative plane obey every rule of complex arithmetic: the order of a product does not matter, $z_1 z_2 = z_2 z_1$, nor does its grouping, $(z_1 z_2) z_3 = z_1 (z_2 z_3)$, and a real number times a General Number only scales it. The formula of Proposition 4.23 shows the first rule, being symmetric in the two factors.

> **In plain words**
>
> Square the Composer Unit, then walk back to the real axis along its own leaning axis, the way every General Number is read. You always land at minus its squared length; the angle only decides how many steps along the leaning axis the walk needs. So the real part of any product is the product of the two real parts, minus $|\mathcal{G}_{\bar\phi}|^2$ times the product of the two Composer coordinates: one fixed minus sign, whatever the numbers. A drop straight down, the shadow, lands a little short instead, at -0.938 for the angle 79.86°, because it also counts the sideways move of the leaning steps. Chapter 6 puts this minus sign to work: it is the sign that turns the pull of two masses into the push of two like charges.

4.7 ONE LENGTH, TWO ANGLES

Figure 4.5 reads the length of the Composer Unit from its angle. Read backward, the same curve answers a new question: given a length, which angles give it? Section 4.3.2 has already met one answer: the angles $\pi/6$ and $\pi/3$ give the Composer Unit the same length, $\frac{1}{2} + \frac{\sqrt{3}}{2} \approx 1.366$.

4.7.1 *The Two Roots*

Theorem 4.25 (One Length, Two Angles)**.** *Let A be a length with $1 < A < \sqrt{2}$. Exactly two angles $\bar\phi \in [0, \pi)$ give the Composer Unit the length $\mathcal{A}_{\bar\phi} = A$:*

$$\bar\phi_1 = \arcsin\left(\frac{A}{\sqrt{2}}\right) - \frac{\pi}{4}, \qquad \bar\phi_2 = \frac{\pi}{2} - \bar\phi_1,$$

with $0 < \bar\phi_1 < \pi/4 < \bar\phi_2 < \pi/2$. We call $\bar\phi_1$ the arcsine root *and $\bar\phi_2$ the* mirror root *of the length A. They are mirror images of each other about the* mirror line *$\bar\phi = \pi/4$, and:*

1. *they swap their sine and cosine:* $\cos\bar\phi_2 = \sin\bar\phi_1$ *and* $\sin\bar\phi_2 = \cos\bar\phi_1$*;*
2. *their doubles have the same sine,* $\sin 2\bar\phi_1 = \sin 2\bar\phi_2 = A^2 - 1$*, so that* $\bar\phi_1 = \frac{1}{2}\arcsin(A^2 - 1)$*;*
3. *the cosine of $2\bar\phi_1$ is positive, and the cosine of $2\bar\phi_2$ is its opposite:* $\cos 2\bar\phi_2 = -\cos 2\bar\phi_1$.

Proof. By the second form of the length (Theorem 4.1), $\mathcal{A}_{\bar\phi} = |\sqrt{2}\sin\theta|$ with $\theta = \bar\phi + \pi/4$, and θ runs over $[\pi/4, 5\pi/4)$. Where $\sin\theta$ is negative, for $\pi < \theta < 5\pi/4$, the length is less than $\sqrt{2}\sin(\pi/4) = 1$, so it cannot equal A. Elsewhere the condition is $\sin\theta = A/\sqrt{2}$, a number

between $1/\sqrt{2}$ and 1, and on $[\pi/4, \pi]$ it has exactly two solutions: $\theta_1 = \arcsin(A/\sqrt{2})$, between $\pi/4$ and $\pi/2$, and $\theta_2 = \pi - \theta_1$, between $\pi/2$ and $3\pi/4$. Taking away $\pi/4$ gives $\bar{\phi}_1$ and $\bar{\phi}_2 = 3\pi/4 - \theta_1 = \pi/2 - \bar{\phi}_1$. Statement 1 is the rule $\cos(\pi/2 - x) = \sin x$. For statement 2, square the length: $A^2 = (\cos\bar{\phi} + \sin\bar{\phi})^2 = 1 + \sin 2\bar{\phi}$ at both roots, and $2\bar{\phi}_1$ lies between 0 and $\pi/2$, where the arcsine gives it back. For statement 3, $2\bar{\phi}_2 = \pi - 2\bar{\phi}_1$ and $\cos(\pi - x) = -\cos x$, while the cosine of $2\bar{\phi}_1$ is positive because $2\bar{\phi}_1 < \pi/2$. □

WORKED EXAMPLE: THE PAIR $\pi/6$ AND $\pi/3$

Take $A = \frac{1}{2} + \frac{\sqrt{3}}{2} \approx 1.36603$, a length between 1 and $\sqrt{2}$.

- *The roots.* $A/\sqrt{2} \approx 0.96593$, the sine of 75°. The arcsine root is $75° - 45° = 30° = \pi/6$, and the mirror root is $90° - 30° = 60° = \pi/3$.
- *Statement 1.* $\cos 60° = \frac{1}{2} = \sin 30°$, and $\sin 60° = \frac{\sqrt{3}}{2} = \cos 30°$.
- *Statement 2.* $A^2 - 1 = 1 + \frac{\sqrt{3}}{2} - 1 = \frac{\sqrt{3}}{2}$, which is $\sin 60°$ and also $\sin 120°$.
- *Statement 3.* $\cos 60° = +\frac{1}{2}$ and $\cos 120° = -\frac{1}{2}$.

Lengths below 1 belong to angles past the upright, between $\pi/2$ and π, and the rest of the book does not need them.

Remark 4.26 (Where the Tip of the Composer Unit Lies). For $0 \le \bar{\phi} \le 3\pi/4$, the tip of $\mathcal{G}_{\bar{\phi}}$ lies on the circle through 0, 1, $1 + i$ and i, whose centre is $\frac{1}{2} + \frac{1}{2}i$ and whose radius is $\sqrt{2}/2$ (Figure 4.13). Indeed, the tip lies $x = \mathcal{A}_{\bar{\phi}} \cos\bar{\phi}$ across and $y = \mathcal{A}_{\bar{\phi}} \sin\bar{\phi}$ up, with $\mathcal{A}_{\bar{\phi}} = \cos\bar{\phi} + \sin\bar{\phi}$ for these angles, and both $x^2 + y^2$ and $x + y$ are equal to $1 + \sin 2\bar{\phi}$; so $x^2 + y^2 = x + y$, the equation of that circle. The roots of a length A are the points where this circle meets the circle of radius A about 0:

- for $1 < A < \sqrt{2}$, two points, mirror images about the mirror line (Theorem 4.25);
- for $A = \sqrt{2}$, one point, $1 + i$, where the two roots meet at $\bar{\phi} = \pi/4$;
- for $A = 1$, the points 1 and i: the angle 0, a line with no plane (Remark 4.13), and the angle $\pi/2$, the complex plane.

4.7.2 *Where the Square Points*

The square term of a product carries the arrow $\mathcal{G}^2_{\bar{\phi}}$, and Theorem 4.20 reads it with the factor $\cos(2\bar{\phi})$. On a line a square never points backward. In a relative plane it can.

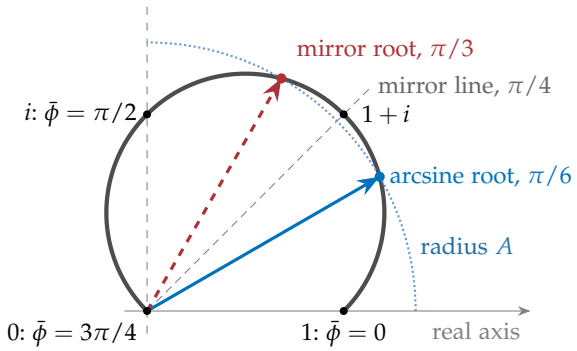


Figure 4.13: Where the tip of the Composer Unit lies (Remark 4.26). As $\bar{\phi}$ runs from 0 to $3\pi/4$, the tip of $\mathcal{G}_{\bar{\phi}}$ moves along the circle through $0, 1, 1+i$ and i (solid arc). The circle of radius A about 0 (dotted) meets that arc at the two roots of the length A (Theorem 4.25). Here $A = \frac{1}{2} + \frac{\sqrt{3}}{2} \approx 1.366$, whose roots are $\pi/6$ and $\pi/3$.

Remark 4.27 (On a Line, Every Square Points Forward). Let the second unit be a real number $\lambda \neq 0$ in the place of $\mathcal{G}_{\bar{\phi}}$, as on the line of Remark 4.13. For $z_1 = a_1 + b_1\lambda$ and $z_2 = a_2 + b_2\lambda$ the product has the same three terms,

$$z_1 z_2 = a_1 a_2 + (a_1 b_2 + b_1 a_2)\,\lambda + b_1 b_2\,\lambda^2,$$

and since $\lambda^2 > 0$, the square term has the sign of $b_1 b_2$. When b_1 and b_2 have the same sign, the square term points forward, the same way as the real term of two positive real parts, whatever the value of λ.

Theorem 4.28 (Where the Square Points). *For $0 \leq \bar{\phi} < \pi$, the square $\mathcal{G}_{\bar{\phi}}^2$ is an arrow of length $\mathcal{A}_{\bar{\phi}}^2$ at the angle $2\bar{\phi}$, with the shadow*

$$\mathrm{Sh}(\mathcal{G}_{\bar{\phi}}^2) = \mathcal{A}_{\bar{\phi}}^2 \cos(2\bar{\phi}).$$

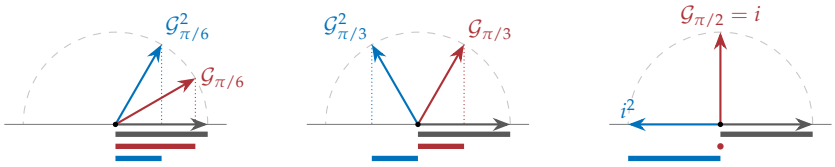
The square points backward, $\mathrm{Sh}(\mathcal{G}_{\bar{\phi}}^2) < 0$, if and only if

$$\frac{\pi}{4} < \bar{\phi} < \frac{3\pi}{4}.$$

This interval is the backward band. *On its lower border, the mirror line $\bar{\phi} = \pi/4$, the shadow of the square vanishes.*

Proof. The first statement is Theorem 4.1 with $n = 2$, read with Theorem 4.20. The length $\mathcal{A}_{\bar{\phi}}^2$ is positive except at $\bar{\phi} = 3\pi/4$, where it vanishes, so $\mathrm{Sh}(\mathcal{G}_{\bar{\phi}}^2)$ is negative exactly when $\cos(2\bar{\phi})$ is, that is, when $2\bar{\phi}$ lies between $\pi/2$ and $3\pi/2$. □

Corollary 4.29 (The Two Roots and the Band). *For a length $1 < A < \sqrt{2}$, the mirror root lies inside the backward band and the arcsine root lies outside it. The square of the Composer Unit has the shadow $A^2\cos(2\bar{\phi}_1) > 0$ at the arcsine root and $-A^2\cos(2\bar{\phi}_1) < 0$ at the mirror root: the same size, with opposite signs.*



(a) $\bar{\phi} = \pi/6$
$\cos\bar{\phi} \approx 0.87$, $\cos 2\bar{\phi} = +\frac{1}{2}$

(b) $\bar{\phi} = \pi/3$
$\cos\bar{\phi} = \frac{1}{2}$, $\cos 2\bar{\phi} = -\frac{1}{2}$

(c) $\bar{\phi} = \pi/2$
$\cos\bar{\phi} = 0$, $\cos 2\bar{\phi} = -1$

Figure 4.14: Where the three terms of a product point, and their shadows (Theorems 4.20 and 4.28). Each arrow has length one and shows a direction only: the real unit (gray), the Composer Unit at $\bar{\phi}$ (dark red) and its square at $2\bar{\phi}$ (blue). The bars under each axis are their shadows, 1, $\cos\bar{\phi}$ and $\cos 2\bar{\phi}$, drawn to the right when they fall forward and to the left when they fall backward. (a) At $\pi/6$ the square leans at $\pi/3$, and its shadow falls forward. (b) At $\pi/3$, the mirror root of the same length, the square leans past the upright, at $2\pi/3$, and its shadow falls backward. (c) At $\pi/2$ the Composer Unit is i, and its square is -1.

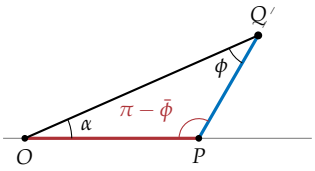
Proof. By Theorem 4.25, $\bar{\phi}_1 < \pi/4 < \bar{\phi}_2 < \pi/2$; the cosine of $2\bar{\phi}_1$ is positive, and the cosine of $2\bar{\phi}_2$ is its opposite. □

> **In plain words**
>
> In the complex plane $i^2 = -1$: the square of the upright unit points straight backward. A leaning Composer Unit keeps part of that behaviour. Its square turns twice as far as the unit, so it points backward exactly when the unit leans past the mirror line, 45°, and stays short of 135°. So two Composer Units of the same length can have squares that point in opposite directions, as the pair $\pi/6$ and $\pi/3$ shows (Figure 4.14).

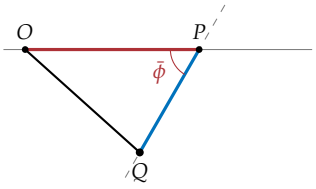
4.8 TRIANGLES AND MAGNITUDES

Section 4.4 drew every General Number as a triangle (Figure 4.8): the real part a runs from the origin O to the corner P, the Composer part $b\mathcal{G}_{\bar{\phi}}$, of length $|b|\,\mathcal{A}_{\bar{\phi}}$, from P to the tip Q, and the base OQ has the length M, the *magnitude* of z. Written out with the General Sinus of Chapter 3, Definition 4.8 reads

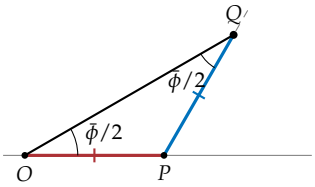
$$a = M\cos(\alpha,\phi) = M\,\frac{\sin\phi}{\sin(\alpha+\phi)},$$
$$b\mathcal{G}_{\bar{\phi}} = M\sin(\alpha,\phi) = M\,\frac{\sin\alpha}{\sin(\alpha+\phi)},$$



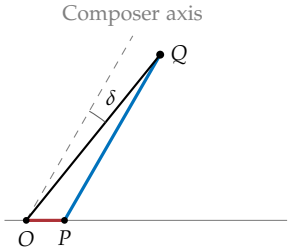
(a) same signs: the corner is $\pi - \bar{\phi}$



(b) opposite signs: the corner is $\bar{\phi}$



(c) equal parts: $\alpha = \phi = \bar{\phi}/2$



(d) a long Composer part: δ is small

Figure 4.15: Four triangles of General Numbers in the relative plane at $\pi/3$ (Remarks 4.30 and 4.31, Proposition 4.32); the real part is dark red and the Composer part blue. (a) $z = 2 + \mathcal{G}_{\pi/3}$: the two parts have the same sign, the corner at P is $\pi - \bar{\phi} = 2\pi/3$, and $\alpha + \phi = \bar{\phi}$. (b) $z = 2 - \mathcal{G}_{\pi/3}$: the signs differ, and the corner is $\bar{\phi} = \pi/3$. (c) Equal parts, $a = b\,\mathcal{A}_{\pi/3}$: the triangle is isosceles, with $\alpha = \phi = \bar{\phi}/2$. (d) A Composer part five times longer than the real part: z lies only $\delta \approx 8.9°$ from the Composer axis.

and Theorem 4.10 finds the studied angle α from the *ratio of the two parts*,

$$\eta = \mathcal{A}_{\bar{\phi}} \frac{b}{a},$$

which the theorem writes with $\mathcal{G}_{\bar{\phi}}$ in place of $\mathcal{A}_{\bar{\phi}}$. This section completes the triangle: the angle at the corner P, three special triangles, how close a General Number can lie to the Composer axis, and all the numbers of one magnitude at once.

4.8.1 *The Corner and Three Special Triangles*

Remark 4.30 (The Corner Angle). Let $z = a + b\mathcal{G}_{\bar{\phi}}$ with $a \neq 0$ and $b \neq 0$. At the corner P, where the real part ends and the Composer part begins, the triangle of z has the angle

$$\angle P = \begin{cases} \pi - \bar{\phi} & \text{when } a \text{ and } b \text{ have the same sign,} \\ \bar{\phi} & \text{when their signs differ.} \end{cases}$$

Indeed, the walk from O to Q turns at P from the real axis onto the Composer axis: by $\bar{\phi}$ when a and b have the same sign, and by $\pi - \bar{\phi}$ when their signs differ; the angle inside the triangle is π minus that

turn. From one General Number to the next only the two sides change, and the corner angle belongs to the relative plane.

Remark 4.31 (Three Special Triangles). Three triangles have shapes of their own.

1. *A real number*, $b = 0$ and $a > 0$: the triangle lies flat on the real axis, with $\alpha = 0$, $\phi = \bar{\phi}$ and $M = a$.
2. *A pure Composer number*, $a = 0$ and $b > 0$: the triangle lies flat on the Composer axis, with $\alpha = \bar{\phi}$, $\phi = 0$ and $M = b\,\mathcal{A}_{\bar{\phi}}$.
3. *Equal parts*, $a = b\,\mathcal{A}_{\bar{\phi}} > 0$, so that $\eta = 1$: the triangle is isosceles, with $\alpha = \phi = \bar{\phi}/2$ and

$$M = a\sqrt{2 + 2\cos\bar{\phi}} = 2a\cos\frac{\bar{\phi}}{2}.$$

With equal lengths and opposite signs, $\eta = -1$, it is isosceles about the corner angle $\bar{\phi}$, and $M = a\sqrt{2 - 2\cos\bar{\phi}} = 2a\sin(\bar{\phi}/2)$.

Proposition 4.32 (Close to the Composer Axis)**.** *Let $z = a + b\mathcal{G}_{\bar{\phi}}$ with $a > 0$ and $b \neq 0$, and let its Composer part be the longer one, $|\eta| = |b|\,\mathcal{A}_{\bar{\phi}}/a > 1$. The angle δ between z and the Composer axis is*

$$b > 0\colon \quad \delta = \arctan\frac{\sin\bar{\phi}}{|\eta| + \cos\bar{\phi}},$$
$$b < 0\colon \quad \delta = \arctan\frac{\sin\bar{\phi}}{|\eta| - \cos\bar{\phi}}.$$

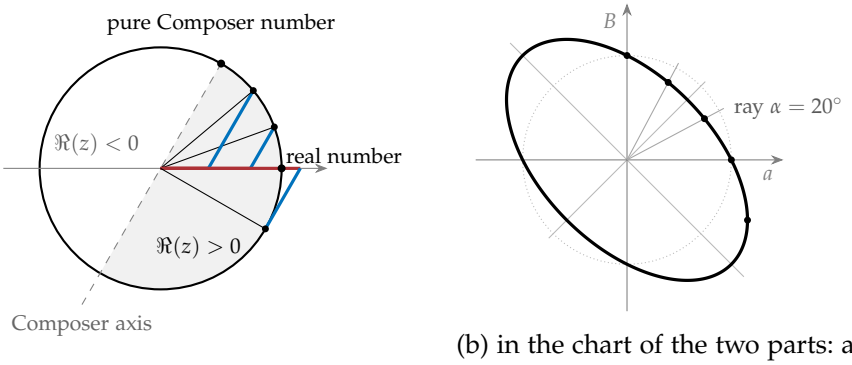
For $b > 0$ it is the projection angle, $\delta = \phi$. When $|\eta|$ is very large, $\delta \approx \sin(\bar{\phi})/|\eta|$ radians.

Proof. The angle δ sits at the tip Q and faces the real part. In a triangle whose sides a and $|b|\,\mathcal{A}_{\bar{\phi}}$ meet at the corner angle P, the angle at Q satisfies $\tan\delta = a\sin P/(|b|\,\mathcal{A}_{\bar{\phi}} - a\cos P)$. With $P = \pi - \bar{\phi}$ for $b > 0$ and $P = \bar{\phi}$ for $b < 0$ (Remark 4.30), dividing by a gives the two formulas, and the condition $|\eta| > 1$ keeps their denominators positive. When $|\eta|$ is very large the fraction is small, its arctangent is the fraction itself, and $\cos\bar{\phi}$ is negligible beside $|\eta|$. □

4.8.2 *All the Numbers of One Magnitude*

Fix the magnitude M and let the General Number turn around 0. The whole family can be drawn in two ways: on the page, and in a chart of its two parts.

Proposition 4.33 (The Circle of One Magnitude)**.** *In a true relative plane, the General Numbers of magnitude M are the points of the circle of radius M about 0. The Composer axis cuts this circle into two halves: on the half that holds the positive real axis the real parts are positive, and on the other half*



(a) on the page: a circle

(b) in the chart of the two parts: an ellipse

Figure 4.16: All the General Numbers of one magnitude M, in the relative plane at $\pi/3$. (a) On the page, the tips lie on the circle of radius M (Proposition 4.33), and the Composer parts (blue) are all parallel to the Composer axis (dashed), which cuts the circle into the half of positive real parts (shaded) and the half of negative ones. The three triangles have $\alpha = 20°$, $40°$ and $-30°$. (b) In the chart of the two parts, the same numbers fill the ellipse $a^2 + B^2 + aB = M^2$ (Theorem 4.34), with the short semi-axis $0.816\,M$ on the diagonal $B = a$ and the long one $1.414\,M$ on $B = -a$; the dotted circle is the case of a right angle, and two rays of Definition 4.35 are drawn. The five dots are the same numbers in both panels.

they are negative. Laid with their real parts along the real axis, the triangles of all these numbers have their tips on the circle and their Composer parts parallel to the Composer axis.

Proof. The magnitude is the distance from 0 to the tip. Sliding a point parallel to the Composer axis never takes it across that axis, and it brings the point to its real part; so a point lies on the side of the positive real axis exactly when its real part is positive. The Composer part of every number is $b\mathcal{G}_{\bar\phi}$, an arrow along the Composer axis. □

Theorem 4.34 (The Ellipse of One Magnitude)**.** *Let $0 < \bar\phi \le \pi/2$ and $M > 0$. Mark each General Number of magnitude M by the pair (a, B) of its real part a and the signed length $B = b\,\mathcal{A}_{\bar\phi}$ of its Composer part, drawn at right angles, a across and B up. These points fill the ellipse*

$$a^2 + B^2 + 2\cos\bar\phi \cdot aB = M^2,$$

whose axes lie on the diagonals $B = a$ and $B = -a$, with the semi-axes

$$\frac{M}{\sqrt{1+\cos\bar\phi}} \text{ on } B = a, \qquad \frac{M}{\sqrt{1-\cos\bar\phi}} \text{ on } B = -a.$$

Their ratio is $\tan(\bar\phi/2)$, the eccentricity e is given by $e^2 = 2\cos\bar\phi/(1+\cos\bar\phi)$, and the area is $\pi M^2/\sin\bar\phi$. As the studied angle α makes one turn, the point

$$a = M\,\frac{\sin(\bar\phi - \alpha)}{\sin\bar\phi}, \qquad B = M\,\frac{\sin\alpha}{\sin\bar\phi}$$

runs once around the ellipse. At $\bar{\phi} = \pi/2$ the ellipse is the circle $a^2 + B^2 = M^2$; as $\bar{\phi}$ falls towards 0 it stretches without end along $B = -a$, and at $\bar{\phi} = 0$ the same equation gives the two parallel lines $a + B = \pm M$.

Proof. The equation is the law of cosines of Theorem 4.9, with the corner angle $\pi - \bar{\phi}$ of Remark 4.30. It has the form $a^2 + 2\cos\bar{\phi} \cdot aB + B^2 = M^2$, and $(2\cos\bar{\phi})^2 - 4 = -4\sin^2\bar{\phi}$ is negative for $0 < \bar{\phi} < \pi$, so the curve is an ellipse. With $u = (a + B)/\sqrt{2}$ and $v = (B - a)/\sqrt{2}$ the chart turns by 45°: $a^2 + B^2 = u^2 + v^2$ and $aB = (u^2 - v^2)/2$, so the equation becomes

$$(1 + \cos\bar{\phi})\, u^2 + (1 - \cos\bar{\phi})\, v^2 = M^2,$$

which gives the two semi-axes. Their ratio and the area are

$$\sqrt{\frac{1 - \cos\bar{\phi}}{1 + \cos\bar{\phi}}} = \tan\frac{\bar{\phi}}{2}, \qquad \frac{\pi M^2}{\sqrt{1 - \cos^2\bar{\phi}}} = \frac{\pi M^2}{\sin\bar{\phi}},$$

and the eccentricity follows from $e^2 = 1 - \tan^2(\bar{\phi}/2)$. The parametric form is the General Sinus reading of Section 4.8 with $\phi = \bar{\phi} - \alpha$. From it, $a + B\cos\bar{\phi} = M\cos\alpha$ and $B\sin\bar{\phi} = M\sin\alpha$; squaring and adding gives the equation again, and as α makes one turn the point $(\cos\alpha, \sin\alpha)$ makes one turn of its circle. At $\bar{\phi} = 0$ the equation reads $(a + B)^2 = M^2$. □

Definition 4.35 (Rings and Rays)**.** The *chart* of a relative plane is the plane of the pairs (a, B) of Theorem 4.34, ruled by two families of lines:

- the *rings*, the ellipses of Theorem 4.34, one for each magnitude M;
- the *rays*, the half-lines from 0 along which the studied angle α stays the same.

A General Number with the magnitude M and the studied angle α is the point where its ring and its ray cross.

> IN PLAIN WORDS
>
> Hold the magnitude fixed, and the General Number can still turn: on the page its tip runs around a circle (Figure 4.16(a)). Keep only its two parts instead, drawn at right angles the way the old habit names a point by two coordinates, and the same family draws an ellipse (Figure 4.16(b)), because the two parts meet at the corner angle $\pi - \bar{\phi}$ and not at a right angle. The equation of the ellipse is the law of cosines, read as a curve, and its shape depends on the angle alone: a circle when the Composer axis stands upright, an oval squeezed along the diagonal of equal parts as the axis leans, a needle as it nears the real axis. Where the two parts have

the same sign the ellipse runs inside the circle of the right angle, and where their signs differ it runs outside; the two curves meet only on the axes. For an angle past the upright, between $\pi/2$ and π, the same equation holds and the two diagonals trade roles. The rings and the rays turn the chart into graph paper. Along a ray the ratio of the two parts is fixed, $\eta = B/a = \sin\alpha/\sin(\bar{\phi} - \alpha)$, and so is the projection angle $\phi = \bar{\phi} - \alpha$: each ray is one value of the general tangent of Chapter 3. The real axis of the chart is the ray $\alpha = 0$, but the upright axis is the ray $\alpha = \bar{\phi}$, not the ray $\alpha = \pi/2$, so the rays fan out where the two parts have the same sign and crowd together where their signs differ. The short radius of every ring lies on the ray $\alpha = \bar{\phi}/2$, where the two parts are equal (Remark 4.31). Only at $\bar{\phi} = \pi/2$ are the rings circles and the rays evenly spread: the chart is then the ordinary polar frame of the complex plane.

4.9 THE ALGEBRA OF GENERAL NUMBERS

Appendix C builds the complex numbers in a few steps: a unit whose square is -1, the four operations and the conjugate, the length and the angle of a number, the rule that multiplying means turning, the formulas of Euler and de Moivre, and the roots of unity. This section takes the same steps in a relative plane. Each law is written with the General Sinus of Chapter 3, and each becomes its law in Appendix C when the Composer axis stands upright, at $\bar{\phi} = \pi/2$.

That last fact needs no extra assumption. At $\bar{\phi} = \pi/2$ the Axiom itself gives the Composer Unit the length 1, so that $\mathcal{G}_{\pi/2} = i$ (Theorem 4.5). And in the polar form of Definition 4.8 the two angles always add up to the angle of the Composer axis, $\alpha + \phi = \bar{\phi}$ (Theorem 4.10); at $\bar{\phi} = \pi/2$ they add up to a right angle, where the general sine and cosine are the ordinary ones (Corollary 3.5).

Nothing in this section can contradict the complex numbers. A true relative plane holds the same points as the complex plane, and Definition 4.17 multiplies them by the same rule (Remark 4.24), so every law of complex arithmetic holds here too. What the section adds is the way each law reads in the leaning frame, where a number is counted by its real part and its Composer coordinate. Three things change: the conjugate keeps the mirror but no longer keeps the real part, the imaginary unit splits into two numbers with two different jobs, and the general sine and cosine take the place of the ordinary ones, bringing terms in $2\cos\bar{\phi}$ with them. The physical chapters of this volume do not need the section. It completes the arithmetic of the chapter, and its last part reads every result in two ways: by its two parts, and by its shadow and its height. Appendix F lists each of its laws beside its complex case. Derraz (2026b, Sections 4 and 5) proves

these laws in full, and notes that the algebra of a relative plane is that of the elliptic generalized complex numbers.

Throughout the section, $\mathbb{G}_{\bar\phi}$ is a true relative plane ($0 < \bar\phi < \pi$ with $\mathcal{G}_{\bar\phi} \neq 0$, Remark 4.13), and $z = a + b\mathcal{G}_{\bar\phi}$, $z_1 = a_1 + b_1\mathcal{G}_{\bar\phi}$ and $z_2 = a_2 + b_2\mathcal{G}_{\bar\phi}$ are General Numbers of that plane. As for complex numbers, we write $|z|$ for the magnitude M of z, the length of its arrow. By Theorem 4.9, with the length $|\mathcal{G}_{\bar\phi}| = \mathcal{A}_{\bar\phi}$ of the Composer Unit written out,

$$|z|^2 = a^2 + 2ab\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b^2|\mathcal{G}_{\bar\phi}|^2.$$

The direction of the Composer Unit, $\vec{u}_{\bar\phi} = \mathcal{G}_{\bar\phi}/|\mathcal{G}_{\bar\phi}|$ (Axiom 4.1), is itself a General Number: one step of length 1 along the Composer axis. The Rule of the Square, divided by $|\mathcal{G}_{\bar\phi}|^2$, gives its square,

$$\vec{u}_{\bar\phi}^{\,2} = 2\cos\bar\phi\;\vec{u}_{\bar\phi} - 1.$$

4.9.1 *The Rules of Arithmetic and the Conjugate*

Remark 4.24 named two rules of the arithmetic, the order and the grouping of a product. Here is the whole list.

Theorem 4.36 (A Relative Plane Is a Field)**.** *In a true relative plane* $\mathbb{G}_{\bar\phi}$*:*

1. *the sum and the product of two General Numbers are General Numbers of the same plane;*
2. *the order and the grouping of a sum, and of a product, do not matter;*
3. *a product distributes over a sum:* $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$*;*
4. 0 *and* 1 *change nothing:* $z + 0 = z$ *and* $1 \cdot z = z$*;*
5. *every* z *has an opposite,* $-z = -a - b\mathcal{G}_{\bar\phi}$*, with* $z + (-z) = 0$*;*
6. *every* $z \neq 0$ *has an inverse* $1/z$*, with* $z \cdot (1/z) = 1$*.*

In the words of algebra, every true relative plane is a field*, as the complex plane is.*

Proof. (1) is the remark that follows Proposition 4.18. (2) A sum is taken part by part, and real numbers add in any order and grouping. A product multiplies the lengths and adds the angles (Definition 4.17), and lengths multiply, and angles add, in any order and grouping as well. (3) Multiplying by z_1 turns the whole page by the angle of z_1 and stretches it by $|z_1|$, and a turn and a stretch carry a sum of arrows to the sum of their images (the proof of Proposition 4.18). (4) and (5) are read off the parts. (6) is Theorem 4.41 below, which writes the inverse out. □

In the complex plane the conjugate of $a + bi$ is $a - bi$: the sign of the second part changes. Seen on the page, that rule is the mirror in the real axis (Appendix C), and it is the mirror, not the change of sign, that carries over to a relative plane.

Definition 4.37 (The Conjugate)**.** The *conjugate* of the General Number $z = a + b\mathcal{G}_{\bar\phi}$ is the General Number

$$\bar z = \left(a + 2b\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi\right) - b\mathcal{G}_{\bar\phi}.$$

In the words of Definition 4.11, the conjugate turns the Composer part around, $\mathrm{Com}(\bar z) = -b\mathcal{G}_{\bar\phi} = -\mathrm{Com}(z)$, as in the complex plane. Its real part, however, is $\Re(\bar z) = a + 2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi$: the real part of z, moved by twice the shadow of the Composer part. Only where $\cos\bar\phi = 0$ does the real part stay. Theorem 4.38 shows why the mirror asks for this.

Theorem 4.38 (The Conjugate)**.** *In a true relative plane $\mathbb{G}_{\bar\phi}$:*

1. *$\bar z$ is the mirror image of z in the real axis;*
2. *$\bar{\bar z} = z$, $\overline{z_1 + z_2} = \bar z_1 + \bar z_2$ and $\overline{z_1 z_2} = \bar z_1\,\bar z_2$;*
3. *a number and its conjugate add up to twice the shadow, $z + \bar z = 2\,\mathrm{Sh}(z)$;*
4. *a number times its conjugate is the square of its magnitude,*

   $$z\bar z = a^2 + 2ab\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b^2|\mathcal{G}_{\bar\phi}|^2 = |z|^2,$$

   a real number, zero only for $z = 0$;
5. *the Composer Unit and its conjugate have the sum $\mathcal{G}_{\bar\phi} + \overline{\mathcal{G}_{\bar\phi}} = 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi$ and the product $\mathcal{G}_{\bar\phi}\,\overline{\mathcal{G}_{\bar\phi}} = |\mathcal{G}_{\bar\phi}|^2$, so they are the two roots of the equation*

   $$x^2 - 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi\; x + |\mathcal{G}_{\bar\phi}|^2 = 0.$$

Proof. (1) The tip of z lies $\mathrm{Sh}(z) = a + b|\mathcal{G}_{\bar\phi}|\cos\bar\phi$ across and $b|\mathcal{G}_{\bar\phi}|\sin\bar\phi$ up (Definition 4.19), and its mirror image lies as far across and as far down. Only the steps along the Composer axis go up or down, so the walk to the mirror image takes $-b$ of them. They carry it across by $-b|\mathcal{G}_{\bar\phi}|\cos\bar\phi$, and the real part must supply the rest, $\mathrm{Sh}(z) + b|\mathcal{G}_{\bar\phi}|\cos\bar\phi = a + 2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi$. (2) A mirror used twice changes nothing, and it carries a sum of arrows to the sum of their images. It keeps every length and reverses every angle, so it carries a product, made by multiplying lengths and adding angles, to the product of the images. (3) Add the parts. (4) By Proposition 4.23, $z\bar z$ has the real part $a(a + 2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi) + |\mathcal{G}_{\bar\phi}|^2 b^2$ and the Composer coordinate $-ab + b(a + 2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi) - 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi\;\, b^2 = 0$. The real part is $|z|^2$ by Theorem 4.9; as the square of a length, it is zero only for $z = 0$. (5) Take $a = 0$ and $b = 1$ in (3) and (4). Then $(x - \mathcal{G}_{\bar\phi})(x - \overline{\mathcal{G}_{\bar\phi}}) = x^2 - (\mathcal{G}_{\bar\phi} + \overline{\mathcal{G}_{\bar\phi}})\,x + \mathcal{G}_{\bar\phi}\,\overline{\mathcal{G}_{\bar\phi}}$ is the left side of the equation. □

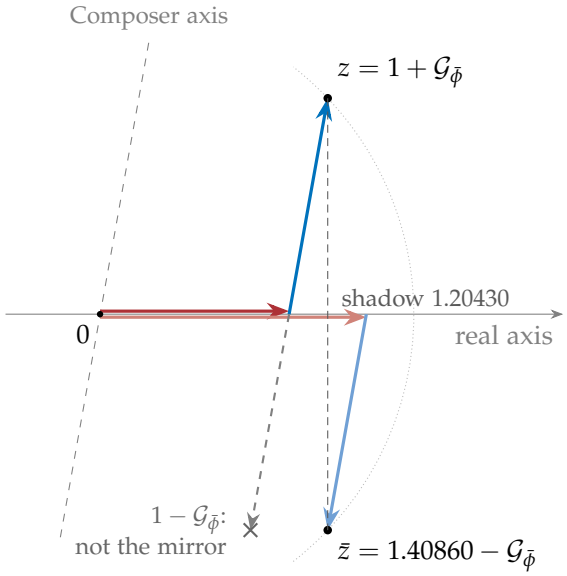


Figure 4.17: The conjugate of $z = 1 + \mathcal{G}_{\bar\phi}$ at $\bar\phi = 79.86°$, where $|\mathcal{G}_{\bar\phi}| = 1.16043$ (Definition 4.37). Each walk runs first along the real axis (dark red, the two drawn a little apart) and then along the Composer axis (blue). The conjugate $\bar z = 1.40860 - \mathcal{G}_{\bar\phi}$ ends at the mirror image of z in the real axis: the same shadow, 1.20430, the opposite height, and the same magnitude, $|z| = 1.65988$ (dotted circle). The skew flip $1 - \mathcal{G}_{\bar\phi}$ (grey) reverses only the walk along the Composer axis; it ends inside the circle, 0.40860 to the left of the mirror image.

Put $x = \mathcal{G}_{\bar\phi}$ in the equation of (5), and it is the Rule of the Square, $\mathcal{G}_{\bar\phi}^2 = 2|\mathcal{G}_{\bar\phi}| \cos\bar\phi \ \mathcal{G}_{\bar\phi} - |\mathcal{G}_{\bar\phi}|^2$ (Proposition 4.22). The rule that gave every product its real part says that the Composer Unit and its mirror image are the two roots of one quadratic equation.

Remark 4.39 (Why Not $a - b\mathcal{G}_{\bar\phi}$). Changing the sign of the Composer part, $a + b\mathcal{G}_{\bar\phi} \mapsto a - b\mathcal{G}_{\bar\phi}$, reverses the walk along the Composer axis, not the height on the page. Where the axis leans, this skew flip is not the mirror. It changes the length,

$$|a - b\mathcal{G}_{\bar\phi}|^2 = |z|^2 - 4ab\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi,$$

and it gives no real product: $(a + b\mathcal{G}_{\bar\phi})(a - b\mathcal{G}_{\bar\phi}) = a^2 + b^2|\mathcal{G}_{\bar\phi}|^2 - 2b^2|\mathcal{G}_{\bar\phi}| \cos\bar\phi \ \mathcal{G}_{\bar\phi}$. The two rules agree for every number only where $\cos\bar\phi = 0$.

IN THE COMPLEX PLANE. At $\bar\phi = \pi/2$, $\bar z = a - bi$, $z + \bar z = 2a$ and $z\bar z = a^2 + b^2$, and the equation of (5) is $x^2 + 1 = 0$, with the roots i and $-i$.

4.9.2 *Magnitude and Division*

Theorem 4.40 (The Magnitude of a Product)**.** *The magnitude of a product is the product of the magnitudes,* $|z_1 z_2| = |z_1|\,|z_2|$*. Write the product as* $z_1 z_2 = c + d\mathcal{G}_{\bar\phi}$*, with* $c = a_1a_2 - |\mathcal{G}_{\bar\phi}|^2 b_1 b_2$ *and* $d = a_1b_2 + b_1a_2 + 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi\, b_1b_2$ *(Proposition 4.23). Then the law is the identity*

$$\begin{aligned}&\left(a_1^2 + 2a_1b_1|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b_1^2|\mathcal{G}_{\bar\phi}|^2\right)\left(a_2^2 + 2a_2b_2|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b_2^2|\mathcal{G}_{\bar\phi}|^2\right)\\ &\qquad = c^2 + 2cd\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi + d^2|\mathcal{G}_{\bar\phi}|^2.\end{aligned}$$

Proof. By Wessel's rule the lengths multiply (Definition 4.17). In the algebra of Theorem 4.38, $|z_1z_2|^2 = (z_1z_2)\,\overline{z_1z_2} = (z_1\bar z_1)(z_2\bar z_2) = |z_1|^2|z_2|^2$, and writing each magnitude with (4) of that theorem gives the identity. □

In the complex plane. At $\bar\phi = \pi/2$ the identity is $(a_1^2+b_1^2)(a_2^2+b_2^2) = (a_1a_2 - b_1b_2)^2 + (a_1b_2 + b_1a_2)^2$, known since Diophantus: a product of two sums of two squares is again a sum of two squares. In a relative plane each sum of two squares becomes the expression $a^2 + 2ab\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b^2|\mathcal{G}_{\bar\phi}|^2$, and the identity keeps its shape.

Theorem 4.41 (Division)**.** *Every General Number* $z \neq 0$ *has the inverse*

$$\frac{1}{z} = \frac{\bar z}{|z|^2} = \frac{(a + 2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi) - b\mathcal{G}_{\bar\phi}}{a^2 + 2ab|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b^2|\mathcal{G}_{\bar\phi}|^2},$$

and for $z_2 \neq 0$ *the quotient is*

$$\frac{z_1}{z_2} = \frac{z_1\bar z_2}{|z_2|^2} = \frac{\left(a_1a_2 + 2a_1b_2|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b_1b_2|\mathcal{G}_{\bar\phi}|^2\right) + \left(b_1a_2 - a_1b_2\right)\mathcal{G}_{\bar\phi}}{a_2^2 + 2a_2b_2|\mathcal{G}_{\bar\phi}|\cos\bar\phi + b_2^2|\mathcal{G}_{\bar\phi}|^2}.$$

In particular, $1/\mathcal{G}_{\bar\phi} = \overline{\mathcal{G}_{\bar\phi}}/|\mathcal{G}_{\bar\phi}|^2 = (2|\mathcal{G}_{\bar\phi}|\cos\bar\phi - \mathcal{G}_{\bar\phi})/|\mathcal{G}_{\bar\phi}|^2$.

Proof. By Theorem 4.38, $z\bar z = |z|^2$, which is not zero, so $z\cdot\bar z/|z|^2 = 1$. Then $z_1/z_2 = z_1\cdot(1/z_2) = z_1\bar z_2/|z_2|^2$, and Proposition 4.23 gives the parts of $z_1\bar z_2$: the real part $a_1(a_2 + 2b_2|\mathcal{G}_{\bar\phi}|\cos\bar\phi) + |\mathcal{G}_{\bar\phi}|^2 b_1b_2$, and the Composer coordinate $-a_1b_2 + b_1(a_2 + 2b_2|\mathcal{G}_{\bar\phi}|\cos\bar\phi) - 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi\, b_1b_2 = b_1a_2 - a_1b_2$. □

> In plain words
>
> To divide, multiply the top and the bottom by the conjugate of the bottom, exactly as in Appendix C; only the conjugate has changed. Notice where the angle has gone. Neither $\cos\bar\phi$ nor $\sin\bar\phi$ enters the Composer coordinate of the top, $b_1a_2 - a_1b_2$, just as neither enters the real part of a product (Proposition 4.23).

In the complex plane. At $\bar\phi = \pi/2$ the quotient is $\left[(a_1a_2 + b_1b_2) + (b_1a_2 - a_1b_2)\,i\right]/(a_2^2 + b_2^2)$, the division of Appendix C, and $1/i = -i$.

WORKED EXAMPLE: THE CONJUGATE AND DIVISION AT 79.86°

Take the angle $\bar{\phi} = 79.86°$ and the length that the Axiom gives there, $|\mathcal{G}_{\bar{\phi}}| = |\cos\bar{\phi} + \sin\bar{\phi}| \approx 0.17605 + 0.98438 = 1.16043$. One step of the Composer Unit goes $|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi} \approx 0.20430$ across and $|\mathcal{G}_{\bar{\phi}}|\sin\bar{\phi} \approx 1.14231$ up, and $|\mathcal{G}_{\bar{\phi}}|^2 \approx 1.34661$.

- *The conjugate of* $z = 1 + \mathcal{G}_{\bar{\phi}}$. The tip of z lies 1.20430 across and 1.14231 up. Definition 4.37 gives $\bar{z} = (1 + 0.40860) - \mathcal{G}_{\bar{\phi}} = 1.40860 - \mathcal{G}_{\bar{\phi}}$, whose tip lies $1.40860 - 0.20430 = 1.20430$ across and 1.14231 down: the mirror image (Figure 4.17). Its Composer part has turned around, from $\mathcal{G}_{\bar{\phi}}$ to $-\mathcal{G}_{\bar{\phi}}$, and its real part has moved from 1 to 1.40860.
- *The magnitude.* $z\bar{z} \approx 1 + 0.40860 + 1.34661 = 2.75521$, so $|z| \approx 1.65988$. On the page, $1.20430^2 + 1.14231^2 \approx 2.75521$ as well.
- *The inverse.* $1/z = (1.40860 - \mathcal{G}_{\bar{\phi}})/2.75521 \approx 0.51125 - 0.36295\,\mathcal{G}_{\bar{\phi}}$.
- *A quotient.* For $(2 - \mathcal{G}_{\bar{\phi}})/(1 + \mathcal{G}_{\bar{\phi}})$ the top is $(2 + 2\cdot 2\cdot 0.20430 - 1.34661) + (-1 - 2)\,\mathcal{G}_{\bar{\phi}} = 1.47059 - 3\mathcal{G}_{\bar{\phi}}$, so the quotient is about $0.53375 - 1.08885\,\mathcal{G}_{\bar{\phi}}$. Multiplied back by $1 + \mathcal{G}_{\bar{\phi}}$ with Proposition 4.23, it gives $2 - \mathcal{G}_{\bar{\phi}}$.
- *The wrong mirror.* The skew flip $1 - \mathcal{G}_{\bar{\phi}}$ has the magnitude $|1 - \mathcal{G}_{\bar{\phi}}| \approx 1.39212$, not 1.65988, and $(1 + \mathcal{G}_{\bar{\phi}})(1 - \mathcal{G}_{\bar{\phi}}) \approx 2.34661 - 0.40860\,\mathcal{G}_{\bar{\phi}}$ is not a real number.

4.9.3 *The Turning Unit and the Polar Form*

In the complex plane the number i does two jobs at once. It is the second step of the frame, the unit of the imaginary axis; and it is the quarter turn, the number whose square is -1. In a relative plane that leans, the two jobs come apart. The second step is the Composer Unit, whose square obeys the Rule of the Square and is not -1. The quarter turn is a new number, made from it.

Definition 4.42 (The Turning Unit)**.** The *Turning Unit* of the relative plane $\mathbb{G}_{\bar{\phi}}$ is the General Number

$$\mathcal{J}_{\bar{\phi}} = \frac{\vec{u}_{\bar{\phi}} - \cos\bar{\phi}}{\sin\bar{\phi}} = -\cot\bar{\phi} + \frac{1}{|\mathcal{G}_{\bar{\phi}}|\sin\bar{\phi}}\,\mathcal{G}_{\bar{\phi}}.$$

Theorem 4.43 (The Turning Unit). *In a true relative plane* $\mathbb{G}_{\bar\phi}$*:*

1. $\mathcal{J}_{\bar\phi}$ *is the arrow of length* 1 *at a right angle to the real axis;*
2. $\mathcal{J}_{\bar\phi}^2 = -1$*, and* $\mathcal{J}_{\bar\phi}$ *and* $-\mathcal{J}_{\bar\phi}$ *are the only General Numbers whose square is* -1*;*
3. *multiplying by* $\mathcal{J}_{\bar\phi}$ *turns every General Number by a right angle and keeps its length;*
4. *at* $\bar\phi = \pi/2$ *the two units are one:* $\mathcal{J}_{\pi/2} = \mathcal{G}_{\pi/2} = i$*.*

Proof. (1) The walk of $\mathcal{J}_{\bar\phi}$ goes $-\cot\bar\phi$ along the real axis and then $1/(|\mathcal{G}_{\bar\phi}|\sin\bar\phi)$ steps along the Composer axis, which carry it $\cot\bar\phi$ across and 1 up. It ends at the point 1 straight above 0. (2) With $\vec{u}_{\bar\phi}^{\,2} = 2\cos\bar\phi\ \vec{u}_{\bar\phi} - 1$,

$$\mathcal{J}_{\bar\phi}^2 = \frac{\vec{u}_{\bar\phi}^{\,2} - 2\cos\bar\phi\ \vec{u}_{\bar\phi} + \cos^2\bar\phi}{\sin^2\bar\phi} = \frac{\cos^2\bar\phi - 1}{\sin^2\bar\phi} = -1.$$

If $w^2 = -1$, then $|w| = 1$ and twice the angle of w is a half turn, so w is the arrow of length 1 at the angle $\pi/2$ or $-\pi/2$: $w = \mathcal{J}_{\bar\phi}$ or $w = -\mathcal{J}_{\bar\phi}$. (3) By Wessel's rule, multiplying by an arrow of length 1 at the angle $\pi/2$ keeps every length and adds a right angle to every angle. (4) At $\bar\phi = \pi/2$, $\cot\bar\phi = 0$, $\sin\bar\phi = 1$ and $|\mathcal{G}_{\bar\phi}| = 1$, so $\mathcal{J}_{\bar\phi} = \mathcal{G}_{\bar\phi} = i$ (Theorem 4.5). □

The Turning Unit gives every relative plane a second frame, an upright one, inside the same plane.

Proposition 4.44 (Shadow and Height). *Call* $\mathrm{Ht}(z) = b\,|\mathcal{G}_{\bar\phi}|\sin\bar\phi$ *the* height *of* z*: how far its tip lies above the real axis. Every General Number of* $\mathbb{G}_{\bar\phi}$ *is*

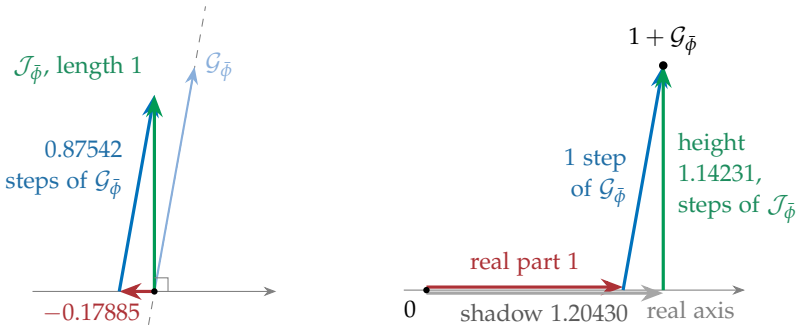
$$z = \mathrm{Sh}(z) + \mathrm{Ht}(z)\,\mathcal{J}_{\bar\phi},$$

and back,

$$a = \mathrm{Sh}(z) - \mathrm{Ht}(z)\cot\bar\phi, \qquad b = \frac{\mathrm{Ht}(z)}{|\mathcal{G}_{\bar\phi}|\sin\bar\phi}.$$

Moreover, $z - \bar z = 2\,\mathrm{Ht}(z)\,\mathcal{J}_{\bar\phi}$ *and* $|z|^2 = \mathrm{Sh}(z)^2 + \mathrm{Ht}(z)^2$*.*

Proof. Write the parts of $\mathcal{J}_{\bar\phi}$ into $\mathrm{Sh}(z) + \mathrm{Ht}(z)\mathcal{J}_{\bar\phi}$. The real part is $a + b|\mathcal{G}_{\bar\phi}|\cos\bar\phi - b|\mathcal{G}_{\bar\phi}|\sin\bar\phi\cot\bar\phi = a$, and the Composer coordinate is $b|\mathcal{G}_{\bar\phi}|\sin\bar\phi/(|\mathcal{G}_{\bar\phi}|\sin\bar\phi) = b$. Solving for a and b gives the formulas back. The conjugate $\bar z$ has the same shadow and the opposite height (Theorem 4.38), so $z - \bar z = 2\,\mathrm{Ht}(z)\mathcal{J}_{\bar\phi}$. The shadow and the height meet at a right angle, so $|z|^2 = \mathrm{Sh}(z)^2 + \mathrm{Ht}(z)^2$ by the theorem of Pythagoras. □



(a) the Turning Unit as a walk (b) one number, two frames

Figure 4.18: The Turning Unit at $\bar\phi = 79.86°$ (Definition 4.42). (a) Its walk goes -0.17885 along the real axis (dark red) and 0.87542 steps of the Composer Unit (blue), and ends one unit straight above 0: the Turning Unit (green) is the quarter turn. (b) The number $1+\mathcal{G}_{\bar\phi}$ in the two frames of its relative plane. In the leaning frame it is 1 step along the real axis and 1 step of the Composer Unit; in the upright frame it is its shadow, 1.20430 (grey), and its height, 1.14231 steps of the Turning Unit (Proposition 4.44).

In plain words

A relative plane carries two frames (Figure 4.18). The leaning frame, 1 and $\mathcal{G}_{\bar\phi}$, is the one in which its numbers are written: the real part and the Composer coordinate are counted in it. The upright frame, 1 and $\mathcal{J}_{\bar\phi}$, sits inside the same plane: there the two coordinates of a number are its shadow and its height, and $\mathcal{J}_{\bar\phi}$ plays the part of i. Sending z to the complex number $\mathrm{Sh}(z) + \mathrm{Ht}(z)\,i$ keeps every sum and every product, and this is the exact sense in which a true relative plane holds the same points as the complex plane. Everything new happens in the leaning frame, and the rest of this section reads each law there.

Theorem 4.45 (The Polar Form with One Angle). *Let $z \neq 0$, and let α be the angle of its arrow, measured counterclockwise from the positive real axis: the studied angle of Definition 4.8, whose projection angle is $\phi = \bar\phi - \alpha$. Then*

$$z = |z|\,\big[\cos(\alpha, \bar\phi - \alpha) + \sin(\alpha, \bar\phi - \alpha)\,\vec{u}_{\bar\phi}\big],$$

that is, the real part is $a = |z|\cos(\alpha, \bar\phi - \alpha)$, and the Composer part $b\mathcal{G}_{\bar\phi}$ has the signed length $b\,|\mathcal{G}_{\bar\phi}| = |z|\sin(\alpha, \bar\phi - \alpha)$, as in Definition 4.8. For every angle α the two values of the General Sinus obey the general Pythagoras identity

$$\cos(\alpha, \bar\phi - \alpha)^2 + 2\cos\bar\phi\,\cos(\alpha, \bar\phi - \alpha)\sin(\alpha, \bar\phi - \alpha) + \sin(\alpha, \bar\phi - \alpha)^2 = 1.$$

Proof. The tip of z lies $|z|\cos\alpha$ across and $|z|\sin\alpha$ up: these are its shadow and its height. Proposition 4.44 gives $b|\mathcal{G}_{\bar\phi}| = |z|\sin\alpha/\sin\bar\phi$

and $a = |z|(\cos\alpha - \sin\alpha\cot\bar{\phi}) = |z|\sin(\bar{\phi}-\alpha)/\sin\bar{\phi}$. With $\alpha+\phi=\bar{\phi}$, the definitions of the General Sinus read $\sin(\alpha,\bar{\phi}-\alpha) = \sin\alpha/\sin\bar{\phi}$ and $\cos(\alpha,\bar{\phi}-\alpha) = \sin(\bar{\phi}-\alpha)/\sin\bar{\phi}$. The identity is (4) of Theorem 4.38 for a number of magnitude 1; in the chart of the two parts it is the ring of magnitude 1 (Theorem 4.34). □

IN THE COMPLEX PLANE. At $\bar{\phi} = \pi/2$ the Turning Unit and the Composer Unit are both i, the shadow and the height are the real and the imaginary parts, the polar form is $z = |z|(\cos\alpha + i\sin\alpha)$, and the general Pythagoras identity is $\cos^2\alpha + \sin^2\alpha = 1$.

> WORKED EXAMPLE: THE TURNING UNIT AT 79.86°
>
> With the numbers of the last example, $\cot\bar{\phi} \approx 0.17885$ and $1/(|\mathcal{G}_{\bar{\phi}}|\sin\bar{\phi}) \approx 0.87542$, so
>
> $$\mathcal{J}_{\bar{\phi}} \approx -0.17885 + 0.87542\,\mathcal{G}_{\bar{\phi}}.$$
>
> Proposition 4.23 checks that its square is -1: the real part is $0.17885^2 - 1.34661 \times 0.87542^2 \approx 0.03199 - 1.03199 = -1$, and the Composer coordinate is $2 \times (-0.17885) \times 0.87542 + 0.40860 \times 0.87542^2 \approx -0.31313 + 0.31313 = 0$. In the upright frame, $1 + \mathcal{G}_{\bar{\phi}} = 1.20430 + 1.14231\,\mathcal{J}_{\bar{\phi}}$: its shadow and its height. And $\mathcal{J}_{\bar{\phi}}(1+\mathcal{G}_{\bar{\phi}}) \approx -1.35770 + 1.05427\,\mathcal{G}_{\bar{\phi}}$, whose tip lies 1.14231 to the left of 0 and 1.20430 up: the tip of $1+\mathcal{G}_{\bar{\phi}}$, turned by a right angle.

4.9.4 *Multiplying Means Turning*

For short, write the two values of the General Sinus at the angle α, with the projection angle $\bar{\phi}-\alpha$, as

$$C_\alpha = \cos(\alpha,\bar{\phi}-\alpha), \qquad S_\alpha = \sin(\alpha,\bar{\phi}-\alpha),$$

so that a number of magnitude 1 at the angle α is $C_\alpha + S_\alpha\,\vec{u}_{\bar{\phi}}$, and $C_{\alpha+\beta} = \cos(\alpha+\beta,\bar{\phi}-\alpha-\beta)$.

Theorem 4.46 (Multiplying Means Turning). *Let* $z_1 = |z_1|\,(C_\alpha + S_\alpha\,\vec{u}_{\bar\phi})$ *and* $z_2 = |z_2|\,(C_\beta + S_\beta\,\vec{u}_{\bar\phi})$. *Then*

$$z_1 z_2 = |z_1|\,|z_2|\,(C_{\alpha+\beta} + S_{\alpha+\beta}\,\vec{u}_{\bar\phi}),$$

and z_1/z_2 *is the same with* $|z_1|/|z_2|$ *and* $\alpha - \beta$. *Equivalently, the General Sinus obeys the* addition formulas along the Composer axis*:*

$$\begin{aligned} C_{\alpha+\beta} &= C_\alpha C_\beta - S_\alpha S_\beta, \\ S_{\alpha+\beta} &= C_\alpha S_\beta + S_\alpha C_\beta + 2\cos\bar\phi\; S_\alpha S_\beta. \end{aligned}$$

Proof. By Wessel's rule (Definition 4.17), $z_1 z_2$ has the magnitude $|z_1|\,|z_2|$ and the angle $\alpha + \beta$, and Theorem 4.45 writes such a number in the form shown. On the other hand, multiply the two brackets out with $\vec{u}_{\bar\phi}^{\,2} = 2\cos\bar\phi\;\vec{u}_{\bar\phi} - 1$:

$$\begin{aligned} (C_\alpha + S_\alpha\vec{u}_{\bar\phi})(C_\beta + S_\beta\vec{u}_{\bar\phi}) &= C_\alpha C_\beta + (C_\alpha S_\beta + S_\alpha C_\beta)\,\vec{u}_{\bar\phi} + S_\alpha S_\beta\,\vec{u}_{\bar\phi}^{\,2} \\ &= (C_\alpha C_\beta - S_\alpha S_\beta) + (C_\alpha S_\beta + S_\alpha C_\beta + 2\cos\bar\phi\; S_\alpha S_\beta)\,\vec{u}_{\bar\phi}. \end{aligned}$$

Two General Numbers are equal when their parts are (Definition 4.12), and comparing the parts gives the two formulas. The quotient follows from the product, since $(z_1/z_2)\,z_2 = z_1$. □

The addition formulas of Appendix A hold the projection angle fixed. The formulas above let it change with the studied angle, so that the two angles always add up to $\bar\phi$. The two families agree.

Remark 4.47 (The Same Formulas from Appendix A). Write $\phi = \bar\phi - \alpha - \beta$, the projection angle of $\alpha + \beta$, and keep the short names C and S.

1. The two formulas of Appendix A for the projection angle, with the shift β, and the reciprocity $\sin(\bar\phi - \beta, \beta) = C_\beta$ and $\cos(\bar\phi - \beta, \beta) = S_\beta$, give

   $$\sin(\alpha,\phi) = \frac{S_\alpha}{C_\beta}, \qquad \cos(\alpha,\phi) = C_\alpha - \frac{S_\alpha S_\beta}{C_\beta},$$

   and in the same way, with the shift α, $\sin(\beta,\phi) = S_\beta/C_\alpha$ and $\cos(\beta,\phi) = C_\beta - S_\alpha S_\beta/C_\alpha$.

2. The second addition formula for the studied angle, $\sin(\alpha + \beta,\phi) = \sin(\alpha,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi+\alpha)$, where $\sin(\beta,\phi+\alpha) = S_\beta$, then gives

   $$S_{\alpha+\beta} = C_\alpha S_\beta + S_\alpha\,\frac{1 - S_\beta^2}{C_\beta},$$

   and the general Pythagoras identity turns $(1 - S_\beta^2)/C_\beta$ into $C_\beta + 2\cos\bar\phi\,S_\beta$. This is the second formula of Theorem 4.46.

3. The consequence $\cos(\alpha+\beta,\phi) = \cos(\alpha,\phi)\cos(\beta,\phi)/\big(1-\sin(\alpha,\phi)\sin(\beta,\phi)\big)$, with the four values of (1), gives $C_{\alpha+\beta} = C_\alpha C_\beta - S_\alpha S_\beta$, the first formula.

The steps divide by C_α and C_β, and the third also by $1-\sin(\alpha,\phi)\sin(\beta,\phi)$, which is zero exactly where $C_{\alpha+\beta}$ is; where one of these is zero, the formulas follow by continuity.

> WORKED EXAMPLE: ONE PRODUCT, TWO WAYS
>
> Stay at 79.86° and multiply $z = 1+\mathcal{G}_{\bar\phi}$ by the Composer Unit.
>
> - *By the parts.* Proposition 4.23 gives $(1+\mathcal{G}_{\bar\phi})\,\mathcal{G}_{\bar\phi} = -|\mathcal{G}_{\bar\phi}|^2 + (1+2|\mathcal{G}_{\bar\phi}|\cos\bar\phi)\,\mathcal{G}_{\bar\phi} \approx -1.34661 + 1.40860\,\mathcal{G}_{\bar\phi}$.
> - *By turning.* The number $1+\mathcal{G}_{\bar\phi}$ has the magnitude 1.65988 and the angle $\alpha \approx 43.487°$, with $C_\alpha \approx 0.60245$ and $S_\alpha \approx 0.69911$; indeed $1.65988 \times 0.60245 \approx 1$, its real part, and $1.65988 \times 0.69911 \approx 1.16043$, the length of its Composer part. The Composer Unit has the magnitude 1.16043 and the angle $\beta = \bar\phi$, where $C_\beta = \cos(\bar\phi,0) = 0$ and $S_\beta = \sin(\bar\phi,0) = 1$. At the angle $\alpha+\beta \approx 123.347°$ the addition formulas give $C_{\alpha+\beta} = -S_\alpha \approx -0.69911$ and $S_{\alpha+\beta} = C_\alpha + 2\cos\bar\phi\ S_\alpha \approx 0.60245 + 0.35211 \times 0.69911 \approx 0.84861$, and the magnitude is $1.65988 \times 1.16043 \approx 1.92618$. So the real part is $1.92618 \times (-0.69911) \approx -1.34661$, and the Composer coordinate is $1.92618 \times 0.84861/1.16043 \approx 1.40860$: the same number.
>
> The extra term $2\cos\bar\phi\,S_\alpha S_\beta \approx 0.24616$ is more than a quarter of $S_{\alpha+\beta}$; without it, the turn would land in the wrong place.

4.9.5 *Euler's and de Moivre's Formulas*

The exponential of a General Number w is defined by the same series as in the complex plane, $e^w = 1 + w + w^2/2! + w^3/3! + \cdots$. It converges because it converges for the arrows of the page, and the rule $e^{w_1+w_2} = e^{w_1}e^{w_2}$ holds for the same reason.

Theorem 4.48 (Euler's Formula in a Relative Plane)**.** *For every real number θ,*

$$e^{\theta \mathcal{J}_{\bar\phi}} = \cos(\theta, \bar\phi - \theta) + \sin(\theta, \bar\phi - \theta)\, \vec{u}_{\bar\phi}.$$

Consequently:

1. *every $z \neq 0$ is $z = |z|\, e^{\alpha \mathcal{J}_{\bar\phi}}$, with α its angle, and $\bar z = |z|\, e^{-\alpha \mathcal{J}_{\bar\phi}}$;*
2. *the Composer Unit is an exponential, $\mathcal{G}_{\bar\phi} = |\mathcal{G}_{\bar\phi}|\, e^{\bar\phi \mathcal{J}_{\bar\phi}}$, and so is the Turning Unit, $\mathcal{J}_{\bar\phi} = e^{(\pi/2)\mathcal{J}_{\bar\phi}}$;*
3. *$e^{\pi \mathcal{J}_{\bar\phi}} + 1 = 0$.*

Proof. The powers of $\mathcal{J}_{\bar\phi}$ run through $1, \mathcal{J}_{\bar\phi}, -1, -\mathcal{J}_{\bar\phi}$ and back, because $\mathcal{J}_{\bar\phi}^2 = -1$. So the series splits exactly as Euler's does, into the series of the cosine and of the sine: $e^{\theta \mathcal{J}_{\bar\phi}} = \cos\theta + \mathcal{J}_{\bar\phi} \sin\theta$. Put in the Turning Unit of Definition 4.42:

$$\begin{aligned} \cos\theta + \sin\theta\, \frac{\vec{u}_{\bar\phi} - \cos\bar\phi}{\sin\bar\phi} &= \frac{\sin\bar\phi\cos\theta - \cos\bar\phi\sin\theta}{\sin\bar\phi} + \frac{\sin\theta}{\sin\bar\phi}\, \vec{u}_{\bar\phi} \\ &= \frac{\sin(\bar\phi - \theta)}{\sin\bar\phi} + \frac{\sin\theta}{\sin\bar\phi}\, \vec{u}_{\bar\phi}, \end{aligned}$$

which is the formula, by the definitions of the General Sinus with $\alpha + \phi = \bar\phi$. (1) is Theorem 4.45 with $\theta = \alpha$; the mirror image has the same magnitude and the opposite angle. (2) At $\theta = \bar\phi$, $\cos(\bar\phi, 0) = 0$ and $\sin(\bar\phi, 0) = 1$, so $e^{\bar\phi \mathcal{J}_{\bar\phi}} = \vec{u}_{\bar\phi} = \mathcal{G}_{\bar\phi}/|\mathcal{G}_{\bar\phi}|$; at $\theta = \pi/2$ the formula gives Definition 4.42 back. (3) At $\theta = \pi$, the π-addition identities of Appendix A give $\cos(\pi, \bar\phi - \pi) = -\cos(0, \bar\phi) = -1$ and $\sin(\pi, \bar\phi - \pi) = -\sin(0, \bar\phi) = 0$. □

Remark 4.49 (The Same Formula from the Derivatives of Appendix A). Along the line where the two angles add up to $\bar\phi$, a change of θ raises the studied angle and lowers the projection angle by the same amount, so $d/d\theta$ is the partial derivative in the first angle minus the partial derivative in the second. The partial derivatives of Appendix A then give

$$\begin{aligned} \frac{d}{d\theta} \sin(\theta, \bar\phi - \theta) &= \frac{\cos(\theta, \bar\phi - \theta)}{\sin\bar\phi} + \sin(\theta, \bar\phi - \theta) \cot\bar\phi, \\ \frac{d}{d\theta} \cos(\theta, \bar\phi - \theta) &= -\cos(\theta, \bar\phi - \theta) \cot\bar\phi - \frac{\sin(\theta, \bar\phi - \theta)}{\sin\bar\phi}. \end{aligned}$$

With $\vec{u}_{\bar\phi}^2 = 2\cos\bar\phi\ \vec{u}_{\bar\phi} - 1$, these two lines say that the unit number $w(\theta) = \cos(\theta, \bar\phi - \theta) + \sin(\theta, \bar\phi - \theta)\, \vec{u}_{\bar\phi}$ obeys $dw/d\theta = \mathcal{J}_{\bar\phi} w$ with $w(0) = 1$, and the only solution of that equation is $e^{\theta \mathcal{J}_{\bar\phi}}$. A unit number turns at the rate of its own quarter turn.

Theorem 4.50 (De Moivre's Formula in a Relative Plane)**.** *For every integer n and every real number θ, $(C_\theta + S_\theta\,\vec{u}_{\bar\phi})^n = C_{n\theta} + S_{n\theta}\,\vec{u}_{\bar\phi}$, that is,*

$$\begin{aligned}&\big[\cos(\theta,\bar\phi-\theta)+\sin(\theta,\bar\phi-\theta)\,\vec{u}_{\bar\phi}\big]^n\\&\qquad=\cos(n\theta,\bar\phi-n\theta)+\sin(n\theta,\bar\phi-n\theta)\,\vec{u}_{\bar\phi}.\end{aligned}$$

Proof. The left side is $(e^{\theta\mathcal{J}_{\bar\phi}})^n = e^{n\theta\mathcal{J}_{\bar\phi}}$; apply Theorem 4.48 with $n\theta$ in place of θ. In words, n turns by θ make one turn by $n\theta$ (Theorem 4.46). □

Corollary 4.51 (The Powers of the Composer Unit in Its Own Frame)**.** *For every integer n,*

$$\begin{aligned}\mathcal{G}_{\bar\phi}^n &= |\mathcal{G}_{\bar\phi}|^n\cos\big(n\bar\phi,(1-n)\bar\phi\big)+|\mathcal{G}_{\bar\phi}|^{n-1}\sin\big(n\bar\phi,(1-n)\bar\phi\big)\,\mathcal{G}_{\bar\phi}\\&=-|\mathcal{G}_{\bar\phi}|^n\,\frac{\sin(n-1)\bar\phi}{\sin\bar\phi}+|\mathcal{G}_{\bar\phi}|^{n-1}\,\frac{\sin n\bar\phi}{\sin\bar\phi}\,\mathcal{G}_{\bar\phi},\end{aligned}$$

and three powers in a row obey the step rule

$$\mathcal{G}_{\bar\phi}^{n+1} = 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi\;\mathcal{G}_{\bar\phi}^n - |\mathcal{G}_{\bar\phi}|^2\,\mathcal{G}_{\bar\phi}^{n-1}.$$

For $n = 2$ the formula is the Rule of the Square. Drawn on the page, it is Theorem 4.1, now for every integer n.

Proof. By Theorem 4.48, $\mathcal{G}_{\bar\phi}^n = |\mathcal{G}_{\bar\phi}|^n e^{n\bar\phi\mathcal{J}_{\bar\phi}}$; apply the formula with $\theta = n\bar\phi$ and write $\vec{u}_{\bar\phi} = \mathcal{G}_{\bar\phi}/|\mathcal{G}_{\bar\phi}|$. The step rule is the Rule of the Square multiplied by $\mathcal{G}_{\bar\phi}^{n-1}$. At $n = 2$, $\cos(2\bar\phi,-\bar\phi) = -1$ and $\sin(2\bar\phi,-\bar\phi) = \sin 2\bar\phi/\sin\bar\phi = 2\cos\bar\phi$, which is Proposition 4.22. □

> IN PLAIN WORDS
>
> Euler's identity, $e^{\pi\mathcal{J}_{\bar\phi}}+1=0$, holds in every relative plane, unchanged: half a turn is half a turn, whatever the frame. What changes is how a turn is read. The unit number at the angle θ is $\cos(\theta,\bar\phi-\theta)$ steps along the real axis and $\sin(\theta,\bar\phi-\theta)$ steps of length one along the Composer axis. And the Composer Unit is now an exponential, $\mathcal{G}_{\bar\phi} = |\mathcal{G}_{\bar\phi}|\,e^{\bar\phi\mathcal{J}_{\bar\phi}}$: the spiral of its powers in Theorem 4.1 is de Moivre's formula, and the Rule of the Square is its second step.

IN THE COMPLEX PLANE. At $\bar\phi = \pi/2$: $e^{i\theta} = \cos\theta + i\sin\theta$, $z = |z|e^{i\alpha}$, $i = e^{i\pi/2}$ and $e^{i\pi}+1 = 0$; de Moivre's formula $(\cos\theta+i\sin\theta)^n = \cos n\theta + i\sin n\theta$; and the powers $i^n = \cos(n\pi/2) + i\sin(n\pi/2)$, with the step rule $i^{n+1} = -i^{n-1}$: the cycle of four of Remark 4.6.

WORKED EXAMPLE: THE CUBE OF THE COMPOSER UNIT, THREE WAYS

At $79.86°$, with $|\mathcal{G}_{\bar\phi}| \approx 1.16043$:

- *By the step rule.* The Rule of the Square gives $\mathcal{G}_{\bar\phi}^2 \approx -1.34661 + 0.40860\,\mathcal{G}_{\bar\phi}$, and then $\mathcal{G}_{\bar\phi}^3 = 0.40860\,\mathcal{G}_{\bar\phi}^2 - 1.34661\,\mathcal{G}_{\bar\phi} \approx -0.55022 - 1.17966\,\mathcal{G}_{\bar\phi}$.
- *By Corollary 4.51.* $|\mathcal{G}_{\bar\phi}|^3 \approx 1.56265$, $\sin 2\bar\phi / \sin\bar\phi = 2\cos\bar\phi \approx 0.35211$ and $\sin 3\bar\phi / \sin\bar\phi \approx -0.87602$, so the real part is $-1.56265 \times 0.35211 \approx -0.55022$ and the Composer coordinate is $1.34661 \times (-0.87602) \approx -1.17966$.
- *On the page.* By Theorem 4.1, $\mathcal{G}_{\bar\phi}^3$ is the arrow of length 1.56265 at $3 \times 79.86° = 239.58°$, whose tip lies 0.79122 to the left of 0 and 1.34753 down. Read back in the leaning frame with Proposition 4.44, these give the same two parts.

4.9.6 *Roots*

Theorem 4.52 (Roots in a Relative Plane)**.** *Let $n \geq 1$ be an integer, and let $z \neq 0$ have the angle α. The equation $w^n = z$ has exactly n solutions in $\mathbb{G}_{\bar\phi}$,*

$$w_k = |z|^{1/n}\left[\cos(\theta_k, \bar\phi - \theta_k) + \sin(\theta_k, \bar\phi - \theta_k)\,\vec{u}_{\bar\phi}\right], \qquad \theta_k = \frac{\alpha + 2\pi k}{n},$$

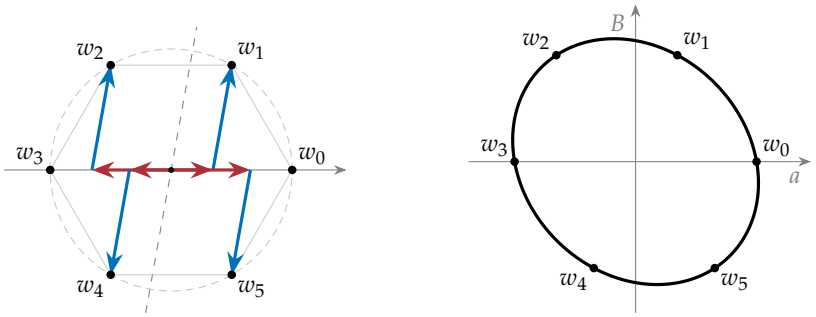
for $k = 0, 1, \ldots, n-1$. For $z = 1$ they are the n-th roots of unity, with $\theta_k = 2\pi k/n$. On the page these are the corners of a regular polygon with n sides inside the circle of radius 1; in the chart of the two parts they lie on the ring of magnitude 1 (Theorem 4.34).

Proof. By Theorem 4.50, $w_k^n = |z|\,e^{(\alpha+2\pi k)\mathcal{J}_{\bar\phi}} = |z|\,e^{\alpha\mathcal{J}_{\bar\phi}} = z$, since a full turn changes nothing: the π-addition identities of Appendix A, used twice, give $\cos(\theta + 2\pi, \bar\phi - \theta - 2\pi) = \cos(\theta, \bar\phi - \theta)$, and the same for the sine. The angles θ_k differ by less than a full turn, so the n numbers are different. There are no others: if $w^n = z$, then $|w| = |z|^{1/n}$, and n times the angle of w is α up to whole turns. □

Corollary 4.53 (The Square Root of the Composer Unit)**.** *The two square roots of the Composer Unit are*

$$\pm\frac{\sqrt{|\mathcal{G}_{\bar\phi}|}}{2\cos(\bar\phi/2)}\,(1 + \vec{u}_{\bar\phi}).$$

They lie on the line of equal parts: the root with the plus sign on the ray $\alpha = \bar\phi/2$ (Remark 4.31), the other on the opposite ray, $\alpha = \bar\phi/2 + \pi$.



(a) on the page (b) in the chart of the two parts

Figure 4.19: The sixth roots of unity at $\bar{\phi} = 79.86°$ (Theorem 4.52). (a) On the page they are the corners of a regular hexagon in the circle of radius 1. Each is reached by a walk along the real axis (dark red) and then in the direction of the Composer axis (blue; the axis itself is dashed): $w_1 = 0.34511 + 0.75814\,\mathcal{G}_{\bar{\phi}}$, $w_2 = -0.65489 + 0.75814\,\mathcal{G}_{\bar{\phi}}$, $w_3 = -1$, and $w_4 = -w_1$, $w_5 = -w_2$. (b) In the chart of the two parts, the real part a across and the length $B = b\,|\mathcal{G}_{\bar{\phi}}|$ of the Composer part up (Theorem 4.34), the same six numbers lie on the ring of magnitude 1, at equal steps of the studied angle.

Proof. Take $z = \mathcal{G}_{\bar{\phi}}$, with the magnitude $|\mathcal{G}_{\bar{\phi}}|$ and the angle $\bar{\phi}$, and $n = 2$. Then $\theta_0 = \bar{\phi}/2$, and the isosceles case of Appendix A, $\sin(\theta,\theta) = \cos(\theta,\theta) = 1/(2\cos\theta)$, gives $\cos(\bar{\phi}/2,\bar{\phi}/2) = \sin(\bar{\phi}/2,\bar{\phi}/2) = 1/(2\cos(\bar{\phi}/2))$. The root with $k = 1$ is the opposite number. □

In the complex plane. At $\bar{\phi} = \pi/2$ the roots are $w_k = |z|^{1/n} e^{i(\alpha+2\pi k)/n}$, the roots of unity are $e^{2\pi i k/n}$ (Appendix C), and the square roots of i are $\pm(1+i)/\sqrt{2}$.

Worked example: roots at 79.86°

- *The square root of* $\mathcal{G}_{\bar{\phi}}$. Here $\sqrt{|\mathcal{G}_{\bar{\phi}}|} \approx 1.07723$ and $\cos(\bar{\phi}/2) = \cos 39.93° \approx 0.76683$, so

$$\sqrt{\mathcal{G}_{\bar{\phi}}} = \pm 0.70240\,(1 + \vec{u}_{\bar{\phi}}) \approx \pm(0.70240 + 0.60529\,\mathcal{G}_{\bar{\phi}}).$$

  Squared with Proposition 4.23, $0.70240 + 0.60529\,\mathcal{G}_{\bar{\phi}}$ gives the real part $0.70240^2 - 1.34661 \times 0.60529^2 \approx 0$ and the Composer coordinate $2 \times 0.70240 \times 0.60529 + 0.40860 \times 0.60529^2 \approx 0.85030 + 0.14970 = 1$: the number $\mathcal{G}_{\bar{\phi}}$.

- *The sixth roots of unity* are, rounded, 1, $0.34511 + 0.75814\,\mathcal{G}_{\bar{\phi}}$, $-0.65489 + 0.75814\,\mathcal{G}_{\bar{\phi}}$, -1, $-0.34511 - 0.75814\,\mathcal{G}_{\bar{\phi}}$ and $0.65489 - 0.75814\,\mathcal{G}_{\bar{\phi}}$ (Figure 4.19). The two roots above the real axis have the same height, so they share the Composer coordinate $\sin 60° / (|\mathcal{G}_{\bar{\phi}}| \sin\bar{\phi}) \approx 0.75814$, and their real parts differ by exactly 1, the side of the hexagon.

4.9.7 *The Parts beside the Shadow and the Height*

A General Number has two pairs of coordinates. Its two parts, the real part $\Re(z) = a$ and the Composer part $\mathrm{Com}(z) = b\mathcal{G}_{\bar\phi}$ of Definition 4.11, are counted in the leaning frame: the tip of z slides back to the real axis along the Composer axis. Its shadow $\mathrm{Sh}(z)$ and its height $\mathrm{Ht}(z)$ are counted in the upright frame: the tip drops straight down (Definition 4.19 and Proposition 4.44). Theorem 4.20 and Proposition 4.23 read a product in both ways on the real axis. This last part does the same for each operation of the section: it gives the law for the two parts beside the law for the shadow and the height.

Proposition 4.54 (From One Pair to the Other)**.** *In a true relative plane* $\mathbb{G}_{\bar\phi}$*:*

1. *the parts follow from the shadow and the height, and back:*

$$\Re(z) = \mathrm{Sh}(z) - \mathrm{Ht}(z)\cot\bar\phi, \qquad \mathrm{Com}(z) = \frac{\mathrm{Ht}(z)}{|\mathcal{G}_{\bar\phi}|\sin\bar\phi}\,\mathcal{G}_{\bar\phi},$$
$$\mathrm{Sh}(z) = \Re(z) + b\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi, \qquad \mathrm{Ht}(z) = b\,|\mathcal{G}_{\bar\phi}|\sin\bar\phi;$$

2. *all four are additive and keep real factors:* $\Re(z_1+z_2) = \Re(z_1)+\Re(z_2)$ *and* $\Re(kz) = k\,\Re(z)$ *for every real number* k*, and the same holds for* Com, Sh *and* Ht*;*

3. *the conjugate has the same coordinates on the mirrored unit,* $\bar z = a + b\,\overline{\mathcal{G}_{\bar\phi}}$*, and the four are read from* z *and* $\bar z$*:*

$$\Re(z) = \frac{\mathcal{G}_{\bar\phi}\bar z - \overline{\mathcal{G}_{\bar\phi}}z}{\mathcal{G}_{\bar\phi} - \overline{\mathcal{G}_{\bar\phi}}}, \qquad \mathrm{Com}(z) = \frac{\mathcal{G}_{\bar\phi}\,(z-\bar z)}{\mathcal{G}_{\bar\phi} - \overline{\mathcal{G}_{\bar\phi}}},$$
$$\mathrm{Sh}(z) = \frac{z+\bar z}{2}, \qquad \mathrm{Ht}(z)\,\mathcal{J}_{\bar\phi} = \frac{z-\bar z}{2},$$

   where $\mathcal{G}_{\bar\phi} - \overline{\mathcal{G}_{\bar\phi}} = 2|\mathcal{G}_{\bar\phi}|\sin\bar\phi\;\mathcal{J}_{\bar\phi}$*;*

4. z *is a real number exactly when* $\bar z = z$*, and a pure Composer number exactly when* $\mathcal{G}_{\bar\phi}\bar z = \overline{\mathcal{G}_{\bar\phi}}z$*.*

Proof. (1) is Proposition 4.44, with $\Re(z) = a$ and $\mathrm{Com}(z) = b\mathcal{G}_{\bar\phi}$. (2) The real part and the Composer coordinate of a sum are the sums of those of its terms (Definition 4.17), and those of kz are ka and kb; the four formulas of (1) are built from a and b with fixed factors. (3) The mirror leaves the real numbers in place and carries sums and products to sums and products (Theorem 4.38), so $\bar z = a + b\,\overline{\mathcal{G}_{\bar\phi}}$; with $\overline{\mathcal{G}_{\bar\phi}} = 2|\mathcal{G}_{\bar\phi}|\cos\bar\phi - \mathcal{G}_{\bar\phi}$, from (5) of that theorem, this is Definition 4.37 again. Hence $z - \bar z = b\,(\mathcal{G}_{\bar\phi} - \overline{\mathcal{G}_{\bar\phi}})$, and $\mathcal{G}_{\bar\phi} - \overline{\mathcal{G}_{\bar\phi}} = 2\big(\mathcal{G}_{\bar\phi} - |\mathcal{G}_{\bar\phi}|\cos\bar\phi\big) = 2|\mathcal{G}_{\bar\phi}|\sin\bar\phi\;\;\mathcal{J}_{\bar\phi}$ by Definition 4.42, which is not zero. Dividing by it gives b, so the Composer part $b\mathcal{G}_{\bar\phi}$ is the second formula, and $\Re(z) = z - \mathrm{Com}(z)$ gives the first. The last two are (3) of Theorem 4.38 and Proposition 4.44. (4) By (3), $\bar z = z$ exactly when $b = 0$, and $\Re(z) = 0$ exactly when $\mathcal{G}_{\bar\phi}\bar z = \overline{\mathcal{G}_{\bar\phi}}z$. □

IN THE COMPLEX PLANE. At $\bar{\phi} = \pi/2$, $\mathcal{G}_{\bar{\phi}} = \mathcal{J}_{\bar{\phi}} = i$ and $\overline{\mathcal{G}_{\bar{\phi}}} = -i$, and the two pairs are one: the real part and the shadow are $\operatorname{Re} z$, and the Composer part is $i \operatorname{Im} z$, the height times i. Then (3) gives the formulas $\operatorname{Re} z = (z + \bar{z})/2$ and $i \operatorname{Im} z = (z - \bar{z})/2$, each of them twice, and (4) says that z is real when $\bar{z} = z$ and imaginary when $\bar{z} = -z$.

Theorem 4.55 (The Parts of a Product). *In a true relative plane* $\mathbb{G}_{\bar{\phi}}$*:*

1. *the parts of a product are*

$$\begin{aligned} \Re(z_1 z_2) &= \Re(z_1)\Re(z_2) - |\mathcal{G}_{\bar{\phi}}|^2 b_1 b_2, \\ \operatorname{Com}(z_1 z_2) &= \operatorname{Sh}(z_1)\operatorname{Com}(z_2) + \operatorname{Sh}(z_2)\operatorname{Com}(z_1); \end{aligned}$$

2. *its shadow and its height are*

$$\begin{aligned} \operatorname{Sh}(z_1 z_2) &= \operatorname{Sh}(z_1)\operatorname{Sh}(z_2) - \operatorname{Ht}(z_1)\operatorname{Ht}(z_2), \\ \operatorname{Ht}(z_1 z_2) &= \operatorname{Sh}(z_1)\operatorname{Ht}(z_2) + \operatorname{Sh}(z_2)\operatorname{Ht}(z_1); \end{aligned}$$

3. *for a square, in particular,*

$$\begin{aligned} \Re(z^2) &= \Re(z)^2 - |\mathcal{G}_{\bar{\phi}}|^2 b^2, \quad & \operatorname{Com}(z^2) &= 2\operatorname{Sh}(z)\operatorname{Com}(z), \\ \operatorname{Sh}(z^2) &= \operatorname{Sh}(z)^2 - \operatorname{Ht}(z)^2, & \operatorname{Ht}(z^2) &= 2\operatorname{Sh}(z)\operatorname{Ht}(z). \end{aligned}$$

Proof. (1) The real part is Proposition 4.23. By the same proposition, and with $\operatorname{Sh}(z_k) = a_k + b_k |\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}$, the Composer coordinate of $z_1 z_2$ is

$$a_1 b_2 + b_1 a_2 + 2|\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}\, b_1 b_2 = \operatorname{Sh}(z_1)\, b_2 + \operatorname{Sh}(z_2)\, b_1,$$

and multiplying by $\mathcal{G}_{\bar{\phi}}$ gives the Composer part. (2) Write both numbers in the upright frame, $z_k = \operatorname{Sh}(z_k) + \operatorname{Ht}(z_k)\, \mathcal{J}_{\bar{\phi}}$ (Proposition 4.44), and multiply out with $\mathcal{J}_{\bar{\phi}}^2 = -1$ (Theorem 4.43):

$$\begin{aligned} z_1 z_2 = {} & \big[\operatorname{Sh}(z_1)\operatorname{Sh}(z_2) - \operatorname{Ht}(z_1)\operatorname{Ht}(z_2)\big] \\ & + \big[\operatorname{Sh}(z_1)\operatorname{Ht}(z_2) + \operatorname{Sh}(z_2)\operatorname{Ht}(z_1)\big]\, \mathcal{J}_{\bar{\phi}}. \end{aligned}$$

Since $\mathcal{J}_{\bar{\phi}}$ is the arrow of length 1 straight up, a number $x + y\mathcal{J}_{\bar{\phi}}$ with real x and y lies x across and y up, so the two brackets are the shadow and the height. (3) Take $z_1 = z_2 = z$. □

IN THE COMPLEX PLANE. At $\bar{\phi} = \pi/2$ the two pairs are one: $|\mathcal{G}_{\bar{\phi}}| = 1$, the real part and the shadow are $\operatorname{Re} z_k$, and $b_k = \operatorname{Ht}(z_k) = \operatorname{Im} z_k$, so (1) and (2) are the same two laws: $\operatorname{Re}(z_1 z_2) = \operatorname{Re} z_1 \operatorname{Re} z_2 - \operatorname{Im} z_1 \operatorname{Im} z_2$ and $\operatorname{Im}(z_1 z_2) = \operatorname{Re} z_1 \operatorname{Im} z_2 + \operatorname{Re} z_2 \operatorname{Im} z_1$. For a square, $\operatorname{Re}(z^2) = a^2 - b^2$ and $\operatorname{Im}(z^2) = 2ab$.

> IN PLAIN WORDS
>
> The two halves of (1) draw on different places. The real part of a product needs only the parts: the product of the real parts, less $|\mathcal{G}_{\bar\phi}|^2$ times the product of the Composer coordinates, with neither $\cos\bar\phi$ nor $\sin\bar\phi$ in it (Proposition 4.23). The Composer part needs the shadows. It is the law of the height in (2), carried onto the Composer axis by the factor $\mathcal{G}_{\bar\phi}/(|\mathcal{G}_{\bar\phi}|\sin\bar\phi)$ of Proposition 4.54. For a square the rule is short: the Composer part of z^2 is the Composer part of z, stretched by twice the shadow of z (Figure 4.20(b)).

Proposition 4.56 (The Parts of the Conjugate, the Inverse and the Quotient)**.** *Let $z \neq 0$ and $z_2 \neq 0$.*

1. *The conjugate has*

$$\begin{aligned} \Re(\bar z) &= 2\,\mathrm{Sh}(z) - \Re(z), & \mathrm{Com}(\bar z) &= -\,\mathrm{Com}(z),\\ \mathrm{Sh}(\bar z) &= \mathrm{Sh}(z), & \mathrm{Ht}(\bar z) &= -\,\mathrm{Ht}(z), \end{aligned}$$

   so the shadow of z lies halfway between the real parts of z and $\bar z$: $\mathrm{Sh}(z) = \big[\Re(z) + \Re(\bar z)\big]/2$.

2. *The inverse has*

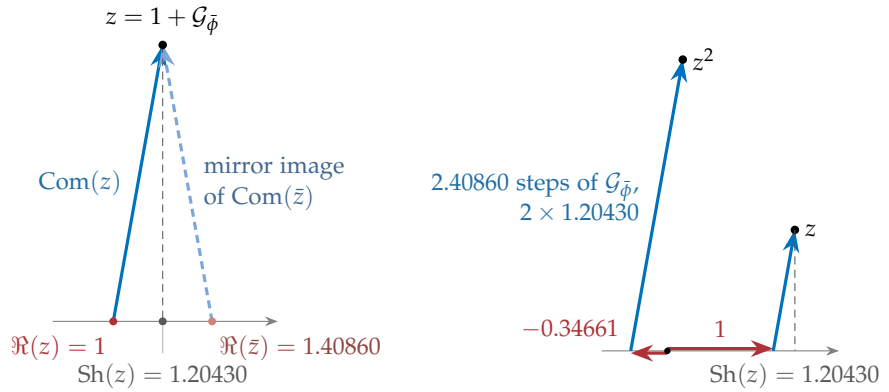
$$\begin{aligned} \Re(1/z) &= \frac{2\,\mathrm{Sh}(z) - \Re(z)}{|z|^2}, & \mathrm{Com}(1/z) &= -\frac{\mathrm{Com}(z)}{|z|^2},\\ \mathrm{Sh}(1/z) &= \frac{\mathrm{Sh}(z)}{|z|^2}, & \mathrm{Ht}(1/z) &= -\frac{\mathrm{Ht}(z)}{|z|^2}. \end{aligned}$$

3. *The quotient has*

$$\begin{aligned} \Re(z_1/z_2) &= \big[\Re(z_1)\Re(\bar z_2) + |\mathcal{G}_{\bar\phi}|^2 b_1 b_2\big]/|z_2|^2,\\ \mathrm{Com}(z_1/z_2) &= \big[\mathrm{Sh}(z_2)\,\mathrm{Com}(z_1) - \mathrm{Sh}(z_1)\,\mathrm{Com}(z_2)\big]/|z_2|^2,\\ \mathrm{Sh}(z_1/z_2) &= \big[\mathrm{Sh}(z_1)\,\mathrm{Sh}(z_2) + \mathrm{Ht}(z_1)\,\mathrm{Ht}(z_2)\big]/|z_2|^2,\\ \mathrm{Ht}(z_1/z_2) &= \big[\mathrm{Sh}(z_2)\,\mathrm{Ht}(z_1) - \mathrm{Sh}(z_1)\,\mathrm{Ht}(z_2)\big]/|z_2|^2. \end{aligned}$$

Proof. (1) By Definition 4.37, $\Re(\bar z) = a + 2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi = 2\big(a + b|\mathcal{G}_{\bar\phi}|\cos\bar\phi\big) - a = 2\,\mathrm{Sh}(z) - \Re(z)$, and $\mathrm{Com}(\bar z) = -b\mathcal{G}_{\bar\phi}$. The mirror keeps the shadow and turns the height around (Theorem 4.38). (2) By Theorem 4.41, $1/z = \bar z/|z|^2$, a real multiple of $\bar z$; apply (1) here and (2) of Proposition 4.54. (3) By the same theorem, $z_1/z_2 = z_1\bar z_2/|z_2|^2$. Apply Theorem 4.55 to z_1 and $\bar z_2$, whose Composer coordinate is $-b_2$ and whose Composer part, shadow and height are given by (1). □

On the page, (1) is an isosceles triangle (Figure 4.20(a)). The walk of z leaves the real axis at $\Re(z)$ and climbs along the Composer axis. The walk of $\bar z$ leaves it at $\Re(\bar z)$ and goes down along the same axis, by



(a) the conjugate (b) the square

Figure 4.20: Two laws of the parts at $\bar\phi = 79.86°$, for $z = 1 + \mathcal{G}_{\bar\phi}$ with $|\mathcal{G}_{\bar\phi}| = 1.16043$. (a) The conjugate (Proposition 4.56). The leg of z along the Composer axis (blue) and the mirror image of the leg of $\bar z$ (dashed; the walk of $\bar z$ itself is drawn in Figure 4.17) are both 1.16043 long. So z is the top of an isosceles triangle, and its shadow, 1.20430, lies halfway between the real parts 1 and 1.40860 of z and $\bar z$. (b) The square (Theorem 4.55). The walk of $z^2 = -0.34661 + 2.40860\,\mathcal{G}_{\bar\phi}$ goes -0.34661 along the real axis (dark red) and then 2.40860 steps of the Composer Unit (blue): twice the shadow of z, times the one step of the Composer part of z.

as many steps; mirrored back into the upper half, this leg also ends at z. The two legs are equally long, $|b|\,|\mathcal{G}_{\bar\phi}|$, so z is the top of an isosceles triangle on the real axis, and the foot of its height, the shadow, is the midpoint of the base.

In the complex plane. At $\bar\phi = \pi/2$ the real part and the shadow are one, so $\Re(\bar z) = \Re(z)$: the two legs of (1) stand on the same point, and the triangle closes up into the upright segment under z. The inverse and the quotient are those of Appendix C, with $\mathrm{Re}(1/z) = a/(a^2 + b^2)$ and $\mathrm{Im}(1/z) = -b/(a^2 + b^2)$.

Corollary 4.57 (Turning by the Two Units)**.** *In a true relative plane* $\mathbb{G}_{\bar\phi}$*:*

1. *a quarter turn, the product by the Turning Unit, gives*

$$\begin{aligned}
\Re(\mathcal{J}_{\bar\phi} z) &= -\,\mathrm{Ht}(z) - \mathrm{Sh}(z)\cot\bar\phi,\\
\mathrm{Com}(\mathcal{J}_{\bar\phi} z) &= \frac{\mathrm{Sh}(z)}{|\mathcal{G}_{\bar\phi}|\sin\bar\phi}\,\mathcal{G}_{\bar\phi},\\
\mathrm{Sh}(\mathcal{J}_{\bar\phi} z) &= -\,\mathrm{Ht}(z), \qquad \mathrm{Ht}(\mathcal{J}_{\bar\phi} z) = \mathrm{Sh}(z);
\end{aligned}$$

2. *the product by the Composer Unit gives*

$$\begin{aligned}
\Re(\mathcal{G}_{\bar\phi} z) &= -|\mathcal{G}_{\bar\phi}|^2 b, \qquad \mathrm{Com}(\mathcal{G}_{\bar\phi} z) = \Re(\bar z)\,\mathcal{G}_{\bar\phi},\\
\mathrm{Sh}(\mathcal{G}_{\bar\phi} z) &= |\mathcal{G}_{\bar\phi}|\big(\mathrm{Sh}(z)\cos\bar\phi - \mathrm{Ht}(z)\sin\bar\phi\big),\\
\mathrm{Ht}(\mathcal{G}_{\bar\phi} z) &= |\mathcal{G}_{\bar\phi}|\big(\mathrm{Sh}(z)\sin\bar\phi + \mathrm{Ht}(z)\cos\bar\phi\big).
\end{aligned}$$

Proof. Apply Theorem 4.55 with $z_1 = \mathcal{J}_{\bar\phi}$ or $z_1 = \mathcal{G}_{\bar\phi}$, and $z_2 = z$. (1) The Turning Unit has the shadow 0 and the height 1 (Theorem 4.43), so (2) of the theorem gives $\mathrm{Sh}(\mathcal{J}_{\bar\phi} z) = -\mathrm{Ht}(z)$ and $\mathrm{Ht}(\mathcal{J}_{\bar\phi} z) = \mathrm{Sh}(z)$, and Proposition 4.54 turns them into the parts. (2) The Composer Unit has the real part 0, the Composer coordinate 1, the shadow $|\mathcal{G}_{\bar\phi}| \cos\bar\phi$ and the height $|\mathcal{G}_{\bar\phi}| \sin\bar\phi$. So $\Re(\mathcal{G}_{\bar\phi} z) = -|\mathcal{G}_{\bar\phi}|^2 b$, and $\mathrm{Com}(\mathcal{G}_{\bar\phi} z) = |\mathcal{G}_{\bar\phi}| \cos\bar\phi \; b\mathcal{G}_{\bar\phi} + \mathrm{Sh}(z)\, \mathcal{G}_{\bar\phi} = \big(a + 2b|\mathcal{G}_{\bar\phi}| \cos\bar\phi\big)\mathcal{G}_{\bar\phi}$, whose coordinate is $\Re(\bar z)$ by Proposition 4.56. The shadow and the height come from (2) of the theorem. □

Multiplying by z sends $\mathcal{G}_{\bar\phi}$ to $\mathcal{G}_{\bar\phi} z = -|\mathcal{G}_{\bar\phi}|^2 b + \Re(\bar z)\, \mathcal{G}_{\bar\phi}$; this pair of parts is the second column of the matrix of z in Appendix F.

IN THE COMPLEX PLANE. At $\bar\phi = \pi/2$ both units are i, and (1) and (2) both say that $\mathrm{Re}(iz) = -\mathrm{Im}\, z$ and $\mathrm{Im}(iz) = \mathrm{Re}\, z$.

Proposition 4.58 (The Signed Area)**.** *In a true relative plane* $\mathbb{G}_{\bar\phi}$*,*

$$\begin{aligned}
\mathrm{Com}(\bar z_1 z_2) &= (a_1 b_2 - b_1 a_2)\, \mathcal{G}_{\bar\phi}, \\
\mathrm{Ht}(\bar z_1 z_2) &= \mathrm{Sh}(z_1)\, \mathrm{Ht}(z_2) - \mathrm{Ht}(z_1)\, \mathrm{Sh}(z_2), \\
\Re(\bar z_1 z_2) &= \Re(\bar z_1)\Re(z_2) + |\mathcal{G}_{\bar\phi}|^2 b_1 b_2, \\
\mathrm{Sh}(\bar z_1 z_2) &= \mathrm{Sh}(z_1)\, \mathrm{Sh}(z_2) + \mathrm{Ht}(z_1)\, \mathrm{Ht}(z_2).
\end{aligned}$$

The height $\mathrm{Ht}(\bar z_1 z_2) = (a_1 b_2 - b_1 a_2)\, |\mathcal{G}_{\bar\phi}| \sin\bar\phi$ *is the signed area of the parallelogram on the arrows* z_1 *and* z_2*, positive when* z_2 *lies counterclockwise from* z_1*, and the shadow* $\mathrm{Sh}(\bar z_1 z_2)$ *is* $|z_1|\, |z_2|$ *times the cosine of the angle between them. So* z_1 *and* z_2 *lie on one line through* 0 *exactly when* $a_1 b_2 = b_1 a_2$*, and at a right angle exactly when* $\mathrm{Sh}(\bar z_1 z_2) = 0$*.*

Proof. Apply Theorem 4.55 to $\bar z_1$ and z_2. By Proposition 4.56, $\bar z_1$ has the Composer coordinate $-b_1$, the shadow $\mathrm{Sh}(z_1)$ and the height $-\mathrm{Ht}(z_1)$, so the Composer coordinate of $\bar z_1 z_2$ is $\mathrm{Sh}(z_1)\, b_2 - \mathrm{Sh}(z_2)\, b_1$. With $\mathrm{Sh}(z_k) = a_k + b_k |\mathcal{G}_{\bar\phi}| \cos\bar\phi$ the two terms in $\cos\bar\phi$ cancel,

$$\mathrm{Sh}(z_1)\, b_2 - \mathrm{Sh}(z_2)\, b_1 = a_1 b_2 - b_1 a_2,$$

and the other three formulas follow in the same way; Proposition 4.54 turns the Composer part into the height. On the page, z_k lies $\mathrm{Sh}(z_k)$ across and $\mathrm{Ht}(z_k)$ up. For two arrows (x_1, y_1) and (x_2, y_2) the number $x_1 y_2 - y_1 x_2$ is the signed area of their parallelogram, and $x_1 x_2 + y_1 y_2$ is the product of their lengths and the cosine of their angle. □

Two laws of the parts carry neither $\cos\bar\phi$ nor $\sin\bar\phi$: the real part of a product, $\Re(z_1 z_2) = a_1 a_2 - |\mathcal{G}_{\bar\phi}|^2 b_1 b_2$ (Proposition 4.23), where the angle enters only through the length $|\mathcal{G}_{\bar\phi}|$, and the Composer part of $\bar z_1 z_2$, which is minus the Composer part of the top $z_1 \bar z_2$ of a quotient (Theorem 4.41). The second has a plain reason. The parallelogram on 1 and $\mathcal{G}_{\bar\phi}$, the cell of the leaning frame, has the area $|\mathcal{G}_{\bar\phi}| \sin\bar\phi$, and the parallelogram on z_1 and z_2 holds $a_1 b_2 - b_1 a_2$ such cells, counted with

their sign: the determinant of the two pairs of coordinates. The angle sits in the size of the cell, not in the count of cells.

IN THE COMPLEX PLANE. At $\bar{\phi} = \pi/2$, $\bar{z}_1 z_2 = (a_1 a_2 + b_1 b_2) + (a_1 b_2 - b_1 a_2)\, i$: its real part is the dot product of the two arrows, its imaginary part is their signed area, and the cell of the frame is the unit square.

Corollary 4.59 (Two Trigonometries). *Let $z \neq 0$ have the angle α, and write $C_\theta = \cos(\theta, \bar{\phi} - \theta)$ and $S_\theta = \sin(\theta, \bar{\phi} - \theta)$, as in Section 4.9.4. For every integer n,*

$$\Re(z^n) = |z|^n\, C_{n\alpha}, \qquad \mathrm{Com}(z^n) = |z|^n\, S_{n\alpha}\, \vec{u}_{\bar{\phi}},$$
$$\mathrm{Sh}(z^n) = |z|^n \cos n\alpha, \qquad \mathrm{Ht}(z^n) = |z|^n \sin n\alpha,$$

and for every General Number z,

$$\Re(e^z) = e^{\mathrm{Sh}(z)}\, C_{\mathrm{Ht}(z)}, \qquad \mathrm{Com}(e^z) = e^{\mathrm{Sh}(z)}\, S_{\mathrm{Ht}(z)}\, \vec{u}_{\bar{\phi}},$$
$$\mathrm{Sh}(e^z) = e^{\mathrm{Sh}(z)} \cos \mathrm{Ht}(z), \qquad \mathrm{Ht}(e^z) = e^{\mathrm{Sh}(z)} \sin \mathrm{Ht}(z).$$

The parts turn with the General Sinus, and the shadow and the height with the ordinary sine and cosine. For $n = 1$ the first line is the polar form of Theorem 4.45.

Proof. By Theorem 4.48, $z = |z|\, e^{\alpha \mathcal{J}_{\bar{\phi}}}$, so $z^n = |z|^n e^{n\alpha \mathcal{J}_{\bar{\phi}}}$. That theorem writes $e^{\theta \mathcal{J}_{\bar{\phi}}}$ in the leaning frame as $C_\theta + S_\theta\, \vec{u}_{\bar{\phi}}$, which gives the parts, and its proof writes it in the upright frame as $\cos\theta + \mathcal{J}_{\bar{\phi}} \sin\theta$, which gives the shadow and the height; take $\theta = n\alpha$. For the exponential, $z = \mathrm{Sh}(z) + \mathrm{Ht}(z)\, \mathcal{J}_{\bar{\phi}}$ (Proposition 4.44), so $e^z = e^{\mathrm{Sh}(z)} e^{\mathrm{Ht}(z)\mathcal{J}_{\bar{\phi}}}$, and the same two forms of $e^{\theta \mathcal{J}_{\bar{\phi}}}$, with $\theta = \mathrm{Ht}(z)$, give the four formulas. □

IN THE COMPLEX PLANE. At $\bar{\phi} = \pi/2$ the General Sinus is the ordinary sine and cosine (Corollary 3.5), and the two lines of each pair are one: $\mathrm{Re}(z^n) = |z|^n \cos n\alpha$ and $\mathrm{Im}(z^n) = |z|^n \sin n\alpha$, and $\mathrm{Re}(e^z) = e^a \cos b$ and $\mathrm{Im}(e^z) = e^a \sin b$.

Remark 4.60 (Parts Longer than the Number). The shadow and the height of a number are never longer than the number: $|\mathrm{Sh}(z)| \leq |z|$ and $|\mathrm{Ht}(z)| \leq |z|$. Its parts can be. With $n = 1$, Corollary 4.59 gives

$$|\Re(z)| = \frac{|z|\,|\sin(\bar{\phi} - \alpha)|}{\sin\bar{\phi}} \leq \frac{|z|}{\sin\bar{\phi}}, \quad |\mathrm{Com}(z)| = \frac{|z|\,|\sin\alpha|}{\sin\bar{\phi}} \leq \frac{|z|}{\sin\bar{\phi}}.$$

The real part is longest for a number at a right angle to the Composer axis, and the Composer part for a number at a right angle to the real axis, such as the Turning Unit: its length is 1, and its Composer part, $\mathcal{G}_{\bar{\phi}}/(|\mathcal{G}_{\bar{\phi}}| \sin\bar{\phi})$, is $1/\sin\bar{\phi}$ long. In the triangle of the walk, the corner between the two parts has the angle $\pi - \bar{\phi}$ or $\bar{\phi}$ (Remark 4.30), and the law of sines gives the same bound. Only at $\bar{\phi} = \pi/2$ is a part never longer than its number; at 79.86° it can be longer by at most $1/\sin\bar{\phi} - 1 \approx 1.6$ percent.

WORKED EXAMPLE: THE PARTS OF $1 + \mathcal{G}_{\bar{\phi}}$ AND ITS RELATIVES AT 79.86°

Take $z = 1 + \mathcal{G}_{\bar{\phi}}$ again, with $|\mathcal{G}_{\bar{\phi}}| \approx 1.16043$, $|\mathcal{G}_{\bar{\phi}}|^2 \approx 1.34661$, $\cot\bar{\phi} \approx 0.17885$ and $1/(|\mathcal{G}_{\bar{\phi}}|\sin\bar{\phi}) \approx 0.87542$. Its parts are $\Re(z) = 1$ and $\mathrm{Com}(z) = \mathcal{G}_{\bar{\phi}}$, its shadow is 1.20430 and its height is 1.14231.

- *The conjugate.* $\Re(\bar{z}) = 2 \times 1.20430 - 1 = 1.40860$, the real part found in Section 4.9.2, and the shadow 1.20430 lies halfway between 1 and 1.40860 (Figure 4.20(a)).
- *The square.* Theorem 4.55 gives $\Re(z^2) = 1 - 1.34661 = -0.34661$ and $\mathrm{Com}(z^2) = 2 \times 1.20430\,\mathcal{G}_{\bar{\phi}} = 2.40860\,\mathcal{G}_{\bar{\phi}}$. The same theorem gives the shadow $1.20430^2 - 1.14231^2 \approx 0.14547$ and the height $2 \times 1.20430 \times 1.14231 \approx 2.7514$, and the walk of z^2 ends there: $-0.34661 + 2.40860 \times 0.20430 \approx 0.14547$ across and $2.40860 \times 1.14231 \approx 2.7514$ up.
- *The inverse.* $\Re(1/z) = 1.40860/2.75521 \approx 0.51125$ and $\mathrm{Com}(1/z) = -\mathcal{G}_{\bar{\phi}}/2.75521 \approx -0.36295\,\mathcal{G}_{\bar{\phi}}$, the inverse of Section 4.9.2. Its shadow is $1.20430/2.75521 \approx 0.43710$ and its height is $-1.14231/2.75521 \approx -0.41460$.
- *The two turns.* Corollary 4.57 gives $\mathcal{G}_{\bar{\phi}}z \approx -1.34661 + 1.40860\,\mathcal{G}_{\bar{\phi}}$, whose Composer coordinate is $\Re(\bar{z})$, and $\mathcal{J}_{\bar{\phi}}z \approx (-1.14231 - 1.20430 \times 0.17885) + 1.20430 \times 0.87542\,\mathcal{G}_{\bar{\phi}} \approx -1.35770 + 1.05427\,\mathcal{G}_{\bar{\phi}}$. These are the products found in Sections 4.9.3 and 4.9.4.
- *An area.* The top of the quotient $(2 - \mathcal{G}_{\bar{\phi}})/(1 + \mathcal{G}_{\bar{\phi}})$ in Section 4.9.2 is $\bar{z}\,(2 - \mathcal{G}_{\bar{\phi}}) \approx 1.47059 - 3\,\mathcal{G}_{\bar{\phi}}$. Its Composer coordinate is $1 \times (-1) - 1 \times 2 = -3$, with no angle in it, and its height is $-3 \times 1.14231 = -3.42693$. So the parallelogram on $1 + \mathcal{G}_{\bar{\phi}}$ and $2 - \mathcal{G}_{\bar{\phi}}$ has the area 3.42693, and $2 - \mathcal{G}_{\bar{\phi}}$ lies clockwise from $1 + \mathcal{G}_{\bar{\phi}}$.
- *A long part.* The Turning Unit $\mathcal{J}_{\bar{\phi}} \approx -0.17885 + 0.87542\,\mathcal{G}_{\bar{\phi}}$ has the length 1, but its Composer part is $1/\sin\bar{\phi} \approx 1.0159$ long.

What the algebra adds

Every rule of Appendix C now has its counterpart in a relative plane, and each counterpart turns into the old rule when the Composer axis stands upright. Three changes run through them all. The conjugate keeps the mirror but no longer keeps the real part. The imaginary unit splits into two: the Composer Unit, the second step of the frame, and the Turning Unit, the quarter turn. And the ordinary sine and cosine give way to the General Sinus, with its two angles adding up to $\bar{\phi}$, which brings the term $2\cos\bar{\phi}$ into the addition formulas. Every result can also be read in two frames. Its shadow and its height obey the complex laws word for word; its real part and its Composer part carry the lean, and two of their laws, for $\Re(z_1 z_2)$ and for $\mathrm{Com}(\bar{z}_1 z_2)$, carry neither $\cos\bar{\phi}$ nor $\sin\bar{\phi}$. Appendix F lists every law of this section beside its complex case, together with a few more: the exponential and the logarithm of a General Number, its matrix, and the passage from one relative plane to another.

WHAT WE HAVE BUILT

At the end of Part i we hold a new kind of number, built on three simple pictures: a plane whose second axis leans, a unit that turns and grows at once, and a triangle inside every number. In the story of Chapter 1, each new kind of number was built because a reasonable question had no answer, and each one turned out, sometimes centuries later, to be exactly the language some part of physics needed. Every word that Part ii uses is now defined. There the words receive their physical task, and with it the question that the rest of the book tries to answer: can a single number carry both the mass and the charge of a particle?

The rest of the book builds on this chapter and on nothing else in mathematics. Table 4.1 lists its tools and the places where Chapters 6 to 9 put them to work: every physical step of those chapters stands on one of them.

What we have so far

- A General Number is $z = a + b\mathcal{G}_{\bar{\phi}}$: a steps along the real axis and b steps of the Composer Unit along an axis that leans at the angle $\bar{\phi}$ (Axiom 4.1). It is the sum of its two parts, $z = \Re(z) + \mathrm{Com}(z)$, and two numbers of one relative plane are equal when both parts are (Definitions 4.11 and 4.12).
- The Composer Unit has a direction $\vec{u}_{\bar{\phi}}$ and a length $\mathcal{A}_{\bar{\phi}} = |\cos(\bar{\phi}) + \sin(\bar{\phi})|$. Because that length depends on the angle, the plane is a non-unified reference (Definition 4.2). Its powers turn by $n\bar{\phi}$ and scale by $\mathcal{A}_{\bar{\phi}}^{n}$ (Theorem 4.1), so they spiral out when $\mathcal{A}_{\bar{\phi}} > 1$ and spiral in when $\mathcal{A}_{\bar{\phi}} < 1$.
- Each angle $\bar{\phi}$ gives a relative plane $\mathbb{G}_{\bar{\phi}}$ (Definition 4.3); the complex plane is the case $\bar{\phi} = \pi/2$, where $\mathcal{G}_{\pi/2} = i$ (Theorem 4.5), and the angles 0 and $3\pi/4$ close the plane into a line (Remark 4.13). The ladder of numbers grows by one rung, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \subset \mathbb{G}$ (Proposition 4.15).
- Every General Number is a triangle, and its two parts are read with the General Sinus, $a = M \cdot \cos(\alpha, \phi)$ and $b\mathcal{G}_{\bar{\phi}} = M \cdot \sin(\alpha, \phi)$ (Definition 4.8); its magnitude and its phase angle follow from Theorems 4.9 and 4.10, and its corner angle belongs to the plane (Remark 4.30).
- A product has a real term, a cross term and a square term, and its shadow on the real axis counts them with the factors 1, $\cos\bar{\phi}$ and $\cos 2\bar{\phi}$ (Proposition 4.18, Theorem 4.20).
- The square of the Composer Unit has the real part $-|\mathcal{G}_{\bar{\phi}}|^2$ in every true plane (Proposition 4.22), so the real part of a product is $a_1a_2 - |\mathcal{G}_{\bar{\phi}}|^2 b_1 b_2$, with one fixed minus sign (Proposition 4.23). General Numbers obey every rule of complex arithmetic (Remark 4.24).
- A length between 1 and $\sqrt{2}$ belongs to two angles, the arcsine root and the mirror root; only the mirror root lies in the backward band $\pi/4 < \bar{\phi} < 3\pi/4$, where the square of the Composer Unit points backward (Theorems 4.25 and 4.28, Corollary 4.29).
- All the numbers of one magnitude lie on a circle on the page, and on an ellipse in the chart of their two parts (Proposition 4.33, Theorem 4.34, Definition 4.35).
- The arithmetic of a relative plane is complete (Section 4.9). Each true relative plane is a field; the conjugate $\bar{z} = (a +$

> $2b|\mathcal{G}_{\bar\phi}|\cos\bar\phi) - b\mathcal{G}_{\bar\phi}$ is the mirror image of z, with $z\bar z = |z|^2$; the Turning Unit $\mathcal{J}_{\bar\phi}$ has $\mathcal{J}_{\bar\phi}^2 = -1$; and Euler's formula $e^{\theta\mathcal{J}_{\bar\phi}} = \cos(\theta, \bar\phi - \theta) + \sin(\theta, \bar\phi - \theta)\,\vec{u}_{\bar\phi}$ gives de Moivre's formula and the roots. Each result reads in two frames, by its two parts and by its shadow and height (Section 4.9.7). At $\bar\phi = \pi/2$ each law is its law in Appendix C, and Appendix F lists them side by side. The physical chapters do not need this last step.

TOOL	HERE	USED IN
the Composer Unit, its length and its powers	Axiom 4.1, Theorem 4.1, Definition 4.2	the unit of the cosmic mass and its size, $\|\mathcal{G}_{\bar\phi}\| = \sqrt{K/G}$ (Section 6.3.1)
the parts of a General Number	Definitions 4.11 and 4.12, Remark 4.16	the mass and the charge part (Definition 6.5); neutral bodies as real numbers (Proposition 8.2); the two coordinates of a field (Proposition 9.4)
sum and product, the three terms	Definition 4.17, Proposition 4.18	the three terms of the law (Theorem 6.6); fusion adds part by part (Proposition 8.2)
the shadow	Definition 4.19, Theorem 4.20	the second reading of the law, and why it fails (Section 6.5.3)
the Rule of the Square and the real part of a product	Propositions 4.22 and 4.23	Newton and Coulomb in one law (Proposition 6.16); the fields and the motion of Chapter 9 (Propositions 9.4 and 9.7)
the rules of arithmetic	Remark 4.24	the tools of field physics (Section 9.3)
two angles of one length, and the backward band	Theorems 4.25 and 4.28, Corollary 4.29	the angle 79.86° from the size of the Composer Unit, and why like charges push (Sections 6.3.2 and 6.6.3)
a length and two angles; the triangle; the phase angle	Definition 4.8, Theorems 4.9 and 4.10	the identity card (r, α, ϕ) of a particle (Remarks 6.11 to 6.15)
the corner, three special triangles, numbers close to the Composer axis	Remarks 4.30 and 4.31, Proposition 4.32	one body, one triangle (Section 7.1); the lean of a charged body (Remark 7.6)
all the numbers of one magnitude: circle, ellipse, rings and rays	Proposition 4.33, Theorem 4.34, Definition 4.35	the cosmic mass ellipse and the Cosmic Mass Frame (Sections 7.2 and 7.3); isotopes on the rings (Definition 8.1)

Table 4.1: The tools of this chapter, and where the physical chapters use them.

Part II

THE FOUNDATIONS OF COSMIC MASS THEORY

5

PROLOGUE: ON THE EXPRESSIVE LIMITS OF PHYSICS LANGUAGE

Before we write the law, a pause. Why should the language of physics ever need to grow? And what, exactly, do we ask it to say in this book?

ROME, 1623

In the autumn of 1618, three comets appeared in the sky over Europe. Everybody watched them, and everybody argued about them. Were they fires in the upper air, just above the clouds, as Aristotle had taught? Or were they bodies far beyond the Moon, as the Danish astronomer Tycho Brahe had argued for an earlier comet?

The following year, Orazio Grassi, a Jesuit mathematician at the Roman College, published a lecture on the comets, and he sided with Tycho. Galileo Galilei, already the most famous scientist in Italy, answered through one of his pupils. Grassi replied under a false name, in a book he called *The Astronomical and Philosophical Balance*. Galileo took his time, and in 1623 he published his answer under the sharpest title he could find: *Il Saggiatore*, *The Assayer*, after the delicate scales that goldsmiths use to weigh gold. Grassi's balance, the title implied, was far too coarse for the job (Galilei, 1623; Drake, 1957).

It is in this quarrel, in the middle of an argument about comets, that Galileo wrote the most famous passage ever written about mathematics and nature. In our own translation from the Italian:

> *Philosophy is written in this very great book which lies continually open before our eyes (I mean the universe), but it cannot be understood unless one first learns to understand the language and to know the characters in which it is written. It is written in the language of mathematics, and its characters are triangles, circles and other geometric figures, without which it is humanly impossible to understand a word of it; without these, one wanders in vain through a dark labyrinth.*

Two details deserve a second look. The passage is often quoted in a shorter and grander form: "Mathematics is the language in which God has written the universe." Galileo never wrote that line. God does not appear in the passage, and its subject is philosophy, in the old sense of natural philosophy. And Galileo's characters are not equations. They are triangles and circles, because in 1623 there was not yet any algebraic physics to speak of. Galileo worked with geometry.

The second detail is almost comic. On the comets themselves, Grassi was right and Galileo was wrong. Comets really are bodies far beyond

the Moon, and Galileo had suggested that they might be tricks of light, a little like a rainbow. He lost the argument on the facts and won it on the language, and it is the language that we remember.

Ten years later, Galileo's own story darkened. In 1633 the Roman Inquisition condemned him, and he spent the rest of his life under house arrest in his villa at Arcetri, in the hills above Florence. He could no longer publish in Italy. Yet his last great book, on motion and on the strength of materials, was carried out of the country and printed in Leiden in 1638, by which time he was blind. The man was silenced. His language was not.

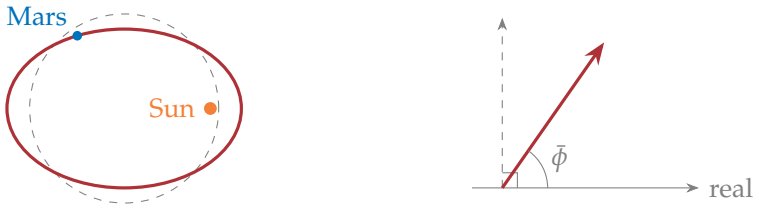
This double fate is where the present chapter begins. The Cosmic Mass Theory reads it in two directions. On the one hand, Galileo defined the relationship between the two schools of thought: mathematics is the language of physics, and without it physics remains speechless. On the other hand, and somehow, Galileo also allows us to doubt the extent of the expressive capacity of mathematics, and to ask whether this language can be extended to new concepts that physics could obey and speak. If nature is a book written in a language, we can read only as much of it as our alphabet allows.

5.1 A LANGUAGE THAT GROWS

Galileo named his alphabet: triangles, circles and other figures. The history of physics since his time is, among other things, the history of that alphabet growing larger. A few moments from that history show the pattern better than any argument can.

KEPLER LETS GO OF THE CIRCLE. Johannes Kepler believed in this alphabet, years before Galileo gave it a name, and it nearly defeated him. For two thousand years, astronomers had described the heavens with circles, and with circles riding on circles. Kepler, working in Prague with the observations that Tycho Brahe had gathered over a lifetime, spent years trying to fit the orbit of Mars with circles. His best circular model missed Tycho's observations by eight minutes of arc, about a quarter of the width of the full Moon. A less careful astronomer would have blamed the observations. Kepler wrote instead that, because these eight minutes could not be ignored, they alone showed the way to reform the whole of astronomy (Kepler, 1609). He let the circle go, and he found the ellipse.

Notice what Kepler did *not* do. He did not throw away geometry. He kept the language and enlarged one of its words. The circle did not disappear either: a circle is simply an ellipse whose two axes are equal, so the old shape survived as a special case of the new one. Part i of this book did exactly the same thing with the right angle (Figure 5.1).



(a) from the circle to the ellipse (b) from the right angle to $\bar{\phi}$

Figure 5.1: Two habits let go. (a) Kepler replaced the circular orbit of Mars (dashed) by an ellipse with the Sun at a focus, because the best circle missed the observations by eight minutes of arc. (b) The General Number Plane replaces the right angle between its two axes (dashed) by an angle $\bar{\phi}$. In both cases the old shape survives as a special case: a circle is an ellipse with equal axes, and the right angle is the case $\bar{\phi} = \pi/2$. Both drawings are exaggerated for clarity.

NEWTON DESCRIBES WITHOUT EXPLAINING. Newton's law of gravitation predicted the orbits of the planets, the tides and the return of comets. It said nothing about *how* one body pulls another across empty space, and to many of his critics this was a scandal. In 1713, in a short essay added to the second edition of the *Principia*, Newton answered with three Latin words that became famous: *hypotheses non fingo*, "I frame no hypotheses" in the English translation of 1729. He had not been able to discover the cause of gravity from the phenomena, and he would not invent one; it was enough that gravity really exists and acts according to the law he had set out (Newton, 1713, 1729). Since that day, much of physics has accepted that a mathematical law which reproduces the measurements is a complete theory, even when nobody can say what lies behind it.

THE EQUATIONS KNOW MORE THAN THEIR AUTHORS. Michael Faraday, the greatest experimenter of the nineteenth century, had almost no training in mathematics. He saw the space around a magnet as filled with lines of force, and hardly anyone took his pictures seriously, because they came without mathematics. In 1857 he wrote to the young James Clerk Maxwell and asked whether mathematicians could not translate their results out of their "hieroglyphics" into ordinary language, so that experimenters might work on them too. Maxwell did more than translate. He gave Faraday's lines a mathematical form, and by 1865 his equations contained something that Faraday's pictures had not: a wave, travelling at a speed that could be computed from two electrical constants measured on a laboratory bench. The speed came out equal to the speed of light (Maxwell, 1865). Light was an electromagnetic wave. Heinrich Hertz, who produced such waves in his laboratory in Karlsruhe in 1887, later summed up

the lesson in a single line: "Maxwell's theory is Maxwell's system of equations" (Hertz, 1893). The theory *was* its language.

KELVIN: NUMBERS, OR NOTHING MUCH. William Thomson, Lord Kelvin, spent much of his life on telegraph cables and on the definition of electrical units. In 1883 he told an audience of engineers that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind (Thomson, 1889). For Kelvin, physics is mathematical not because nature is secretly made of numbers, but because without numbers nothing can be checked, and nothing can be built.

HELMHOLTZ AND EINSTEIN: EVEN GEOMETRY CAN CHANGE. In 1870 Hermann von Helmholtz asked his listeners to imagine thinking beings who live on a surface, flat creatures on a sphere, say. Their straightest lines would be great circles, the angles of their triangles would add up to more than two right angles, and their geometry would feel as natural to them as ours does to us. The rules of geometry, Helmholtz concluded, are not laws of thought fixed in advance; we learn them from the world we live in (von Helmholtz, 1870). Forty-five years later, Albert Einstein wrote gravity itself as the curvature of space and time, with a geometry that mathematicians had built decades earlier for its own sake. In a lecture of 1921 he drew the famous line: as far as the laws of mathematics refer to reality, they are not certain, and as far as they are certain, they do not refer to reality (Einstein, 1921).

WIGNER AND HAMMING: A MIRACLE, OR A CHOICE? In 1960 the physicist Eugene Wigner gave this whole story its name: *the unreasonable effectiveness of mathematics in the natural sciences* (Wigner, 1960). Mathematicians build their ideas for their own beauty, he observed, and then those ideas turn out, again and again, to describe the world with absurd precision. He called it a miracle that we neither understand nor deserve. Twenty years later the mathematician and engineer Richard Hamming gave a more down-to-earth answer, and one of his sentences could stand as the motto of this book (Hamming, 1980):

> *Mathematics does not always work. When we found that scalars did not work for forces, we invented a new mathematics, vectors. And going further we have invented tensors.*

> **IN PLAIN WORDS**
>
> The language of physics is not fixed. When it is too small for what physics must say, somebody enlarges it: the ellipse, the calculus, vectors, curved geometry, and, in Chapter 1, the complex

> numbers. Each time, the old words survive inside the new ones as special cases. That is how the General Sinus and the General Numbers of Part i were built: the right triangle and the complex plane are still there, as the cases $\alpha + \phi = \pi/2$ and $\bar{\phi} = \pi/2$.

5.2 WHEN THE LANGUAGE IS TOO SMALL

The history above is a history of successes, and it is easy to tell it backwards, knowing how each story ended. At the moment when the language was too small, however, it was never obvious that the fault lay with the language. There were always two ways to read the difficulty.

The first reading says: *the world is not as simple as we hoped*. Nature follows one rule here and another rule there, and our task is to map where each rule applies. The second reading says: *our language is not yet rich enough*. Nature may follow one rule everywhere, and what looks like a change of rule is only the place where our words run out.

Kepler faced exactly this choice. He could have given Mars a rule of its own, a circle with special corrections valid for Mars alone, and for a while he tried several such repairs. What he found in the end was one shape, the ellipse, that worked for Mars and for every other planet.

The Cosmic Mass Theory makes the same choice on purpose. Doubting the expressive capacity of mathematics has shaped the philosophical method of the theory, rather than hastening to doubt the consistency of Existence, the cosmos itself. To see what this means in practice, we need one more scene.

MANCHESTER, 1911

In Ernest Rutherford's laboratory in Manchester, Hans Geiger and Ernest Marsden had fired tiny, fast alpha particles at a sheet of gold foil far thinner than a page. Almost all of them went straight through. A very few bounced back, which Rutherford later compared to firing a heavy shell at a sheet of tissue paper and seeing it come back at you. In 1911 he explained why: nearly all the mass of an atom, and all of its positive charge, is packed into a tiny *nucleus* at its centre (Rutherford, 1911). By 1932, when James Chadwick discovered the neutron (Chadwick, 1932), the picture was clear. A nucleus is made of protons and neutrons, pressed together within a few femtometres, a few millionths of a billionth of a metre.

And here Coulomb's law raised an alarm. Protons all carry positive charge, and like charges repel. At the distance between two protons in a nucleus, about 10^{-15} m, Coulomb's law gives a push of roughly 230 newtons. That is about the weight of a large suitcase, pushing apart two particles that each weigh about 10^{-26} newtons. By Coulomb's law alone, no nucleus with more than one proton could hold together for

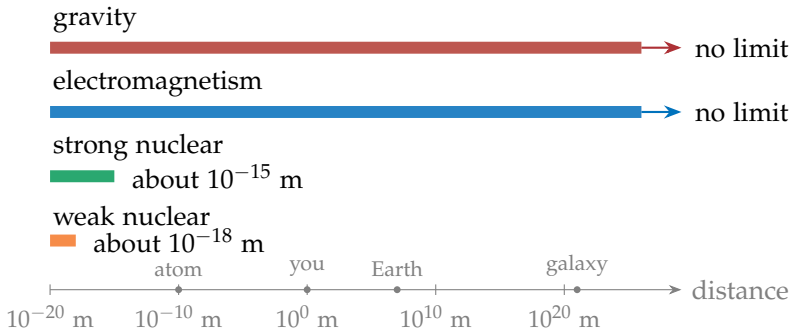


Figure 5.2: How far the four interactions reach, on a scale where each step is a factor of ten. Gravity and electromagnetism weaken as $1/r^2$ but never stop. The strong nuclear force acts across distances about the size of a nucleus, and the weak nuclear force across even shorter ones. The ranges are rough orders of magnitude.

an instant. Yet such nuclei exist, and most of the nuclei around us are perfectly stable.

Physics answered by adding a new force. The *strong nuclear force* attracts protons and neutrons to one another, more strongly than the electric repulsion pushes the protons apart, but only across very short distances: beyond the size of a nucleus it fades away almost completely. In 1935 Hideki Yukawa gave it a theory. The force, he proposed, is carried by a new particle, and the mass of that particle sets how far the force can reach (Yukawa, 1935). The particle, the pion, was found in 1947. A second new force, the *weak nuclear force*, was needed to explain a kind of radioactivity in which a neutron turns into a proton, giving out an electron and an almost invisible particle, the antineutrino (Fermi, 1934). It reaches even less far. With these two forces, physics can explain why nuclei hold together, how the Sun shines by fusing hydrogen into helium, and why very heavy nuclei can split.

So physics now counts four fundamental interactions. Two of them, gravity and electromagnetism, reach as far as you like, weakening with the square of the distance but never stopping. The other two act only inside domains the size of a nucleus, or smaller (Figure 5.2).

5.3 TWO NECESSITIES

The Cosmic Mass Theory reads this history in its own way, and it is this reading that sets the present book in motion. At this point, in this reading, physics changed its path in a structural way. It began to define a range of *intermittent domains*: places where a force must show itself and places where it must not, domains where a force can exist and domains where it cannot. The strong force, in particular, came with an existential condition: it could act only, and exclusively, within the domain of the nucleus, since otherwise its own existence would be impossible.

Such an approach, the theory argues, amounts to accepting a *necessity of inconsistency of the cosmos*: the necessity that the laws of force change from one domain of nature to another. And it asks two questions, which are the real starting point of this book. *What if the reason lay in the way we interpreted these results, given the limited expressive capacity of mathematics? Could the judgment that the cosmos is inconsistent be a kind of haste, like the haste that Galileo faced?*

Against the necessity of inconsistency, the Cosmic Mass Theory sets a double hypothesis:

1. the *necessity of consistency of the cosmos*: the same laws hold everywhere, at every scale;
2. the *necessity of extending the expressive capacity of mathematics*: past experiments may have been read imprecisely, through a language that could not yet say what they were showing.

The two hypotheses hold each other up. As that paper puts it, if a unified mathematical framework for the fundamental forces is possible, then the only reason for its existence lies in the necessity of consistency of the cosmos.

It is only fair to say how most physicists see the same history, and the theory itself is careful not to turn its reading into an accusation. The strong and weak forces are not boundaries drawn by hand. Today they are described by detailed theories that have passed an enormous number of experimental tests, and the short reach of each force comes out of its theory: in Yukawa's picture, from the mass of the particle that carries it. Nothing in this book disputes those results. The theory states its own position with a memorable image. If nine hundred and ninety-nine articles support the approach of intermittent domains of existence of forces, and the cosmic mass theory is the thousandth, then its aim is not criticism. Its aim is to broaden the diversity of the methods and approaches used to explain causes, and to avoid fanaticism towards any school or any scholar, whatever their scientific standing, since theories, like all other intellectual productions, exist to be discussed.

> **In plain words**
>
> There are two ways to react when the known laws do not seem to fit a new fact. One is to say that nature changes its rules from place to place, and to draw the borders of each rule. The other is to say that the rules may be the same everywhere, and that our mathematics is not yet rich enough to show it. This book takes the second road. It does so as a hypothesis to be tested, not as a verdict on the physics of the nucleus.

5.4 INTRINSIC AND EXTRINSIC

The double hypothesis leads the theory to sort the forces in its own way. Gravity and electrostatics are called *intrinsic* forces: they come from properties that every particle carries with it, at every moment and everywhere, namely its mass and its charge. The strong and weak nuclear forces are called *extrinsic*: in this reading, they belong to the context in which they were born, the nucleus. In the same spirit, mass and charge are treated as intrinsic concepts, while heat, radiation and energy are treated as extrinsic ones.

This sorting is a choice of starting point, not a verdict on the nuclear forces. It tells us where the Cosmic Mass Theory begins: with the two quantities that every particle carries, and with the two laws, Newton's and Coulomb's, that are written with them. That paper also hoped to bring heat and radiation into the same single number. This book is more modest. It keeps to the two intrinsic quantities, and it asks the most direct question that they allow.

5.5 WHAT THIS BOOK ASKS, AND WHAT IT DOES NOT

It is important to be exact about the scope of the theory, because a phrase like "one law for gravity and electricity" can easily be heard as more than it is. Table 5.1 sets out the boundaries.

In one sentence: the Cosmic Mass Theory takes mass and charge, the two intrinsic quantities of a particle, and gives them one number; from that number it writes one law, meant to describe both attraction and repulsion, and then it opens that law up to see what is inside. Part iii then follows the same number into the nuclei, and into bodies in motion and their fields.

5.6 THE LADDER OF CORRECTNESS

Why take such a step at all, when the two classical laws work so well on their own? The theory gives an answer that is closer to an attitude than to an argument. We may not yet have reached a high enough level of correctness in the way we read nature, it says, and so the door remains open to climb higher on the *ladder of correctness*. Experiments may remain pointless if mathematics stays speechless in front of the numbers they measure.

The theory describes physics as the expressed perception of Existence. It begins with the alphabetic language of words, and it reaches its highest form when it can speak mathematics. This is why the growth of mathematics is not a side issue for physics. Each new word in the mathematical language is a new way of perceiving. The complex plane did not only describe rotation; it made rotation something

	IN THIS BOOK	NOT ATTEMPTED
Inputs	two measured quantities per particle: its mass in kilograms [kg] and its charge in coulombs [C]	any other property of particles
Behaviours	attraction and repulsion between two particles, along the line that joins them (Chapter 6); the masses of nuclei and the energy of fusion (Chapter 8); bodies in motion and their fields (Chapter 9)	the strong and weak nuclear forces themselves; a unification of the four fundamental forces
Laws	Newton's law and Coulomb's law as the real part of one law, which also contains a new term, the Coupling Force, not read as a force; the Closure Law of the nuclei; one law of fields and motion that holds Newton's gravity, Maxwell's equations and the force of Lorentz	heat; quantum versions of the laws; gravity where general relativity is needed
Mathematics	the General Sinus and the General Number Plane, which contain classical trigonometry and the complex numbers as special cases	replacing the mathematics of physics; the classical tools remain valid
Status	a proposal, stated with all its assumptions, to be tested; its new predictions are the work of Volume 2	an established result of physics

Table 5.1: The scope of this book. The Cosmic Mass Theory starts from the two intrinsic quantities of a particle and the two behaviours they produce.

that engineers can see, add and multiply without a second thought, as Steinmetz showed for alternating currents in 1893 (Steinmetz, 1893).

Part i has already built the new words. The General Sinus lets a triangle lean. The General Number Plane lets the second axis of the number plane lean. It is time to give these words their first physical sentence, and to ask the question that the whole book has been preparing: *can a single number carry both the mass and the charge of a particle, and can one law, written with that number, hold for every pair of particles?*

What we have so far

- Galileo wrote that nature is a book in the language of mathematics, with triangles, circles and other figures as its characters. We can read only as much of that book as our alphabet allows.

- Physics has enlarged its alphabet many times: Kepler's ellipse, Newton's law that describes without explaining, Maxwell's equations, the geometries of Helmholtz and Einstein, vectors and tensors. Each time, the old words survived as special cases.
- When the facts and the laws disagree, we can doubt the world, and give each force its own domain, or doubt the language. The Cosmic Mass Theory chooses the second road: the necessity of consistency of the cosmos, and the necessity of a more expressive mathematics.
- The scope of this book is deliberately narrow: mass and charge as inputs, attraction and repulsion as behaviours, and one law that contains the laws of Newton and Coulomb together with one new term; Part iii carries the same number into the nuclei and into bodies in motion (Table 5.1).

6

THE INTERACTION LAW

Mass and charge enter the General Number Plane: one number for each particle, one law for every pair of particles, a triangle that becomes a new identity card for matter, and one way of reading the law, in its own plane, in which like charges push apart with the full strength of Coulomb's law.

LONDON, 1849

On 19 March 1849, Michael Faraday wrote five words in capital letters in his laboratory notebook: "ALL THIS IS A DREAM." He had been thinking about gravity. Eighteen years earlier he had discovered that a moving magnet drives an electric current around a coil of wire, and he had spent his life finding links between powers of nature that others believed to be separate: electricity and magnetism, magnetism and light. Now he wondered whether gravity, too, was tied to electricity. If a body falling towards the Earth could stir up electricity, the two would be two faces of a single power.

The dream did not stop him. "Still examine it by a few experiments," he went on in the same entry. "Nothing is too wonderful to be true, if it be consistent with the laws of nature; and in such things as these, experiment is the best test of such consistency." He wound hundreds of feet of copper wire into a coil, connected it to a sensitive galvanometer, and let the coil fall again and again from a height of about eleven metres onto a cushion, watching the needle for the smallest kick (Figure 6.1).

The needle did not move. In his report to the Royal Society, Faraday wrote: "Here end my trials for the present. The results are negative. They do not shake my strong feeling of the existence of a relation between gravity and electricity, though they give no proof that such a relation exists" (Faraday, 1851).

This chapter takes up Faraday's feeling with the tools of Part i. We will not drop a coil; we will look instead at the two laws themselves.

Physics sorts the forces of nature into four kinds: gravitational, electromagnetic, weak and strong. Two of them, gravity and electricity, are written in laws of the same shape (Figure 1.1). Yet one law speaks of mass and the other of charge, and a particle such as the proton carries both at once. Could the two laws be one law, read in two ways?

This chapter answers with one new object, the **Cosmic Mass**, which writes the mass and the charge of a particle as a single General Number. It is built with the **Composer Unit** $\mathcal{G}_{\bar{\phi}}$ of Chapter 4, the second unit of a number plane whose second axis leans; for the cosmic mass

[First draft, November 9, 2026]

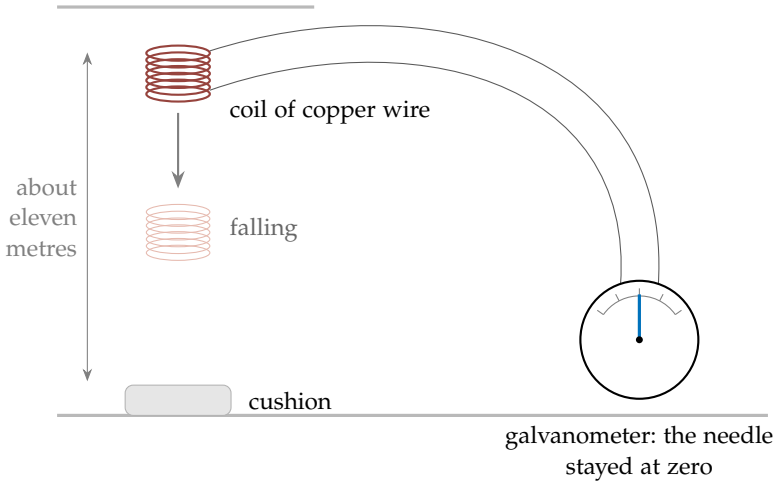


Figure 6.1: Faraday's test of 1849, as he described it (not to scale). A coil of copper wire falls from a height of about eleven metres onto a cushion, and long wires join it to a sensitive galvanometer. If falling under gravity stirred up electricity, the needle would kick. It stayed at zero (Faraday, 1851).

that axis leans at 79.86°. One law built on cosmic masses, read in the plane in which it lives, gives back Newton's gravity and Coulomb's

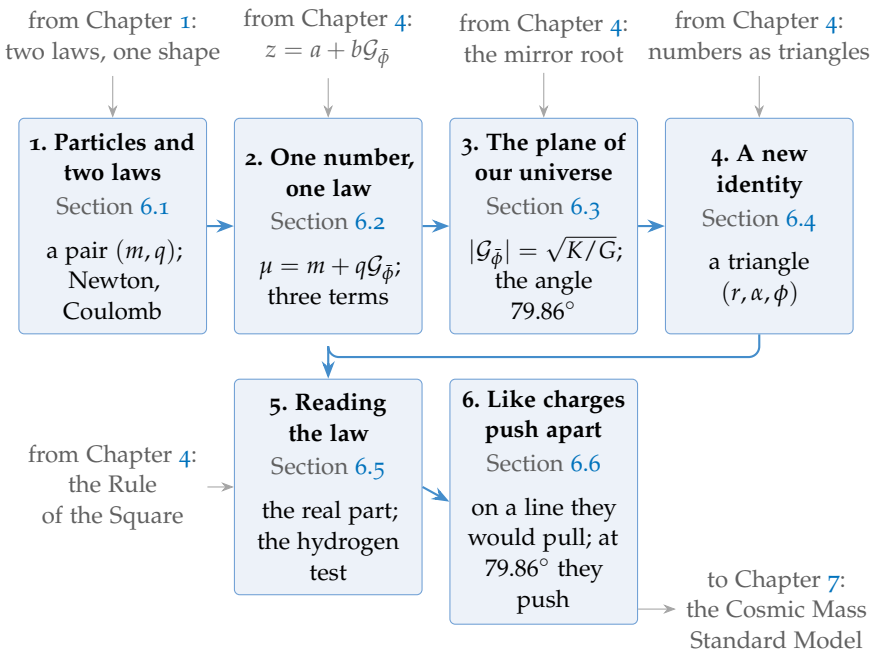


Figure 6.2: The road through the chapter. Each box is one step, with the section where it is taken. Ideas come in from Chapters 1 and 4, and the last step shows why like charges push apart at the angle 79.86°, on which Chapter 7 builds.

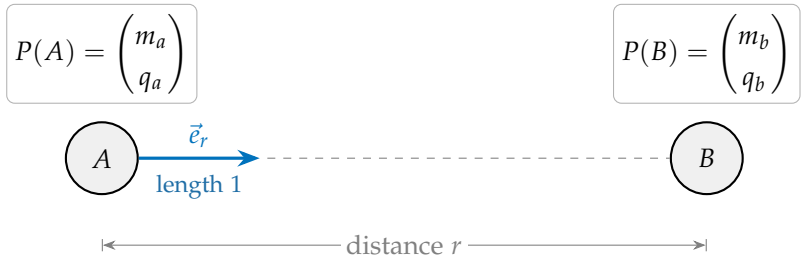


Figure 6.3: Two particles and their signatures (Definition 6.1 and Remark 6.2). Each particle carries an identity card with two lines, its mass and its charge. The arrow $\vec{e}_r$ has length one and points from A to B; the distance between the two particles is r.

law, each at full strength, and its third term, the **Coupling Force**, finds its place in the Composer coordinate.

The chapter is a concrete attempt to rethink the algebraic and geometric foundation of force and of the identity of a particle, not an abstract exercise. Every formal statement is followed by plain words, and every calculation is written out step by step.

THE ROAD THROUGH THIS CHAPTER. The chapter climbs in six steps, shown in Figure 6.2. The first three build the cosmic mass, its law and the plane in which it lives; the fourth gives every particle a new identity card; the last two read the law in its own plane, test the reading on the hydrogen atom, and show why like charges push apart. Every step takes its mathematics from Chapter 4, and Table 4.1 lists where.

6.1 TWO NUMBERS, TWO LAWS

6.1.1 *A Particle and Its Signature*

Let us begin by considering two particles labeled $P(A)$ and $P(B)$. Each particle is characterized not only by its mass and charge but by the simultaneous presence of both as essential attributes.

Definition 6.1 (Particle Signature)**.** We define each particle as a column vector consisting of its mass and electric charge:

$$P(A) = \begin{pmatrix} m_a \\ q_a \end{pmatrix}, \qquad P(B) = \begin{pmatrix} m_b \\ q_b \end{pmatrix}$$

where:

- $m_a, m_b \in \mathbb{R}$ are the masses in kilograms $[kg]$
- $q_a, q_b \in \mathbb{R}$ are the electric charges in Coulombs $[C]$

Remark 6.2 (Unit Direction Vector). Let $\vec{e}_r$ be the normalized direction vector that points from particle A to particle B. It satisfies:

$$|\vec{e}_r| = 1$$

and defines the directional component of all force vectors in the system.

> IN PLAIN WORDS
>
> A particle signature is an identity card with two lines: the mass on top, the charge below. Here are three of them, with rounded values (Appendix D.3 gives more digits):
>
> $$P(\text{electron}) = \begin{pmatrix} 9.109 \times 10^{-31} \\ -1.602 \times 10^{-19} \end{pmatrix},$$
> $$P(\text{proton}) = \begin{pmatrix} 1.673 \times 10^{-27} \\ 1.602 \times 10^{-19} \end{pmatrix},$$
> $$P(\text{neutron}) = \begin{pmatrix} 1.675 \times 10^{-27} \\ 0 \end{pmatrix},$$
>
> with masses in kilograms and charges in coulombs. The arrow $\vec{e}_r$ of Remark 6.2 has length one, so it carries only a direction: the line from A to B (Figure 6.3). In every force of this chapter, $\vec{e}_r$ says *where* the force acts, and the rest of the formula says *how strongly*.

6.1.2 *The Laws of Newton and Coulomb*

Definition 6.3 (Newtonian Gravitation)**.** The gravitational force acting between two particles at a distance r apart is:

$$\vec{F}_N = G \cdot \frac{m_a m_b}{r^2} \cdot \vec{e}_r$$

where G is the gravitational constant.

Definition 6.4 (Coulomb Electrostatics)**.** The electrostatic force acting between the two charged particles is:

$$\vec{F}_C = K \cdot \frac{q_a q_b}{r^2} \cdot \vec{e}_r$$

where K is the Coulomb constant.

While these two laws are treated independently, the algebraic structure of their force vectors reveals a parallelism: both forces are radially directed, follow inverse-square laws, and scale with scalar products of intrinsic properties.

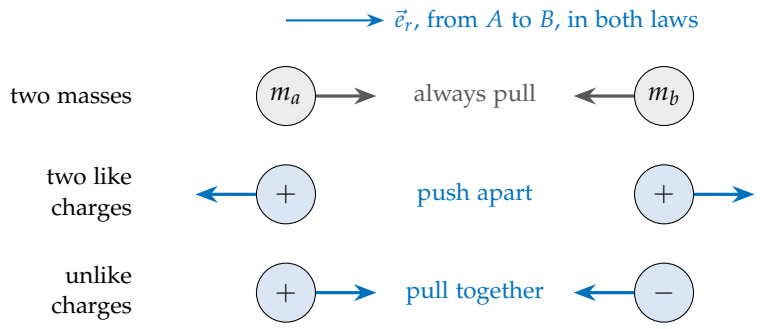


Figure 6.4: Pull and push. Two masses always pull together. Two like charges push apart, and unlike charges pull together. Definitions 6.3 and 6.4 write both laws along the same arrow $\vec{e}_r$, and leave the direction of each force to the nature of the interaction.

> **In plain words**
>
> These are the two laws of Figure 1.1, now written in the formal language of the chapter. Two differences between them will matter in what follows. The first is size: the ratio of the two constants is $K/G \approx 1.35 \times 10^{20}$, a number that will return in Section 6.3. The second is behaviour: two masses always pull together, while two like charges push apart (Figure 6.4). Both formulas are written with the same arrow $\vec{e}_r$; they give the line of the force and its strength, and the difference between pulling and pushing is left to the nature of each interaction. Section 6.6 will show that the cosmic mass gives this difference a reason.

6.2 ONE NUMBER, ONE LAW

6.2.1 *The Cosmic Mass*

Definition 6.5 (Cosmic Mass)**.** The Cosmic Mass of a particle is defined as:

$$\mu = m + q \cdot \mathcal{G}_{\bar{\phi}}$$

This scalar belongs to the General Number Plane $\mathbb{G}$ and has physical units of $[kg]$, where:

- $m \in \mathbb{R}$: classical mass
- $q \in \mathbb{R}$: electric charge
- $\mathcal{G}_{\bar{\phi}}$: Composer Unit with unit $\left[\frac{kg}{C}\right]$

This unified representation binds the two classical quantities, mass and charge, into a single mathematical object. This enables a single formulation for what were previously two separate physical laws.

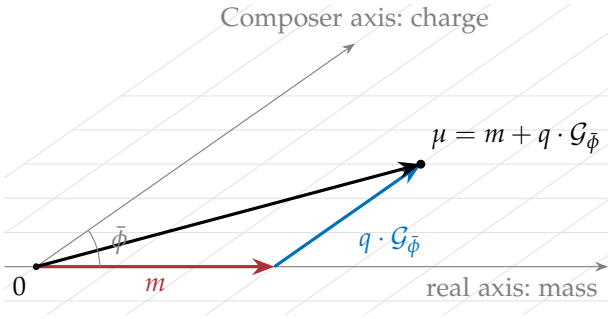


Figure 6.5: The cosmic mass of a particle as a General Number (Definition 6.5): m steps along the real axis, which carries mass, and q steps of the Composer Unit along the Composer axis, which carries charge. The angle $\bar{\phi}$ is drawn at 35° and the two parts with comparable lengths, so that the construction is easy to see; Section 6.3 finds the angle of our universe.

> **In plain words**
>
> The Cosmic Mass is a General Number in the sense of Definition 4.11. Its real part is the mass, $\Re(\mu) = m$, measured along the real axis, which this chapter therefore calls the *mass axis*. Its Composer coordinate is the charge q, and its Composer part, $\mathrm{Com}(\mu) = q\mathcal{G}_{\bar{\phi}}$, is q steps of the Composer Unit along the Composer axis, which carries charge (Figure 6.5). We call m the *mass part* and $q\mathcal{G}_{\bar{\phi}}$ the *charge part* of the cosmic mass. A neutral body has no Composer part, so its cosmic mass is a real number, $\mu = m$; a charge without mass would be a pure Composer number (Remark 4.16).
>
> The units work out just as they should. A charge in coulombs times a Composer Unit in kilograms per coulomb gives kilograms, $[C] \cdot [kg/C] = [kg]$, so both parts of μ are measured in kilograms. That is why the Cosmic Mass can be called a mass.

From here on, every word of Chapter 4 has a physical name, and Table 6.1 sets the two vocabularies side by side.

6.2.2 *The Cosmic Interaction Law*

Theorem 6.6 (Cosmic Interaction Law). *Let $\mu_a = m_a + q_a\mathcal{G}_{\bar{\phi}}$ and $\mu_b = m_b + q_b\mathcal{G}_{\bar{\phi}}$ be the Cosmic Masses of particles A and B. Then the total interaction force is given by:*

$$\vec{F}_G = G \cdot \frac{\mu_a \mu_b}{r^2} \cdot \vec{e}_r$$

CHAPTER 4: MATHEMATICS		CHAPTERS 6 TO 9: PHYSICS
General Number $z = a + b\mathcal{G}_{\bar{\phi}}$	4.11	cosmic mass $\mu = m + q\mathcal{G}_{\bar{\phi}}$
real axis; Composer axis	4.1	mass axis; Composer axis, or charge axis
real part $\Re(z) = a$	4.11	mass m, the mass part
Composer coordinate b	4.11	charge q
Composer part $\mathrm{Com}(z) = b\mathcal{G}_{\bar{\phi}}$	4.11	charge part $q\mathcal{G}_{\bar{\phi}}$, in kilograms
length $\mathcal{A}_{\bar{\phi}}$ of the Composer Unit	4.1	size $\sqrt{K/G}$, read as the length 1.16043
real number; pure Composer number	4.16	neutral body; a charge without mass
real term; cross term; square term	4.18	gravity; Coupling Force; Coulomb's force
real part of a product	4.23	every measured force, energy and momentum (Definitions 6.17 and 9.1)
Rule of the Square, $\Re(\mathcal{G}_{\bar{\phi}}^2) = -\lvert\mathcal{G}_{\bar{\phi}}\rvert^2$	4.22	the minus sign between gravity and Coulomb's force
shadow on the real axis	4.19	not used to read forces
mirror root	4.25	the angle 79.86° of the Composer axis
backward band, $\pi/4 < \bar{\phi} < 3\pi/4$	4.28	$\mathcal{G}_{\bar{\phi}}^2$ points back along the mass axis: like charges push
magnitude M; angles α and ϕ	4.8	cosmic magnitude r; charge and mass phase angles
ratio of the two parts, $\eta = \mathcal{A}_{\bar{\phi}}\, b/a$	4.10	$\eta = \mathcal{G}_{\bar{\phi}}\, q/m$
equal parts, $\eta = 1$	4.31	equal parts; the balanced body
angle δ from the Composer axis	4.32	the lean of a charged body
ellipse of one magnitude	4.34	cosmic mass ellipse
rings and rays	4.35	Cosmic Mass Frame
sum, part by part	4.17	fusion: the charges add, the masses add less the loss

Table 6.1: One object, two vocabularies. On the left, the words of Chapter 4, with the number of the statement that defines each one; on the right, the names they take for the cosmic mass in Chapters 6 to 9.

Expanding this, we obtain:

$$\begin{aligned}\vec{F}_G = {} & G \cdot \frac{m_a m_b}{r^2} \cdot \vec{e}_r \\ & + \mathcal{G}_{\bar{\phi}}^2 \cdot G \cdot \frac{q_a q_b}{r^2} \cdot \vec{e}_r \\ & + \mathcal{G}_{\bar{\phi}} \cdot G \cdot \frac{m_a q_b + q_a m_b}{r^2} \cdot \vec{e}_r\end{aligned}$$

	m_b	$q_b\,\mathcal{G}_{\bar{\phi}}$
m_a	$m_a m_b$ gravity	$m_a q_b\,\mathcal{G}_{\bar{\phi}}$ Coupling Force
$q_a\,\mathcal{G}_{\bar{\phi}}$	$q_a m_b\,\mathcal{G}_{\bar{\phi}}$ Coupling Force	$q_a q_b\,\mathcal{G}_{\bar{\phi}}^2$ Coulomb-like

Figure 6.6: Multiplying out the two brackets, as at school (Theorem 6.6 and Proposition 4.18). The product $(m_a + q_a\mathcal{G}_{\bar{\phi}})(m_b + q_b\mathcal{G}_{\bar{\phi}})$ is the area of the whole rectangle. Mass times mass gives gravity (grey), charge times charge gives the Coulomb-like term with $\mathcal{G}_{\bar{\phi}}^2$ (blue), and the two cross products, each mass with the other particle's charge, give the Coupling Force (red). The pieces are not drawn to scale.

> IN PLAIN WORDS
>
> The law has exactly the shape of Newton's law, with cosmic masses in place of masses. Its three lines are the three terms of a product of two General Numbers (Proposition 4.18): the real term gives gravity, the cross term the Coupling Force and the square term the Coulomb-like force (Figure 6.6):
>
> $$(m_a + q_a\mathcal{G}_{\bar{\phi}})(m_b + q_b\mathcal{G}_{\bar{\phi}}) = m_a m_b + (m_a q_b + q_a m_b)\,\mathcal{G}_{\bar{\phi}} + q_a q_b\,\mathcal{G}_{\bar{\phi}}^2.$$

Remark 6.7. The force decomposes naturally into three components:

- Gravitational force (mass-mass)
- Coulomb-like force (charge-charge), scaled by $\mathcal{G}_{\bar{\phi}}^2$
- A novel cross-coupling term (mass-charge), scaled by $\mathcal{G}_{\bar{\phi}}$

This third term introduces a possible new fundamental interaction, which we call the **Coupling Force**.

> IN PLAIN WORDS
>
> Three things can be read straight off the law.
>
> - For two neutral particles, $q_a = q_b = 0$, the second and third lines vanish and only Newton's gravity is left. Nothing changes for uncharged matter.
> - The Coupling Force vanishes whenever $m_a q_b + q_a m_b = 0$. For example, between a particle and its antiparticle, which have the same mass and opposite charges, the Coupling Force is exactly zero. (The Composer coordinate of their

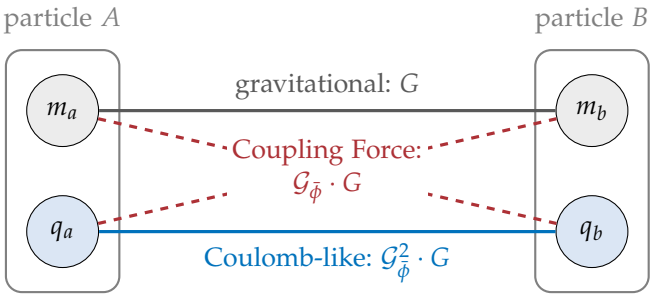


Figure 6.7: Three components of one law (Theorem 6.6 and Remark 6.7). Each pairing has its own constant: G for mass with mass, $\mathcal{G}_{\bar{\phi}}^2 \cdot G$ for charge with charge, and $\mathcal{G}_{\bar{\phi}} \cdot G$ for each mass with the other particle's charge.

product, found in Section 6.5.2, is still not zero: it also holds a share of the term of two charges.)

- The second line looks like Coulomb's law, with $\mathcal{G}_{\bar{\phi}}^2 \cdot G$ in the place of K. The next section asks what the Composer Unit must be for the two to agree.

6.3 THE RELATIVE PLANE OF OUR UNIVERSE

The Composer Unit has a size and an angle. This section finds both from the constants of nature, first the size and then the angle (Figure 6.8).

6.3.1 *The Size of the Composer Unit*

To determine the projection angle $\bar{\phi}$, we propose using the well-known numerical values of the two universal constants:

$$G = 6.67430 \times 10^{-11} \quad (\mathrm{Nm^2Kg^{-2}}),$$
$$K = 8.98755 \times 10^{9} \quad (\mathrm{Nm^2C^{-2}})$$

By imposing that the coefficient of the Coulomb force term in the expanded Cosmic Force equation matches the Coulomb constant, i.e.,

$$\mathcal{G}_{\bar{\phi}}^2 \cdot G = K \Longrightarrow \mathcal{G}_{\bar{\phi}}^2 = \frac{K}{G} \quad (\mathrm{Kg^2C^{-2}}) \quad ,$$
$$\mathcal{G}_{\bar{\phi}} = \sqrt{\frac{K}{G}} \quad (\mathrm{KgC^{-1}})$$
$$\mathcal{G}_{\bar{\phi}}^2 = 1.3466 \times 10^{20} \quad (\mathrm{Kg^2C^{-2}}) \quad ,$$
$$\mathcal{G}_{\bar{\phi}} = 1.16043 \times 10^{10} \quad (\mathrm{KgC^{-1}})$$

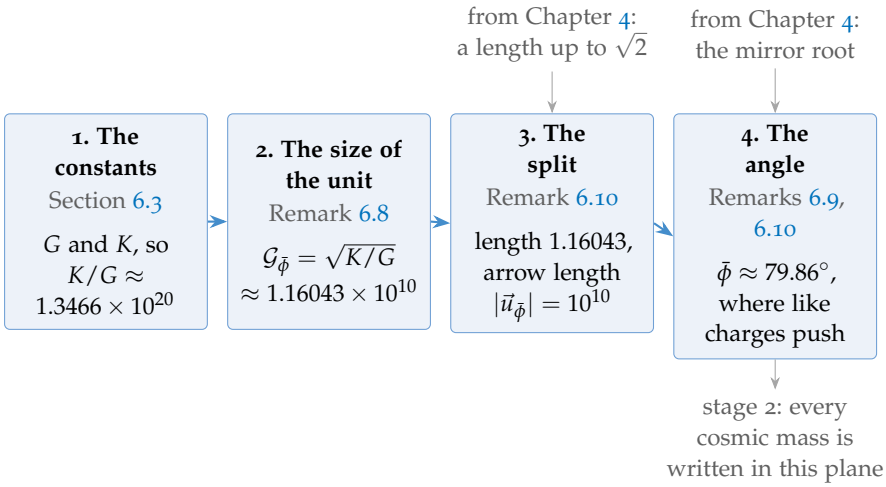


Figure 6.8: From the constants of nature to the angle of the Composer axis (stage 1 of Section 6.3). The constants G and K fix the size of the Composer Unit in kilograms per coulomb. Remark 6.10 splits that size into a length and a power of ten, and Remark 6.9 turns the length into an angle. The book takes the mirror root of Theorem 4.25, 79.86°, the angle at which like charges push apart (Section 6.6.3). Every later calculation is done in the relative plane with that angle (stage 2).

In words: the Composer Unit is about 1.16043×10^{10} kilograms per coulomb in size. The box below checks the arithmetic, and Remark 6.8 shows what this size is made of.

> **Worked example: checking the numbers**
>
> Divide the two constants: $K/G = (8.98755 \times 10^{9})/(6.67430 \times 10^{-11}) \approx 1.3466 \times 10^{20}$. Its square root is $\sqrt{1.3466 \times 10^{20}} \approx 1.16043 \times 10^{10}$. The units follow the same division: newton square metres per coulomb squared, divided by newton square metres per kilogram squared, leave kilograms squared per coulomb squared, and the square root leaves kilograms per coulomb, the unit that Definition 6.5 gives to the Composer Unit.

Remark 6.8 (Simplified Composer Unit Expression).

$$\mathcal{G}_{\bar{\phi}} = c\sqrt{\frac{\mu_0}{4\pi G}},$$

making the speed of light c a direct factor in the Composer Unit amplitude.

This form further emphasizes the connection between electromagnetic constants and gravitational constant G in setting the geometry of the General Number Plane.

In plain words

Why does the speed of light appear? In the international system of units, the Coulomb constant can be written $K = \mu_0 c^2/(4\pi)$, where μ_0 is the magnetic constant and c the speed of light. Put this into $\mathcal{G}_{\bar{\phi}} = \sqrt{K/G}$, and the square root splits into $c\sqrt{\mu_0/(4\pi G)}$.
That the speed of light hides inside the electric constants was one of the great surprises of the nineteenth century. In 1856 Wilhelm Weber and Rudolf Kohlrausch compared two ways of measuring the same electric charge and found that the ratio between them was a speed. Maxwell noticed that, in his units, it came out almost exactly equal to the speed of light, and concluded that light is an electromagnetic wave (Maxwell, 1865).

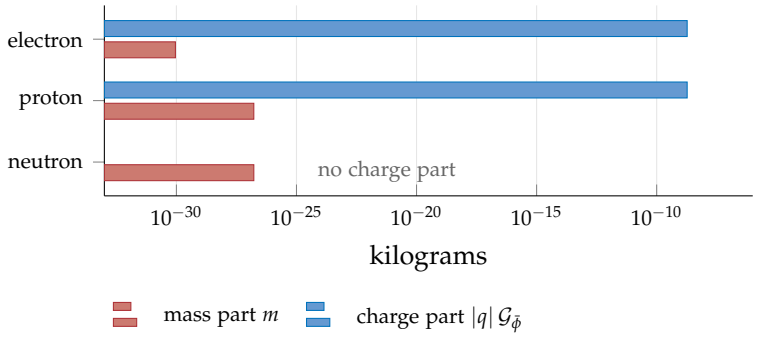
Worked example: the two parts of a cosmic mass

How large is each part of a real particle's cosmic mass? Take the electron, with $m = 9.109 \times 10^{-31}$ kg and $q = -1.602 \times 10^{-19}$ C. With the size of the Composer Unit found above, 1.16043×10^{10} kg/C, the charge part has size

$$|q| \times 1.16043 \times 10^{10} \approx 1.859 \times 10^{-9} \text{ kg}.$$

The charge part is about 2×10^{21} times larger than the mass part. For the proton, the same calculation gives a ratio of about 1.1×10^{18}. Square that ratio and you get about 1.2×10^{36}: the factor by which the electric push between two protons beats their gravitational pull, the number we met in Chapter 1. This is no accident. The condition $\mathcal{G}_{\bar{\phi}}^2 \cdot G = K$ was chosen precisely so that, inside the Cosmic Interaction Law, the charge part of a cosmic mass weighs as much as the charge does in Coulomb's law.
Figure 6.9 draws these sizes on a scale of powers of ten, together with the three terms of Theorem 6.6 for two protons one metre apart. Read as ordinary forces along $\vec{e}_r$, the gravitational term is about 1.9×10^{-64} newtons, the Coulomb-like term about 2.3×10^{-28} newtons, and the Coupling Force about 4.2×10^{-46} newtons. The Coupling Force sits halfway between the other two, up to a factor of two, because its coefficient is the geometric mean of the two constants: $\mathcal{G}_{\bar{\phi}} \cdot G = \sqrt{K/G} \cdot G = \sqrt{KG}$. How the three terms combine into one measured force is the subject of Section 6.5.1.

(a) the two parts of a cosmic mass



(b) the three terms for two protons one metre apart

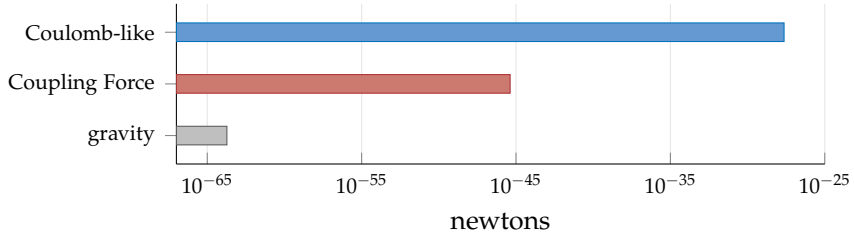


Figure 6.9: Sizes on a scale of powers of ten. (a) The two parts of the cosmic masses of the electron, the proton and the neutron, in kilograms. The charge part $|q|\,\mathcal{G}_{\bar{\phi}} \approx 1.859 \times 10^{-9}$ kg is the same for the electron and the proton, and the neutron has none. (b) The three terms of Theorem 6.6 for two protons one metre apart, in newtons, if each is read as an ordinary force along $\vec{e}_r$. The Coupling Force sits halfway between gravity and the Coulomb-like term, up to a factor of two.

6.3.2 *From the Size to an Angle*

The Composer Unit is not a universal constant but rather a parametrized unit whose behavior depends on a geometric angle $\bar{\phi}$. As $\bar{\phi}$ varies from 0 to π, the amplitude $\mathcal{A}_{\bar{\phi}} = \left|\sqrt{2}\sin\left(\bar{\phi} + \frac{\pi}{4}\right)\right|$ varies between 0 and $\sqrt{2}$, modulating both the scalar strength and directional behavior of the Composer axis.

Remark 6.9 (Expressions for the Projection Angle $\bar{\phi}$). Based on the amplitude definitions of the Composer Unit $\mathcal{G}_{\bar{\phi}}$ and its square $\mathcal{G}^2_{\bar{\phi}}$, we can express the projection angle $\bar{\phi}$ as follows:

$$\bar{\phi} = \arcsin\left(\frac{\mathcal{G}_{\bar{\phi}}}{\sqrt{2}}\right) - \frac{\pi}{4}, \qquad \text{and} \qquad \bar{\phi} = \frac{1}{2}\arcsin\left(\mathcal{G}^2_{\bar{\phi}} - 1\right).$$

These inverse relations enable solving for $\bar{\phi}$ given the amplitude values of $\mathcal{G}_{\bar{\phi}}$ or $\mathcal{G}^2_{\bar{\phi}}$, facilitating the determination of the Subgeneral Plane configuration.

> **IN PLAIN WORDS**
>
> These are the formulas of Theorem 4.25, written with the symbol $\mathcal{G}_{\bar{\phi}}$ for the length of the Composer Unit. Each turns a length into an angle.

Remark 6.10 (Numerical Value of the Projection Angle $\bar{\phi}$). Using the physical constants to determine the Composer Unit modulus and considering the direction vector magnitude as

$$|\vec{u}_{\bar{\phi}}| = 10^{10},$$

we obtained the normalized Composer Unit value:

$$\mathcal{G}_{\bar{\phi}} \approx 1.16043 \cdot \vec{u}_{\bar{\phi}}.$$

Applying the formula

$$\bar{\phi} = \arcsin\left(\frac{1.16043}{\sqrt{2}}\right) - \frac{\pi}{4},$$

we calculate the projection angle to be

$$\bar{\phi} \approx 0.1770 \text{ radians} \approx 10.14^{\circ}.$$

This value of $\bar{\phi}$ defines the specific skew angle of the Composer axis relative to the real axis for the physical universe modeled within the General Number Plane framework. It quantifies the departure from orthogonality and sets the geometric basis for unifying gravitational and electromagnetic interactions.

> **IN PLAIN WORDS**
>
> Remark 6.10 splits the size of the Composer Unit, 1.16043×10^{10}, into two factors. The power of ten becomes the length of the direction arrow, $|\vec{u}_{\bar{\phi}}| = 10^{10}$, and the number 1.16043 is read as the length $\mathcal{A}_{\bar{\phi}}$ of the Composer Unit, which Axiom 4.1 calls its amplitude. The number fits that role, since the length lies between 0 and $\sqrt{2} \approx 1.414$ (Figure 4.5). The formula of Remark 6.9 then turns the length into an angle: $1.16043/\sqrt{2} \approx 0.82055$, its arcsine is about 55.14°, and taking away 45° leaves 10.14°.

The same length belongs to a second angle. By Theorem 4.25, a length between 1 and $\sqrt{2}$ is reached at two angles, mirror images of each other about 45°, and the arcsine button of a calculator returns only the first. The second, the mirror root, is

$$\bar{\phi} = \frac{\pi}{2} - 0.1770 \approx 1.3938 \text{ radians} \approx 79.86^{\circ},$$

and it is the angle of the Composer axis in this book, from here to the last page.

The book takes this angle because it shows why like charges push apart and do not pull. The term of two charges in the law of Theorem 6.6 carries the square $\mathcal{G}_{\bar{\phi}}^2$ (Remark 6.7), and the square leans at twice the angle of the Composer Unit (Theorem 4.1). At 79.86° it leans at 159.72°: past the upright, pointing backward, against the mass axis along which two masses pull. This is the backward band of Theorem 4.28, and the mirror root lies inside it (Corollary 4.29). So in this plane the arrow of two like charges points against the arrow of two masses: where masses pull, like charges push. Section 6.6.3 checks this with both readings of the law.

Once the angle $\bar{\phi}$ is set, all future computations of the General scalar

$$\mu = m + q\mathcal{G}_{\bar{\phi}}$$

and cosmic interactions are carried out within this fixed Relative Plane.

In summary, the characterization of the General Number Plane occurs in two stages:

1. **Angle Selection:** Compute or assume a specific $\bar{\phi}$ based on physical constants or modeling choices.
2. **Activation:** Use $\mathcal{G}_{\bar{\phi}}$ as the base Composer Unit in all scalar-vector definitions and operations.

> IN PLAIN WORDS
>
> Remember the fan of relative planes in Figure 4.6: each blade is a Composer axis at its own angle, with its own length. The General Number Plane is the whole fan, but a physical theory must work in one blade, and the two stages above choose it (Figure 6.8): the blade at 79.86°, sized by the constants of nature.

6.4 A NEW IDENTITY CARD FOR PARTICLES

In the relative plane, every particle is one point: the tip of its cosmic mass, reached by walking the mass along the mass axis and then the charge part along the Composer axis (Figure 6.5). The five remarks of this section describe that point with one length and two angles, as a triangle, exactly as Section 4.8 describes every General Number. They use the numbers of the angle 79.86°: $\sin\bar{\phi} \approx 0.98438$, $\cos\bar{\phi} \approx 0.17605$ and $2\cos\bar{\phi} \approx 0.35210$.

6.4.1 *Mass and Charge as Two Projections*

Remark 6.11 (Mass and Charge Representation in the General Number and Cosmic Mass Theory). Within the General Number Plane and Cosmic Mass framework, the intrinsic properties of a particle, mass

and electric charge, can be expressed parametrically in terms of a scalar magnitude r and the generalized trigonometric functions depending on angles α and ϕ, the charge entering through the Composer part $q\mathcal{G}_{\bar{\phi}}$ of the cosmic mass, as in Definition 4.8:

$$m\ [\mathrm{Kg}] = r \cdot \cos(\alpha, \phi) = r \cdot \frac{\sin(\phi)}{\sin(\alpha + \phi)} \approx r \cdot \frac{\sin(79.86^\circ - \alpha)}{0.98438},$$
$$q\mathcal{G}_{\bar{\phi}}\ [\mathrm{Kg}] = r \cdot \sin(\alpha, \phi) = r \cdot \frac{\sin(\alpha)}{\sin(\alpha + \phi)} \approx r \cdot \frac{\sin(\alpha)}{0.98438}.$$

This representation highlights the geometric interpretation of mass and charge as projections of a unified scalar r along directions defined by the angles α and ϕ, consistent with the skewed geometry of the General Number Plane.

Remark 6.12 (Generalized Tangent and the Charge-to-Mass Ratio). From the parametric expressions of mass and charge in the General Number Plane, the generalized tangent function naturally represents the charge-to-mass ratio, measured through the Composer Unit:

$$\tan(\alpha, \phi) = \frac{q\mathcal{G}_{\bar{\phi}}}{m} = \frac{\sin(\alpha)}{\sin(\phi)} = \frac{\sin(\alpha)}{\sin(79.86^\circ - \alpha)}.$$

This elegantly connects the particle's intrinsic properties with the geometric framework, showing that the ratio of electric charge to mass is governed by the ratio of the generalized sine functions of angles α and ϕ. Thus, the tangent function here provides a direct physical interpretation in the unified Cosmic Mass theory.

In plain words

Remark 6.11 is the General Sinus reading of the triangle of Section 4.8, written for the cosmic mass: the mass is the real part, $r \cdot \cos(\alpha, \phi)$, and the charge part is the Composer part, $r \cdot \sin(\alpha, \phi)$, with the cosmic magnitude r, in kilograms, in the place of the magnitude M. It does for a particle what polar coordinates do for a point: in a perpendicular frame, a point is given by two coordinates or by one length and one angle, $x = r\cos\theta$ and $y = r\sin\theta$; in the leaning frame, one length and two angles take their place. The number 0.98438 in the denominators is $\sin(\alpha + \phi) = \sin(\bar{\phi}) = \sin(79.86^\circ)$, the sine of the angle of the Composer axis.

Remark 6.12 reads the same triangle as a ratio. Divide the charge part by the mass: the length r and the common denominator cancel, and what is left is the ratio of the sines of the two angles, the general tangent. A particle whose charge part is much larger than its mass has a large ratio, and its triangle leans towards the Composer axis; a neutral particle has the ratio zero and lies on the mass axis.

6.4.2 *The Identity Triangle*

Remark 6.13 (New Particle Identity Parameters). In the framework of the General Number Plane and Cosmic Mass theory, the traditional particle descriptors mass m and charge q can be equivalently expressed using the parameters r, α, and ϕ, where

$$\begin{pmatrix} m \\ q\mathcal{G}_{\bar{\phi}} \end{pmatrix} = r \cdot \begin{pmatrix} \cos(\alpha, \phi) \\ \sin(\alpha, \phi) \end{pmatrix}, \qquad \text{with} \quad \alpha + \phi = \bar{\phi} \approx 79.86^\circ.$$

Here, r can be interpreted as the **Cosmic Mass Magnitude**, representing the overall intrinsic scalar magnitude of the particle's identity. The angle ϕ is the **Mass Phase Angle**, characterizing the portion of this magnitude projected onto the mass axis, while α is the **Charge Phase Angle**, representing the complementary projection onto the charge axis.

Together, the triplet (r, α, ϕ) forms a new, geometrically grounded particle identity that generalizes and unifies the classical scalar attributes (m, q) within the General Number Plane.

> IN PLAIN WORDS
>
> Think of (r, α, ϕ) as a new identity card, printed in the geometry of the relative plane. The magnitude r says how much identity the particle has in all; the two angles say how that identity is shared between mass and charge. Because $\alpha + \phi = \bar{\phi}$ is fixed, only two of the three numbers are free, just as the old card had two lines, a mass and a charge.

Remark 6.14 (New Particle Identity and Triangle Representation). We can represent the **New Particle Identity Parameters** using a triangle, where in this triangle:

- The side opposite to angle α represents the **charge part** $q\mathcal{G}_{\bar{\phi}}$,
- The side opposite to angle ϕ represents the **mass** m,
- The base of the triangle is r, the **Cosmic Magnitude**.

Using the **Law of Cosines**, the base r can be computed by:

$$r^2 = m^2 + q^2\mathcal{G}_{\bar{\phi}}^2 + 2mq\cos(\bar{\phi})\mathcal{G}_{\bar{\phi}} = m^2 + q^2\mathcal{G}_{\bar{\phi}}^2 + 0.35210 \cdot mq\mathcal{G}_{\bar{\phi}}.$$

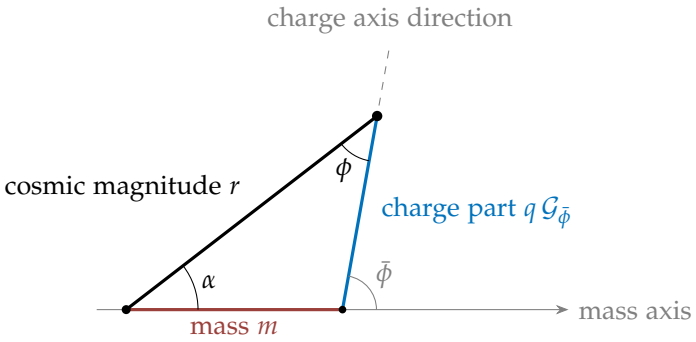


Figure 6.10: The particle identity triangle (Remarks 6.13 and 6.14), the triangle of a General Number of Figure 4.8 with $a = m$ and $b = q$. The side opposite the charge phase angle α is the charge part $q\,\mathcal{G}_{\bar{\phi}}$, the side opposite the mass phase angle ϕ is the mass m, and the base is the cosmic magnitude r. The Composer axis, which carries charge, makes the angle $\bar{\phi} = \alpha + \phi = 79.86°$ with the mass axis.

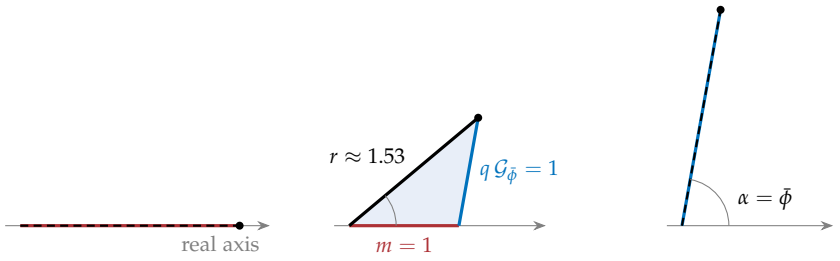
6.4.3 *The Charge Phase Angle*

Remark 6.15. The angle α in the particle identity triangle can be expressed in terms of the charge-to-mass ratio, measured through the Composer Unit, $\eta = \mathcal{G}_{\bar{\phi}} \frac{q}{m}$, using the following general formula:

$$\alpha = k\pi + \beta \quad \text{where}$$
$$\beta = \arctan\left(\frac{\eta \sin(\bar{\phi})}{1 + \eta\cos(\bar{\phi})}\right) = \arctan\left(\frac{0.98438 \cdot \eta}{1 + 0.17605 \cdot \eta}\right)$$

with $k \in \mathbb{Z}$, and $\bar{\phi} = 79.86°$.

> IN PLAIN WORDS
>
> Figure 6.10 is the triangle of a General Number from Figure 4.8, now carrying a mass and a charge, and Remarks 6.14 and 6.15 are Theorems 4.9 and 4.10 written for the cosmic mass, with $a = m$ and $b = q$. In Remark 6.15, η is the ratio of the two parts of Section 4.8, the charge part over the mass part: it finds the angle at the origin from that ratio. The three numbers that appear in the last two remarks all come from the angle of the Composer axis: $2\cos(79.86°) \approx 0.35210$, $\sin(79.86°) \approx 0.98438$ and $\cos(79.86°) \approx 0.17605$.



(a) a neutral particle: $\alpha = 0$, $r = m$

(b) equal parts: $\alpha = \phi \approx 39.93°$

(c) a charge without mass: $m = 0$, $r = q\,\mathcal{G}_{\bar\phi}$

Figure 6.11: The triangle of a cosmic mass drawn to scale in the relative plane of the angle $\bar\phi \approx 79.86°$ (Remarks 6.11 to 6.15). (a) A neutral particle: the triangle closes onto the real axis, $\alpha = 0$ and $r = m$. (b) A particle whose two parts are equal, $m = q\,\mathcal{G}_{\bar\phi} = 1$ kg: the angles α and ϕ are equal, each about 39.93°, and $r \approx 1.53$ kg. (c) A charge without mass: the triangle closes onto the Composer axis, $\alpha = \bar\phi$ and $r = q\,\mathcal{G}_{\bar\phi}$. The neutron is case (a); the proton and the electron lie extremely close to case (c).

Worked example: a particle whose two parts are equal

Take a particle whose mass part and charge part are equal, $m = 1$ kg and $q\mathcal{G}_{\bar\phi} = 1$ kg, so that $q \approx 8.62 \times 10^{-11}$ C and $\eta = 1$. Use the three numbers above.

- Magnitude, by Remark 6.14: $r^2 \approx 1 + 1 + 0.35210 = 2.35210$, so $r \approx 1.53366$ kg.
- Charge phase angle, by Remark 6.15 with $k = 0$: $\beta = \arctan\big(0.98438/(1 + 0.17605)\big) \approx \arctan(0.83702) \approx 39.93°$, so $\alpha \approx 39.93°$ and $\phi = 79.86° - 39.93° \approx 39.93°$.
- Check with the General Sinus, by Remark 6.11: $m = r\sin(\phi)/0.98438 \approx 1.53366 \cdot 0.64185/0.98438 \approx 1.000$ kg, and in the same way $q\mathcal{G}_{\bar\phi} \approx 1.000$ kg.

When the two parts are equal, the two angles are equal too, each half of $\bar\phi$, as for every General Number with equal parts (Remark 4.31). Figure 6.11 draws this triangle to scale, between the two ends of the next example.

Worked example: the two ends of the triangle

A neutral particle. The neutron has $q = 0$: its cosmic mass is a real number (Remark 4.16), and $\eta = 0$. Remark 6.15 gives $\beta = \arctan(0) = 0$, and with $k = 0$ the charge phase angle is

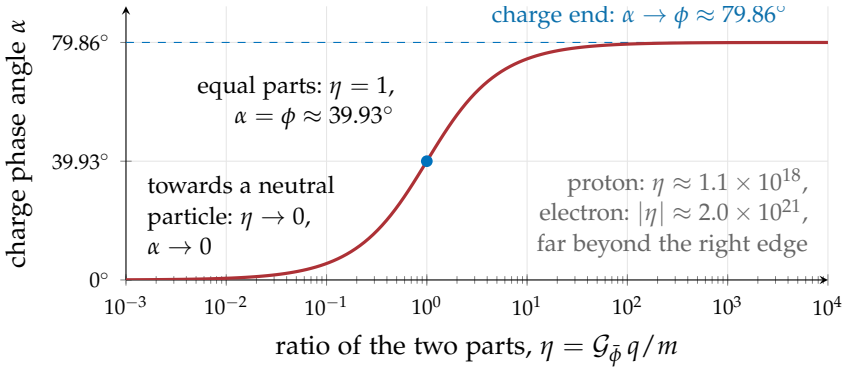


Figure 6.12: From a ratio to an angle (Remark 6.15 with $k = 0$), on a scale of powers of ten. At $\eta = 1$ the two parts are equal and α is half of $\bar{\phi}$. As η grows, α climbs towards $\bar{\phi} \approx 79.86°$, the charge end. The proton lies far beyond the right edge of the picture, and so does the electron, on the side flipped by $k\pi$.

$\alpha = 0$, so the mass phase angle is $\phi = \bar{\phi} = 79.86°$. Remark 6.11 then gives $m = r \cdot \sin(\bar{\phi}) / \sin(\bar{\phi}) = r$. For a neutral particle, the cosmic magnitude is simply the mass, and Remark 6.14 agrees, since $r^2 = m^2 + 0 + 0$.

A charge without mass. At the other end, imagine a particle with charge but no mass. Then $m = 0$, so $\sin(\phi) = 0$ in Remark 6.11, which gives $\phi = 0$ and $\alpha = \bar{\phi}$. The particle sits entirely on the charge side, and $r = q\mathcal{G}_{\bar{\phi}}$, the size of its charge part. Remark 6.15 agrees in the limit of a very large ratio η, where β approaches $\arctan(0.98438/0.17605) \approx 79.86°$. The two ends are the first two special triangles of Remark 4.31, a real number and a pure Composer number.

Every real particle lies somewhere between these two ends, and its charge phase angle α tells how far its identity has turned from pure mass towards pure charge (Figure 6.12). For a negative charge, the triangle flips to the other side of the mass axis; the term $k\pi$ in Remark 6.15 is there to allow for such flips. Measured through the Composer Unit, the ratios η of the electron and the proton are enormous, about 2.0×10^{21} and 1.1×10^{18} in size, the ratios of the two parts found in Section 6.3.1. So Remark 6.15 places both of them within less than a billionth of a billionth of a radian of the charge end: α is extremely close to $\bar{\phi}$ for the proton, and to $\bar{\phi} + \pi$ for the electron, whose charge is negative.

Every number on this card is computed at the angle 79.86°. The next section reads the law in its own plane, and the last one shows why, at this angle, like charges push apart.

6.5 READING THE LAW IN ITS OWN PLANE

PARIS, 1733

In 1733 Charles François de Cisternay du Fay, a chemist in charge of the royal garden of plants in Paris, was working through the new experiments on electricity that Stephen Gray had begun in England. He rubbed a glass tube and let it chase a scrap of gold leaf through the air. The leaf flew to the tube, touched it, and was then pushed away, and it kept its distance as long as the tube followed it. Du Fay expected a rubbed lump of copal, a hard resin like amber, to push the leaf away as well. It pulled the leaf towards itself.

From this surprise he drew a rule. In a letter to the Duke of Richmond he wrote that "there are two distinct electricities, very different from one another," and he called them vitreous, from glass, and resinous, from resin: "a body of the vitreous electricity, for example, repels all such as are of the same electricity; and on the contrary, attracts all those of the resinous electricity" (Du Fay, 1733). In 1747 Benjamin Franklin gave the two kinds the names that are still in use, positive and negative (Franklin, 1751), and in 1785 Coulomb measured how the push between two like charges falls off with distance (Chapter 1).

Du Fay's rule, with Coulomb's strength, is the first fact that the law must meet, and the simplest atom is the place to test it. But first the law must be read.

6.5.1 *Two Readings of One Product*

The Cosmic Interaction Law of Theorem 6.6 gives a General Number multiplied by the arrow $\vec{e}_r$. A laboratory, however, measures one ordinary number along $\vec{e}_r$: so many newtons, pulling or pushing. Before the law can be compared with nature, we need a rule that turns the General Number $\mu_a\mu_b$ into one real number.

Chapter 4 offers two such rules, both readings on the real axis, which carries mass (Remark 4.21).

- The *shadow* drops the tip of the number straight down onto the real axis, as a lamp overhead would (Definition 4.19).
- The *real part* walks back to the real axis along the Composer axis, the way the number was built: a steps along the real axis, then b steps of the Composer Unit (Definition 4.11).

Two neutral masses must pull together, so in both readings a value forward along the mass axis means a pull, and a value backward means a push. For two neutral bodies the product $\mu_a\mu_b = m_a m_b$ is a real number, and both readings give Newton's law exactly. They part as soon as charge enters.

By Theorem 4.20, the shadow counts each of the three terms of the law with the shadow of its own direction: 1 for gravity, along the mass axis; $\cos\bar{\phi}$ for the Coupling Force, which carries the Composer Unit; and $\cos 2\bar{\phi}$ for the term of two charges, which carries the square $\mathcal{G}_{\bar{\phi}}^2$, an arrow of length $|\mathcal{G}_{\bar{\phi}}|^2$ at the angle $2\bar{\phi}$ (Theorem 4.1). Since $G\,|\mathcal{G}_{\bar{\phi}}| = \sqrt{KG}$ and $G\,|\mathcal{G}_{\bar{\phi}}|^2 = K$,

$$G\ \mathrm{Sh}(\mu_a\mu_b) = G\,m_a m_b + \sqrt{KG}\,\cos\bar{\phi}\,(m_a q_b + q_a m_b) + K\cos 2\bar{\phi}\ q_a q_b.$$

Read this way, the electric term keeps the share $\cos 2\bar{\phi}$ of Coulomb's strength, and the Coupling Force acts along the line with the share $\cos\bar{\phi}$.

The real part needs one more step. The first two terms of the product are already written as a real number and a Composer part, but the third carries $\mathcal{G}_{\bar{\phi}}^2$, an arrow that leans at $2\bar{\phi}$. Chapter 4 supplies its real part: the Rule of the Square (Proposition 4.22) writes the square as $\mathcal{G}_{\bar{\phi}}^2 = 2\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\ \mathcal{G}_{\bar{\phi}} - |\mathcal{G}_{\bar{\phi}}|^2$, with the real part $-|\mathcal{G}_{\bar{\phi}}|^2$. Figure 4.12 shows the two readings of this square side by side at 79.86°: the walk back along the Composer axis lands at -1 times the squared length, and the drop straight down at -0.938 times it.

6.5.2 *The Law Read by Its Real Part*

Proposition 6.16 (Newton and Coulomb in One Law)**.** *In the relative plane of the Composer Unit, the product of two cosmic masses is*

$$\mu_a\mu_b = \left(m_a m_b - |\mathcal{G}_{\bar{\phi}}|^2\,q_a q_b\right) + \left(m_a q_b + q_a m_b + 2\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\ q_a q_b\right)\mathcal{G}_{\bar{\phi}},$$

and its real part, multiplied by G, is

$$G\,\Re(\mu_a\mu_b) = G\,m_a m_b - K\,q_a q_b.$$

The Coupling Force has no real part, and the angle $\bar{\phi}$ appears only in the Composer coordinate.

Proof. This is Proposition 4.23 with $a = m$ and $b = q$: multiply out the two brackets (Proposition 4.18), $\mu_a\mu_b = m_a m_b + (m_a q_b + q_a m_b)\,\mathcal{G}_{\bar{\phi}} + q_a q_b\,\mathcal{G}_{\bar{\phi}}^2$, and replace $\mathcal{G}_{\bar{\phi}}^2$ by the Rule of the Square. The real part is the first bracket, and $G\,|\mathcal{G}_{\bar{\phi}}|^2 = K$ by the size of the Composer Unit (Section 6.3.1). □

Read by its real part, one law gives both of the laws of Section 6.1.2, each with its full constant, and with opposite signs (Table 6.2). The minus sign is the one that Definitions 6.3 and 6.4 left to the nature of each interaction (Figure 6.4): two masses pull, and the term of two charges, for like charges, counts the other way and pushes. No factor of the angle remains in any force. The Coupling Force of Remark 6.7 has no real part at all: read in its own plane, it is not a force along

TERM OF THE LAW	SHADOW	REAL PART
gravity, $G\,m_a m_b$	1	1
Coupling Force, $\sqrt{KG}\,(m_a q_b + q_a m_b)$	$\cos\bar{\phi}$	0
two charges, $K\,q_a q_b$	$\cos 2\bar{\phi}$	-1
at 79.86°	1, 0.17605, -0.93801	1, 0, -1

Table 6.2: Two readings of the law: the factor with which each of its three terms counts along the line between the two bodies. A positive value means a pull, a negative value a push. The real part gives Newton and Coulomb at full strength; the shadow gives Coulomb's law only the share $\cos 2\bar{\phi} \approx -0.938$ of its strength, and a Coupling Force along the line with the share $\cos\bar{\phi} \approx 0.176$.

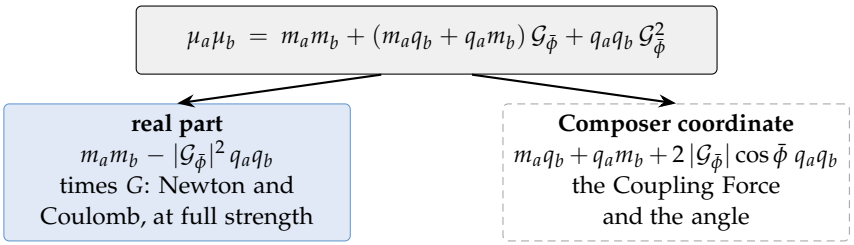


Figure 6.13: The product of two cosmic masses splits in two (Proposition 6.16). Its real part is the force along the line: gravity and Coulomb's law, each at full strength. Its Composer coordinate holds the Coupling Force and the angle of the Composer axis, and it is not read as a force. Beside the Coupling Force it also holds a share of the term of two charges, by far the larger part for charged particles (Section 6.5.2).

the line. It lives, with the angle, in the Composer coordinate of the product (Figure 6.13).

It does not live there alone. By Theorem 4.55, the Composer coordinate of a product is built from the shadows $\mathrm{Sh}(\mu) = m + |\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\;q$ of the two factors (Definition 4.19):

$$m_a q_b + q_a m_b + 2\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\;q_a q_b = \mathrm{Sh}(\mu_a)\,q_b + \mathrm{Sh}(\mu_b)\,q_a.$$

Beside the Coupling term it holds a share of the term of two charges, $2\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\;\;q_a q_b$, and for charged particles that share is by far the larger: the Coupling term is 5×10^{-18} of it for two protons and 3×10^{-21} of it for two electrons. Read like a force, multiplied by G/r^2, the Composer part of the product of two charged particles is an arrow about $2\cos\bar{\phi} \approx 0.352$ times as long as Coulomb's force between them. Only between a neutral body and a charge is the whole Composer coordinate the Coupling term.

6.5.3 *The Hydrogen Test*

Without Du Fay's rule there would be no atoms: the electron of a hydrogen atom is held in place by the pull of the proton. The pull must have the right sign, and it must also have the right size, because the size sets how tightly the electron is bound.

> **WORKED EXAMPLE: THE HYDROGEN ATOM**
>
> In a hydrogen atom the electron and the proton are about $a_0 = 5.29 \times 10^{-11}$ m apart, and their charges are $+e$ and $-e$, with $e = 1.602 \times 10^{-19}$ C.
>
> - *Nature.* Coulomb's law gives a pull of size Ke^2/a_0^2. Step by step: $e^2 \approx 2.566 \times 10^{-38}$ C^2; then $Ke^2 \approx 8.98755 \times 10^9 \times 2.566 \times 10^{-38} \approx 2.307 \times 10^{-28}$ N m^2; and $a_0^2 \approx 2.798 \times 10^{-21}$ m^2. Dividing, the pull is about 8.24×10^{-8} N.
> - *The law, read by its real part.* With $q_a q_b = -e^2$, Proposition 6.16 gives the electric part $-K(-e^2)/a_0^2 = +Ke^2/a_0^2$: a pull of 8.24×10^{-8} N.
> - *The law, read by its shadow.* The electric term counts with $\cos 2\bar{\phi}$. At 79.86° it is a pull of $0.93801 \times 8.24 \times 10^{-8} \approx 7.73 \times 10^{-8}$ N, only 93.8% of the strength.
> - *How tightly the electron is held.* In Bohr's model of the atom, the energy needed to free the electron grows with the square of the strength of the pull (Bohr, 1913). With the full strength of Coulomb's law it is 13.598 eV, the measured value (Sansonetti and Martin, 2005). With 93.8% of the strength it would be $0.93801^2 \times 13.598 \approx 11.96$ eV, short by 1.6 eV.

A second test: the neutron. The shadow also keeps the share $\cos\bar{\phi}$ of the Coupling Force. Between a neutral body of mass m and a charge, that share acts exactly like an electric force on a body of charge

$$Q = \cos\bar{\phi}\,\sqrt{G/K}\,m,$$

so every neutral body would carry an apparent charge in proportion to its mass. For the neutron, at 79.86°, $Q \approx 0.17605 \times 8.617 \times 10^{-11}$ C/kg $\times\, 1.675 \times 10^{-27}$ kg $\approx 1.59 \times 10^{-19}\,e$. The charge of the neutron has been measured, by bending beams of slow neutrons in strong electric fields and by testing the neutrality of a gas, and found to be $(-0.2 \pm 0.8) \times 10^{-21}\,e$ (Baumann et al., 1988; Bressi et al., 2011; Particle Data Group et al., 2024). The shadow misses by about two hundred standard errors. Read by its real part, the Coupling Force is zero, and so is the apparent charge.

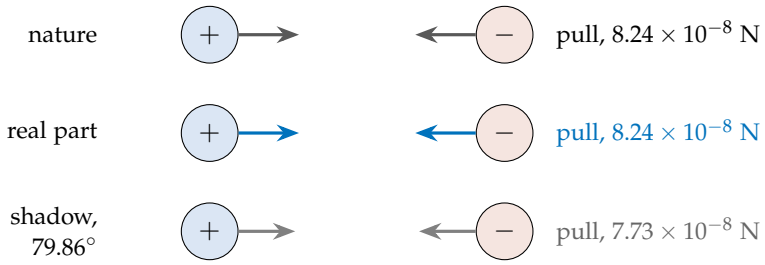


Figure 6.14: The hydrogen test (Section 6.5.3). In nature the proton and the electron pull each other together with about 8.24×10^{-8} N. Read by its real part, the law gives exactly that pull. Read by its shadow, it gives $\cos 2\bar{\phi}$ times that size: a pull of only 93.8% of the strength.

Result. The shadow fails the hydrogen test on the strength, and it gives the neutron a charge that measurements rule out. The real part passes all three checks: the sign, the strength and the neutral neutron. It is also the reading that the identity card already uses for a single body, whose mass is its real part, $m = \Re(\mu) = r\cos(\alpha, \phi)$ (Remark 6.11): the General Sinus projects along the Composer axis, not straight down. From here on, the book reads the law the same way.

Both failures of the shadow have one cause. By Proposition 4.54, the shadow of a General Number is its real part plus the shadow of its Composer part, $\mathrm{Sh}(z) = \Re(z) + \mathrm{Sh}\big(\mathrm{Com}(z)\big)$. For the law, the shadow is therefore the real part plus $\cos\bar{\phi} \approx 0.176$ times the signed length of the Composer part: it lets the Composer coordinate of the product into the force. Between the proton and the electron that coordinate is almost entirely the share of the two charges of Section 6.5.2, which takes $2\cos^2\bar{\phi} \approx 0.062$ of the pull away; between the neutron and a charge it is the Coupling term, which gives the apparent charge.

Definition 6.17 (Reading the Law)**.** The force that the Cosmic Interaction Law exerts along $\vec{e}_r$ is the real part of the law,

$$F = G \cdot \frac{\Re(\mu_a \mu_b)}{r^2} = G \cdot \frac{m_a m_b}{r^2} - K \cdot \frac{q_a q_b}{r^2},$$

a pull when it is positive and a push when it is negative. The Composer coordinate of $\mu_a \mu_b$ is not read as a force.

> IN PLAIN WORDS
>
> A force is one number along one line, but the law gives a General Number, so it must be brought down to the mass axis. There are two ways: drop it straight down, like a shadow, or walk it back along the leaning axis, the way it was built. Dropping gives the electric force only 93.8% of its strength at our angle, loosens the hydrogen atom by 1.6 eV, and gives the neutron a charge that

> nobody has found. Walking gives Coulomb's law exactly, holds the hydrogen atom as tightly as it is held, and leaves the neutron neutral. The book walks.

6.5.4 *The Law Read from Two Cards*

The identity card of Section 6.4 and the reading of Definition 6.17 meet in one formula. By Euler's formula in a relative plane (Theorem 4.48), a cosmic mass with the card (r, α, ϕ) is the exponential $\mu = r\, e^{\alpha \mathcal{J}_{\bar{\phi}}}$, where $\mathcal{J}_{\bar{\phi}}$ is the Turning Unit (Definition 4.42): the magnitude r and the charge phase angle α are the length and the angle of its arrow. Multiplying two cosmic masses multiplies their magnitudes and adds their angles (Theorem 4.46). In this subsection the distance between two bodies is written d, the r of Definition 6.17, so that it does not clash with their magnitudes.

Proposition 6.18 (The Force from Two Cards)**.** *Let two bodies the distance d apart have the identity cards (r_a, α_a, ϕ_a) and (r_b, α_b, ϕ_b) of Remark 6.13. Then*

$$\mu_a \mu_b = r_a r_b\, e^{(\alpha_a + \alpha_b)\mathcal{J}_{\bar{\phi}}},$$

and the force of Definition 6.17 is

$$F = G\,\frac{r_a r_b}{d^2}\,\cos(\alpha_a + \alpha_b, \bar{\phi} - \alpha_a - \alpha_b) = G\,\frac{r_a r_b}{d^2}\cdot\frac{\sin(\bar{\phi} - \alpha_a - \alpha_b)}{\sin\bar{\phi}}.$$

With angles counted modulo 360°, it is a pull when $\alpha_a + \alpha_b$ lies between $\bar{\phi} - 180^\circ$ and $\bar{\phi}$, a push when it lies between $\bar{\phi}$ and $\bar{\phi} + 180^\circ$, and zero exactly when $\alpha_a + \alpha_b$ is $\bar{\phi}$ or $\bar{\phi} + 180^\circ$, that is, when $G\, m_a m_b = K\, q_a q_b$.

Proof. By Theorem 4.48, every nonzero General Number is $|z|\, e^{\alpha \mathcal{J}_{\bar{\phi}}}$, with α the angle of its arrow. For a cosmic mass $|\mu| = r$, and α is the charge phase angle, because the polar form of Theorem 4.45 is the card of Remark 6.11. The rule $e^{w_1 + w_2} = e^{w_1} e^{w_2}$ gives the product. With $\theta = \alpha_a + \alpha_b$, Theorem 4.45 gives the real part of $r_a r_b\, e^{\theta \mathcal{J}_{\bar{\phi}}}$ as $r_a r_b \cos(\theta, \bar{\phi} - \theta)$, and the definition of the General Sinus gives $\cos(\theta, \bar{\phi} - \theta) = \sin(\bar{\phi} - \theta)/\sin\bar{\phi}$. Multiplied by G/d^2, this is the force. Its sign is the sign of $\sin(\bar{\phi} - \theta)$, which gives the two bands, and it is zero exactly when $\sin(\bar{\phi} - \theta) = 0$. The same real part is $m_a m_b - |\mathcal{G}_{\bar{\phi}}|^2 q_a q_b$ (Proposition 6.16), so the force is zero exactly when $G\, m_a m_b = K\, q_a q_b$. □

Read on the cards, Du Fay's rule needs only the angles. Two neutral bodies have $\alpha = 0$: their angles add up to 0°, and they pull. A proton lies within 10^{-16} degree of the angle $\bar{\phi}$, and an electron within as little of $\bar{\phi} + 180^\circ$ (Section 6.4.3). Two protons, or two electrons, add up to about $2\bar{\phi} = 159.72^\circ$ and push; a proton and an electron add up to about 339.72° and pull. These are the angles of Figure 6.15, now for

every pair of bodies. A pair neither pulls nor pushes when its angles add up to $\bar{\phi}$, as for two positive bodies whose two parts are equal, each on the ray $\alpha = \bar{\phi}/2 \approx 39.93°$ (Figure 6.11(b)), or to $\bar{\phi} + 180°$, as for two such bodies with negative charges. The proposition is a reading of the law, not a new fit: the force is still $G\,m_a m_b - K\,q_a q_b$ over d^2, and at another angle the cards would change and give the same force.

> WORKED EXAMPLE: THREE PAIRS, READ FROM THEIR CARDS
>
> Put each pair 1 m apart.
>
> - *Two protons.* Each card has $r \approx e\,|\mathcal{G}_{\bar{\phi}}| \approx 1.85921 \times 10^{-9}$ kg, and the angles add up to $2\bar{\phi} \approx 159.72°$, less about 10^{-16} degree, so the factor $\sin(\bar{\phi} - \alpha_a - \alpha_b)/\sin\bar{\phi}$ is -1 to many digits. The force is $-G\,(1.85921 \times 10^{-9})^2 \approx -2.307 \times 10^{-28}$ N: a push of Coulomb's strength, since $G\,|\mathcal{G}_{\bar{\phi}}|^2 = K$.
> - *A proton and an electron.* The angles add up to 339.72°, the factor is $+1$, and the same product of magnitudes gives a pull of 2.307×10^{-28} N.
> - *A neutron and a proton.* The neutron has $r = m_n$ and $\alpha = 0$, so the sum is the angle of the proton, short of $\bar{\phi}$ by $\delta \approx 5.07 \times 10^{-17}$ degree. The factor is $\sin\delta/\sin\bar{\phi}$, and $r_p \sin\delta/\sin\bar{\phi} = m_p$ by Remark 6.11, so the force is $G\,m_n m_p \approx 1.87 \times 10^{-64}$ N: gravity alone.

6.6 WHY LIKE CHARGES PUSH APART

Du Fay found the rule, and Coulomb measured it, but neither said why: the classical laws take the sign from experiment and write it in by hand (Figure 6.4).

Physics has since found a deeper reason, in a different kind of theory. In quantum field theory a force is carried by a particle that the two bodies exchange, and the sign of the force between two like sources follows from the spin of that particle: the exchange of a particle of spin one, such as the photon, pushes like charges apart, while the exchange of a particle of spin zero or spin two, the spin expected of the graviton, pulls like sources together (Zee, 2010). The Cosmic Interaction Law is a classical law between two bodies, and it gives the same fact a reason of another kind, a geometric one.

6.6.1 *On a Line, Like Charges Would Pull*

Suppose that mass and charge had been put into one ordinary number, $\mu = m + \lambda q$, with a real number λ, in kilograms per coulomb, in

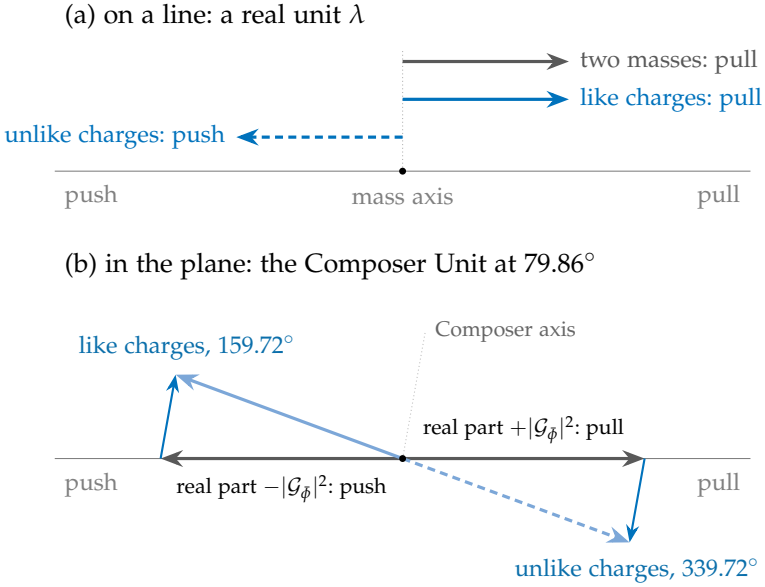


Figure 6.15: On a line and in the plane (Sections 6.6.1 and 6.6.2). The products of the law for pairs of bodies, read along the mass axis: to the right is a pull, to the left a push. (a) With a real number in the place of the Composer Unit, every product lies on the mass axis (drawn side by side): like charges pull as masses do, and unlike charges push, the opposite of nature. (b) With the Composer Unit at 79.86°, the product of two like charge parts leans at 159.72°. Walked back along the Composer axis, it reaches the mass axis at $-|\mathcal{G}_{\bar{\phi}}|^2$: a push of Coulomb's full strength. For unlike charges the product turns half a turn, to 339.72°, and the walk reaches $+|\mathcal{G}_{\bar{\phi}}|^2$: a pull. The lengths are not to scale.

the place of the Composer Unit. The product of two such numbers multiplies out exactly as in Figure 6.6:

$$\mu_a \mu_b = m_a m_b + \lambda \left(m_a q_b + q_a m_b \right) + \lambda^2 \, q_a q_b .$$

Whatever the number λ, positive or negative, its square is positive (Remark 4.27). For two like charges $q_a q_b$ is positive too, so the last term has the sign of the first: one law built on this product would pull two like charges together exactly as it pulls two masses, and it would push a proton away from an electron. The choice $\lambda^2 = K/G$ would give the electric term the strength of Coulomb's law, but no choice of λ can give it the sign. Turning the whole law around does not help either, since two masses would then push apart.

A real number in the place of the Composer Unit makes the relative plane a line, no plane (Remark 4.13): the two parts of a cosmic mass simply add, and charge acts as extra mass. On a line, one product of cosmic masses can carry the sign of gravity or the sign of electricity, but never both (Figure 6.15(a)). To carry both, the law needs a unit whose square points the other way, back along the mass axis. The Composer Unit, leaning at 79.86°, is such a unit.

6.6.2 *In the Plane, Like Charges Push*

The Rule of the Square (Proposition 4.22) gives the square of the Composer Unit the real part $-|\mathcal{G}_{\bar{\phi}}|^2$, a negative number, and the law read by its real part turns it into the electric term of Proposition 6.16. Step by step:

1. By Proposition 6.16, the electric part of the force along $\vec{e}_r$ is $-K\,q_a q_b / r^2$, where a negative value is a push (Definition 6.17).
2. For like charges $q_a q_b$ is positive, so the electric part is a push; for unlike charges it is negative, and the same expression is a pull.
3. The constant is Coulomb's constant K itself, so the push and the pull have the full strength of Coulomb's law.

One fact gives both halves of Du Fay's rule, at Coulomb's strength (Figure 6.15(b)): the square of the Composer Unit points back along the mass axis. Proposition 6.18 reads the same rule off the cards of any two bodies.

> **In plain words**
>
> Picture the mass axis as a floor. The term of two masses is a stick lying flat on the floor and pointing forward; the term of two charges is a stick that leans at 159.72°, twice the angle of the Composer axis. Walk its tip back to the floor along the leaning axis: the walk lands behind the starting point, at minus the squared length, and behind means a push. Make one of the charges negative, and the walk lands in front: a pull. On a line the stick of two charges would lie flat and point forward, and like charges would pull as masses do.

> **Worked example: two protons**
>
> Two protons one metre apart push each other apart, in nature, with $Ke^2 \approx 2.31 \times 10^{-28}$ N. Read by its real part, the law gives $-Ke^2/r^2 \approx -2.31 \times 10^{-28}$ N: a negative number, a push, at Coulomb's full strength. The shadow, the drop straight down, gives $\cos 159.72^\circ \approx -0.93801$ times as much, -2.16×10^{-28} N: a push too, but at 93.8% of the strength, the shortfall that the hydrogen test of Section 6.5.3 detects.

6.6.3 *Why the Angle Is 79.86 Degrees*

Read by its real part, the force between two bodies contains the size of the Composer Unit, through $G\,|\mathcal{G}_{\bar{\phi}}|^2 = K$, but not its angle. The real part of a product has no angle in it (Proposition 4.23); the angle lives in the Composer coordinate, with the Coupling term, and in the corner

NUMBER	VALUE
angle $\bar{\phi}$ of the Composer Unit	$79.86°$
$\cos\bar{\phi}$; $\sin\bar{\phi}$	0.17605; 0.98438
length $\lvert\mathcal{G}_{\bar{\phi}}\rvert = \sqrt{K/G}$	1.16043×10^{10} kg/C
angle $2\bar{\phi}$ of $\mathcal{G}_{\bar{\phi}}^2$	$159.72°$
$\cos 2\bar{\phi}$; $\sin 2\bar{\phi}$	-0.93801; 0.34660
real part of $\mathcal{G}_{\bar{\phi}}^2$, over $\lvert\mathcal{G}_{\bar{\phi}}\rvert^2$	-1
Composer coordinate of $\mathcal{G}_{\bar{\phi}}^2$, over $\lvert\mathcal{G}_{\bar{\phi}}\rvert$: $2\cos\bar{\phi}$	0.35210
shadow of $\mathcal{G}_{\bar{\phi}}^2$, over $\lvert\mathcal{G}_{\bar{\phi}}\rvert^2$: $\cos 2\bar{\phi}$	-0.93801
corner of a particle triangle whose two parts have the same sign, $180° - \bar{\phi}$	$100.14°$

Table 6.3: The Composer Unit of our universe in numbers (Sections 6.3.1 and 6.6.3). Chapters 7 to 9 use these numbers.

of every particle triangle, $\alpha + \phi = \bar{\phi}$ (Remark 6.13). So no measured force fixes the angle. The size of the Composer Unit gives it, as the mirror root of Section 6.3.2, $79.86°$, and the book holds it there from this point to the end.

The reason for this angle is the one given in Section 6.3.2: at $79.86°$ the push of like charges can be seen in the plane itself. The square of the Composer Unit leans at $159.72°$, inside the backward band of Theorem 4.28 (Corollary 4.29), so the arrow of two like charges points back along the mass axis, against the arrow of two masses. The two readings of the law then agree in sign: the real part of the square is $-\lvert\mathcal{G}_{\bar{\phi}}\rvert^2$, and its shadow, $\lvert\mathcal{G}_{\bar{\phi}}\rvert^2 \cos 159.72° \approx -0.938\,\lvert\mathcal{G}_{\bar{\phi}}\rvert^2$, falls behind the origin too (Figure 4.12). This is a choice of coherence, not a fit to data, and no number measured in this book depends on it. Table 6.3 gathers the numbers of the Composer Unit at this angle, which the next three chapters use.

LOOKING BACK, AND AHEAD

Faraday looked for a link between gravity and electricity, and the cosmic mass offers one, though not the kind he looked for. It is not a new force. Read in its own plane, the law has no part along the line between the mass of one body and the charge of the other: the Coupling Force of Remark 6.7 lives in the Composer coordinate, beside, for charged particles, a far larger share of the term of two charges, and Faraday's needle, which stayed at zero, is not asked to move. The link is the law itself. One number holds the mass and the charge of a body, one product of two such numbers holds gravity and electricity, and the minus sign that pushes like charges apart, which Du Fay

found and Coulomb measured, comes from the plane: the square of the Composer Unit has a negative real part (Proposition 4.22). Read on the identity cards, the same law multiplies the magnitudes of the two bodies and adds their angles (Proposition 6.18). Read this way, the law says what Newton and Coulomb say, and no more; its test is that it agrees with them, which it does exactly.

The angle of the Composer axis is held at 79.86°, where the plane itself shows the push of like charges (Section 6.6.3). With the angle fixed, every body of ordinary matter has one triangle, whose corner between the mass side and the charge side is set by the angle alone. The next chapter starts there. It draws all the triangles of one size at once, finds that they line up on one ellipse, and builds from the ellipses a frame in which the electron, the proton, the neutron, the atoms, the ions and the nuclei all have their place: the Cosmic Mass Standard Model of Chapter 7.

7

THE COSMIC MASS STANDARD MODEL

One fixed angle gives every body one triangle; all the triangles of one size make one ellipse; the ellipses make one frame; and in that frame every body of ordinary matter, from the electron to the nucleus of oganesson, has its own place.

BELFAST, AUGUST 1874

In August 1874 the British Association for the Advancement of Science held its yearly meeting in Belfast, and on one of its mornings George Johnstone Stoney read a short paper to the section of mathematics and physics. Stoney was forty-eight. As a young man he had worked at Birr Castle beside the great telescope of the Earl of Rosse; since 1857 he had been the secretary of the Queen's University of Ireland, an administrative post that left physics to his spare time.

His starting point was Faraday's law of electrolysis. When a current splits water or salt, the amount of matter set free is fixed by the amount of electricity that passes. Stoney read the law the other way round. "For each chemical bond which is ruptured within an electrolyte," he wrote, "a certain quantity of electricity traverses the electrolyte which is the same in all cases" (Stoney, 1894). Electricity, then, comes in equal portions, and nature itself supplies the size of one portion. From the number of molecules in a gas, known only roughly at the time, he estimated it at "a twentieth (that is 10^{-20}) of the quantity of electricity which was at that time called the ampere", about sixteen times too small, as Robert Millikan's oil drops would show forty years later (Millikan, 1913).

Stoney did something bolder with his portion of electricity. He put it beside two other constants that nature supplies, the speed of light and the constant of gravitation, and built from the three of them a whole system of units of length, time and mass "in which there shall be nothing arbitrary" (Stoney, 1894). His paper waited seven years for print (Stoney, 1881). In 1891 he gave his portion of electricity a name, the *electron*, and six years later J. J. Thomson found the particle that carries it (Thomson, 1897).

Written with today's value of the charge, Stoney's unit of mass is $e\sqrt{K/G} \approx 1.85921 \times 10^{-9}$ kg, a little less than two micrograms. Chapter 6 met this number without naming it, as the charge part of the electron and as the charge part of the proton (Figure 6.9). This chapter gives it its name and puts it to work. It is the step of the charge side of every particle triangle, in the same way that the dalton

is the step of the mass side, and with it the triangles of all ordinary matter can be drawn in one frame.

Chapter 6 ended with the law read in its own plane: Newton's gravity and Coulomb's law, each at full strength, in the relative plane whose Composer axis leans at 79.86°, the angle at which like charges push apart (Section 6.6.3). This chapter starts from that fixed angle, builds a geometric frame from it one step at a time, and then places in that frame the 306 bodies of ordinary matter for which mass and charge are known: the electron, the proton and the neutron, the 118 atoms of the periodic table, 67 common ions, and the 118 bare nuclei.

Remark 7.1 (The Angle Held Fixed). In this chapter the angle of the Composer axis is held fixed at $\bar{\phi} \approx 79.86°$, with $\sin\bar{\phi} \approx 0.98438$, $\cos\bar{\phi} \approx 0.17605$ and $2\cos\bar{\phi} \approx 0.35210$ (Table 6.3).

We call the result the *Cosmic Mass Standard Model*. The name borrows a phrase that physics already uses for its theory of elementary particles and forces, so it is worth saying at once what is meant here. The model of this chapter is a standard frame: one set of coordinates, the same for every body, in which each body of ordinary matter has a fixed place, much as each element has its fixed place in the periodic table. It does not replace the Standard Model of particle physics and makes no claim about new particles. It shows the bodies we know, all at once, in the geometry of the cosmic mass.

THE ROAD THROUGH THIS CHAPTER. The chapter climbs in six steps, shown in Figure 7.1, and each step takes one tool of Chapter 4. Section 7.1 draws one body as one triangle, the triangle of a General Number (Theorem 4.9, Remark 4.30), and names the Stoney mass. Section 7.2 draws all the triangles of one size at once and finds that they fill an ellipse (Theorem 4.34). Section 7.3 stacks the ellipses of all sizes into a polar-like frame, the rings and rays of Definition 4.35. Section 7.4 shows where real matter falls in that frame, close to the axes (Proposition 4.32), and why two lenses are needed to see it. Section 7.5 computes the identity cards of all 306 bodies, and Section 7.6 plots them together: the Cosmic Mass Standard Model.

7.1 STEP 1: ONE BODY, ONE TRIANGLE

Remarks 6.13 and 6.14 give every body a new identity card, the triplet (r, α, ϕ), and draw it as a triangle (Figure 7.2). Its three sides are the mass m, laid along the mass axis from the origin O to the corner P; the charge part $q\mathcal{G}_{\bar{\phi}}$, laid from P to the tip Q along the direction of the Composer axis; and the cosmic magnitude r, the base OQ. Its three angles are:

- the *charge phase angle* α at O, which faces the charge side $q\mathcal{G}_{\bar{\phi}}$;

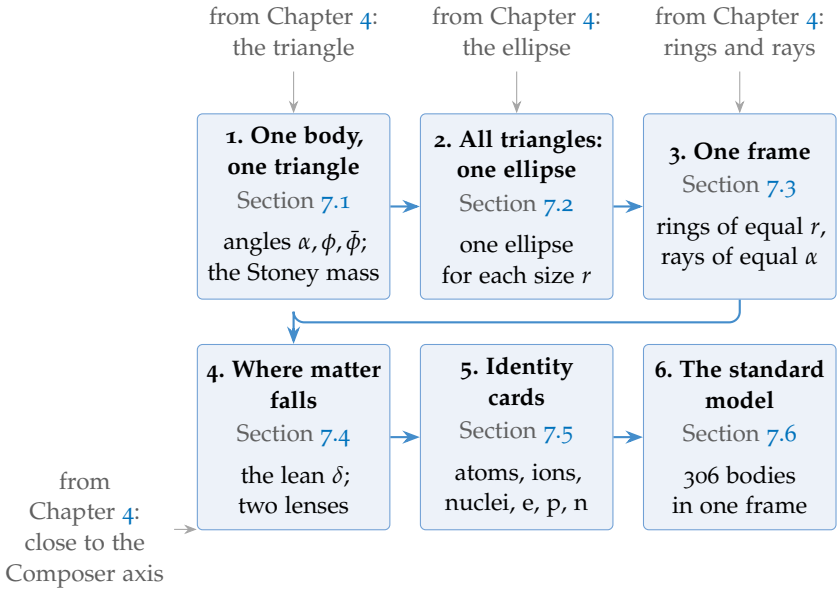


Figure 7.1: The road through the chapter, from one fixed angle to one frame for all ordinary matter. Each box is one step, with the section where it is taken, and each step stands on a tool of Chapter 4.

- the *mass phase angle* ϕ at Q, which faces the mass side m;
- the angle at the corner P, where the two sides meet. Because the charge side always leaves the mass axis at the angle $\bar{\phi}$, this corner is always $180^\circ - \bar{\phi} \approx 100.14^\circ$ when m and $q\mathcal{G}_{\bar{\phi}}$ have the same sign, and $\bar{\phi} \approx 79.86^\circ$ when their signs differ (Remark 4.30).

The three angles of a triangle add up to 180°, so $\alpha + \phi = \bar{\phi}$ (Remark 6.13). This is the whole of the geometry. From one body to the next only the two sides change; the corner angle never does.

7.1.1 *The Stoney Mass*

The mass side of a triangle can take any value, but the charge side cannot. Every free body in nature carries a whole number of elementary charges, $q = z\,e$, where z is the charge number: $z = -1$ for the electron, $z = +1$ for the proton, $z = +3$ for the iron ion Fe^{3+}, and $z = Z$, the atomic number, for a bare nucleus. (Quarks carry thirds of e, but no quark has ever been seen on its own.) The charge side $q\mathcal{G}_{\bar{\phi}} = z\,e\,\mathcal{G}_{\bar{\phi}}$ is therefore always a whole number of steps of one fixed length.

Definition 7.2 (Stoney Mass)**.** The **Stoney mass** is the charge part of one elementary charge, the size of $e\,\mathcal{G}_{\bar{\phi}}$ with $\mathcal{G}_{\bar{\phi}} = \sqrt{K/G}$ (Section 6.3):

$$m_S := e\sqrt{\frac{K}{G}} \approx 1.602177 \times 10^{-19}\ \mathrm{C} \times 1.16043 \times 10^{10}\ \mathrm{kg/C} \approx 1.85921 \times 10^{-9}\ \mathrm{kg}.$$

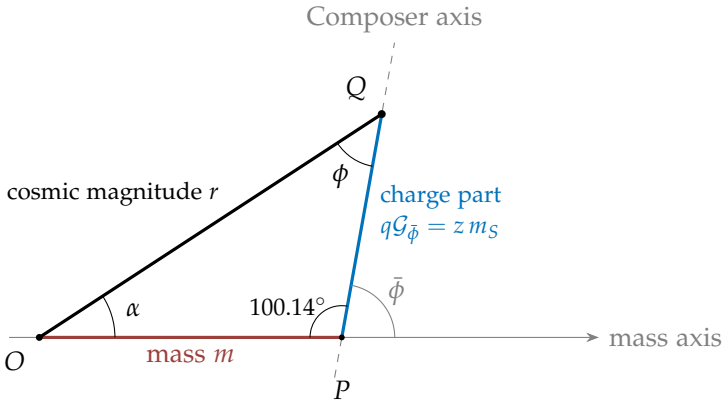


Figure 7.2: One body, one triangle, drawn to scale at 79.86°. The mass m runs along the mass axis from O to P, the charge part $q\mathcal{G}_{\bar{\phi}} = z\, m_S$ runs from P to Q along the direction of the Composer axis, and the cosmic magnitude r closes the triangle. The corner at P is always 100.14° when the two parts have the same sign. The drawing uses $m = 2$ and $q\mathcal{G}_{\bar{\phi}} = 1.5$ in any unit of mass, which give $r \approx 2.703$, $\alpha \approx 33.11°$ and $\phi \approx 46.75°$.

A body of charge number z has the charge part $q\mathcal{G}_{\bar{\phi}} = z\, m_S$, laid along the Composer axis.

Remark 7.3 (The Stoney Mass and the Planck Mass). The Stoney mass has two meanings.

1. *Balance.* The Composer Unit was sized so that $\mathcal{G}_{\bar{\phi}}^2 G = K$ (Section 6.3). Multiplied by e^2, this reads $G\, m_S^2 = Ke^2$: two bodies of mass m_S, each carrying one elementary charge, attract each other by gravity exactly as strongly as they repel each other electrically. They are a balanced pair: in the laws of Newton and Coulomb, two such bodies can rest at any distance from each other. More generally, two bodies of positive mass are a balanced pair exactly when their charge phase angles add up to $\bar{\phi}$, or to $\bar{\phi} + 180°$ when both charges are negative (Proposition 6.18); a body of mass m_S with one elementary charge has two equal parts and lies on the ray $\alpha = \bar{\phi}/2 \approx 39.93°$.

2. *The Planck mass.* With the reduced Planck constant $\hbar$ and the speed of light c, the Planck mass is $m_P = \sqrt{\hbar c / G} \approx 2.17643 \times 10^{-8}$ kg, and the fine-structure constant is $\alpha_{\text{fs}} = Ke^2/(\hbar c) \approx 1/137.036$. Then

$$\begin{aligned} m_S^2 &= \frac{Ke^2}{G} = \frac{Ke^2}{\hbar c} \cdot \frac{\hbar c}{G} \\ &= \alpha_{\text{fs}}\, m_P^2, \\ m_S &= \sqrt{\alpha_{\text{fs}}}\, m_P \approx 0.0854245\, m_P. \end{aligned}$$

   One Planck mass is $1/\sqrt{\alpha_{\text{fs}}} \approx 11.706$ Stoney masses.

The fine-structure constant is written α_{fs} here, so that it is not confused with the charge phase angle α.

BERLIN, MAY 1899

Twenty-five years after Stoney's paper, Max Planck was working on a problem that had nothing to do with electricity: how much light, of each color, a hot body gives off. In May 1899 he presented to the Prussian Academy of Sciences a paper on the radiation of heat in which a new constant of nature appeared; he wrote it b, and we now write it h. At the end of the paper he noticed what Stoney had noticed before him. With his new constant, the speed of light, the constant of gravitation and a constant of heat, it was possible to set up units of length, mass, time and temperature which, "independent of special bodies or substances, necessarily retain their meaning for all times and for all civilizations, even extraterrestrial and non-human ones" (Planck, 1899).

Planck did not use the electric charge, and Stoney had not used Planck's constant, which did not yet exist. Their two units of mass therefore differ, and the number that links them was named only later. In 1916 Arnold Sommerfeld, explaining the fine splitting of the lines of hydrogen, met a pure number made of e, $\hbar$ and c, about $1/137$ (Sommerfeld, 1916). That number, the fine-structure constant, is exactly the bridge of Remark 7.3: the Stoney mass is the Planck mass times its square root. Planck's own unit of mass, written with h instead of $\hbar$, was $\sqrt{hc/G} \approx 5.46 \times 10^{-8}$ kg; today's Planck mass, 2.17643×10^{-8} kg, is about 22 micrograms, roughly the mass of a small grain of sand.

In the frame of this chapter the Planck mass will appear as one ring among others, at 11.706 Stoney masses, between the nuclei of sodium ($Z = 11$) and magnesium ($Z = 12$). Every nucleus from magnesium on has a cosmic magnitude larger than the Planck mass. This is a fact about sizes in the frame, not a new physical threshold: nothing happens to a nucleus at the Planck ring.

WORKED EXAMPLE: THE IDENTITY CARD OF Fe^{3+}

The iron ion Fe^{3+} is an iron atom that has lost three electrons. Its charge number is $z = 3$.

- *Mass.* The atomic mass of iron is 55.845 u. Three electrons, each 0.000548580 u, are taken away: $m = 55.845 - 3 \times 0.000548580 \approx 55.843354$ u, or, with $1\ \text{u} = 1.66053907 \times 10^{-27}$ kg, $m \approx 9.27301 \times 10^{-26}$ kg.
- *Charge part.* By Definition 7.2, $q\mathcal{G}_{\bar{\phi}} = 3\, m_S \approx 5.57763 \times 10^{-9}$ kg, about sixty million billion times the mass.

- *Magnitude.* Remark 6.14 at 79.86° gives $r^2 = m^2 + (q\mathcal{G}_{\bar{\phi}})^2 + 0.35210\, m\, q\mathcal{G}_{\bar{\phi}}$. The mass changes r by less than one part in 10^{17}, so $r \approx 5.57763 \times 10^{-9}$ kg $= 3\, m_S$.
- *Angles.* The tiny mass tips the triangle away from the charge axis by an angle δ, which Remark 7.6 below gives as $\delta \approx 5.0374 \times 10^{-17\circ} \times 55.843354/3 \approx 9.3768 \times 10^{-16}$ degree. So $\alpha = 79.86^\circ - 9.3768 \times 10^{-16\circ}$ and $\phi = 9.3768 \times 10^{-16}$ degree, and indeed $\alpha + \phi = 79.86^\circ$.

In plain words

Charge comes in whole numbers of e, so the charge side of every triangle is a whole number of Stoney masses: one for the electron and the proton, three for Fe^{3+}, ninety-two for the nucleus of uranium. The Stoney mass is to the charge side what the dalton is to the mass side, the step of the ladder. It is also the mass of a speck whose gravity exactly balances the electric push of one elementary charge, and it is a fixed fraction, about one twelfth, of the Planck mass.

7.2 Step 2: All Triangles at Once: The Cosmic Mass Ellipse

One angle is fixed and two sides are free. What shape holds all the triangles that this angle allows? To find out, we keep the size of the triangle fixed and let its shape change.

7.2.1 *Eight Frames of a Film*

Fix the base of the triangle, the cosmic magnitude r, and take it as the unit of length, $r = 1$. Its two base angles are α and ϕ, and they always add up to $\bar{\phi} \approx 79.86^\circ$. Now let α grow, step by step, from 0 to 79.86°, so that ϕ shrinks from 79.86° to 0. Figure 7.3 shows eight frames of this film. By Remark 6.11, each frame has the two sides

$$m = \frac{\sin\phi}{\sin\bar{\phi}} = \frac{\sin(79.86^\circ - \alpha)}{0.98438}, \qquad q\mathcal{G}_{\bar{\phi}} = \frac{\sin\alpha}{\sin\bar{\phi}} = \frac{\sin\alpha}{0.98438}.$$

At the start of the film the charge side is almost nothing and the triangle lies nearly flat on its base: a nearly neutral body. At the end the mass side is almost nothing: a nearly pure charge. In between, the two sides trade length, and in frame 4, near $\alpha = \phi = 39.93^\circ$, they are equal.

Watch the corner where the two sides meet. Because its angle never changes, it always sees the base under the same angle, 100.14°, and an old theorem of Euclid says that all such points lie on one arc of

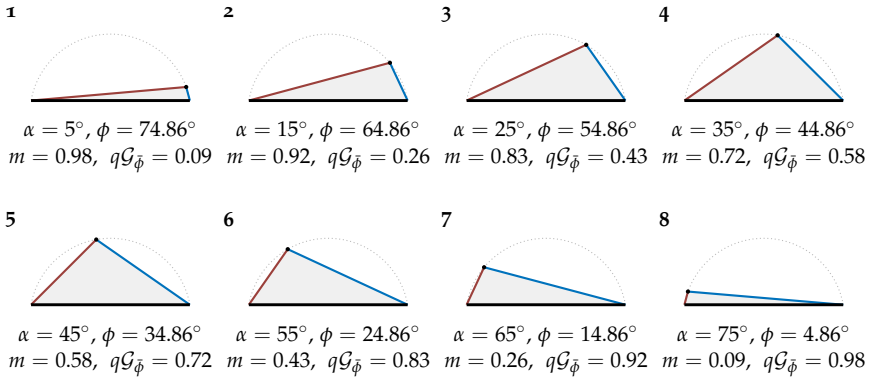


Figure 7.3: Eight frames of a film: triangles with the same base $r = 1$ (black) and the same corner angle 100.14°, for $\alpha = 5°, 15°, \ldots, 75°$. The mass side m is dark red and the charge side $q\mathcal{G}_{\bar\phi}$ blue, both in units of the base. The corner where they meet always lies on the dotted arc, part of a circle of radius $1/(2\sin\bar\phi) \approx 0.508$ through the two ends of the base.

a circle through the two ends of the base. The radius of that circle is $1/(2\sin\bar\phi) \approx 0.50793$ times the base, and its centre lies 0.08942 base-lengths below the middle of the base. With the base from $(0,0)$ to $(1,0)$, the corner moves on the circle $x^2 + y^2 = x - y\cot\bar\phi$. The moving corner does not draw an ellipse; the ellipse appears in the next picture, when the two sides are read as a pair of numbers.

7.2.2 *The Circle of the Tips*

In the relative plane itself, the same family looks even simpler (Figure 7.4). Lay every triangle with its mass side along the mass axis, from the origin O to the corner P. By Proposition 4.33, the tips of all the triangles of magnitude r lie on one circle, and the charge sides PQ, all parallel to the Composer axis, are parallel to one another. Each point Q of the circle is exactly one triangle: if Q sits at the charge phase angle α, the two sides are those of Remark 6.11. Three places on the circle are special:

- At $\alpha = 0$ the charge side vanishes and the triangle lies flat on the mass axis: a neutral body, such as the neutron or an atom, with $r = m$.
- At $\alpha = \bar\phi$ the mass side vanishes and the triangle lies flat on the Composer axis: a pure positive charge. The proton sits at this end, to within about 5×10^{-17} degree (Remark 7.6).
- At $\alpha = \bar\phi + 180° \approx 259.86°$ lies the pure negative charge. The electron sits at this end, to within about 3×10^{-20} degree.

The Composer axis cuts the circle into two halves. On the half that holds the positive mass axis, from the electron end through the neutral

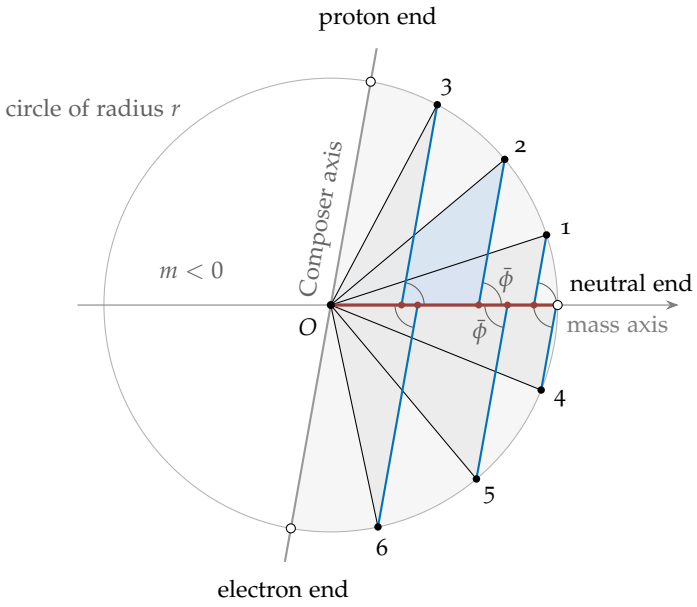


Figure 7.4: All the particle triangles of one magnitude r at 79.86°, in the relative plane: the mass side m (dark red) from O to the corner P, the charge side $q\mathcal{G}_{\bar{\phi}}$ (blue) from P to the tip Q, and the base r (black) from O to Q. The tips lie on the circle of radius r, and all the charge sides are parallel to the Composer axis: the gray arcs all measure $\bar{\phi}$. Triangles 1 to 3 have a positive charge part, 4 to 6 a negative one, and triangle 2 has two equal parts. The shaded half, $m > 0$, holds every real body.

end to the proton end, m is positive: this half holds every real body. A tip on the other half would need a negative mass.

7.2.3 *The Ellipse*

Now forget the drawing and keep only the two sides, as the old identity card kept only a mass and a charge. Mark each triangle as one point, with the mass m across and the charge part $q\mathcal{G}_{\bar{\phi}}$ up, at right angles. The points of all the triangles of one magnitude fill one curve.

Corollary 7.4 (The Cosmic Mass Ellipse)**.** *Let the angle $\bar{\phi}$ of the Composer axis be fixed, with $0 < \bar{\phi} < 180°$. For a fixed cosmic magnitude r, the pairs $(m, q\mathcal{G}_{\bar{\phi}})$ of all particle triangles of magnitude r form the ellipse*

$$m^2 + (q\mathcal{G}_{\bar{\phi}})^2 + 2\cos\bar{\phi} \cdot m\, q\mathcal{G}_{\bar{\phi}} = r^2,$$

which at 79.86° reads $m^2 + (q\mathcal{G}_{\bar{\phi}})^2 + 0.35210 \cdot m\, q\mathcal{G}_{\bar{\phi}} = r^2$. Its axes lie on the diagonals $q\mathcal{G}_{\bar{\phi}} = m$ and $q\mathcal{G}_{\bar{\phi}} = -m$, with the semi-axes

$$b = \frac{r}{\sqrt{1+\cos\bar{\phi}}} \approx 0.92212\, r \quad \text{on the diagonal } q\mathcal{G}_{\bar{\phi}} = m,$$

$$a = \frac{r}{\sqrt{1-\cos\bar{\phi}}} \approx 1.10167\, r \quad \text{on the diagonal } q\mathcal{G}_{\bar{\phi}} = -m,$$

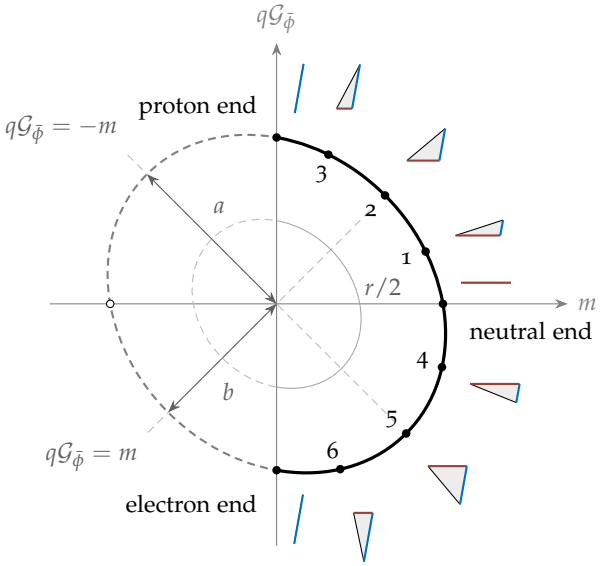


Figure 7.5: The cosmic mass ellipse. Each triangle of magnitude r is marked by its two sides, m across and $q\mathcal{G}_{\bar\phi}$ up, at right angles, and together they fill the ellipse $m^2 + (q\mathcal{G}_{\bar\phi})^2 + 0.35210\, m\, q\mathcal{G}_{\bar\phi} = r^2$ (Corollary 7.4), with the short semi-axis $b \approx 0.922\, r$ on the diagonal $q\mathcal{G}_{\bar\phi} = m$ and the long one $a \approx 1.102\, r$ on $q\mathcal{G}_{\bar\phi} = -m$. The small triangles, drawn with their true angles and the same base r, show which triangle each point stands for; the numbers match Figure 7.4. The inner ellipse holds the triangles of magnitude $r/2$; the dashed half needs a negative mass.

so that $b/a = \tan(\bar\phi/2) \approx 0.83702$. *The eccentricity* e *of the ellipse is given by* $e^2 = 2\cos\bar\phi/(1+\cos\bar\phi)$, *so* $e \approx 0.54717$, *and its area is* $\pi r^2/\sin\bar\phi \approx 3.19144\, r^2$. *As the charge phase angle* α *makes one turn,* $0° \le \alpha < 360°$, *the point*

$$m = r\,\frac{\sin(\bar\phi - \alpha)}{\sin\bar\phi}, \qquad q\mathcal{G}_{\bar\phi} = r\,\frac{\sin\alpha}{\sin\bar\phi}$$

runs once around the ellipse.

Proof. This is Theorem 4.34 for the cosmic mass: the real part is the mass m, the signed length of the Composer part is the charge part $q\mathcal{G}_{\bar\phi}$, and the magnitude is r; the equation is the law of cosines of Remark 6.14. At the mirror root, $\cos\bar\phi \approx 0.17605$ and $\sin\bar\phi \approx 0.98438$ (Remark 7.1) give $1/\sqrt{1.17605} \approx 0.92212$, $1/\sqrt{0.82395} \approx 1.10167$, $\tan 39.93° \approx 0.83702$, $e^2 = 0.35210/1.17605$ and $\pi/0.98438 \approx 3.19144$. □

Figure 7.5 draws the ellipse with the triangles of Figure 7.4 around it. Six readings follow.

1. *The four ends.* The ellipse crosses the axes at $\pm r$: the neutral end $(r, 0)$ and the two pure charge ends $(0, \pm r)$, where the proton and the electron sit.

2. *The arc of real triangles.* Triangles whose two base angles are both positive, $0 < \alpha < 79.86°$, fill the arc from the neutral end to the proton end; the eight frames of Figure 7.3 are eight points of it. The rest of the ellipse belongs to triangles with a side of negative length: a negative charge part, as for anions and the electron, on the arc below the mass axis, and a negative mass on the dashed half, where no known body lies.

3. *The short axis* lies on the diagonal $q\mathcal{G}_{\bar{\phi}} = m$: two equal parts with the same sign, $\eta = 1$. The triangle is isosceles, with $\alpha = \phi = \bar{\phi}/2 \approx 39.93°$ and $r = m\sqrt{2 + 2\cos\bar{\phi}} \approx 1.53366\,m$ (Remark 4.31). This is the balanced body of Remark 7.3, whose charge part equals its mass: a speck of one Stoney mass, 1.86 micrograms, carrying one elementary charge.

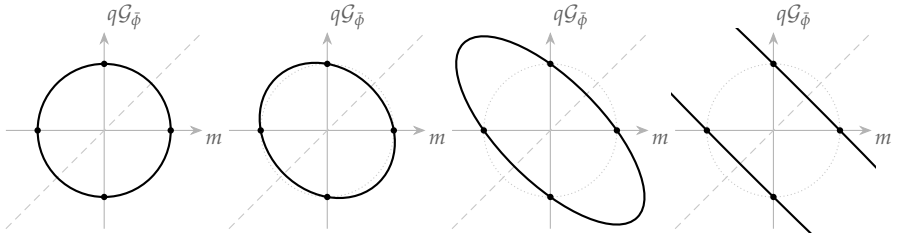
4. *The long axis* lies on the diagonal $q\mathcal{G}_{\bar{\phi}} = -m$: two equal parts with opposite signs, $\eta = -1$. The triangle is again isosceles, with the angle $\bar{\phi} \approx 79.86°$ at the corner and $r = m\sqrt{2 - 2\cos\bar{\phi}} \approx 1.28370\,m$ (Remark 4.31). Opposite parts partly cancel, so for the same r they can be longer, and the ellipse reaches further along this diagonal.

5. *The shape belongs to the angle.* A different magnitude only scales the ellipse, so all sizes give nested copies of the same shape (the copy for $r/2$ is drawn). One shape, set by $\bar{\phi}$ alone, holds every triangle that the angle allows, and the shape gives the angle back: $\bar{\phi} = 2\arctan(b/a)$.

6. *Where real matter falls.* Neutral bodies sit at the right end, each on the ellipse of its own size. Ions and nuclei sit at the top and bottom ends, on the ellipses of size $|z|\,m_S$, pushed off the ends only by their masses, which are 10^{16} to 10^{22} times smaller. The long stretches of the ellipse in between belong to bodies whose charge part is comparable to their mass, with a charge-to-mass ratio near $\sqrt{G/K} \approx 8.6 \times 10^{-11}$ C/kg.

7.2.4 *How the Angle Shapes the Ellipse*

Since the ellipse depends on nothing but $\bar{\phi}$ (Theorem 4.34), it shows at once what another angle would mean (Figure 7.6 and table 7.1). All the ellipses pass through the same four ends, $(\pm r, 0)$ and $(0, \pm r)$; the angle only decides how they bulge between them. At $\bar{\phi} = 90°$ the middle term vanishes and the ellipse is a circle: every triangle is a right triangle, and the relative plane is the complex plane (Theorem 4.5). At the mirror root the circle is slightly squeezed along the diagonal of equal signs, and at the arcsine root it becomes a thin needle along the diagonal of opposite signs.



(a) $\bar{\phi} = 90°$ right angle (b) $\bar{\phi} = 79.86°$ mirror root (c) $\bar{\phi} = 45°$ mirror line (d) $\bar{\phi} = 10.14°$ arcsine root

Figure 7.6: How the angle of the Composer axis shapes the ellipse of all triangles of one magnitude r (Table 7.1). The dots are the four ends that every ellipse shares, the dotted circle is the case $\bar{\phi} = 90°$, and the dashed line is the diagonal $q\mathcal{G}_{\bar{\phi}} = m$. In (d) the needle leaves the frame and reaches $8.0\,r$ along $q\mathcal{G}_{\bar{\phi}} = -m$.

$\bar{\phi}$	CASE	b/r	a/r	b/a	ECCENTRICITY
90°	complex plane	1	1	1	0
79.86°	mirror root	0.92212	1.10167	0.83702	0.54717
45°	mirror line	0.76537	1.84776	0.41421	0.91018
10.14°	arcsine root	0.70988	8.00141	0.08872	0.99606
0°	a line, no plane	0.70711	none	0	1

Table 7.1: The ellipse of all triangles of magnitude r for five angles of the Composer axis: the semi-axes b (on $q\mathcal{G}_{\bar{\phi}} = m$) and a (on $q\mathcal{G}_{\bar{\phi}} = -m$), their ratio $b/a = \tan(\bar{\phi}/2)$, and the eccentricity. At 0° the ellipse opens into two parallel lines.

> IN PLAIN WORDS
>
> One angle, one shape. Every particle triangle has the same corner angle; only its two sides are free. Hold the base fixed and the corner slides along an arc of a circle. Read the two sides as a pair of numbers instead, and all the triangles of one size line up on an ellipse, whose shape is set by the angle alone: a circle at 90°, a slightly squeezed oval at the mirror root, a needle at the arcsine root. Its equation is the law of cosines, read as a curve.

7.3 STEP 3: THE POLAR-LIKE FRAME

Polar coordinates name a point of the plane by a distance and an angle: circles of equal distance, rays of equal angle. The identity card (r, α, ϕ) names a body in the same way, and the ellipses of Corollary 7.4 are its circles. The frame is the chart of rings and rays of Definition 4.35, drawn for the cosmic mass.

Definition 7.5 (The Cosmic Mass Frame)**.** The **Cosmic Mass Frame** is the plane of the two parts of the cosmic mass, the mass m across and the charge part $q\mathcal{G}_{\bar\phi}$ up, at right angles, ruled by two families of lines:

- the *rings*, the ellipses $m^2 + (q\mathcal{G}_{\bar\phi})^2 + 2\cos\bar\phi \cdot m\, q\mathcal{G}_{\bar\phi} = r^2$ of Corollary 7.4, one for each cosmic magnitude r;
- the *rays* of equal charge phase angle α, straight lines from the origin.

A body with the identity card (r, α, ϕ) is the point where its ring and its ray cross.

Along a ray the ratio of the two parts stays fixed, $q\mathcal{G}_{\bar\phi}/m = \eta = \sin\alpha / \sin(79.86^\circ - \alpha)$, the general tangent of Remark 6.12, and so does the mass phase angle $\phi = 79.86^\circ - \alpha$. The ring r and the ray α cross at the point

$$m = r\,\frac{\sin(79.86^\circ - \alpha)}{0.98438}, \qquad q\mathcal{G}_{\bar\phi} = r\,\frac{\sin\alpha}{0.98438}$$

of Remark 6.11. The frame is therefore the General Sinus drawn as graph paper: the old card (m, q) and the new card (r, α, ϕ) are two names for the same point. In the relative plane the same frame is the ordinary polar frame, with circles and evenly spaced rays (Figure 7.4 shows one of its circles); drawing the two parts at right angles turns the circles into ellipses and moves the rays. Figure 7.7 draws it.

Five readings follow.

1. *The two axes are rays.* The mass axis is the ray $\alpha = 0$, and 180° on its negative side. The charge axis is the ray $\alpha = 79.86^\circ$, and 259.86° below the origin; the rays just beyond it lean into the half of negative mass.
2. *The rays are not evenly spread.* Between the positive mass and charge axes, 79.86 degrees of α fill a right angle; in the next quarter, 100.14 degrees do. The rays fan out a little where m and $q\mathcal{G}_{\bar\phi}$ have the same sign and crowd a little where their signs differ.
3. *Reading along the axes and the diagonals.* On the mass axis $r = |m|$, and on the charge axis $r = |q\mathcal{G}_{\bar\phi}|$, so the numbers on the mass axis also name the rings. The short axis of every ring lies on the ray $\alpha = \bar\phi/2 \approx 39.93^\circ$, where $\eta = 1$, and the long axis on the ray $\alpha \approx -50.07^\circ = 309.93^\circ$, where $\eta = -1$.

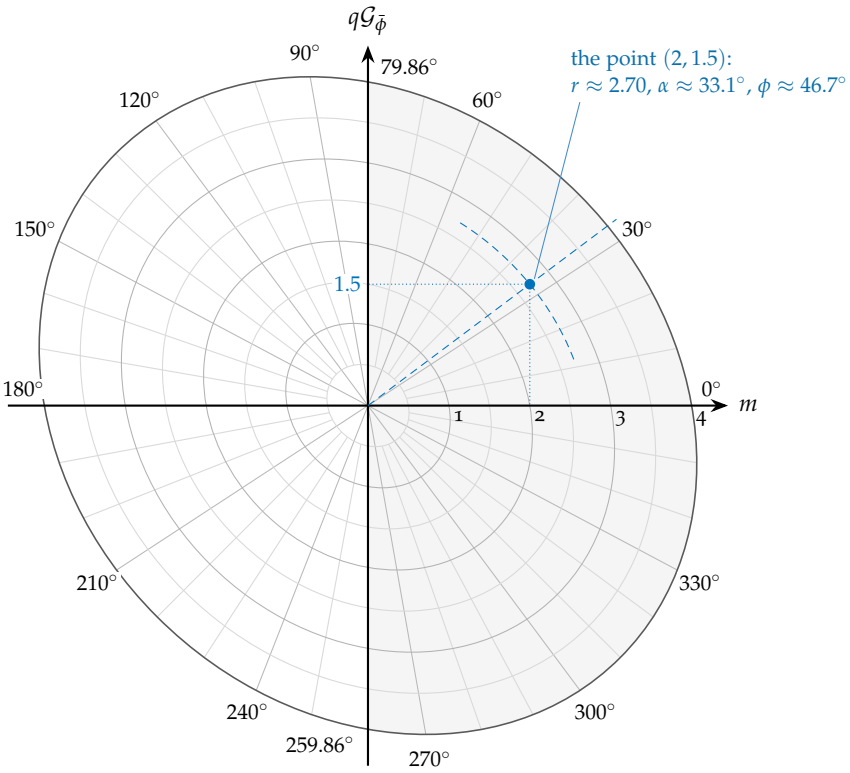


Figure 7.7: The Cosmic Mass Frame at 79.86° (Definition 7.5), with m across and $q\mathcal{G}_{\bar{\phi}}$ up, in any unit of mass. Rings: the magnitudes $r = 0.5$ to 4, numbered on the mass axis, where $r = m$. Rays: the charge phase angle α, every 10° and labeled every 30°; along each ray $\phi = 79.86° - \alpha$ is fixed too. The blue point $(2, 1.5)$ reads $r \approx 2.70$ and $\alpha \approx 33.1°$. The shaded half, $m > 0$, holds every real body.

4. *Reading a point.* The blue point $(m, q\mathcal{G}_{\bar{\phi}}) = (2, 1.5)$, in any unit of mass, lies between the rings 2.5 and 3 and between the rays 30° and 40°. The remarks give its card exactly:

$$r = \sqrt{4 + 2.25 + 0.35210 \cdot 3} \approx 2.703, \qquad \eta = 0.75,$$
$$\alpha = \arctan \frac{0.98438 \cdot 0.75}{1 + 0.17605 \cdot 0.75} \approx 33.11°,$$
$$\phi = 79.86° - \alpha \approx 46.75°.$$

   It is the triangle of Figure 7.2.

5. *Real bodies.* Neutral bodies lie on the ray $\alpha = 0$, positive charges near the ray 79.86° and negative charges near the ray 259.86°. With the microgram as the unit, one elementary charge sits on the charge axis at 1.86, the Stoney mass, so this frame has the right size for specks of dust that carry one or two elementary charges. Atoms, 10^{15} to 10^{18} times lighter, crowd at the origin.

> **In plain words**
>
> Stack the ellipses of all sizes and add the rays of equal angle, and they become graph paper: rings instead of circles, rays instead of spokes. Any body can be found on it in two ways, by its old card, a mass and a charge, or by its new card, a magnitude and two angles. Both lead to the same point.

7.4 STEP 4: WHERE MATTER FALLS

The frame is ready. Where in it do the bodies of ordinary matter lie? Two kinds of answer follow from the circle of the tips (Section 7.2.2).

A *neutral body*, an atom or a neutron, has no charge part. Its triangle lies flat on the mass axis, so it sits on the ray $\alpha = 0$, with $\phi = 79.86^\circ$ and $r = m$: its ring is its mass.

A *charged body*, an ion, a nucleus, a proton or an electron, has a charge part $z\,m_S$ that dwarfs its mass. Its ring is therefore, to many digits, $r = |z|\,m_S$, and its triangle lies almost flat on the Composer axis. Almost, but not quite: the small mass tips it away from the charge axis by a tiny angle, which we call the *lean*.

Remark 7.6 (The Lean of a Charged Body). Let a body have the mass m and the charge number $z \neq 0$, and let $|\eta| = |z|\,m_S/m$ be the size of its charge-to-mass ratio measured through the Composer Unit (Remark 6.15). The **lean** δ of its cosmic mass from the charge axis, and its two angles, are

$$\begin{aligned} z > 0\colon\quad & \delta = \arctan\frac{\sin\bar{\phi}}{|\eta| + \cos\bar{\phi}}, && \alpha = \bar{\phi} - \delta, \quad \phi = \delta; \\ z < 0\colon\quad & \delta = \arctan\frac{\sin\bar{\phi}}{|\eta| - \cos\bar{\phi}}, && \alpha = \bar{\phi} + 180^\circ + \delta, \\ & && \phi = -180^\circ - \delta. \end{aligned}$$

For every body of ordinary matter $|\eta|$ lies between 10^{15} and 10^{22}, so the arctangent equals its argument to many digits, and

$$\begin{aligned} \delta &\approx \frac{180^\circ}{\pi} \cdot \frac{\sin\bar{\phi}\cdot 1\,\mathrm{u}}{m_S} \cdot \frac{m}{|z|} \\ &\approx 5.0374 \times 10^{-17}\ \text{degree} \times \frac{m}{|z|}, \end{aligned}$$

with m in daltons. The lean is proportional to the mass per charge, $m/|z|$: one step of 10^{-15} degree of lean is 19.85 daltons of mass per elementary charge.

The formulas are Proposition 4.32 read for a cosmic mass, with $|\eta| = |z|\,m_S/m$, that is, Remark 6.15 written for a small mass: the lean is the mass phase angle ϕ of a positive charge, and its distance from -180° for a negative one. Two bodies with the same mass per

charge lie on the same ray, whatever their charge; they differ only in their ring. The heavier a body is for its charge, the more it leans. The electron, with 0.000549 daltons per charge, leans least, by 2.8×10^{-20} degree; the francium ion Fr^{+}, with 223 daltons per charge, leans most, by 1.12×10^{-14} degree.

CAMBRIDGE, 1919

When the war ended, Francis William Aston went back to the Cavendish Laboratory in Cambridge, where J. J. Thomson had invited him in 1910. Before the war, sorting beams of charged atoms with Thomson's method, Aston had seen that neon seemed to come in two kinds, of mass about 20 and 22; he called the second meta-neon, but could not prove that it was a separate kind of atom. In 1919 he finished an instrument to settle the question, which he called a positive ray spectrograph. A beam of ions passes first through an electric field and then through a magnetic field, arranged so that all the ions with the same ratio of mass to charge come together at one place on a photographic plate, whatever their speed. Each kind of ion leaves one sharp line (Aston, 1919).

Neon gave two lines, at 20 and 22. Chlorine, whose atomic weight is 35.45, gave lines at 35 and 37 and nothing in between. Element after element came apart into isotopes whose masses were very nearly whole numbers, and Aston went on to identify 212 of the isotopes found in nature. In 1922 he received the Nobel Prize in Chemistry for the mass spectrograph and for the whole-number rule.

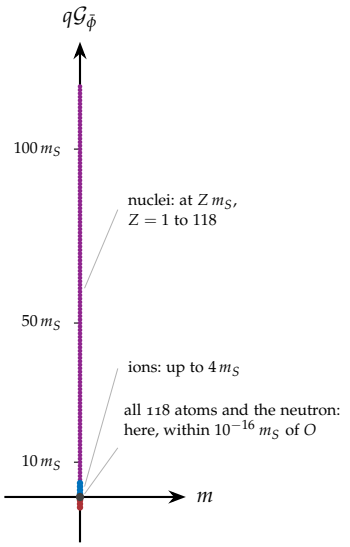
Aston's instrument sorts ions by exactly the quantity that sets the lean in the Cosmic Mass Frame: the mass per charge. Every line on his plate is a ray of the frame.

7.4.1 *Sixteen Orders of Magnitude*

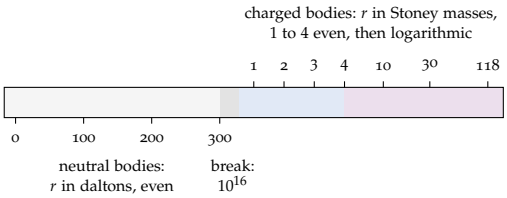
Now try to draw all of ordinary matter in the frame at its true scale. The masses of atoms lie between 1.7×10^{-27} and 4.9×10^{-25} kg, while the charge part of a single elementary charge is 1.86×10^{-9} kg, sixteen orders of magnitude larger. If the ring of one Stoney mass were drawn a metre from the origin, even the heaviest atom would lie less than 3×10^{-16} metre from it, a small fraction of the width of a proton; and the leans, at most 10^{-14} degree, are invisible. Drawn true to scale, the whole of ordinary matter is a dot at the origin and a short row of dots up and down the charge axis (Figure 7.8a).

To see ordinary matter, the frame needs two lenses. Both are monotonic: they keep the order of the numbers, so rings stay rings, rays stay rays, and every body keeps its place relative to its neighbours.

- *The radial lens* (Figure 7.8b) gives the inner part of the frame to neutral bodies, with r counted in daltons from 0 to 300 u, and the outer part to charged bodies, with r counted in Stoney masses.



(a) true scale



(b) the radial lens of Figure 7.9: distance from O along any ray

Figure 7.8: (a) Ordinary matter in the Cosmic Mass Frame at true scale, in Stoney masses. All the atoms and the neutron fall on the origin, the ions sit within four Stoney masses of it, and the bare nuclei climb the charge axis one Stoney mass at a time, up to 118; no lean can be seen. (b) The radial lens used in Figure 7.9: the distance from the origin along any ray, with neutral bodies measured in daltons, a break of sixteen orders of magnitude, and charged bodies measured in Stoney masses, evenly from ring 1 to ring 4 and logarithmically from ring 4 to ring 118.

Between the two lies a break of sixteen orders of magnitude. The rings of the ions, 1 to 4, are evenly spaced; beyond ring 4 the rings of the nuclei come closer together, by the logarithm of $|z|$, so that ring 118 fits on the page.

- *The angular lens* keeps the true angles on the half $m < 0$, where no body lies. Next to the charge axis it magnifies the lean 6.4×10^{15} times, so that positive bodies fan out clockwise from 79.86° and negative ones counterclockwise from 259.86°, each at an angle proportional to its mass per charge. Two thin sectors beside the mass axis hold all the remaining angles, squeezed.

> **IN PLAIN WORDS**
>
> Matter uses the frame in a very lopsided way. Everything without charge lies on one line, the mass axis, and everything with charge lies on another, the charge axis, tipped away from it by angles smaller than anything a protractor could ever measure. To see the bodies, we magnify: the masses of atoms and the charges of ions on separate scales, and the tiny tips by a factor of six thousand million million. The tip of a charged body turns out to be its mass per charge, the very number that a mass spectrometer reads.

7.5 STEP 5: THE IDENTITY CARDS OF MATTER

Every body of ordinary matter now has a recipe for its card. Only two numbers are needed, its mass m and its charge number z; Remark 6.14 then gives r, and Remark 7.6 gives α and ϕ. The bodies fall into five kinds, and the masses come from the following sources.

- *The electron, the proton and the neutron:* their masses and charges from CODATA 2018 (Tiesinga et al., 2021), as in Appendix D.3.
- *Atoms* ($z = 0$, 118 bodies): the standard atomic weights of 2024 (Prohaska et al., 2022; Commission on Isotopic Abundances and Atomic Weights, 2024); for the fourteen elements whose standard weight is an interval, the conventional value; and for the elements that have no standard weight, the mass number of their longest-lived isotope, as periodic tables print it, good to about 0.1%.
- *Cations and anions* (67 bodies): the common monatomic ions of chemistry, with charges from -3 to $+4$. An ion of charge number z is its atom with z electrons removed, or $|z|$ added:

  $$m_{\text{ion}} = A - z\,m_e,$$

  where A is the atomic mass and $m_e \approx 0.000548580$ u. A cation ($z > 0$) is lighter than its atom and an anion ($z < 0$) heavier. The energy that binds the moved electrons, below 10^{-7} u, is left out.
- *Nuclei* (118 bodies): each atom with all its Z electrons removed, so that $z = Z$. The electrons were bound to the atom, and the energy of that binding, B_e, is given back as mass when they leave:

  $$m_{\text{nucleus}} = A - Z\,m_e + B_e/c^2,$$
  $$B_e \approx \left(14.4381\,Z^{2.39} + 1.55468 \times 10^{-6}\,Z^{5.35}\right) \text{ eV},$$

  a standard fit of the total binding energy of the electrons (Lunney et al., 2003). The correction is small: 0.00082 u for uranium and at most 0.0016 u, for oganesson.

A nucleus is here the average nucleus of the natural element, because the atomic mass is an average over the element's isotopes. That is why the hydrogen nucleus, 1.007451 u, is a little heavier than the proton, 1.007276 u: natural hydrogen holds some deuterium. The ion H^+ and the hydrogen nucleus are the same body; their masses differ only by the 14 eV that bound the one electron.

MANCHESTER, 1913

In 1913 Henry Moseley, one of the youngest researchers in Ernest Rutherford's laboratory in Manchester, set out to measure the X-rays of the elements. Two years earlier Rutherford had found that every atom has a small, heavy nucleus with a positive charge (Chapter 5). Moseley made each element in turn the target of a beam of cathode rays and measured the X-rays it gave off by reflecting them from a crystal, the new method of William and Lawrence Bragg. For a dozen elements from calcium to zinc, the frequency of the strongest line rose in perfectly regular steps from one element to the next. "We have here a proof that there is in the atom a fundamental quantity," he wrote, and: "This quantity can only be the charge on the central positive nucleus" (Moseley, 1913).

That quantity is the atomic number Z. Before Moseley, the number of an element was only its place in Mendeleev's table; after him it was a measured charge. His X-rays put cobalt before nickel, against the order of their atomic weights, as chemists had always placed them, and in the second part of the work, finished in Oxford in 1914, they left exactly four places empty below gold, at $Z = 43, 61, 72$ and 75, for elements still to be found (Moseley, 1914). On 10 August 1915 Moseley was killed at Gallipoli, aged twenty-seven.

Moseley's number is the ring of a nucleus in the Cosmic Mass Frame: a bare nucleus has the charge number $z = Z$ and lies on the ring $Z\, m_S$.

Table 7.2 shows a few cards, and Appendix E gives all 306. The same cards come with this book as a workbook, *Identity-Cards-data.xlsx*, in which every number is a live formula: the mass of each ion and each nucleus is calculated from the mass of its atom, so that a new atomic weight updates its ions and its nucleus at once; the columns give the mass in daltons and in kilograms, the charge part, the magnitude r, the three angles α, ϕ and $\bar{\phi}$, and the lean δ. A spreadsheet keeps about fifteen digits, while α for a charged body differs from 79.86° or 259.86° only from its sixteenth to its twenty-second digit; the lean is therefore computed on its own, with the formula of Remark 7.6, in which no digits cancel.

BODY	z	m (u)	r (kg)	α (degree)	ϕ (degree)
electron	-1	0.000548579909	1.85921×10^{-9}	$259.86 + 2.76342\times10^{-20}$	$-180 - 2.76342\times10^{-20}$
proton	$+1$	1.00727647	1.85921×10^{-9}	$79.86 - 5.07406\times10^{-17}$	5.07406×10^{-17}
neutron	0	1.00866492	1.67493×10^{-27}	0	79.86
H	0	1.008	1.67382×10^{-27}	0	79.86
H^+	$+1$	1.00745142	1.85921×10^{-9}	$79.86 - 5.07494\times10^{-17}$	5.07494×10^{-17}
H nucleus	$+1$	1.00745144	1.85921×10^{-9}	$79.86 - 5.07494\times10^{-17}$	5.07494×10^{-17}
He nucleus	$+2$	4.00150492	3.71842×10^{-9}	$79.86 - 1.00786\times10^{-16}$	1.00786×10^{-16}
C	0	12.011	1.99447×10^{-26}	0	79.86
O^{2-}	-2	16.0000972	3.71842×10^{-9}	$259.86 + 4.02995\times10^{-16}$	$-180 - 4.02995\times10^{-16}$
Na^+	$+1$	22.9892207	1.85921×10^{-9}	$79.86 - 1.15806\times10^{-15}$	1.15806×10^{-15}
Cl^-	-1	35.4505486	1.85921×10^{-9}	$259.86 + 1.78579\times10^{-15}$	$-180 - 1.78579\times10^{-15}$
Fe	0	55.845	9.27328×10^{-26}	0	79.86
Fe^{3+}	$+3$	55.8433543	5.57763×10^{-9}	$79.86 - 9.37685\times10^{-16}$	9.37685×10^{-16}
Fe nucleus	$+26$	55.8307743	4.83394×10^{-8}	$79.86 - 1.08170\times10^{-16}$	1.08170×10^{-16}
U	0	238.02891	3.95256×10^{-25}	0	79.86
U nucleus	$+92$	237.979259	1.71047×10^{-7}	$79.86 - 1.30304\times10^{-16}$	1.30304×10^{-16}
Og nucleus	$+118$	293.936858	2.19387×10^{-7}	$79.86 - 1.25481\times10^{-16}$	1.25481×10^{-16}

Table 7.2: A few identity cards of matter at 79.86°, out of the 306 of Appendix E. The magnitude r of a charged body is $|z|\,m_S$ to many digits; the angles are written with the lean δ of Remark 7.6, which is far too small to be written as a plain decimal.

WORKED EXAMPLE: SALT, AND THE NUCLEUS OF URANIUM

Na^+. Sodium has the atomic mass 22.98976928 u. The ion has lost one electron: $m = 22.98976928 - 0.00054858 = 22.98922070$ u. Its ring is $r = m_S \approx 1.85921 \times 10^{-9}$ kg, and its lean is $\delta \approx 5.0374 \times 10^{-17} \times 22.98922 \approx 1.15806 \times 10^{-15}$ degree. So $\alpha = 79.86° - 1.15806 \times 10^{-15}°$ and $\phi = 1.15806 \times 10^{-15}$ degree.

Cl^-. Chlorine, 35.45 u, has gained one electron: $m = 35.45054858$ u. Its ring is again m_S, on the negative side, and its lean is $\delta \approx 1.78579 \times 10^{-15}$ degree, so $\alpha = 259.86° + 1.78579 \times 10^{-15}°$ and $\phi = -180° - 1.78579 \times 10^{-15}°$. In a grain of salt, each Na^+ and each Cl^- lean the same way, towards positive mass, by amounts in the ratio of their masses.

The uranium nucleus. Uranium, 238.02891 u, loses all 92 electrons, $92 \times 0.00054858 \approx 0.05047$ u, and gains back their binding energy, about 763 keV or 0.00082 u: $m \approx 237.97926$ u. Its ring is $92\,m_S \approx 1.71047 \times 10^{-7}$ kg, almost eight Planck masses, and its lean is $\delta \approx 5.0374 \times 10^{-17} \times 237.97926/92 \approx 1.30304 \times 10^{-16}$ degree, about nine times less than the lean of Na^+: the uranium nucleus carries far more charge for its mass.

7.6 STEP 6: THE COSMIC MASS STANDARD MODEL

Figure 7.9 places all 306 bodies in one Cosmic Mass Frame, with the two lenses of Section 7.4.1. This is the Cosmic Mass Standard Model: one frame, built from one fixed angle, in which every body of ordinary matter has its card and its place.

Seven readings follow.

1. *Three directions.* Ordinary matter uses only three directions of the plane: the mass axis, for all neutral bodies; the positive charge axis, for the proton, the cations and the nuclei; and the negative charge axis, for the electron and the anions.

2. *The lean is the mass per charge.* Along each charge axis the bodies fan out by their mass per charge (Remark 7.6). The frame works like Aston's spectrograph: bodies with the same mass per charge share one ray, and Figure 7.10 shows the ions with their names, ring by ring.

3. *Cations and anions lean the same way.* Both lean towards positive mass, clockwise from 79.86° for positive charges and counter-clockwise from 259.86° for negative ones, because every mass is positive.

4. *The nuclear spine.* The 118 nuclei climb the positive charge axis ring by ring and lean only 0.5 to 1.3×10^{-16} degree, because a nucleus carries much more charge for its mass than any ion. Their lean grows with Z (Figure 7.11). Light nuclei, from helium to calcium, hold about as many neutrons as protons, so their mass per charge is about 2 daltons; heavy nuclei hold more neutrons than protons, and the mass per charge of uranium is 2.59 daltons. The excess of neutrons in heavy nuclei, which keeps them together against the push of their protons, shows in the frame as a slow bend of the spine away from the charge axis.

5. *The atoms.* Neutral atoms lie on the mass axis at $r = m$, from hydrogen at 1.008 u to oganesson at 294 u (Figure 7.12). The neutron lies beside hydrogen.

6. *Three rings to remember.* Ring 1 is the Stoney mass, the home of the proton, the electron and every singly charged ion. Ring 11.706 is the Planck mass, between the nuclei of sodium and magnesium. Ring 92 is the uranium nucleus, almost eight Planck masses.

7. *The empty middle.* Between the axes lie bodies whose charge part is comparable to their mass, with a charge-to-mass ratio near $\sqrt{G/K} \approx 8.6 \times 10^{-11}$ C/kg. On the ray $\alpha = 39.93°$ the two parts are equal: the balanced body of Remark 7.3, which in the laws of Newton and Coulomb can rest at any distance from its twin. The next reading finds balanced partners for every charged body.

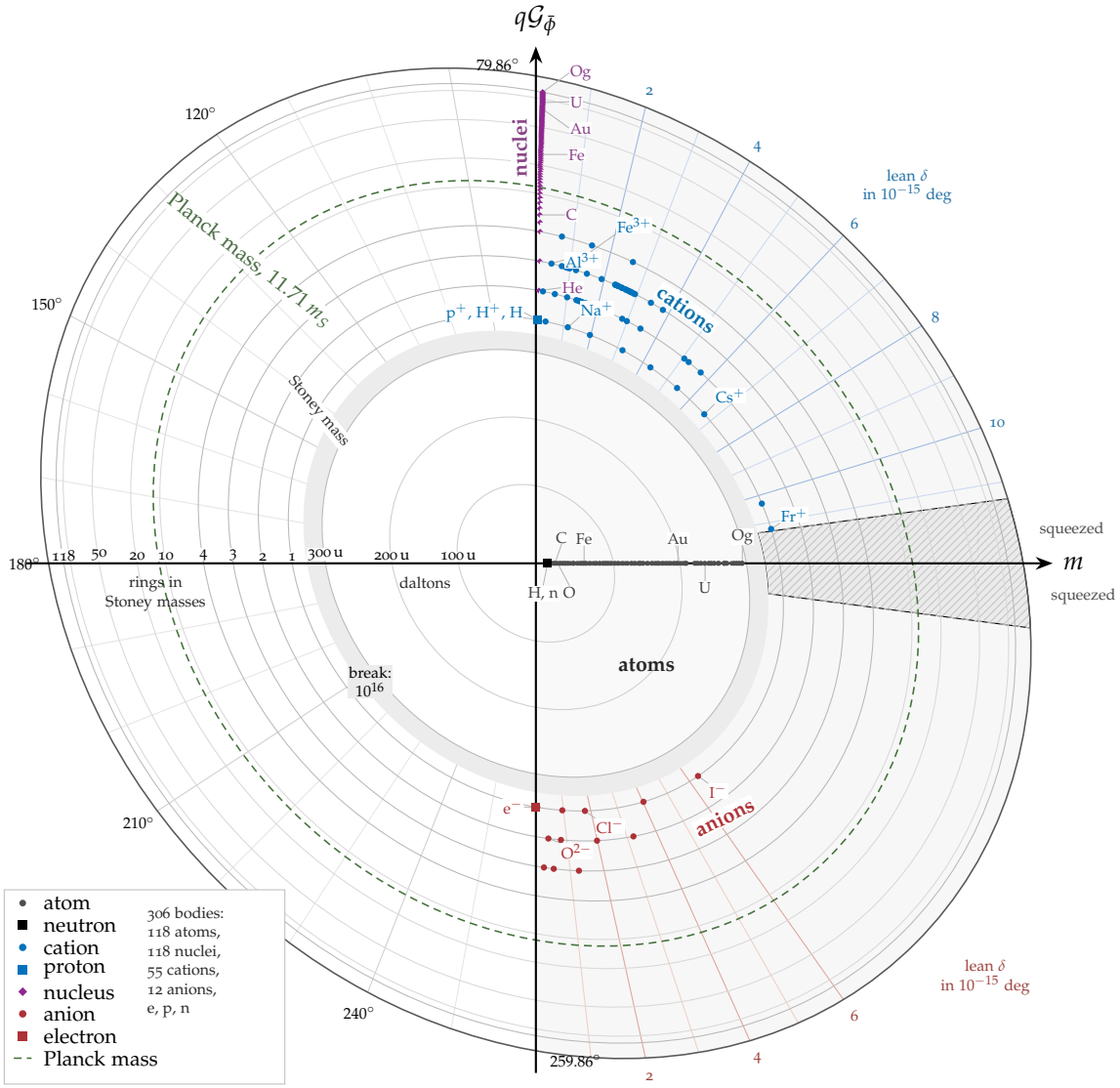


Figure 7.9: The Cosmic Mass Standard Model: all 306 bodies of Appendix E in the Cosmic Mass Frame at 79.86°, with m across and $q\mathcal{G}_{\bar{\phi}}$ up. Two lenses make them visible (Section 7.4.1). Radially, the inner zone counts neutral bodies in daltons, the gray band is a break of sixteen orders of magnitude, and the outer zone counts charged bodies in Stoney masses, rings 1 to 4 evenly and rings 4 to 118 logarithmically; the dashed green ring is the Planck mass. Angularly, the lean of a charged body from the charge axis is magnified 6.4×10^{15} times and labeled in 10^{-15} degree, true angles are kept on the half $m < 0$, and the hatched sectors hold the remaining angles, squeezed. Gray dots: atoms, on the mass axis. Blue: cations, and the proton as a square. Red: anions, and the electron as a square. Violet diamonds: bare nuclei, on the rings $Z = 1$ to 118. Figures 7.10 to 7.12 show the crowded parts with every name.

STEP	WHAT IS BUILT	THE FORMULA
1	one body, one triangle	$\alpha + \phi = \bar{\phi}$; corner $100.14°$; $q\mathcal{G}_{\bar{\phi}} = z\, m_S$
2	all triangles of one size	$m^2 + (q\mathcal{G}_{\bar{\phi}})^2 + 0.35210\, m\, q\mathcal{G}_{\bar{\phi}} = r^2$
3	the frame	rings of equal r, rays of equal α
4	where matter falls	$r = m$ or $r = \lvert z\rvert\, m_S$; $\delta \approx 5.0374 \times 10^{-17°} \times m/\lvert z\rvert$
5	the cards	$m_{\text{ion}} = A - z\, m_e$; $m_{\text{nucleus}} = A - Z\, m_e + B_e/c^2$
6	the standard model	306 bodies on three directions of one frame

Table 7.3: The Cosmic Mass Standard Model in six lines, all at the angle $79.86°$ of the Composer axis.

In this region a part can even be longer than the ring: the mass exceeds the ring by up to 1.6%, for a body whose arrow stands at right angles to the Composer axis, $\alpha = \bar{\phi} - 90°$, with a negative charge part of 0.17605 times its mass (Remark 4.60). No atom, ion or nucleus comes anywhere near; that region of the frame belongs to charged grains, planets, stars and black holes.

One more reading: pairs on the frame. The frame also reads the force between any two bodies (Proposition 6.18): add the angles of their two rays, and the product of their two rings sets the scale. The sum decides the sign, a pull between $-100.14°$ and $79.86°$ and a push between $79.86°$ and $259.86°$. Two neutral bodies add up to $0°$ and pull; two ions, nuclei or particles of the same sign add up to about $159.72°$ and push, and of opposite signs to about $339.72°$, and pull. A neutral body and an ion add up to an angle that lies the lean δ of the ion inside the band of pulls, and what is left is $r_a r_{\text{ion}} \sin\delta / \sin\bar{\phi} = m_a m_{\text{ion}}$: their gravity alone, because the lean carries the mass of the ion, $r \sin\delta = m \sin\bar{\phi}$ (Remarks 6.11 and 7.6). In the frame itself, the border between the pulling and the pushing partners of a body is a straight line through 0: its own ray mirrored in a diagonal of the frame, $q\mathcal{G}_{\bar{\phi}} = m$ for a positive body and $q\mathcal{G}_{\bar{\phi}} = -m$ for a negative one. The bodies on that line are its balanced partners.

7.6.1 *The Standard Model in Six Lines*

Table 7.3 gathers the road of this chapter. Each line uses only what the lines above it established.

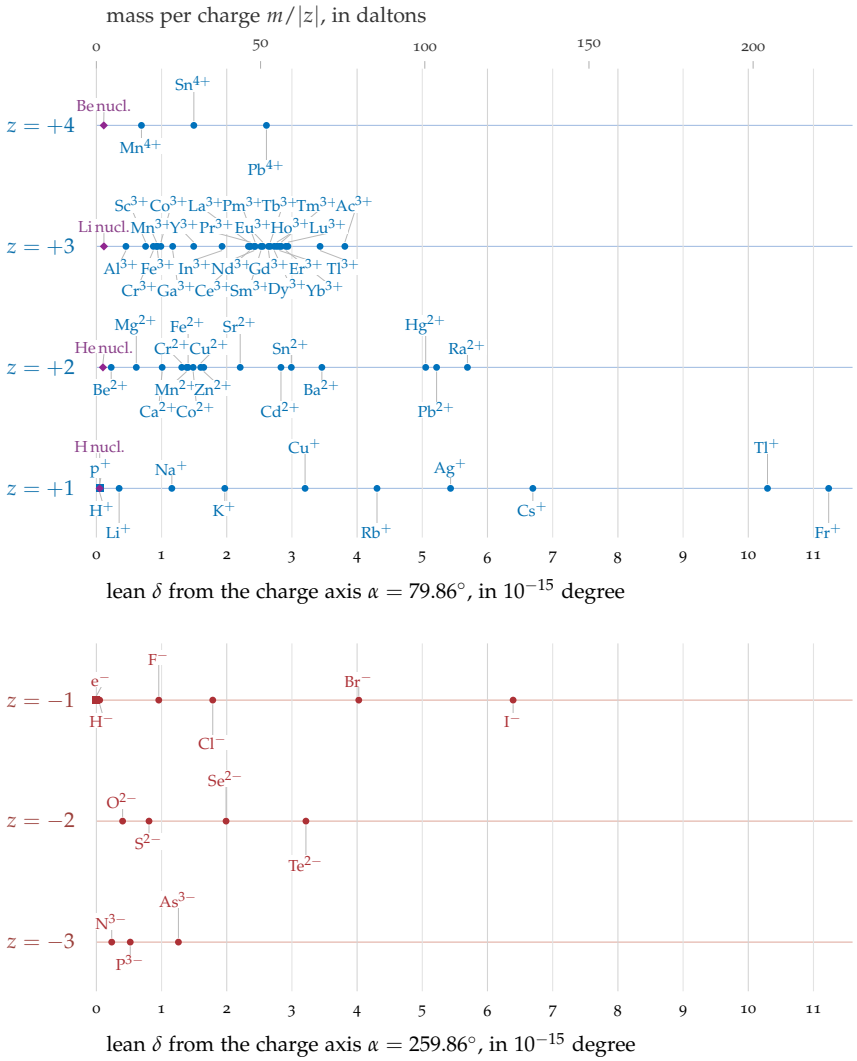


Figure 7.10: The ions of Figure 7.9, every one named. The two magnified sectors next to the charge axis are unrolled: each ring z becomes a line, and the lean δ runs across, in 10^{-15} degree, from the charge axis on the left. The scale on top reads the same lean as mass per charge. Blue: cations and the proton (square); red: anions and the electron (square); violet: the bare nuclei of hydrogen to beryllium, which share the rings 1 to 4 with the ions.

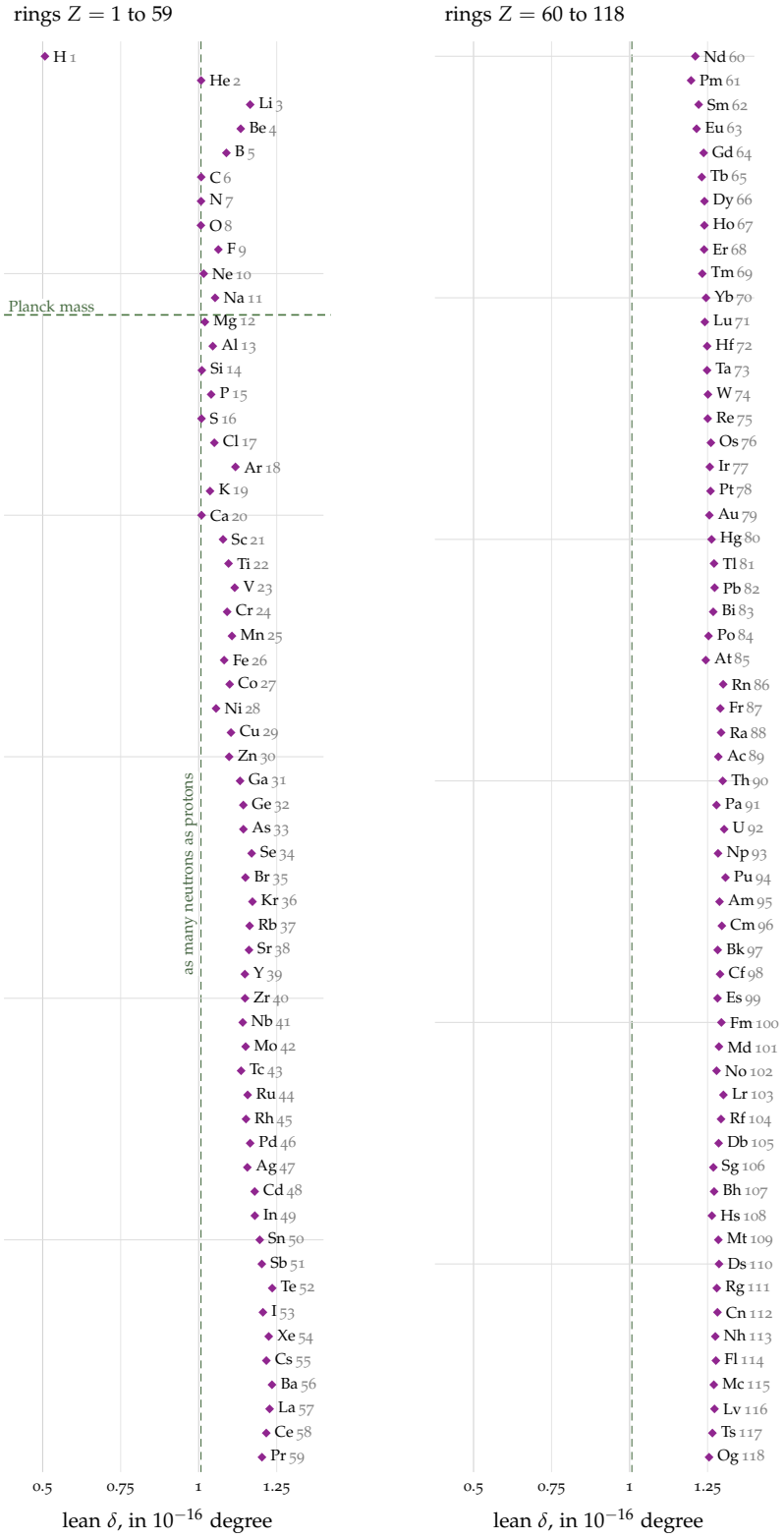


Figure 7.11: The nuclear spine of Figure 7.9, unrolled: each nucleus at its ring Z, down the page, and at its lean δ, across, in 10^{-16} degree. The dashed green line marks two daltons per charge, a nucleus with as many neutrons as protons; light nuclei lie on it, heavy ones to its right. The dashed green bar is the Planck mass, between sodium and magnesium.

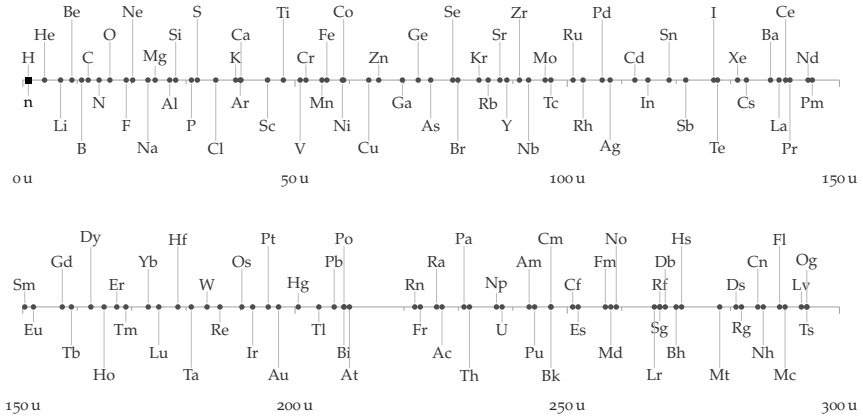


Figure 7.12: The mass axis of Figure 7.9, unrolled: the 118 atoms and the neutron (square) at $r = m$, in daltons, in two lines from 0 to 150 u and from 150 to 300 u.

7.6.2 *What the Model Is, and What It Is Not*

The Cosmic Mass Standard Model is a frame and a catalogue. It takes two numbers that physics has measured for every body, the mass and the charge, and writes them as one magnitude and two angles in a single geometry, built from one fixed angle. Everything in it can be checked with a spreadsheet, and the workbook that comes with this book does so for all 306 bodies.

It is not a new law of nature. It predicts no new particle, and it moves no measured mass or charge. What it adds is a language: a single picture in which neutral matter, ions and nuclei stand side by side; a reading of the charge-to-mass ratio as an angle, exactly the quantity that Thomson's tube and Aston's spectrograph measured; and a family of ellipses whose shape is fixed by the angle of the Composer axis alone.

One question remains open, and Chapter 2 names it as the work of Volume 2: the physical meaning of the Composer coordinate of a product, which holds the Coupling term and a share of the term of two charges, by far the larger for charged particles (Section 6.5.2), but is not read as a force (Definition 6.17). The frame itself rests on the angle 79.86° of the Composer axis (Remark 7.1): its corner angle of 100.14°, its coefficient 0.35210 and its lean of 5.0374×10^{-17} degree per dalton all come from that angle. Until the Composer coordinate is tied to a measurement, the model is a more expressive way to see the bodies we know, which is exactly what it sets out to be.

In plain words

Start with one angle, 79.86°, the angle at which like charges push apart (Chapter 6). Draw every body as a triangle whose corner always has the same angle. All the triangles of one size line up on an ellipse; the ellipses of all sizes, with the rays of equal angle, make graph paper. On that paper, every neutral body sits on one line, every positive body near a second line and every negative body near a third, each tipped by its mass per charge. That picture, with all 306 bodies in it, is the Cosmic Mass Standard Model.

Part III

FROM NUCLEI TO FIELDS

8

THE CLOSURE LAW

Six neighbouring nuclei, added with alternating signs, close on themselves: their charge parts cancel exactly, and their binding almost. From that one fact the mass of a nucleus can be read off its neighbours, and the energy of a fusion can be known before anyone weighs it.

CARDIFF, AUGUST 1920

On 24 August 1920 the British Association for the Advancement of Science met in Cardiff, and Arthur Stanley Eddington opened its section of mathematics and physics with an address on the inside of the stars. Eddington was thirty-seven, Plumian Professor of Astronomy at Cambridge, and a year earlier he had photographed the eclipse of the Sun from the island of Príncipe to test Einstein's theory of gravity. His question in Cardiff was older: what keeps the stars shining? If the Sun drew its heat from its slow contraction under gravity, as Kelvin had computed, it could not be much older than a few tens of millions of years, and geology needed far more. "Lord Kelvin's date of the creation of the sun," Eddington said, "is treated with no more respect than Archbishop Ussher's."

His answer came from another laboratory of his own university. At the Cavendish, Francis Aston had just weighed the atoms one by one with his new spectrograph (Section 7.4), and Eddington read the result to his audience: "the mass of the helium atom is less than the sum of the masses of the four hydrogen atoms which enter into it." "There is a loss of mass in the synthesis amounting to about 1 part in 120," he went on, "the atomic weight of hydrogen being 1·008 and that of helium just 4." If only five per cent of a star's mass were hydrogen, slowly built up into heavier elements, the energy set free would more than suffice, and "we need look no further for the source of a star's energy." He saw at once what else it meant. It would bring nearer, he said, "our dream of controlling this latent power for the well-being of the human race", and he added: "or for its suicide" (Eddington, 1920).

Today's masses give the same picture with more digits. Four hydrogen atoms weigh $4 \times 1.007825 = 4.031300$ u, and one helium atom 4.002603 u; the loss is 0.028697 u, seven parts in a thousand, or 26.73 MeV.

Eddington could name the loss because Aston had weighed both sides of the reaction. Chapter 7 is in the same position. It gives every body of ordinary matter a card and a place in one frame, but it takes every mass from a table: it "moves no measured mass or charge"

(Section 7.6.2). This chapter asks a question that Eddington could not answer: can the mass lost in a fusion be known before it is weighed?

The answer is yes, within a stated error, whenever the product has measured neighbours. The key is a closed loop of six neighbouring nuclei, added with alternating signs. Around such a loop the charge parts of the cosmic masses cancel exactly, the protons and neutrons cancel exactly, and so do whole families of binding terms; the rest of the binding, where it varies smoothly, nearly cancels. What is left is small, and its size can be measured. Solved for one of the six nuclei, the loop predicts its mass from the other five, and with it the energy of every fusion that makes it. We call this statement, with its constant and its error law, the *Closure Law*. The loops themselves are old: they are the mass relations of Garvey and Kelson, known to nuclear physics since 1966 (Garvey and Kelson, 1966). Their use for fusion in cosmic masses was first set out as a recipe with no fitted constant, with its own calculator; we call it the *plain recipe*. The law itself, with one constant, was first stated with a workbook that computes it; we call it the *first version*. Both were tested on a table in which some of the masses were estimates, and this chapter repeats every test on measured masses only. What the cosmic mass adds to the old relations is their reading in the Cosmic Mass Frame, their use for fusion, one correction, an error law with its range, and a list of predictions that can fail.

THE ROAD THROUGH THIS CHAPTER. The chapter climbs in seven steps, shown in Figure 8.1. Section 8.1 opens each nucleus of Chapter 7 into its isotopes, and Section 8.2 writes fusion on their cards. Section 8.3 closes the loops and proves what they cancel, and Section 8.4 states the law with its error. Section 8.5 weighs fusion with it, Section 8.6 shows where it stops, and Section 8.7 registers predictions that can prove it wrong. As in Chapters 6 and 7, the mathematics comes from Chapter 4, and each step names the statement it uses.

8.1 STEP 1: ISOTOPES ON THE FRAME

Each nucleus of Chapter 7 is the average nucleus of its element, because the atomic weight is an average over the element's isotopes (Section 7.5). Aston's spectrograph took that average apart: chlorine, of atomic weight 35.45, gave lines at 35 and 37 and nothing in between. Fusion happens between such single isotopes, not between averages, and the averages give absurd answers if they are used for it. Two average hydrogen nuclei of Chapter 7 weigh $2 \times 1.007451 = 2.014903$ u, and the average helium nucleus 4.001505 u, so the "fusion" of two hydrogen nuclei into helium would have to create 1.99 u out of nothing. The four nucleons of helium are not in two hydrogen nuclei. Two deuterons, each a proton and a neutron, weigh 4.027106 u; fused into a helium-4 nucleus of 4.001506 u, they lose 0.025600 u, or 23.85 MeV.

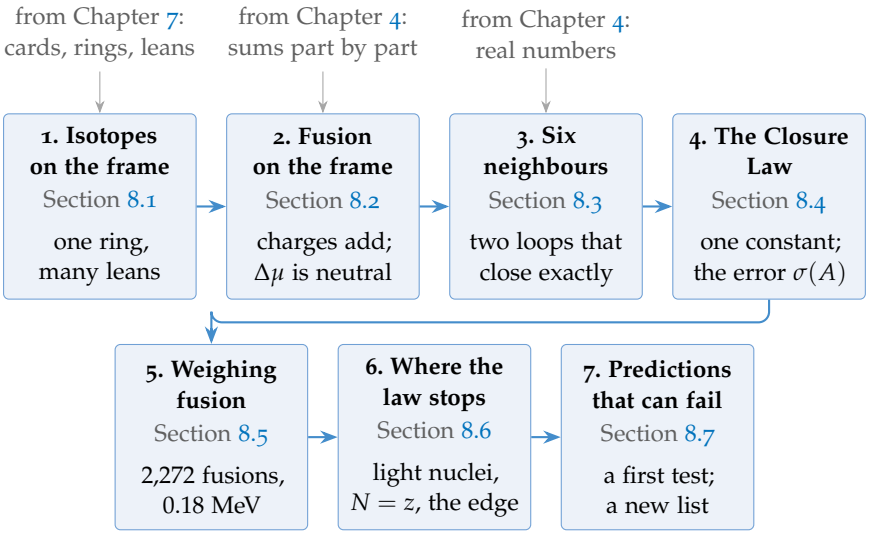


Figure 8.1: The road through the chapter, from the cards of single isotopes to predicted masses. Each box is one step, with the section where it is taken.

So this chapter works with single isotopes. An isotope is fixed by two whole numbers: its charge number z, which for a bare nucleus is the atomic number Z (Section 7.1.1), and its number of neutrons N. Their sum $A = z + N$ is the *mass number*. By Aston's whole-number rule, the atomic mass of a single isotope, in daltons, lies within a small fraction of a dalton of A. In Chapter 7 the letter A stood for the atomic mass of an element; in this chapter it always means the whole number $z + N$, and the atomic mass of the isotope is written $M(z, N)$. The mass of its bare nucleus follows by the rule of Section 7.5,

$$m(z, N) = M(z, N) - z\, m_e + B_e/c^2,$$

with the electron binding energy B_e of that section.

The atomic masses come from the Atomic Mass Evaluation of 2020 and its companion NUBASE2020 (Wang et al., 2021; Kondev et al., 2021). Leaving out the free neutron, they give the ground states of 3,557 nuclei. For 2,549 of them the mass is derived from measurements; for the other 1,008 it is an estimate, drawn by the evaluators from the smooth trends of the measured masses and marked as such. *In this chapter only the measured masses are used as data.* The estimates appear only where they are compared with a prediction, and never as a test of one.

Definition 8.1 (Isotope Card)**.** The **isotope card** of the nucleus with $z \geq 1$ protons and N neutrons is the identity card (r, α, ϕ) (Remark 6.13) of its cosmic mass (Definition 6.5)

$$\mu(z, N) = m(z, N) + z\, e\, \mathcal{G}_{\bar{\phi}},$$

whose charge part, $z\, e\, \mathcal{G}_{\bar{\phi}} = z\, m_S$, is z Stoney masses laid along the Composer axis (Definition 7.2). Its angles are $\alpha = \bar{\phi} - \delta$ and $\phi = \delta$,

with the lean δ of Remark 7.6, and the General Sinus reads its two parts exactly (Remark 6.11):

$$r \sin \delta = m \sin \bar{\phi}, \qquad r \sin \alpha = z\, m_S \sin \bar{\phi}.$$

At 79.86° (Remark 7.1), its ring is $r = z\, m_S$ to many digits, and its lean is

$$\delta \approx 5.0374 \times 10^{-17} \text{ degree} \times \frac{m(z,N)}{z},$$

for m in daltons.

The two exact relations are the reading of the two parts of the triangle, $m = r \cos(\alpha, \phi) = r \sin \phi / \sin \bar{\phi}$ and $z\, m_S = r \sin(\alpha, \phi) = r \sin \alpha / \sin \bar{\phi}$, with $\phi = \delta$. The first says that, on the frame, the mass of a nucleus is its ring times the sine of its lean, up to the fixed factor $\sin \bar{\phi}$.

It is also the mass that is measured. Chapter 6 reads every force by its real part (Definition 6.17), and the same reading, carried to bodies in motion, makes the real part of a cosmic mass its inertia (Chapter 9). A nucleus is weighed by how its ions turn or bend in a magnetic field, so what is weighed is the real part of its cosmic mass, $m = \Re(\mu)$, and so is every mass in this chapter.

Figure 8.2 shows what happens to the spine of Chapter 7 when every nucleus is opened into its isotopes. The ring does not change, since all the isotopes of an element carry the same charge; only the lean does. Each added neutron moves the nucleus a little further along its ring, by about 5×10^{-17} degree divided by z, so the isotopes of an element spread into a short row: long for the light elements, whose few protons share each added neutron, and short for the heavy ones. The stable isotopes sit near the dashed line for the light elements and drift to the right, towards more neutrons, as the charge grows.

Worked example: the two cards of chlorine

Chlorine-35 has the atomic mass 34.968853 u. Its nucleus, with 17 electrons removed and their binding given back, has $m = 34.959540$ u. Its ring is $17\, m_S \approx 3.16066 \times 10^{-8}$ kg, and its lean is $\delta \approx 5.0374 \times 10^{-17} \times 34.959540/17 \approx 1.03591 \times 10^{-16}$ degree. Chlorine-37 has $m = 36.956590$ u, the same ring, and the lean 1.09509×10^{-16} degree. The average chlorine nucleus of Appendix E, 35.440688 u, leans by 1.05017×10^{-16} degree: between its two isotopes, and closer to chlorine-35, which makes up three quarters of natural chlorine. The two lines on Aston's plate are two cards on one ring.

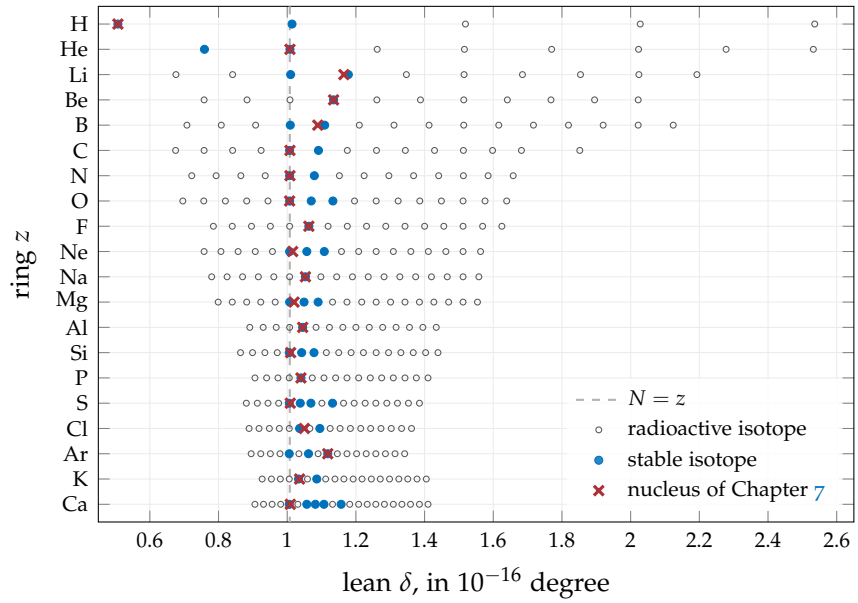


Figure 8.2: The first twenty rings of the nuclear spine of Figure 7.11, opened into their isotopes. Each ring z carries every isotope of the element whose mass has been measured, at its lean $\delta \approx 5.0374 \times 10^{-17\circ} \times m/z$; filled dots are stable, open dots radioactive. The cross is the average nucleus of Chapter 7, which lies among its stable isotopes. The dashed line is two daltons per charge, a nucleus with as many neutrons as protons.

> **In plain words**
>
> Chapter 7 drew one dot for each element. In this chapter each dot opens into a short row of dots on the same circle, one for each isotope. Isotopes of one element share their charge, so they share their circle; they differ only in mass, and the heavier ones tip a little further. Reactions happen between those single dots, so from now on every card is the card of one isotope.

8.2 STEP 2: FUSION ON THE FRAME

In a fusion, two nuclei join into one: z_1 protons and N_1 neutrons meet z_2 protons and N_2 neutrons, and the product has $z_1 + z_2$ protons and $N_1 + N_2$ neutrons. The products considered in this chapter are single nuclei, with nothing else emitted, so the counts of protons and of neutrons are kept. The masses are not: the product is lighter than its two parts, by the mass that leaves as energy.

Proposition 8.2 (Fusion on the Frame)**.** *Let two nuclei with the cosmic masses μ_1 and μ_2 fuse into one nucleus with the cosmic mass μ. Then:*

1. *the charge parts add exactly, $z\, m_S = z_1 m_S + z_2 m_S$, so the product lies on the ring of the sum of the two charges, to many digits;*
2. *the loss of cosmic mass is a real number,*

$$\Delta\mu = \mu_1 + \mu_2 - \mu = m_1 + m_2 - m = \Delta m,$$

   and the energy released is $Q = \Delta m\, c^2$;
3. *on the cards, exactly, $r \sin\delta = r_1 \sin\delta_1 + r_2 \sin\delta_2 - \Delta m \sin\bar{\phi}$.*

At 79.86°, with the lean of Remark 7.6 and Δm in daltons, statement 3 reads

$$z\,\delta \approx z_1\delta_1 + z_2\delta_2 - 5.0374 \times 10^{-17}\ \textit{degree} \times \Delta m.$$

Proof. General Numbers of one relative plane are added part by part (Definition 4.17). The charge $q = z\,e$ is kept, so the Composer parts of $\mu_1 + \mu_2$ and of μ are equal, and the difference $\Delta\mu$ has no Composer part: it is a real number (Remark 4.16), the difference of the real parts. Statement 3 is statement 2 written with the exact relation $r \sin\delta = m \sin\bar{\phi}$ of Definition 8.1, and the last formula is statement 2 written with $z\,\delta \approx 5.0374 \times 10^{-17\circ} \times m$. Statements 1 to 3 use no value of $\bar{\phi}$. □

In the dictionary of Table 6.1, a real number is a neutral body. The mass lost in a fusion is therefore a neutral body in the frame of Chapter 7: like an atom or a neutron, it has no charge part and lies on the mass axis, and when energy is released it sits on the ray $\alpha = 0$ with its ring $r = \Delta m$ (Section 7.4). The last formula says how the lean of the product follows from the leans of its parts: weighted by their charges, the leans add, less the loss.

The first two statements are the two coordinates of one conservation law. In the Unified Cosmic Law of Chapter 9, the source of the fields is the density of cosmic mass, and it flows without being made or destroyed, $\partial_t \rho_\mu + \nabla \cdot \mathbf{J}_\mu = 0$. Its Composer coordinate is the conservation of charge, which holds exactly: that is statement 1. Its real coordinate is the conservation of mass, which a fusion keeps only when the energy that leaves is counted as mass, Q/c^2: that is statement 2, where the energy released is the real part of the loss, $Q = \Re(\Delta\mu)\, c^2 = \Delta m\, c^2$, as Chapter 6 reads every energy.

Figure 8.3 draws the two coordinates at work, as the plain recipe does: the two cosmic masses add tip to tail, and the tip slides back along the mass axis by the loss, at the same height, which holds the charge. Walked back along the Composer axis, the way Chapter 6 reads a real part, the sum of the inputs and the product land Δm apart; dropped straight down, as the shadow would read them, they land elsewhere, but still Δm apart. The loss is a real number, its own

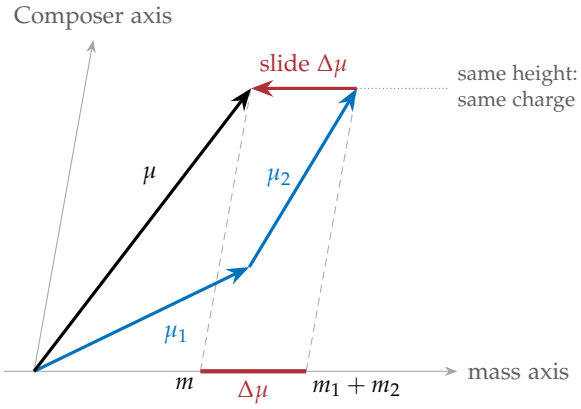


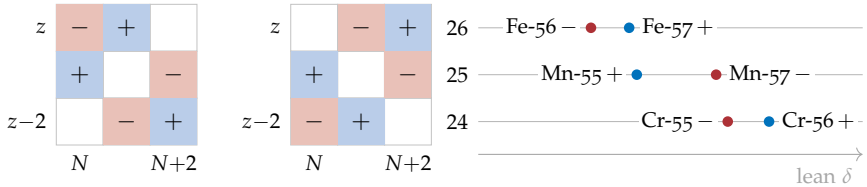
Figure 8.3: Fusion on the frame, as the plain recipe draws it. The inputs add tip to tail; the tip then slides back along the mass axis by the loss $\Delta\mu$, at the same height, since the charge is kept. Walked back along the Composer axis, the sum and the product meet the mass axis at their real parts, $\Delta\mu$ apart. Not to scale: a nucleus has a charge part about 10^{18} times its mass part.

real part and its own shadow (Definition 4.19), so the two readings of Chapter 6 cannot differ on the energy of a fusion.

The cards also give what a fusion must overcome first: the push between the two nuclei. By Proposition 6.18, the force between two nuclei the distance d apart is $G\,r_1 r_2 \sin(\bar{\phi} - \alpha_1 - \alpha_2)/(d^2 \sin\bar{\phi})$. Each angle lies within 10^{-15} degree of $\bar{\phi}$, so the sine is $-\sin\bar{\phi}$ to many digits and the force is a push; with the rings $r_1 = z_1 m_S$ and $r_2 = z_2 m_S$ and $G\,m_S^2 = Ke^2$ (Remark 7.3) the energy of the push is $z_1 z_2\, Ke^2/d$, the Coulomb barrier. For a deuteron and a triton 4 fm apart it is 0.360 MeV. Sums of cards keep the books of a fusion; products of cards give the barrier before it.

The frame keeps the books of a fusion exactly, but it cannot say how large the loss is. That number depends on how tightly the product is bound, and nothing on the two cards of the inputs records it. Two simple guesses show how far off one can be. On the 2,272 fusions of Section 8.5, taking $\Delta m = 0$ misses the energy by 21.5 MeV on average, root mean square, since that is the typical size of Q itself. The liquid-drop formula of Weizsäcker (von Weizsäcker, 1935), with its five constants fitted to the measured masses with $A \geq 16$, misses by 2.5 MeV and gets the sign of Q wrong 66 times. The next two steps find a far better source of information: not the two inputs, but the neighbours of the product.

> IN PLAIN WORDS
>
> When two nuclei join, their charges simply add, and the frame records that exactly: the new nucleus sits on the circle of the total charge. Their masses do not quite add. Put the arrows of the two



(a) loop A (b) loop B (c) loop A at Fe-56

Figure 8.4: The two closure loops of Definition 8.3. (a, b) Each square is one nucleus, with protons up and neutrons across; blue squares count with the sign + and red ones with −. (c) Loop A anchored at iron-56, drawn on the spine of Figure 8.2: the six nuclei lie on three neighbouring rings, two on each, at their leans (from 1.083 to 1.174×10^{-16} degree).

> nuclei tip to tail: the tip then slides back a little along the line of neutral matter, and that slide is the mass that is lost and leaves as energy. It has no charge part at all: on the frame it is a neutral body. The frame can draw the loss once it is known, but it cannot tell how big it will be. For that we need more than the two nuclei that fuse.

8.3 STEP 3: SIX NEIGHBOURS CLOSE

Lay out the nuclei on a map, with the number of protons z going up and the number of neutrons N going across, so that every nucleus is one square (Figure 8.8 in Section 8.6 draws the whole map). Take six neighbouring squares and give them alternating signs.

Definition 8.3 (The Two Closure Loops)**.** For any function f on the map of nuclei, let

$$\begin{aligned}\mathcal{L}_A f(z,N) &= f(z{-}2,N{+}2) - f(z,N) + f(z,N{+}1)\\ &\qquad - f(z{-}1,N{+}2) + f(z{-}1,N) - f(z{-}2,N{+}1),\\ \mathcal{L}_B f(z,N) &= f(z,N{+}2) - f(z{-}2,N) + f(z{-}2,N{+}1)\\ &\qquad - f(z{-}1,N{+}2) + f(z{-}1,N) - f(z,N{+}1).\end{aligned}$$

Each loop has three nuclei with the sign + and three with − (Figure 8.4). Every row of the map that the loop touches holds one + and one −, and so does every column; that is the whole secret of what follows.

Theorem 8.4 (Exact Closure)**.** *For either loop $\mathcal{L} = \mathcal{L}_A$ or $\mathcal{L}_B$, at every point of the map:*

1. *The Composer parts cancel: $\mathcal{L}[\mu]$ is a real number.*
2. *With the binding mass $\beta = z\,m_p + N\,m_n - m$, one has exactly $\mathcal{L}[\mu] = \mathcal{L}[m] = -\mathcal{L}[\beta]$.*

3. *$\mathcal{L}$ sends to zero every function of the form $f(z) + g(N) + k\,zN$, whatever f, g and k.*

4. *In addition, $\mathcal{L}_A$ sends to zero every function of A alone, and $\mathcal{L}_B$ every function of $N - z$ alone.*

Proof. Write the six points as $(z + a_i, N + b_i)$ with signs s_i. For $\mathcal{L}_A$ the triples (a, b, s) are $(-2, 2, +)$, $(0, 0, -)$, $(0, 1, +)$, $(-1, 2, -)$, $(-1, 0, +)$, $(-2, 1, -)$.

For $f(z)$: the values $f(z-2)$, $f(z)$ and $f(z-1)$ each occur once with each sign, so they cancel. For $g(N)$: the same holds for $g(N+2)$, $g(N)$ and $g(N+1)$. For zN: since $\sum s_i = \sum s_i a_i = \sum s_i b_i = 0$, the sum reduces to $\sum s_i a_i b_i$, whose three non-zero terms, in the order of the list, are $-4 + 2 + 2 = 0$. This proves (3) for $\mathcal{L}_A$. The numbers $a + b$ are $0, 0, 1, 1, -1, -1$ with signs $+, -, +, -, +, -$, which proves the first half of (4).

For $\mathcal{L}_B$ the triples are $(0, 2, +)$, $(-2, 0, -)$, $(-2, 1, +)$, $(-1, 2, -)$, $(-1, 0, +)$, $(0, 1, -)$. The values of a and of b again pair off, and $\sum s_i a_i b_i = -2 + 2 = 0$. The numbers $b - a$ are $2, 2, 3, 3, 1, 1$ with signs $+, -, +, -, +, -$, which proves the second half of (4).

Statement (1) follows from (3), since the Composer part is $z\,m_S$, linear in z. Statement (2) follows from (3) as well, since $z\,m_p + N\,m_n$ is of the form $f(z) + g(N)$. Nothing used the value of $\bar{\phi}$. □

Theorem 8.4 is mathematics and has no error. It says what the loop removes: the charge, the bricks, and all the binding that depends on z alone, on N alone, on their product, on A alone (loop A) or on $N - z$ alone (loop B). The volume and surface energies of the liquid drop are functions of A, which is why the loop removes so much. In the words of Chapter 4, the six cosmic masses are added part by part (Definition 4.17); their Composer parts cancel, and what is left is a real number, a neutral body.

Remark 8.5 (The Closure on the Cards). Two readings of Theorem 8.4 follow at once.

1. The loop gives the same number whether atomic or nuclear masses are used, because the electrons, $z\,m_e - B_e/c^2$, form a function of z alone.

2. On the isotope cards (Definition 8.1), the loop of the masses is the loop of ring times lean:

$$\mathcal{L}[m] = \frac{1}{\sin\bar{\phi}}\,\mathcal{L}[r\sin\delta], \qquad \mathcal{L}[z\,\delta] \approx 5.0374 \times 10^{-17}\ \text{degree} \times \mathcal{L}[m],$$

   the second at 79.86°, for $\mathcal{L}[m]$ in daltons. The closure of six masses is the closure of six charge-weighted leans on the frame.

The two loops were found in 1966 by Gerald Garvey and Igal Kelson, as relations among the masses of neighbouring nuclei (Garvey and

Kelson, 1966), and three years later Garvey and his colleagues used them to build a table of predicted masses (Garvey et al., 1969). They are not a discovery of this book. What the cosmic mass adds is the first statement of the theorem: the charge parts cancel visibly, and the loop of six cosmic masses is a real number (Remark 4.16), a neutral body.

Worked example: one loop around iron-56

Loop A anchored at iron-56 uses chromium-56 (+), iron-56 (−), iron-57 (+), manganese-57 (−), manganese-55 (+) and chromium-55 (−). Their nuclear masses are 55.927514, 55.920710, 56.921166, 56.924605, 54.924363 and 54.927702 u. With the signs,

$$\mathcal{L}_A[m] = 0.0000259 \text{ u} = 0.024 \text{ MeV}.$$

Each of the six nuclei is bound by 480 to 500 MeV, and the loop cancels these bindings to 0.024 MeV. On the frame, the six charge-weighted leans $z\,\delta$, each about 2.8×10^{-15} degree, add up with their signs to 1.3×10^{-21} degree. This loop is a lucky one: loop B at iron-56 leaves -0.37 MeV, and for nuclei with $A \geq 40$ the loops leave 0.21 MeV on average (Section 8.4).

In plain words

Pick six neighbouring nuclei that form a little ring on the map, and count three of them as plus and three as minus. Every row and every column of the ring holds one plus and one minus. So anything that depends only on the row, the number of protons, cancels, and so does anything that depends only on the column, the number of neutrons. That removes the charge, the protons and neutrons themselves, and most of the energy that binds them. What survives is a small remainder, and the next step measures it.

8.4 Step 4: the closure law

What Theorem 8.4 leaves is the leftover $\varepsilon = \mathcal{L}[\mu]$, a real number. On the 3,977 loops whose six nuclei all have measured masses (1,920 of kind A and 2,057 of kind B), its root mean square is 0.54 MeV, and 0.24 MeV for the loops with $A \geq 40$. That size is set by a few large leftovers: 92% of the loops close within 0.5 MeV, and 96.5% within 1 MeV. Nearly all the large leftovers come from one place. The 214 loops that touch the line $N = z$ leave 2.1 MeV on average, the other 3,763 only 0.25 MeV, and 39 of the 40 largest leftovers touch that line. Eugene Wigner's symmetry between protons and neutrons, of 1937, gives the nuclei with $N = z$ extra binding (Wigner, 1937); mass formulas write

it as a term proportional to $|N - z|/A$, which has a sharp corner on that line. A sharp corner is not smooth, so no loop can cancel it, and we remove it with one constant:

$$\tilde{m}(z, N) = m(z, N) - \frac{w}{c^2}\,\frac{|N - z|}{A}, \qquad w = 22.75 \text{ MeV}.$$

The first version found w by least squares on the loops of its table. Fitted on the loops of measured nuclei alone, it comes out at 22.9 MeV, and the predictions below hardly change for any value between 20 and 35 MeV. Mass formulas use values of 30 to 47 MeV for the same coefficient (Satuła et al., 1997); the loops find a Wigner term of the same order, somewhat smaller. With the corner removed, the leftover falls to 0.42 MeV, and to 0.21 MeV for $A \geq 40$, while the loops that touch the line $N = z$ still leave 1.5 MeV.

A leftover that is small on average is a tool for prediction. Solved for any one of its six nuclei, a loop gives that nucleus's mass from the other five.

Worked example: one loop weighs magnesium-24

Loop A anchored at magnesium-24 holds Ne-24 (+), Mg-24 (−), Mg-25 (+), Na-25 (−), Na-23 (+) and Ne-23 (−). Setting its sum to zero and solving for Mg-24 gives

$$m_{\text{Mg-24}} \approx m_{\text{Ne-24}} + m_{\text{Mg-25}} - m_{\text{Na-25}} + m_{\text{Na-23}} - m_{\text{Ne-23}}.$$

With the nuclear masses of the five neighbours, in daltons,

23.988129, 24.979260, 24.983924, 22.983740 and 22.988985,

the loop gives 23.978219 u. The measured mass, which the loop never used, is 23.978465 u: the loop misses by 0.000246 u, or 0.23 MeV, one part in a hundred thousand of the mass, with no fitted number and no nuclear theory, only the addition and subtraction of five cards. The six nuclei are bound by 183 to 206 MeV each, and with the signs of the loop their bindings add up to the miss, 0.23 MeV. For two carbon-12 nuclei fusing into magnesium-24, this one loop gives $Q \approx 14.16$ MeV, against the measured 13.93 MeV. This is the first example of the plain recipe, and its numbers do not change on the masses of the 2020 evaluation. None of the five neighbours has $N = z$, so the cusp constant moves this estimate by only 0.01 MeV.

Definition 8.6 (The Closure Estimate)**.** Let T be a nucleus. Each loop of either kind that has T in one of its six positions, and five nuclei of measured mass in the other five, gives one estimate of $\tilde{m}(T)$, by setting its sum to zero; there are at most twelve. The **closure estimate** of the mass of T is the trimmed mean of these estimates, with a quarter of them dropped at each end, at least one, and the rest averaged (with

one or two estimates, their mean), plus the corner $\frac{w}{c^2}\frac{|N-z|}{A}$ of T. The mass of T itself is never used.

Law 8.7 (The Cosmic Mass Closure Law)**.** For nuclei with measured neighbours,

$$\mathcal{L}_A[\tilde{m}] \approx 0, \qquad \mathcal{L}_B[\tilde{m}] \approx 0,$$

with a leftover whose size falls with the size of the nucleus. The closure estimate of the mass of a nucleus with $A \geq 8$ has the error

$$\sigma(A) = \sqrt{\left(\frac{9.1}{A}\right)^2 + 0.063^2} \ \text{MeV},$$

in the sense that about 78% of the measured nuclei fall within $\sigma(A)$ of it and 95% within $2\sigma(A)$.

Fusion form. For a fusion $\mu_1 + \mu_2 \to \mu_{\text{out}}$ whose product has measured neighbours,

$$\begin{aligned} \Delta\mu &= m_1 + m_2 - m_{\text{out}}, \\ m_{\text{out}} &= \langle \text{estimates from the loops} \rangle + \frac{w}{c^2}\frac{|N-z|}{A}, \end{aligned}$$

where the bracket is the trimmed mean of Definition 8.6, so that m_{out} is the closure estimate of the product, and the error of $Q = \Delta\mu\, c^2$ is the error $\sigma(A)$ of the product.

Every quantity in the law can be read off the cards: $z = r\sin(\alpha,\phi)/m_S$ and $m = r\cos(\alpha,\phi)$ (Remark 6.11), and on the frame the closure is that of the charge-weighted leans (Remark 8.5). The other way round, once m_{out} is known, the whole card of the product follows from its two parts, at the angle the book holds, $\bar{\phi} = 79.86°$:

$$\begin{aligned} q\mathcal{G}_{\bar{\phi}} &= z\, m_S, \quad \text{where } z = z_1 + z_2, \\ r_{\text{out}} &= \sqrt{m_{\text{out}}^2 + (q\mathcal{G}_{\bar{\phi}})^2 + 2\cos\bar{\phi}\; m_{\text{out}}\, q\mathcal{G}_{\bar{\phi}}}, \\ \tan\phi_{\text{out}} &= \frac{m_{\text{out}}\sin\bar{\phi}}{q\mathcal{G}_{\bar{\phi}} + m_{\text{out}}\cos\bar{\phi}}, \\ \alpha_{\text{out}} &= \bar{\phi} - \phi_{\text{out}}. \end{aligned}$$

Its angle ϕ_{out} is the lean δ of the product (Definition 8.1).

Table 8.1 gives the test. Every measured nucleus with at least one closed loop of measured neighbours was predicted from those neighbours, and its own mass was used only to measure the miss. The error falls from more than 1 MeV for the lightest nuclei to about 0.1 MeV for the heavy ones. The cusp constant matters most where it should, for the nuclei with $N = z$, whose error it nearly halves. For nuclei with $A \geq 60$ and all twelve estimates the error is 79 keV, the size that Barea

NUCLEI	NUMBER	NO CUSP	WITH CUSP
all that can be predicted	2,481	0.34 MeV	0.25 MeV
light, $A < 16$	64	1.88 MeV	1.27 MeV
$16 \leq A < 40$	182	0.40 MeV	0.33 MeV
$A \geq 40$	2,235	0.12 MeV	0.11 MeV
with $N = z$	38	1.83 MeV	1.01 MeV
with all twelve estimates	1,385	0.20 MeV	0.16 MeV

Table 8.1: Root mean square error of the closure estimate on the 2,481 measured nuclei that have at least one closed loop of measured neighbours. Each nucleus is predicted from its neighbours alone. If w is refitted five times, each time without a fifth of the nuclei, and each nucleus is predicted with the constant fitted without it, the first line changes from 0.247 to 0.249 MeV.

and colleagues found for the same relations with an earlier evaluation, 80 keV (Barea et al., 2005).

Figure 8.5 shows where $\sigma(A)$ comes from. Up to $A \approx 100$ the error falls nearly in proportion to $1/A$: on measured masses the first seven bins give $10.3/A^{1.03}$ MeV, close to the $8.5/A^{0.96}$ MeV of the first version. Beyond, it falls more slowly, towards a floor near 0.06 MeV that the heavy nuclei do not get below, whatever their size; the second term of $\sigma(A)$ is that floor. The errors also have long tails. For $A \geq 40$, 68% of the misses are below 0.08 MeV and 95% below 0.21 MeV, but the largest is 0.95 MeV, for rubidium-73, which has only three estimates. Over all nuclei with $A \geq 8$, 1.3% miss by more than $3\sigma(A)$, where a bell curve would have 0.3%. So $\sigma(A)$ is a typical size, not a hard limit.

> IN PLAIN WORDS
>
> The ring of six never closes perfectly; a small remainder is left. Most of the remainder comes from one place, the nuclei with equal numbers of protons and neutrons, which are bound a little more tightly than their neighbours. One constant takes most of that extra binding out. After that, the remainder is small enough to use: hide one of the six nuclei, and the other five tell you its mass, to about $9/A$ MeV, and never much better than 0.06 MeV.

8.5 STEP 5: WEIGHING FUSION

The fusion form of the law answers the question of the opening. The two inputs are measured nuclei, the product is predicted from its neighbours, and the loss of mass follows. Since the inputs are known exactly, the error of Q is the error of the predicted product, with the opposite sign.

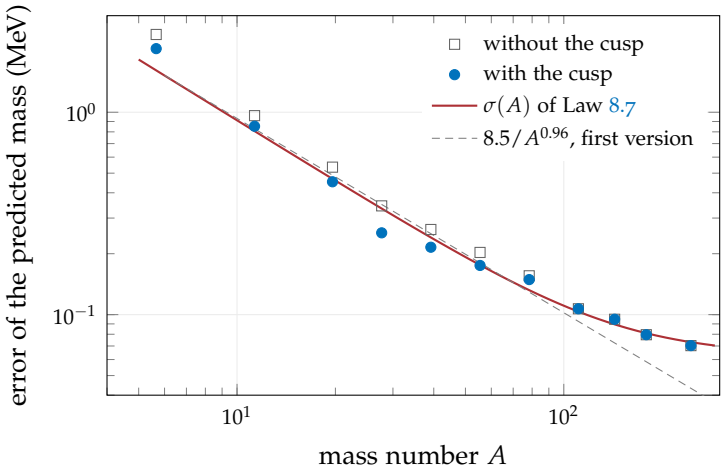


Figure 8.5: The error law. Root mean square error of the closure estimate in bins of the mass number, with each nucleus predicted from its neighbours alone. The solid line is $\sigma(A)$ of Law 8.7; the dashed line is the law of the first version, fitted to light and middle nuclei on a table that mixed measured and estimated masses.

Worked example: carbon and carbon make magnesium

Two carbon-12 nuclei, each of mass 11.996710 u, fuse into magnesium-24. The twelve loops around magnesium-24 all have measured nuclei in their other five places, and give twelve estimates, in daltons:

23.978164, 23.978226, 23.978227, 23.978457, 23.978581, 23.978601, 23.978858, 23.978896, 23.980119, 23.980175, 23.980258, 23.980274.

The trimmed mean drops the three lowest and the three highest and averages the middle six: 23.978919 u. Magnesium-24 has $N = z$, so its own corner is zero. The predicted loss is $2 \times 11.996710 - 23.978919 = 0.014501$ u, and $Q \approx 13.51$ MeV. The measured mass is 23.978465 u, which gives $Q = 13.93$ MeV: the prediction misses by 0.42 MeV. The twelve estimates fall into two groups. Eight lie within about 0.4 MeV of the measured mass; the other four, all from loops of kind B that reach the line $N = z$ from one side, are 1.5 to 1.7 MeV too high. The trimmed mean keeps one of those four, which makes half of the miss. Nuclei with $N = z$ are the hard case of the law.

Figure 8.6 sets these twelve estimates beside the twelve that the same loops give with no constant. Without it, eight of them fall 1.5 to 2.4 MeV short of the measured energy, and the median that the plain recipe takes, 12.30 MeV, lies among them. The constant lifts four of the eight, those from loops of kind A, to within 0.4 MeV. The four from loops of kind B do not move: they reach the line $N = z$ from

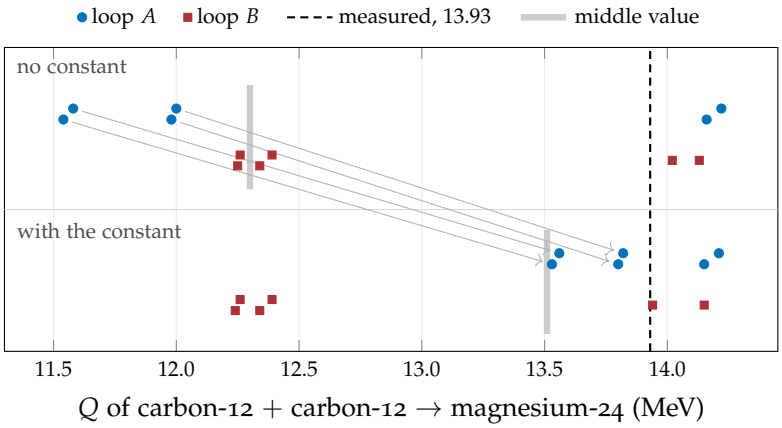


Figure 8.6: The twelve estimates of the energy of carbon-12 + carbon-12 → magnesium-24, one from each loop around magnesium-24, with no constant (top) and with the cusp constant (bottom); circles come from loops of kind A, squares from loops of kind B. With no constant, eight estimates fall 1.5 to 2.4 MeV short of the measured value, and the median of the plain recipe falls among them. The constant lifts the four from loops of kind A (gray arrows); the four from loops of kind B do not move. The gray bars are the middle values: the median of the plain recipe, 12.30 MeV, and the trimmed mean of Law 8.7, 13.51 MeV.

one side only, where $|N - z|$ has no corner, so the constant changes almost nothing in them.

To test the fusion form on a large scale, take the 288 nuclei found in nature, from the proton to uranium-238, and every fusion of two of them whose product is also found in nature. There are 2,272 such fusions; 1,395 of them release energy and the others need it. They land on 264 different natural nuclei. Table 8.2 compares the four ways of predicting their energy that this chapter has met, and Figure 8.7 shows the closure prediction against the measured value.

PREDICTION OF Δm	ERROR IN Q	WRONG SIGNS
no loss, $\Delta m = 0$	21.5 MeV	
liquid-drop formula, five constants	2.5 MeV	66
plain recipe, no constant	0.28 MeV	1
closure estimate, Law 8.7	0.18 MeV	1

Table 8.2: Four ways to predict the energy of the 2,272 fusions of two natural nuclei into a natural nucleus: root mean square error in Q, and the number of fusions whose predicted Q has the wrong sign. The liquid-drop formula has its five constants fitted to all the measured masses with $A \geq 16$. The plain recipe takes the median of the loop estimates, with no constant; the closure estimate takes their trimmed mean, with the cusp constant.

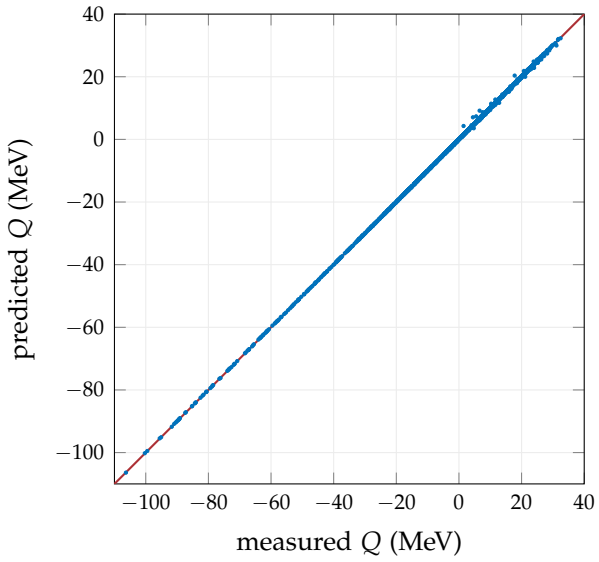


Figure 8.7: The 2,272 fusions of two natural nuclei into a natural nucleus: the energy Q predicted by Law 8.7, with the product's own mass never used, against the measured Q. Points on the line are exact.

The closure estimate misses the energy by 0.18 MeV, more than ten times better than the liquid drop. Only one sign is wrong: fluorine-19 and yttrium-89 make cadmium-108 with $Q = -0.003$ MeV, which the law predicts as $+0.04$ MeV. The number of fusions flatters the test a little, because heavy products are reached by many pairs and are predicted best. Counted once per product, the 264 products are predicted with an error of 0.31 MeV. The plain recipe, with no constant at all, misses by 0.28 MeV, the figure that the guide found on its own table, and gets the same one sign wrong. The constant earns its place on the light and middle products: it brings the error from 2.1 to 1.3 MeV for products with $A < 16$, and from 0.60 to 0.32 MeV for $16 \leq A < 40$; for the heavier products the two give 0.1 MeV or less.

Table 8.3 looks at twenty-three fusions one by one: the captures of helium that build carbon, oxygen and the heavier elements in stars, fusions of two heavier nuclei, such as carbon with carbon, and five light ones. Every sign is right, but the light products remain poor: helium-4 and helium-4 to beryllium-8, deuterium and helium-4 to lithium-6, helium-3 and helium-4 to beryllium-7. The law is a law of medium and heavy nuclei. The other eighteen fusions are all predicted a little low, by 0.1 to 1.2 MeV, and for one reason: each makes an even nucleus with $N = z$, and the closure estimate of such a nucleus comes out too heavy (Section 8.6).

FUSION	MEASURED	PREDICTED	MISS
light fusions			
H-2 + H-2 → He-4	23.85	24.86	1.02
He-4 + He-4 → Be-8	−0.09	−3.27	−3.18
H-2 + He-4 → Li-6	1.47	4.28	2.81
He-3 + He-4 → Be-7	1.59	3.24	1.65
H-2 + C-12 → N-14	10.27	11.39	1.12
captures of helium-4			
Be-8 + He-4 → C-12	7.37	6.61	−0.76
C-12 + He-4 → O-16	7.16	6.68	−0.48
O-16 + He-4 → Ne-20	4.73	3.55	−1.17
Ne-20 + He-4 → Mg-24	9.31	8.89	−0.42
Mg-24 + He-4 → Si-28	9.98	9.37	−0.61
Si-28 + He-4 → S-32	6.94	6.71	−0.24
S-32 + He-4 → Ar-36	6.64	6.11	−0.53
Ar-36 + He-4 → Ca-40	7.04	6.84	−0.20
Ca-40 + He-4 → Ti-44	5.12	4.44	−0.69
Ti-44 + He-4 → Cr-48	7.69	7.58	−0.12
Cr-48 + He-4 → Fe-52	7.93	7.52	−0.41
Fe-52 + He-4 → Ni-56	7.99	7.57	−0.42
fusions of two heavier nuclei			
C-12 + C-12 → Mg-24	13.93	13.51	−0.42
C-12 + O-16 → Si-28	16.75	16.14	−0.61
O-16 + O-16 → S-32	16.53	16.30	−0.24
Ne-20 + Ne-20 → Ca-40	20.75	20.55	−0.20
Mg-24 + Mg-24 → Cr-48	14.94	14.82	−0.12
Si-28 + Si-28 → Ni-56	10.90	10.47	−0.42

Table 8.3: Twenty-three fusions of the stars and of the laboratory: the twelve captures of helium-4 that climb from beryllium-8 to nickel-56, the fusions of carbon, oxygen, neon, magnesium and silicon with themselves and of carbon with oxygen, and five light ones. Q in MeV: measured, predicted by Law 8.7, and the miss, predicted minus measured. The error is 1.10 MeV, root mean square, and every sign is right.

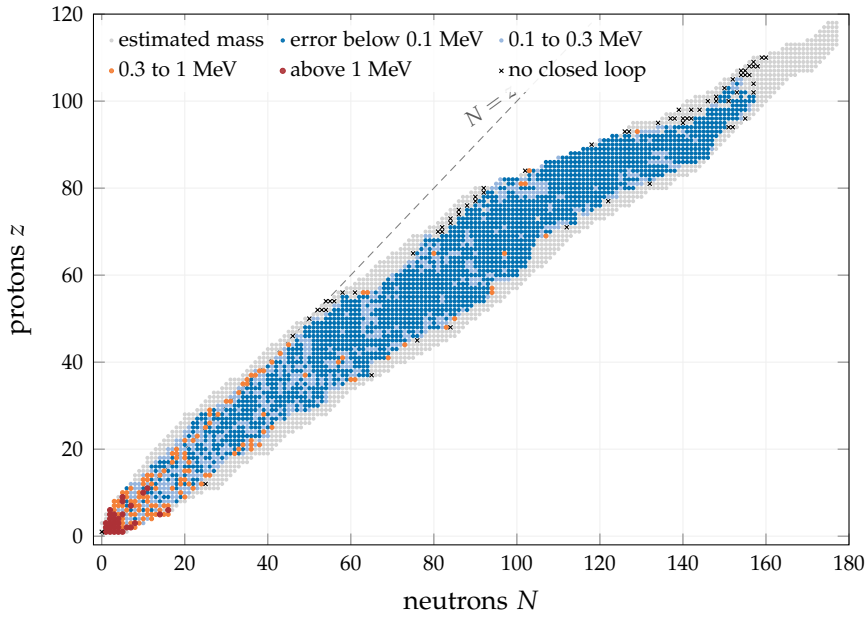


Figure 8.8: The map of nuclei, with each measured nucleus coloured by the miss of its closure estimate, made from its neighbours alone. Gray dots are nuclei whose masses the 2020 evaluation gives only as estimates; crosses are measured nuclei with no closed loop of measured neighbours. The dashed line is $N = z$.

In plain words

To know how much energy a fusion gives, you need the mass of what it makes. The law reads that mass off the product's neighbours, without ever weighing the product. With no adjustable number at all, only the adding and subtracting of the masses of neighbours, the energy of more than two thousand fusions of natural nuclei comes out within about a quarter of an MeV; one constant, for the nuclei with as many neutrons as protons, brings it to a fifth. Only one sign is wrong, for a fusion that gives almost exactly nothing. For the lightest nuclei the law is much rougher, and there it should not be trusted.

8.6 Step 6: where the law stops

A law is only as good as the range in which it has been tested, and this one has three edges.

Light nuclei. Below $A = 16$ the error is above 1 MeV (Table 8.1). A nucleus of a few nucleons is not a smooth drop, and the loops have little to cancel against. Figure 8.8 shows the misses above 1 MeV gathered at the bottom left of the map.

The line $N = z$. Even with the cusp constant, nuclei with as many neutrons as protons are missed by 1.0 MeV, ten times the error of

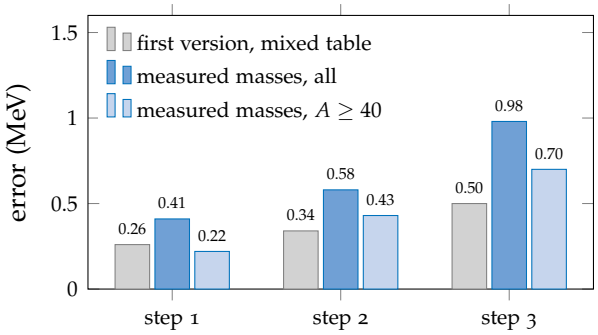


Figure 8.9: Predicting beyond the edge of the map, one layer at a time: root mean square error in each of three steps outward. Gray: the first version of the law, on a table in which many edge masses were estimates. Blue: measured masses only, for all the nuclei reached and for those with $A \geq 40$.

their neighbours of the same size. The misses are not random. From carbon-12 to krypton-72, all sixteen even nuclei on the line come out too heavy, by 0.1 to 1.2 MeV; the next one, strontium-76, at the edge of the measured map, comes out 0.7 MeV too light. The odd nuclei on the line, from nitrogen-14 to potassium-38, all come out too light. The extra binding on the line $N = z$ is not one smooth corner: it differs between even and odd nuclei, and one constant cannot describe it.

The edge of the measured map. Inside the map, every nucleus has neighbours on all sides; at its edge it has them on one side only, and 68 measured nuclei have no closed loop at all. To see how the law does beyond the edge, we peeled three layers off the map of measured nuclei, keeping 1,400 inside, refitted w on those alone, and predicted outward one layer at a time, using the predicted masses as inputs for the next layer. Only part of each layer can be reached this way: 187, 112 and 66 nuclei.

Figure 8.9 gives the result. With measured masses the errors are 0.41, 0.58 and 0.98 MeV in the three steps, and 0.22, 0.43 and 0.70 MeV for $A \geq 40$. The first version of the law, tested on a table in which 565 of the 3,083 nuclei were estimates, had found 0.26, 0.34 and 0.50 MeV, and had warned that this might flatter it: estimates drawn from smooth trends agree with the loops more easily than real nuclei do. The warning was right. One step past the edge the error is about twice $\sigma(A)$: in units of $2\sigma(A)$, 76% of the nuclei of the first step fall within one unit and 98% within two. After three steps the law should not be trusted. Morales and colleagues found the same growth when they carried the Garvey and Kelson relations outward one layer at a time (Morales et al., 2009).

> In plain words
>
> The law works best where it has neighbours on every side: in the middle of the map, for medium and heavy nuclei. It works badly for the lightest nuclei, for nuclei with equal numbers of protons and neutrons, and as soon as it steps outside the measured region, where each step roughly doubles the error. Testing it on estimated masses made it look better than it is, and only measured masses give the honest figure.

8.7 STEP 7: PREDICTIONS THAT CAN FAIL

A statement earns the name of a law of physics only when it risks being wrong. The Closure Law can be put at risk in the simplest way: by predicting masses that have not been measured, and waiting.

8.7.1 *A First Test, Already Passed*

The first version registered the predicted masses of 67 nuclei in its workbook, *Closure-Law-Predictor.xlsx*, dated 2 November 2026. They were made from the isotope table of the `mendeleev` library (Mentel, 2014): these 67 nuclei lay just outside the map used there, with no measured half-life in that table, and their own table masses were not used. Their stated error was the larger of 0.26 MeV and $8.5/A$ MeV.

The 2020 evaluation shows that 24 of the 67 masses had in fact been measured. For these the prediction can be checked now. Table 8.4 gives the miss of each, in MeV and in units of its stated error combined with the error of the measurement. Twenty fall within one stated error and 22 within two; a bell curve would give about 16 and 23. The two larger misses are boron-18, by 1.03 MeV or 2.0 errors, and rubidium-73, by 0.60 MeV or 2.3 errors, a light nucleus and a nucleus next to the line $N = z$.

This is a fair test, though not yet a prospective one: the masses were evaluated in 2021, before the list was made, but the list did not use them. The law could have failed it, and did not.

8.7.2 *A New List*

Table 8.5 registers the masses of 61 nuclei whose ground-state masses the 2020 evaluation gives only as estimates. Each is predicted from measured neighbours alone, with at least three closure estimates, and since each lies just past the edge of the measured map, its stated error is $2\sigma(A)$ (Section 8.6). The masses are those of the bare nuclei, in daltons; adding $z\,m_e - B_e/c^2$ gives the atomic mass. Compared with the estimates of the evaluation they differ by 0.20 MeV, root mean square, and by 0.17 MeV for $A \geq 40$; this is a comparison

NUCLEUS	MISS (MEV)	IN σ	NUCLEUS	MISS (MEV)	IN σ
Li-12	1.12	1.6	Tb-144	−0.03	0.1
B-16	0.50	1.0	Ta-161	−0.01	0.0
B-18	1.03	2.0	Re-170	−0.00	0.0
Ne-16	−0.49	0.9	Re-193	0.03	0.1
Na-19	0.08	0.2	Tl-187	0.19	0.7
Cl-30	0.50	1.8	Tl-191	0.08	0.3
Sc-39	0.21	0.8	Pb-193	−0.01	0.0
Br-69	−0.13	0.5	Po-220	0.03	0.1
Rb-73	−0.60	2.3	Fr-206	−0.09	0.3
Pm-137	0.01	0.0	Cm-237	−0.02	0.1
Eu-138	−0.18	0.7	Cm-250	0.08	0.3
Eu-162	0.02	0.1	Bk-248	−0.02	0.1

Table 8.4: The 24 nuclei of the first registered list whose masses turn out to be measured in the 2020 evaluation: the miss, predicted minus measured, in MeV and in units of the stated error combined with that of the measurement.

of two predictions, not a test. The list is also given in a data file that accompanies this chapter, with the number of estimates for each nucleus.

What would refute the law. One step past the edge, the measured errors put about three quarters of the nuclei within one stated error and all but one or two in a hundred within two (Section 8.6). If, once these nuclei are measured, clearly fewer than two thirds of them land within one stated error, or more than a few miss by more than two, then the error law is wrong past the edge. A miss that grows with the distance from the measured map would show where the law ends. The most likely places for it to fail are the nuclei far from stability, where the shells can change, and the nuclei near $N = z$. The next step is to deposit this list in a public archive, so that the test is on record before the measurements come.

> IN PLAIN WORDS
>
> A good law sticks its neck out. The first version of this one predicted the masses of 67 nuclei; 24 of them turn out to have been measured already, and 22 of those 24 land within two stated errors, about as many as chance allows. Here are 61 more, not yet measured. When they are, the law will either pass again or show where it breaks.

nucleus	mass (u)	σ (MeV)	nucleus	mass (u)	σ (MeV)
C-21	21.046346	0.88	W-189	188.921574	0.16
F-13	13.040375	1.41	Re-167	166.921899	0.17
Al-22	22.012791	0.84	Os-165	164.935516	0.17
Al-38	38.010770	0.50	Ir-170	169.933202	0.17
P-26	26.003572	0.71	Ir-198	197.930594	0.16
K-34	33.988205	0.55	Pt-169	168.944495	0.17
Sc-38	37.983340	0.50	Au-174	173.941996	0.16
Ti-59	58.960693	0.33	Au-204	203.935009	0.15
V-42	41.978655	0.45	Tl-178	177.950992	0.16
Fe-68	67.938111	0.30	Tl-212	211.954495	0.15
As-88	87.927827	0.24	Bi-185	184.952502	0.16
Mo-112	111.915362	0.21	Th-209	208.968810	0.15
Rh-91	90.912332	0.24	U-217	216.974964	0.15
Rh-120	119.912457	0.20	U-220	219.974985	0.15
Ag-95	94.910108	0.23	Np-228	227.986014	0.15
In-99	98.907463	0.22	Np-232	231.989910	0.15
In-133	132.911583	0.19	Am-236	235.998195	0.15
Sn-103	102.900646	0.22	Am-237	236.998839	0.15
Te-107	106.906348	0.21	Am-246	246.018433	0.15
Xe-111	110.911830	0.21	Bk-242	242.009703	0.15
Ce-152	151.905315	0.17	Bk-252	252.031983	0.15
Pm-162	161.917300	0.17	Es-248	248.022037	0.15
Gd-139	138.903422	0.18	Es-249	249.023089	0.15
Er-173	172.905535	0.16	Es-250	250.025338	0.15
Tm-149	148.915546	0.18	Md-253	253.032816	0.15
Tm-150	149.912565	0.17	Md-254	254.035216	0.14
Yb-153	152.911361	0.17	Md-256	256.039459	0.14
Lu-182	181.916586	0.16	No-251	251.033886	0.15
Hf-157	156.919331	0.17	Lr-257	257.043914	0.14
Ta-178	177.906232	0.16	Rf-255	255.045216	0.14
W-161	160.927188	0.17			

Table 8.5: Registered predictions of the Closure Law, 2 November 2026: the mass of the bare nucleus, in daltons, of 61 nuclei whose masses were not measured in the 2020 evaluation, each predicted from measured neighbours only, with its stated error $2\sigma(A)$ in MeV.

STATEMENT	STANDING
Proposition 8.2: fusion on the frame	proved
Theorem 8.4: the exact cancellations	proved
the cusp constant $w = 22.75$ MeV	fitted; tested on nuclei held out
the error law $\sigma(A)$	measured on 2,481 nuclei, each held out
fusion energies	tested on 2,272 fusions: 0.18 MeV, and 0.28 MeV with no constant
prediction past the edge	tested; the error doubles at each step
the first registered list	22 of the 24 measured within two errors
the new registered list	waiting for measurement
the origin of nuclear binding	not explained

Table 8.6: What is proved, what is measured, and what is still open.

8.7.3 *The Standing of the Law*

Table 8.6 sorts the statements of this chapter by what supports them: a proof, a measurement, or nothing yet.

Credit. The loops are the mass relations of Garvey and Kelson, published in 1966 (Garvey and Kelson, 1966; Garvey et al., 1969), and the corner removed by the cusp constant is the trace of Wigner's symmetry of 1937 (Wigner, 1937). The plain recipe first wrote them in cosmic masses and used them for fusion, and the first version added the constant and a first error law. The accuracy of the relations on measured masses, and the growth of their errors beyond the edge, have been studied before (Barea et al., 2005; Morales et al., 2009), and nuclear physics has global mass models that predict far from the measured map, where the loops cannot go. What the cosmic mass adds is the statement of the loops in its own terms, with the charge parts cancelling visibly and the leftover a neutral body; the reading of the closure on the cards, as a closure of charge-weighted leans; the fusion form; an error law with its floor, tested on measured masses only; and a registered list. These are modest additions to an old law, and that is a respectable thing for them to be.

What the chapter does show is that the frame of Chapter 7 can carry a law that predicts. Chapter 7 could only place the bodies whose masses were already known. Extended to isotopes, the same frame states a regularity of nuclear masses in its own terms, as the closure of six charge-weighted leans, and with it predicts masses that have not yet been measured and the energy of fusions that have not yet been weighed. The predictive power comes from nuclear physics, written

in the language of the cosmic mass; the measurements to come will decide how far it reaches.

The reading that this chapter shares with Chapter 6, every measured mass, energy and force a real part, does not stop at bodies at rest. Chapter 9 carries it to bodies in motion and to the fields between them, where it gives Newton's mechanics, Newton's gravity with a small magnetic part, and Maxwell's equations, all from one action. Its masses are the real parts of cosmic masses, and they include the masses that this chapter predicts.

> In plain words
>
> Isotopes sit on the rings of the frame, each tipped by its mass. When two nuclei fuse, the charges add exactly and the lost mass is a neutral body. Six neighbouring nuclei, counted with alternating signs, cancel almost everything: the charge, the protons and neutrons, and nearly all the binding. Hide one, and the other five give its mass, to about $9/A$ MeV, and with it the energy of every fusion that makes it. The law is old nuclear physics in a new frame, and it is now on record with predictions that can prove it wrong.

9

THE UNIFIED COSMIC LAW

One number for each body, one reading for every measurement, and one action. From them come Newton's mechanics, Newton's gravity with a small magnetic part, Maxwell's equations and the force of Lorentz. For slow bodies the law says nothing that physics did not already know, and it fails where general relativity is needed; what it adds is the form.

LONDON, DECEMBER 1864

On 8 December 1864 a paper by James Clerk Maxwell, then thirty-three and professor of natural philosophy at King's College London, was read to the Royal Society: *A Dynamical Theory of the Electromagnetic Field*. It gathered electricity and magnetism into one set of equations for a field that fills space, and from those equations it found waves whose speed came out close to the measured speed of light. "This velocity is so nearly that of light," Maxwell wrote, "that it seems we have strong reason to conclude that light itself (including radiant heat, and other radiations if any) is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field according to electromagnetic laws" (Maxwell, 1865).

In the same paper he turned to gravity. "After tracing to the action of the surrounding medium both the magnetic and the electric attractions and repulsions, and finding them to depend on the inverse square of the distance, we are naturally led to inquire whether the attraction of gravitation, which follows the same law of the distance, is not also traceable to the action of a surrounding medium." The answer troubled him. Like charges push apart, but like masses pull together, and in a field of his kind that one change of sign turns the energy of gravity below zero: "The intrinsic energy of the field of gravitation must therefore be less wherever there is a resultant gravitating force." The medium would have to hold "an enormous intrinsic energy" when undisturbed, and "the presence of dense bodies" would "diminish this energy wherever there is a resultant attraction." Maxwell stopped there: "As I am unable to understand in what way a medium can possess such properties, I cannot go any further in this direction in searching for the cause of gravitation."

Twenty-nine years later Oliver Heaviside wrote the equations of gravity in Maxwell's form, with a second field, like the magnetic one, made by masses in motion, and waves that cross space at a finite speed (Heaviside, 1893). In 1915 general relativity replaced them. For slow bodies it gives equations of Heaviside's form, but with the magnetic effects of gravity four times larger than his.

[First draft, November 9, 2026]

This book took the same road from the other end. Chapter 6 put mass and charge into one number for each body, the cosmic mass, and found one law for the pull between two bodies at rest: read by its real part, it is Newton's law and Coulomb's law at once (Proposition 6.16). Chapter 8 used the same reading for the masses of nuclei. This chapter asks how far the reading goes: can one number for each body, and one rule for reading it, give the laws of bodies in motion and of the fields between them?

For slow bodies, the answer is yes. From the cosmic mass, the Rule of the Square and the reading of Chapter 6, this chapter builds, one step at a time, two fields that carry General Numbers, four equations for them, one law of motion, the conservation of the source, of energy and of momentum, and a single action from which all of it follows. Opened into its two coordinates, the law is Newton's mechanics, Newton's gravity with Heaviside's magnetic part, Maxwell's equations and the force of Lorentz. The minus sign of the Rule of the Square does the work that Maxwell could not place: from one formula it gives the energy of charge fields a plus sign and the energy of gravity a minus sign. It does not remove his difficulty. The waves of the gravity part carry negative energy, and that is one of the places where the law fails. The chapter sets out the law step by step, and it ends by running the law, with one program, on every body it can reach.

THE ROAD THROUGH THIS CHAPTER. The chapter climbs in seven steps, shown in Figure 9.1. Section 9.1 recalls the one number and the one reading, with the two facts of Chapter 4 on which the whole chapter stands: the real part of a product (Proposition 4.23) and the rules of arithmetic (Remark 4.24). Section 9.2 turns forces into fields and states the law, and Section 9.3 splits it into Maxwell's equations and Heaviside's gravity. Section 9.4 opens the law of motion, Section 9.5 shows what the law keeps, and Section 9.6 draws all of it from one action. Section 9.7 runs the law, with one program, on every body it can reach.

9.1 STEP 1: ONE NUMBER, ONE READING

Every body has one cosmic mass (Definition 6.5),

$$\mu = m + q\,\mathcal{G}_{\bar\phi},$$

where m is its mass, q its charge, and $\mathcal{G}_{\bar\phi}$ the Composer Unit, an arrow that leans at the angle $\bar\phi = 79.86^\circ$ and has the length

$$|\mathcal{G}_{\bar\phi}| = \sqrt{\frac{K}{G}} = 1.16043 \times 10^{10}\ \mathrm{kg/C},$$
$$|\mathcal{G}_{\bar\phi}|^2 = \frac{K}{G} = 1.34659 \times 10^{20}\ \mathrm{kg^2/C^2}.$$

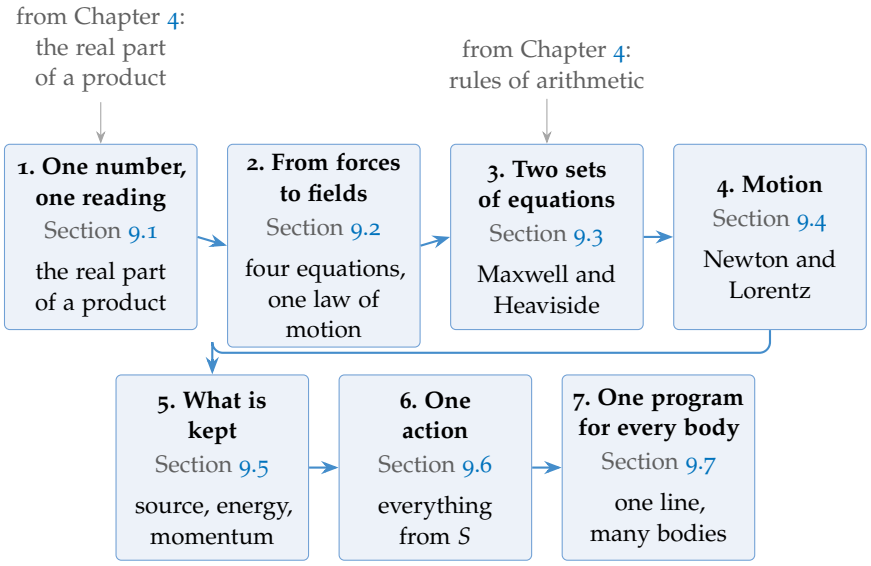


Figure 9.1: The road through the chapter, from one number for one body to one action for all bodies and fields. Each box is one step, with the section where it is taken, and the mathematics comes from Chapter 4.

The two parts are both in kilograms, so they can stand in one number. For a proton, $m = 1.6726 \times 10^{-27}$ kg and $q = e$, and the charge part has the size $e\,|\mathcal{G}_{\bar{\phi}}| = 1.8592 \times 10^{-9}$ kg, the Stoney mass of Definition 7.2.

A General Number $z = a + b\,\mathcal{G}_{\bar{\phi}}$ has two coordinates: its real part $\Re(z) = a$ and its Composer coordinate b, the number of steps of the Composer Unit in its Composer part $\mathrm{Com}(z) = b\,\mathcal{G}_{\bar{\phi}}$ (Definition 4.11). Chapter 6 chose one of the two readings that Chapter 4 offers for bringing a General Number down to the real axis, the real part rather than the shadow, and it is the only choice made in this chapter.

Definition 9.1 (The Real Reading)**.** Every measured force, energy and momentum is the real part of a General Number.

This is the reading of Definition 6.17, carried from forces to energies and momenta. It rests on the square of the Composer Unit, which the Rule of the Square of Chapter 4 writes in the form $a + b\,\mathcal{G}_{\bar{\phi}}$ (Proposition 4.22):

$$\mathcal{G}_{\bar{\phi}}^2 = 2\,|\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}\;\mathcal{G}_{\bar{\phi}} - |\mathcal{G}_{\bar{\phi}}|^2.$$

With it, Chapter 4 gives every product of two General Numbers of one true relative plane its real part (Proposition 4.23),

$$\Re(z_1 z_2) = a_1 a_2 - |\mathcal{G}_{\bar{\phi}}|^2 b_1 b_2,$$

with no angle in it and one fixed minus sign, and General Numbers obey every rule of complex arithmetic (Remark 4.24): every true relative plane is a field (Theorem 4.36).

These rules are all that the usual identities of vector calculus need. For smooth vector fields $\mathbf{F}$ and $\mathbf{H}$, and a constant vector $\mathbf{v}$,

(a) $\nabla \cdot (\nabla \times \mathbf{F}) = 0$,
(b) $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$,
(c) $\nabla \cdot (\mathbf{F} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{H})$,
(d) $\nabla(\mathbf{v} \cdot \mathbf{F}) - (\mathbf{v} \cdot \nabla)\mathbf{F} = \mathbf{v} \times (\nabla \times \mathbf{F})$.

Their proofs use only sums, products, and derivatives taken in a fixed order. By Remark 4.24 and Theorem 4.36, they hold word for word for fields whose values are General Numbers, and Maxwell's whole toolbox can be used on them. Here the divergence $\nabla \cdot \mathbf{F}$ measures how much a field flows out of a point, and the curl $\nabla \times \mathbf{F}$ how much it circles around one.

Put two cosmic masses into Proposition 4.23, with $a = m$ and $b = q$, and multiply by G. Since $G\,|\mathcal{G}_{\bar\phi}|^2 = K$,

$$G\,\Re(\mu_a \mu_b) = G\,(m_a m_b - |\mathcal{G}_{\bar\phi}|^2 q_a q_b) = G\,m_a m_b - K\,q_a q_b,$$

which is Proposition 6.16: the force between two bodies at rest is this number divided by r^2, counted positive for a pull. Two masses pull, like charges push, unlike charges pull, all from one expression. The Composer coordinate of the product, $m_a q_b + q_a m_b + 2\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi\; q_a q_b$, holds the old Coupling term, a share of the term of two charges that is far larger for charged particles (Section 6.5.2), and the whole of the angle. Under the Real Reading it is not a force, so a neutron feels no force from a charged plate, as measured (Section 6.5.3).

Worked example: one line for every pair

Two protons 10^{-10} m apart, a typical distance inside a molecule, pull each other by gravity with $Gm_p^2/r^2 = 1.87 \times 10^{-44}$ N and push each other electrically with $Ke^2/r^2 = 2.31 \times 10^{-8}$ N, which is 1.24×10^{36} times more; the line above gives the push, -2.31×10^{-8} N. The Earth and the Moon, 3.844×10^8 m apart, pull each other with $GMm/r^2 = 1.98 \times 10^{20}$ N. Both numbers come from the same formula, with different values of m and q.

In plain words

Each body carries one number with two parts, its mass and its charge, both written in kilograms. Two such numbers multiply by a rule in which the angle of the Composer axis drops out of the real part, the part that is measured. That part is the pull of gravity minus the push of the charges, Newton's law and Coulomb's law in one line. The rest of the product, where the angle lives, is not a force. Because the numbers obey the ordinary rules of arithmetic, every tool of field physics works on them unchanged.

9.2 STEP 2: FROM FORCES TO FIELDS

Newton's law and Coulomb's law act at a distance. Faraday and Maxwell replaced the distance by a field: each body fills space around it with a field, and each other body feels the field at its own place. In this chapter vectors are written in bold, and a field gives one vector at every point, as a weather map gives a wind at every town.

Definition 9.2 (The Cosmic Field of a Body at Rest)**.** A body of cosmic mass μ_s at rest at the origin makes, at the distance r in the outward direction $\hat{\mathbf{r}}$, the **cosmic field**

$$\boldsymbol{\mathcal{E}} = -\frac{G\mu_s}{r^2}\,\hat{\mathbf{r}}.$$

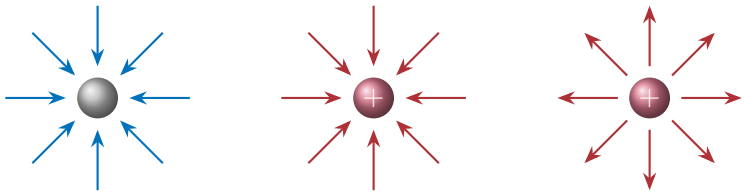
Its value at each point is a General Number vector, with two coordinates: $\boldsymbol{\mathcal{E}} = \mathbf{g} + \mathbf{e}\,\mathcal{G}_{\bar{\phi}}$.

A test body μ feels the force $\Re(\mu\boldsymbol{\mathcal{E}}) = -G\,\Re(\mu\mu_s)\,\hat{\mathbf{r}}/r^2$. By Proposition 6.16 it points inward, a pull, when $\Re(\mu\mu_s) > 0$, and outward, a push, when it is negative: the field gives back the law of Chapter 6.

Around a source, Figure 9.2 draws the field in its two coordinates. For a mass, the real coordinate $\mathbf{g}$ points inward. For a positive charge, the Composer coordinate $\mathbf{e}$ points inward too, since it comes from the same formula; but the force it gives a second positive charge, $-|\mathcal{G}_{\bar{\phi}}|^2 q\,\mathbf{e}$, points outward, because the real part of a product counts the Composer coordinates with the minus sign of Proposition 4.23.

Surround the source with a sphere of radius r. The field has the size $G\mu_s/r^2$ and points inward, and the sphere has the area $4\pi r^2$, so the total flow of the field out of the sphere is

$$-\frac{G\mu_s}{r^2} \times 4\pi r^2 = -4\pi G\mu_s.$$



(a) a mass: $\mathbf{g}$ points in

(b) a charge: $\mathbf{e}$ points in

(c) the same charge: $\mathbf{E} = -|\mathcal{G}_{\bar{\phi}}|^2\mathbf{e}$ points out

Figure 9.2: The cosmic field $\boldsymbol{\mathcal{E}} = -G\mu_s\hat{\mathbf{r}}/r^2$ of one source at rest, in its two coordinates. (a) The real coordinate $\mathbf{g}$ of a mass points in, toward the source. (b) The Composer coordinate $\mathbf{e}$ of a positive charge points in as well, by the same formula. (c) Maxwell's electric field of the same charge, $\mathbf{E} = -|\mathcal{G}_{\bar{\phi}}|^2\mathbf{e}$ (Theorem 9.5), points out. The minus sign of the Rule of the Square turns the arrows round: masses pull, and like charges push.

The radius has gone. Any closed surface around the source gives the same value, sources outside it add nothing, and the total flow out of a surface equals the sum of the divergence inside it. With a density of cosmic mass $\rho_\mu = \rho_m + \rho_q \mathcal{G}_{\bar{\phi}}$, made of a density of mass and a density of charge, this is Gauss's law for the cosmic field, $\nabla \cdot \boldsymbol{\mathcal{E}} = -4\pi G \rho_\mu$.

A moving source makes a second field, just as a current of charge makes a magnetic field. Its source is the *cosmic current*, the flow of cosmic mass,

$$\mathbf{J}_\mu = \mathbf{J}_m + \mathbf{J}_q \mathcal{G}_{\bar{\phi}},$$

where $\mathbf{J}_m$ is a flow of mass and $\mathbf{J}_q$ a flow of charge, an ordinary electric current. For one body of velocity $\mathbf{v}$, the cosmic current is $\mu\,\mathbf{v}$ at the place of the body. The law of this chapter is then the following.

Law 9.3 (The Unified Cosmic Law)**.** Two fields with General Number values, the cosmic field $\boldsymbol{\mathcal{E}}$ and the second field $\boldsymbol{\mathcal{B}}$, obey

$$\text{(F1)}\quad \nabla \cdot \boldsymbol{\mathcal{E}} = -4\pi G \rho_\mu, \qquad \text{(F2)}\quad \nabla \cdot \boldsymbol{\mathcal{B}} = 0,$$

$$\text{(F3)}\quad \nabla \times \boldsymbol{\mathcal{E}} = -\partial_t \boldsymbol{\mathcal{B}}, \qquad \text{(F4)}\quad \nabla \times \boldsymbol{\mathcal{B}} = -\frac{4\pi G}{c^2}\mathbf{J}_\mu + \frac{1}{c^2}\partial_t \boldsymbol{\mathcal{E}},$$

and a body of cosmic mass μ and velocity $\mathbf{v}$ moves by

$$\text{(M)}\qquad \Re(\mu)\,\frac{d\mathbf{v}}{dt} = \Re\big[\mu\,(\boldsymbol{\mathcal{E}} + \mathbf{v} \times \boldsymbol{\mathcal{B}})\big].$$

In words: sources make $\boldsymbol{\mathcal{E}}$ (F1); $\boldsymbol{\mathcal{B}}$ has no point sources (F2); a changing $\boldsymbol{\mathcal{B}}$ makes $\boldsymbol{\mathcal{E}}$ (F3); currents and a changing $\boldsymbol{\mathcal{E}}$ make $\boldsymbol{\mathcal{B}}$ (F4). The constants are not free. The factor $4\pi G$ is the Gauss flow found above, and c is the speed of the waves of Theorem 9.6, which is the speed of light. In the law of motion, the force is the real part of the cosmic mass times the field it feels, and the inertia is the real part of the cosmic mass, $\Re(\mu) = m$: the only part that resists a push, and the part that a spectrograph weighs (Section 8.1).

> **In plain words**
>
> Instead of bodies pulling each other across empty space, each body fills the space around it with a field, and every other body feels the field where it is. One field, written as one number with two parts at each point, does the work of the gravity field and of the electric field together; a second field, made by moving bodies, does the work of the magnetic field. Four short equations say how bodies make the two fields, and a fifth says how a body moves in them.

9.3 STEP 3: TWO SETS OF EQUATIONS

Write each field and each source in its two coordinates: $\boldsymbol{\mathcal{E}} = \mathbf{g} + \mathbf{e}\,\mathcal{G}_{\bar{\phi}}$, $\boldsymbol{\mathcal{B}} = \mathbf{B}_g + \mathbf{b}\,\mathcal{G}_{\bar{\phi}}$, $\rho_\mu = \rho_m + \rho_q \mathcal{G}_{\bar{\phi}}$ and $\mathbf{J}_\mu = \mathbf{J}_m + \mathbf{J}_q \mathcal{G}_{\bar{\phi}}$.

REAL COORDINATE	COMPOSER COORDINATE
$\nabla \cdot \mathbf{g} = -4\pi G \rho_m$	$\nabla \cdot \mathbf{e} = -4\pi G \rho_q$
$\nabla \cdot \mathbf{B}_g = 0$	$\nabla \cdot \mathbf{b} = 0$
$\nabla \times \mathbf{g} = -\partial_t \mathbf{B}_g$	$\nabla \times \mathbf{e} = -\partial_t \mathbf{b}$
$\nabla \times \mathbf{B}_g = -\frac{4\pi G}{c^2}\mathbf{J}_m + \frac{1}{c^2}\partial_t \mathbf{g}$	$\nabla \times \mathbf{b} = -\frac{4\pi G}{c^2}\mathbf{J}_q + \frac{1}{c^2}\partial_t \mathbf{e}$

Table 9.1: The four field equations of Law 9.3, split into their two coordinates (Proposition 9.4). Each equation splits in two, and the angle appears in neither set.

Proposition 9.4 (The Split)**.** *Each of the field equations (F1) to (F4) holds separately for the real coordinate and for the Composer coordinate, as in Table 9.1.*

Proof. Every operation in (F1) to (F4) is a sum, a product by a real number, or a derivative in space or time; no field multiplies a field, and no field is multiplied by $\mathcal{G}_{\tilde{\phi}}$. Each of these operations acts on the two coordinates one at a time: the real part and the Composer part of a sum are the sums of the parts, and a real factor comes out of each (Proposition 4.54). Two General Numbers are equal exactly when both coordinates are equal (Definition 4.12), so each equation is two equations, one for each coordinate. □

The Composer set is Maxwell's. Its fields are scaled and turned round by the same factor that turns a Composer coordinate into a force.

Theorem 9.5 (Maxwell's Equations)**.** *Define the electric and magnetic fields by*

$$\mathbf{E} = -|\mathcal{G}_{\tilde{\phi}}|^2\, \mathbf{e}, \qquad \mathbf{B} = -|\mathcal{G}_{\tilde{\phi}}|^2\, \mathbf{b}.$$

Then the Composer set of Table 9.1 is Maxwell's set of equations,

$$\nabla \cdot \mathbf{E} = 4\pi K \rho_q, \qquad \nabla \cdot \mathbf{B} = 0,$$
$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \qquad \nabla \times \mathbf{B} = \frac{4\pi K}{c^2}\mathbf{J}_q + \frac{1}{c^2}\partial_t \mathbf{E},$$

with the magnetic constant $\mu_0 = 4\pi K/c^2$.

Proof. Multiply each Composer equation by $-|\mathcal{G}_{\tilde{\phi}}|^2$, and use $G\,|\mathcal{G}_{\tilde{\phi}}|^2 = K$:

$$\begin{aligned} \nabla \cdot \mathbf{e} = -4\pi G \rho_q \quad &\Rightarrow \quad \nabla \cdot \mathbf{E} = 4\pi K \rho_q, \\ \nabla \cdot \mathbf{b} = 0 \quad &\Rightarrow \quad \nabla \cdot \mathbf{B} = 0, \\ \nabla \times \mathbf{e} = -\partial_t \mathbf{b} \quad &\Rightarrow \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \\ \nabla \times \mathbf{b} = -\tfrac{4\pi G}{c^2}\mathbf{J}_q + \tfrac{1}{c^2}\partial_t \mathbf{e} \quad &\Rightarrow \quad \nabla \times \mathbf{B} = \tfrac{4\pi K}{c^2}\mathbf{J}_q + \tfrac{1}{c^2}\partial_t \mathbf{E}. \end{aligned}$$

With $K = 1/(4\pi\varepsilon_0)$ and $\mu_0 = 4\pi K/c^2$, these are Maxwell's equations in their usual form. □

The minus sign in the definition turns the signs of Gauss's law round, which is why the field of a positive charge points out while the field of a mass points in (Figure 9.2). The magnetic constant $\mu_0 = 4\pi K/c^2$ is Remark 6.8 read the other way: there, the length of the Composer Unit was written $c\sqrt{\mu_0/(4\pi G)}$.

> WORKED EXAMPLE: ONE POINT CHARGE
>
> A charge q at rest at the origin has, by Definition 9.2, the Composer coordinate $\mathbf{e} = -Gq\,\hat{\mathbf{r}}/r^2$. Maxwell's field is then $\mathbf{E} = -|\mathcal{G}_{\bar{\phi}}|^2\mathbf{e} = G\,|\mathcal{G}_{\bar{\phi}}|^2 q\,\hat{\mathbf{r}}/r^2 = Kq\,\hat{\mathbf{r}}/r^2$, pointing out: Coulomb's field. A second charge q' feels $q'\mathbf{E}$, the push of Coulomb's law for like charges.

The real set holds Newton's gravity. For bodies at rest, (F3) gives the field no curl, and (F1) gives the field of a mass M, $\mathbf{g} = -GM\hat{\mathbf{r}}/r^2$, Newton's pull. The field $\mathbf{B}_g$ is new relative to Newton. It is the magnetic part of gravity that Heaviside wrote in 1893, made by moving masses and felt by moving masses. For slow bodies it is tiny, and it is where this law parts from general relativity.

Theorem 9.6 (Waves at the Speed c)**.** *In empty space both fields obey the wave equation with the speed c,*

$$\nabla^2\boldsymbol{\mathcal{E}} = \frac{1}{c^2}\,\partial_t^2\boldsymbol{\mathcal{E}}, \qquad \nabla^2\boldsymbol{\mathcal{B}} = \frac{1}{c^2}\,\partial_t^2\boldsymbol{\mathcal{B}},$$

in both coordinates: the waves of the gravity part travel at the speed of light.

Proof. With no sources, take the curl of (F3) and use identity (b), then (F4) and (F1):

$$\begin{aligned} \nabla\times(\nabla\times\boldsymbol{\mathcal{E}}) &= -\partial_t(\nabla\times\boldsymbol{\mathcal{B}}) && \text{curl of (F3)}\\ \nabla(\nabla\cdot\boldsymbol{\mathcal{E}}) - \nabla^2\boldsymbol{\mathcal{E}} &= -\frac{1}{c^2}\partial_t^2\boldsymbol{\mathcal{E}} && \text{identity (b), and (F4)}\\ \nabla^2\boldsymbol{\mathcal{E}} &= \frac{1}{c^2}\partial_t^2\boldsymbol{\mathcal{E}} && \text{(F1) with no source.} \end{aligned}$$

The same steps, starting from the curl of (F4) and using (F3) and (F2), give the equation for $\boldsymbol{\mathcal{B}}$. By Proposition 9.4, each equation holds for both coordinates. □

On 17 August 2017, the gravitational waves of two merging neutron stars and the gamma rays of the same event arrived within two seconds of each other after a journey of about 130 million years, which fixes the speed of gravity to that of light within 3 parts in 10^{15} (Abbott et al., 2017). The law passes this test, as general relativity does.

> **IN PLAIN WORDS**
>
> Split into its two parts, the law becomes two copies of the same equations. The charge part is Maxwell's electromagnetism, word for word, once its fields are scaled by the square of the length of the Composer Unit and turned round by its minus sign. The mass part is Newton's gravity, with a small magnetic-like field for moving masses that Heaviside wrote down in 1893. Both parts carry waves, and both waves travel at the speed of light.

9.4 STEP 4: MOTION

Newton's second law says that mass times acceleration equals force. In the law of motion (M), the force is the real part of the cosmic mass times the field it feels, and the mass is the real part of the cosmic mass.

Proposition 9.7 (Newton and Lorentz in One Law of Motion)**.** *Opened into its coordinates, the law of motion (M) reads*

$$m\,\frac{d\mathbf{v}}{dt} = m\,(\mathbf{g} + \mathbf{v}\times\mathbf{B}_g) + q\,(\mathbf{E} + \mathbf{v}\times\mathbf{B}).$$

Proof. By Proposition 4.23, applied to each component, with $\mu = m + q\,\mathcal{G}_{\bar\phi}$ and $\boldsymbol{\mathcal{E}} = \mathbf{g} + \mathbf{e}\,\mathcal{G}_{\bar\phi}$,

$$\Re(\mu\boldsymbol{\mathcal{E}}) = m\,\mathbf{g} - |\mathcal{G}_{\bar\phi}|^2 q\,\mathbf{e} = m\,\mathbf{g} + q\,\mathbf{E}.$$

In the same way, with $\mathbf{v}$ real, $\Re(\mu\,\mathbf{v}\times\boldsymbol{\mathcal{B}}) = m\,\mathbf{v}\times\mathbf{B}_g - |\mathcal{G}_{\bar\phi}|^2 q\,\mathbf{v}\times\mathbf{b} = m\,\mathbf{v}\times\mathbf{B}_g + q\,\mathbf{v}\times\mathbf{B}$. The left side of (M) is $\Re(\mu)\,d\mathbf{v}/dt = m\,d\mathbf{v}/dt$. □

The second bracket is the force of Lorentz, at full strength. The first is gravity, with its small magnetic-like term. Three of Newton's statements follow at once. With no fields the right side is zero, so $d\mathbf{v}/dt = 0$ and a body keeps its velocity: the first law. The left side is mass times acceleration and the right side is the total force: the second law. For a neutral body, $q = 0$ and $\mathbf{v}\times\mathbf{B}_g$ is tiny, so the mass cancels and $d\mathbf{v}/dt = \mathbf{g}$: every body falls alike. The cancellation works because the same m, the real part of μ, stands on both sides. The satellite MICROSCOPE compared the fall of titanium and platinum in orbit and found them equal to within about 3 parts in 10^{15} (Touboul et al., 2022). The law passes the test, but it passes it by construction, as Newton's law does: it does not explain why the mass that resists a push is the mass that gravity pulls.

Bodies with the same charge per mass, started alike, therefore move alike in every field: divided by m, the law reads $d\mathbf{v}/dt = \mathbf{g} + \mathbf{v}\times\mathbf{B}_g + (q/m)(\mathbf{E} + \mathbf{v}\times\mathbf{B})$, and such bodies lie on one ray of the frame of Chapter 7 (Remark 6.15). In an electric field the accelerations of two bodies differ by $(m_a q_b - q_a m_b)\,\mathbf{E}/(m_a m_b)$, where $m_a q_b - q_a m_b$, the Composer coordinate of $\bar\mu_a \mu_b$ (Proposition 4.58), is zero exactly when the two cosmic masses lie on one line through 0.

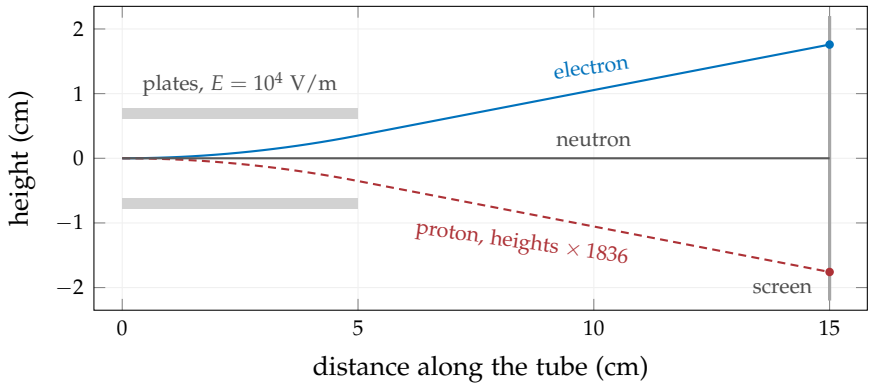


Figure 9.3: Three bodies in Thomson's tube, moved by the law of motion (M) with nothing changed but m and q. The electron reaches the screen 17.59 mm high. The proton is turned the other way by 0.0096 mm; enlarged 1836 times, its path (dashed) is the mirror of the electron's. The neutron goes straight. The paths come from the program of Section 9.7.

Worked example: Thomson's tube, by hand

An electron enters a tube at 2.5×10^7 m/s, between two plates 5 cm long with the field $E = 10^4$ V/m between them; a screen stands 10 cm beyond the plates. Inside the plates the electron feels only $q\mathbf{E}$, so its sideways acceleration is $eE/m_e = 1.7588 \times 10^{15}$ m/s^2. It spends $0.05/(2.5 \times 10^7) = 2 \times 10^{-9}$ s in the plates and rises $\frac{1}{2}at^2 = 3.518$ mm. It leaves with the sideways speed $at = 3.518 \times 10^6$ m/s, a tilt of 0.14071, so it rises another $0.10 \times 0.14071 = 14.07$ mm on the way to the screen: 17.59 mm in all. A proton entering at the same speed is turned the other way and 1836 times less, and a neutron goes straight (Figure 9.3). This is the experiment with which J. J. Thomson measured the ratio e/m of the electron in 1897 (Thomson, 1897).

Worked example: a circle in a magnetic field

With only a magnetic field, the force $q\,\mathbf{v} \times \mathbf{B}$ is perpendicular to the velocity, so the speed stays fixed and the path bends into a circle. Setting $mv^2/R = qvB$ gives the radius $R = mv/(qB)$, and one turn takes $2\pi R/v = 2\pi m/(qB)$, whatever the speed. An electron in a field of 1 T makes 2.7992×10^{10} turns per second, and at 10^6 m/s its circle has the radius 5.69 μm.

WORKED EXAMPLE: A DRIFT IN CROSSED FIELDS

A steady motion with no acceleration needs $\mathbf{E} + \mathbf{v} \times \mathbf{B} = 0$. With $\mathbf{E}$ along y and $\mathbf{B}$ along z, this gives $v_x = E/B$. A proton set free at rest in $E = 1000$ V/m and $B = 1$ T does not run away along $\mathbf{E}$: it rolls along x in loops, and its average speed is $E/B = 1000$ m/s (Figure 9.4(c)).

WORKED EXAMPLE: TWO CIRCULAR ORBITS

On a circle, the mass times v^2/R equals the pull.

The Moon. With the Earth held at rest, $v^2/R = GM/R^2$ gives $v = \sqrt{GM/R} = 1018$ m/s and a period of 27.45 days. The Earth does not stay at rest: the two bodies turn about their common centre, and with the Moon's own mass counted the period is 27.28 days. The measured sidereal month is 27.32 days; most of what remains is the pull of the Sun, which this two-body estimate leaves out.

Hydrogen. For an electron on a circle of radius a_0 around a proton, $m_e v^2/a_0 = Ke^2/a_0^2$ gives $v = 2.1877 \times 10^6$ m/s, which is $0.0072974\,c$, the fine-structure constant. The energy needed to free the electron is $Ke^2/(2a_0) = 13.61$ eV, and 13.598 eV with the motion of the proton counted, as measured (Sansonetti and Martin, 2005). The radius a_0 itself comes from quantum theory, not from this law: the law moves the electron once its orbit is given, but it does not say why the orbit has that size.

IN PLAIN WORDS

One rule moves every body: its mass times its acceleration equals the real part of its cosmic mass times the fields it feels. Opened up, the rule is gravity plus the electric and magnetic forces, each at its known strength. It sends an electron up a tube, a proton the other way and a neutron straight on; it bends an electron into a circle in a magnet; it keeps the Moon and, in the classical picture, the electron of hydrogen on their orbits. Only two numbers change from one body to the next: the mass and the charge.

9.5 STEP 5: WHAT IS KEPT

A law of fields must say what is kept: the source, the energy and the momentum. Each follows from Law 9.3 in a few lines.

Theorem 9.8 (Conservation of the Source)**.** *The density and the current of cosmic mass obey*

$$\partial_t \rho_\mu + \nabla \cdot \mathbf{J}_\mu = 0.$$

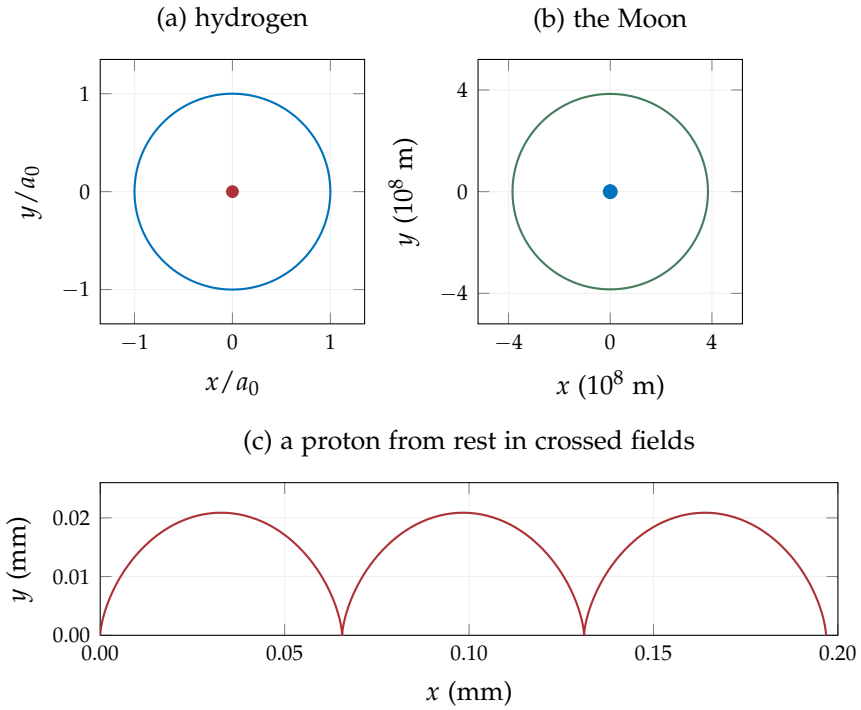


Figure 9.4: Three motions from one law of motion. (a) An electron on the classical orbit of hydrogen around a proton, at the Bohr radius a_0. (b) The Moon around the Earth, held at rest. (c) A proton from rest in $E = 1000$ V/m and $B = 1$ T, rolling along x at the average speed $E/B = 1000$ m/s; heights enlarged three times. An electric orbit, a gravitational orbit and a magnetic drift, computed by the same line of the program of Section 9.7.

Proof. Take the divergence of (F4). By identity (a), the left side is zero:

$$0 = -\frac{4\pi G}{c^2}\nabla \cdot \mathbf{J}_\mu + \frac{1}{c^2}\partial_t(\nabla \cdot \boldsymbol{\mathcal{E}}).$$

Replace $\nabla \cdot \boldsymbol{\mathcal{E}}$ by (F1) and divide by $-4\pi G/c^2$. □

The Composer coordinate of Theorem 9.8 is the conservation of charge, which holds exactly. The real coordinate says that mass is conserved, which holds for slow bodies, where energies are small. In a fusion it fails as it stands: the loss of cosmic mass $\Delta\mu$ is a real number that leaves as energy (Proposition 8.2), so the real coordinate holds only when the energy that leaves is counted as mass, Q/c^2. The law itself does not count it; this is the first sign that it is a law for slow bodies.

Theorem 9.9 (The Balance of Energy). *Let*

$$u = -\frac{1}{8\pi G}\,\Re\big[\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mathcal{E}} + c^2\,\boldsymbol{\mathcal{B}}\cdot\boldsymbol{\mathcal{B}}\big], \qquad \mathbf{S} = -\frac{c^2}{4\pi G}\,\Re\big[\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}}\big].$$

Then

$$\partial_t u + \nabla\cdot\mathbf{S} = -\Re\big[\boldsymbol{\mathcal{E}}\cdot\mathbf{J}_\mu\big].$$

Proof. Start from identity (c), with $\mathbf{F} = \boldsymbol{\mathcal{E}}$ and $\mathbf{H} = \boldsymbol{\mathcal{B}}$, and use (F3) and (F4):

$$\begin{aligned}\nabla\cdot(\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}}) &= \boldsymbol{\mathcal{B}}\cdot(\nabla\times\boldsymbol{\mathcal{E}}) - \boldsymbol{\mathcal{E}}\cdot(\nabla\times\boldsymbol{\mathcal{B}})\\ &= -\boldsymbol{\mathcal{B}}\cdot\partial_t\boldsymbol{\mathcal{B}} + \frac{4\pi G}{c^2}\,\boldsymbol{\mathcal{E}}\cdot\mathbf{J}_\mu - \frac{1}{c^2}\,\boldsymbol{\mathcal{E}}\cdot\partial_t\boldsymbol{\mathcal{E}}\\ &= -\tfrac{1}{2}\,\partial_t(\boldsymbol{\mathcal{B}}\cdot\boldsymbol{\mathcal{B}}) - \frac{1}{2c^2}\,\partial_t(\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mathcal{E}}) + \frac{4\pi G}{c^2}\,\boldsymbol{\mathcal{E}}\cdot\mathbf{J}_\mu.\end{aligned}$$

Take the real part and multiply by $-c^2/(4\pi G)$. □

In words, the energy u in a small volume changes because energy flows out through its surface, carried by $\mathbf{S}$, and because the field does work on the bodies inside, the right side. That the right side is indeed the work can be checked on one body: its power is $\mathbf{v}\cdot\Re[\mu(\boldsymbol{\mathcal{E}} + \mathbf{v}\times\boldsymbol{\mathcal{B}})]$, and since $\mathbf{v}\cdot(\mathbf{v}\times\boldsymbol{\mathcal{B}}) = 0$ this is $\Re[\mu\,\mathbf{v}\cdot\boldsymbol{\mathcal{E}}]$, which is $\Re[\boldsymbol{\mathcal{E}}\cdot\mathbf{J}_\mu]$ for that body. The energy balances exactly.

Proposition 9.10 (One Energy, Two Faces). *Opened into its coordinates, the energy density of Theorem 9.9 is*

$$u = -\frac{g^2 + c^2B_g^2}{8\pi G} + \frac{E^2 + c^2B^2}{8\pi K}\,:$$

negative for the gravity part and positive for the Maxwell part.

Proof. By Proposition 4.23, and with $\mathbf{e} = -\mathbf{E}/|\mathcal{G}_{\bar\phi}|^2$,

$$\Re[\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mathcal{E}}] = g^2 - |\mathcal{G}_{\bar\phi}|^2 e^2 = g^2 - \frac{E^2}{|\mathcal{G}_{\bar\phi}|^2}.$$

Multiply by $-1/(8\pi G)$, use $G\,|\mathcal{G}_{\bar\phi}|^2 = K$, and do the same for $\boldsymbol{\mathcal{B}}$. □

> WORKED EXAMPLE: THE SIZES OF THE TWO FACES
>
> At the surface of the Earth, $g = 9.81\ \mathrm{m/s^2}$ gives the gravity part an energy density of $-g^2/(8\pi G) = -5.74\times10^{10}\ \mathrm{J/m^3}$. An electric field of 100 V/m, the fair-weather field of the open air, gives the Maxwell part $+E^2/(8\pi K) = +4.43\times10^{-8}\ \mathrm{J/m^3}$: the energy of Maxwell's theory, $\varepsilon_0 E^2/2$.

The minus sign of the gravity part is the sign that stopped Maxwell in 1864: the energy of the field of gravitation is "less wherever there

is a resultant gravitating force." Here it needs no medium with an enormous energy of its own. It comes from the same minus sign of the Rule of the Square that makes like charges push apart, and one formula gives the two faces. For gravity at rest, the negative sign is correct: it is the energy of Newton's theory, the one that binds the planets to the Sun. For the waves of the gravity part, it is a known weakness of every theory of gravity written in Maxwell's form (Misner et al., 1973, Section 7.2).

Theorem 9.11 (The Balance of Momentum)**.** *Let* $\mathbf{f} = \Re[\rho_\mu \boldsymbol{\mathcal{E}} + \mathbf{J}_\mu \times \boldsymbol{\mathcal{B}}]$ *be the force on the matter in a unit volume and* $\mathbf{p}_f = \mathbf{S}/c^2 = -\frac{1}{4\pi G}\Re[\boldsymbol{\mathcal{E}} \times \boldsymbol{\mathcal{B}}]$ *the momentum of the field in a unit volume. Then* $\mathbf{f} + \partial_t \mathbf{p}_f = \nabla \cdot T$*, with the stress*

$$T_{ij} = -\frac{1}{4\pi G}\,\Re\Big[\mathcal{E}_i\mathcal{E}_j + c^2\mathcal{B}_i\mathcal{B}_j - \tfrac{1}{2}\,\delta_{ij}(\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mathcal{E}} + c^2\,\boldsymbol{\mathcal{B}}\cdot\boldsymbol{\mathcal{B}})\Big].$$

Proof. Replace the sources by (F1) and (F4), $\rho_\mu = -\nabla\cdot\boldsymbol{\mathcal{E}}/(4\pi G)$ and $\mathbf{J}_\mu = -\frac{c^2}{4\pi G}\nabla\times\boldsymbol{\mathcal{B}} + \frac{1}{4\pi G}\partial_t\boldsymbol{\mathcal{E}}$:

$$\mathbf{f} = -\frac{1}{4\pi G}\,\Re\big[\boldsymbol{\mathcal{E}}\,\nabla\cdot\boldsymbol{\mathcal{E}} + c^2(\nabla\times\boldsymbol{\mathcal{B}})\times\boldsymbol{\mathcal{B}} - \partial_t\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}}\big].$$

By (F3), $\partial_t\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}} = \partial_t(\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}}) - \boldsymbol{\mathcal{E}}\times\partial_t\boldsymbol{\mathcal{B}} = \partial_t(\boldsymbol{\mathcal{E}}\times\boldsymbol{\mathcal{B}}) + \boldsymbol{\mathcal{E}}\times(\nabla\times\boldsymbol{\mathcal{E}})$. Collecting terms, and adding $c^2\,\boldsymbol{\mathcal{B}}\,\nabla\cdot\boldsymbol{\mathcal{B}} = 0$ by (F2),

$$\begin{aligned}\mathbf{f} + \partial_t\mathbf{p}_f = -\frac{1}{4\pi G}\,\Re\big[\,&\boldsymbol{\mathcal{E}}\,\nabla\cdot\boldsymbol{\mathcal{E}} + (\nabla\times\boldsymbol{\mathcal{E}})\times\boldsymbol{\mathcal{E}} \\ &+ c^2\,\boldsymbol{\mathcal{B}}\,\nabla\cdot\boldsymbol{\mathcal{B}} + c^2(\nabla\times\boldsymbol{\mathcal{B}})\times\boldsymbol{\mathcal{B}}\big].\end{aligned}$$

For any field, $(\nabla\cdot\mathbf{F})\mathbf{F} + (\nabla\times\mathbf{F})\times\mathbf{F}$ is the divergence of $F_iF_j - \frac{1}{2}\delta_{ij}F^2$, so the right side is the divergence of T. □

The momentum of the bodies and of the fields together changes only by what flows through the boundary; in a closed region far from all fields, the total is constant. This is what replaces Newton's third law. For two bodies at rest, $\Re(\mu_a\mu_b) = \Re(\mu_b\mu_a)$, so the forces are equal and opposite. For two moving bodies, the magnetic forces between them need not balance at each moment, as in Maxwell's theory; the difference is stored in the field, and the total momentum is still kept.

> IN PLAIN WORDS
>
> Three things are kept. Charge is kept exactly, and mass is kept as long as no energy leaves, which is why this law cannot weigh a fusion by itself. Energy is kept: what a field gives to the bodies it loses itself, and one formula counts the energy of gravity as negative and the energy of electric and magnetic fields as positive. Momentum is kept, provided the momentum of the fields is counted along with the momentum of the bodies.

9.6 STEP 6: ONE ACTION

A body could take many paths between two places. For each path, add up over time a quantity L, the Lagrangian; the sum $S = \int L\,dt$ is the *action*. The path that the body takes is the one for which a small change of the path does not change the action, to first order, and the rule that says so is the equation of Euler and Lagrange,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\mathbf{x}}} = \frac{\partial L}{\partial \mathbf{x}}.$$

> WORKED EXAMPLE: A FALLING BALL
>
> Take $L = \frac{1}{2}m\dot{y}^2 - m\,g\,y$, the energy of motion less the energy of height. The equation of Euler and Lagrange gives $m\ddot{y} = -mg$: free fall.

The same rule can hold the whole law. Equations (F2) and (F3) are solved at once by two potentials with General Number values, Φ and $\boldsymbol{\mathcal{A}}$:

$$\boldsymbol{\mathcal{B}} = \nabla \times \boldsymbol{\mathcal{A}}, \qquad \boldsymbol{\mathcal{E}} = -\nabla\Phi - \partial_t \boldsymbol{\mathcal{A}}.$$

Indeed (F2) is identity (a), and $\nabla \times \boldsymbol{\mathcal{E}} = -\nabla \times \partial_t \boldsymbol{\mathcal{A}} = -\partial_t \boldsymbol{\mathcal{B}}$, because the curl of a gradient is zero.

Definition 9.12 (The Action of the Unified Law)**.** For bodies of cosmic masses μ and velocities $\mathbf{v}$, in fields given by the potentials Φ and $\boldsymbol{\mathcal{A}}$, the action is

$$S = \int dt \Big\{ \sum_{\text{bodies}} \Big[\tfrac{1}{2}\Re(\mu)\, v^2 - \Re\big(\mu(\Phi - \mathbf{v}\cdot\boldsymbol{\mathcal{A}})\big) \Big] - \frac{1}{8\pi G}\int \Re\big[\boldsymbol{\mathcal{E}}\cdot\boldsymbol{\mathcal{E}} - c^2\,\boldsymbol{\mathcal{B}}\cdot\boldsymbol{\mathcal{B}}\big]\,dV \Big\}.$$

Theorem 9.13 (Everything from One Action)**.** *The action of Definition 9.12 is stationary*

1. *under every small change of the path of a body, exactly when the body obeys the law of motion (M);*
2. *under every small change of Φ, exactly when (F1) holds;*
3. *under every small change of $\boldsymbol{\mathcal{A}}$, exactly when (F4) holds.*

Together with the potentials, which give (F2) and (F3), the action holds the whole of Law 9.3.

Proof. (1) For one body, $L = \frac{1}{2}mv^2 - \Re(\mu\Phi) + \Re(\mu\,\mathbf{v}\cdot\boldsymbol{\mathcal{A}})$. Then

$$\frac{\partial L}{\partial \mathbf{v}} = m\mathbf{v} + \Re(\mu\boldsymbol{\mathcal{A}}),$$
$$\frac{d}{dt}\frac{\partial L}{\partial \mathbf{v}} = m\frac{d\mathbf{v}}{dt} + \Re\big(\mu(\partial_t\boldsymbol{\mathcal{A}} + (\mathbf{v}\cdot\nabla)\boldsymbol{\mathcal{A}})\big),$$
$$\frac{\partial L}{\partial \mathbf{x}} = -\Re(\mu\nabla\Phi) + \Re\big(\mu\nabla(\mathbf{v}\cdot\boldsymbol{\mathcal{A}})\big).$$

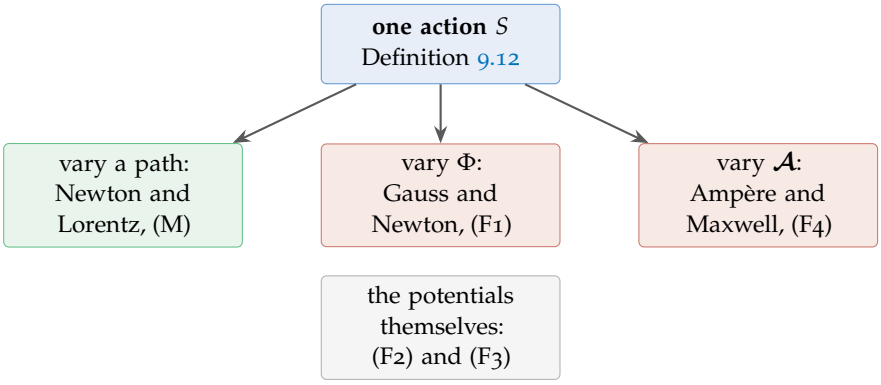


Figure 9.5: One action and three ways to vary it (Theorem 9.13). Varying the path of a body gives its law of motion; varying the two potentials gives the two field equations with sources. The other two field equations hold for any potentials.

Setting the second line equal to the third, and using identity (d),

$$m\frac{d\mathbf{v}}{dt} = \Re\big[\mu\big(-\nabla\Phi - \partial_t\boldsymbol{\mathcal{A}} + \mathbf{v}\times(\nabla\times\boldsymbol{\mathcal{A}})\big)\big] = \Re\big[\mu(\boldsymbol{\mathcal{E}} + \mathbf{v}\times\boldsymbol{\mathcal{B}})\big].$$

(2) Change Φ by a small $\delta\Phi$; then $\delta\boldsymbol{\mathcal{E}} = -\nabla\delta\Phi$. Collecting the bodies into the density ρ_μ, and integrating by parts in the second line,

$$\begin{aligned}\delta S &= \int \Re\Big[-\rho_\mu\,\delta\Phi + \frac{1}{4\pi G}\,\boldsymbol{\mathcal{E}}\cdot\nabla\delta\Phi\Big]dV\,dt\\ &= -\int \Re\Big[\delta\Phi\Big(\rho_\mu + \frac{1}{4\pi G}\,\nabla\cdot\boldsymbol{\mathcal{E}}\Big)\Big]dV\,dt.\end{aligned}$$

This vanishes for every $\delta\Phi$ exactly when the bracket is zero, by statement 3 of Proposition 4.23: that is (F1). (3) Change $\boldsymbol{\mathcal{A}}$ by a small $\delta\boldsymbol{\mathcal{A}}$; then $\delta\boldsymbol{\mathcal{B}} = \nabla\times\delta\boldsymbol{\mathcal{A}}$ and $\delta\boldsymbol{\mathcal{E}} = -\partial_t\delta\boldsymbol{\mathcal{A}}$. After integrating by parts in time and in space,

$$\delta S = \int \Re\Big[\delta\boldsymbol{\mathcal{A}}\cdot\Big(\mathbf{J}_\mu + \frac{c^2}{4\pi G}\,\nabla\times\boldsymbol{\mathcal{B}} - \frac{1}{4\pi G}\,\partial_t\boldsymbol{\mathcal{E}}\Big)\Big]dV\,dt,$$

and statement 3 of Proposition 4.23 again gives $\nabla\times\boldsymbol{\mathcal{B}} = -\frac{4\pi G}{c^2}\mathbf{J}_\mu + \frac{1}{c^2}\partial_t\boldsymbol{\mathcal{E}}$, which is (F4). □

Figure 9.5 sums up the step. Opened with Proposition 4.23, the action is the sum of two known actions: the action of Newton's gravity with Heaviside's magnetic part, whose field term is $-\frac{1}{8\pi G}\int(g^2 - c^2B_g^2)\,dV$, and the action of Maxwell's theory, whose field term is $\frac{1}{8\pi K}\int(E^2 - c^2B^2)\,dV$ and whose charge term is $q(\mathbf{v}\cdot\mathbf{A} - V)$, the source of Lorentz's force, with Maxwell's potentials V and $\mathbf{A}$ equal to $-|\mathcal{G}_{\bar\phi}|^2$ times the Composer coordinates of Φ and $\boldsymbol{\mathcal{A}}$. The cosmic mass writes the two as one.

> IN PLAIN WORDS
>
> There is one quantity, the action, from which the whole law follows. Ask which path makes it least sensitive to small changes, and the answer is the law of motion. Ask which fields do the same, and the answer is the field equations. Gravity and electromagnetism were each known to come from an action of their own; here the two actions are the two parts of one, written with one number for each body.

9.7 STEP 7: ONE PROGRAM FOR EVERY BODY

Since every body is one number and every field is one number at each point, the force on any body is one line,

$$\mathbf{F} = \sum_{\text{sources } s} \frac{G\,\Re(\mu\,\mu_s)}{r_s^2}\,\hat{\mathbf{u}}_s + \Re\big[\mu\,(\boldsymbol{\mathcal{E}} + \mathbf{v}\times\boldsymbol{\mathcal{B}})\big],$$

the pull of each source at rest, along the unit vector $\hat{\mathbf{u}}_s$ toward it, and the force of the applied fields. The program that accompanies this chapter is this line and a step in time. Only the two numbers m and q change from one body to the next. The program multiplies each pair of General Numbers as arrows on the page, at the angle $\bar{\phi} = 79.86°$, lengths times lengths and angles plus angles, and reads the real part back in the leaning frame, so the angle enters every product.

Table 9.2 collects the results. Each is a check on the reading rather than a discovery: where the law is Newton, Maxwell and Lorentz, it gives what they give. Two lines bear on the reading itself. Read by its shadow, the law of Chapter 6 kept a share $\cos\bar{\phi}$ of the Coupling term, which would give a neutron an apparent charge in an electric field (Section 6.5.3). Read by its real part, the Coupling term is no force, and a neutron goes straight between charged plates. In the same way, the spin of a neutral planet is a current of mass, not of charge: it makes only the field $\mathbf{B}_g$ of gravity, and no magnetic field. Mars and Venus, whose cores run no dynamo, have no global magnetic field, as the spacecraft that visited them found. Two deuterons that fuse push each other with the full strength of Coulomb's law, as nuclear physics measures it, and not with 93.8% of it.

> IN PLAIN WORDS
>
> One short program moves every body of this chapter: an electron in a tube, a proton in crossed fields, the Moon, the electron of hydrogen. The force on each is one line, the real part of its cosmic mass times the fields it feels, and only its mass and its charge change from one body to the next. The numbers come out as measured, because for slow bodies this law is the laws of Newton, Maxwell and Lorentz written as one.

CASE	QUANTITY	THIS LAW	STANDARD
Thomson's tube	spot of the electron	17.59 mm	17.59 mm
electron in 1 T	turns per second	2.7992×10^{10}	2.7992×10^{10}
proton in crossed fields	drift speed	1000.000 m/s	E/B
hydrogen, classical orbit	speed$/c$	0.0072974	$\alpha = 0.0072974$
hydrogen, at the radius a_0	energy to free the electron	13.598 eV	13.598 eV
the Moon, two bodies	period	27.28 days	27.32 days
neutron in an electric field	force	0	no charge seen
Mars and Venus	field from their spin	0	no global field
deuteron and deuteron	electric push	full	full
waves of either part	speed	c	c

Table 9.2: One law, many bodies. The values of this law come from the program of this step. The standard values are measured, or, for Thomson's tube, the electron in 1 T and the drift, given by the laws of Newton, Maxwell and Lorentz with measured constants. The hydrogen energy includes the motion of the proton and takes the radius a_0 from quantum theory; the Moon's period includes the motion of the Earth.

Part IV

OPEN QUESTIONS

10

WHAT IS THE COUPLING FORCE? AN OPEN DISCUSSION

Two bodies give one product. Its real part is the force that Newton and Coulomb measured. Its Composer coordinate holds the Coupling term, which no measurement has yet read. This chapter gathers readings of that term, the author's two and several more, and gives each a status: what holds as mathematics, what measurement already excludes, and what stays open.

ZÜRICH, SPRING 1918

On 1 March 1918 Hermann Weyl wrote to Albert Einstein that he had succeeded "to derive electricity and gravitation from a common source" (O'Raifeartaigh and Straumann, 2000). In his geometry a measuring rod carried from place to place could change its length, and the rule of the change behaved like the electromagnetic potential (Weyl, 1918). Einstein objected that an atom is a clock: atoms carried along different paths would then shine in different colours, yet the lines of a spectrum are sharp. The idea survived by changing its meaning. Fock and London found the same factor in the phase of the quantum wave of a charged particle (Fock, 1926; London, 1927), and in 1929 Weyl rewrote his theory with a change of phase instead of length (Weyl, 1929a,b; Jackson and Okun, 2001). This is gauge invariance, on which the theories of the electromagnetic, the weak and the strong forces are built today. Weyl's factor failed as one thing and succeeded as another.

The Coupling term of this book stands where Weyl's factor stood in 1918. Every measured force is read by the real part of a product (Definition 6.17). The other part, which holds the Coupling term, is left unread: read along the line between two bodies, any sizeable share of it would give the neutron a charge that it does not have (Section 6.5.3). This chapter discusses what that part could be. The author's two readings are stated as *Views*, a statement type of this chapter: readings put forward for discussion, neither proved nor yet tested. Each reading, with those the chapter adds, gets a status: what holds as mathematics, what measurement excludes, and what stays open.

TERM	TWO PROTONS	HYDROGEN	EARTH, ELECTRON
gravity	1.87×10^{-64}	3.63×10^{-47}	8.93×10^{-30}
two charges	2.31×10^{-28}	8.24×10^{-8}	0
Coupling term	4.15×10^{-46}	7.41×10^{-26}	1.82×10^{-8}
charge share	8.12×10^{-29}	2.90×10^{-8}	0

Table 10.1: The four terms of one product, read like forces: multiplied by G/d^2, in newtons, with the Composer coordinate measured by its length. They are gravity $G\,m_a m_b/d^2$, two charges $K\,q_a q_b/d^2$, the Coupling term $\sqrt{GK}\,(m_a q_b + q_a m_b)/d^2$ and the charge share $2\cos\bar{\phi}\,K\,q_a q_b/d^2$. The pairs are two protons 1 m apart, the proton and the electron of hydrogen at the radius a_0, and the Earth, taken as neutral, and an electron at the ground. Only the first two terms are forces (Definition 6.17).

10.1 ONE PRODUCT, TWO QUESTIONS

For two bodies with $\mu_a = m_a + q_a\mathcal{G}_{\bar{\phi}}$ and $\mu_b = m_b + q_b\mathcal{G}_{\bar{\phi}}$, Proposition 6.16 gives the product

$$\mu_a\mu_b = \underbrace{\left(m_a m_b - |\mathcal{G}_{\bar{\phi}}|^2\, q_a q_b\right)}_{\text{real part}} + \underbrace{\left(m_a q_b + q_a m_b + 2\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\; q_a q_b\right)}_{\text{Composer coordinate}}\,\mathcal{G}_{\bar{\phi}}.$$

Both parts are in square kilograms, the Composer coordinate once it is multiplied by $|\mathcal{G}_{\bar{\phi}}| = \sqrt{K/G}$.

- *The real part is the force.* Multiplied by G/d^2, it is Newton's gravity and Coulomb's law, each at full strength (Definition 6.17). Every measured force, energy and momentum of Chapters 6 to 9 is read here.
- *The Composer coordinate is not read as a force.* It holds the Coupling term $m_a q_b + q_a m_b$ of Remark 6.7, which pairs the mass of each body with the charge of the other, and $2\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\; q_a q_b$, a share of the term of two charges moved there by the Rule of the Square (Proposition 4.22), which this chapter calls the *charge share* (Section 6.5.2).

Table 10.1 gives the four terms read like forces. The Coupling term vanishes between two neutral bodies and between a particle and its antiparticle (Section 2.4), and it is not a force along the line: read that way, even its share $\cos\bar{\phi}$ would give the neutron a charge some two hundred standard errors beyond what is measured (Section 6.5.3). For charged particles the charge share is much larger: 2×10^{17} times the Coupling term for two protons, and more than a third of Coulomb's pull in hydrogen. The author's two views ask two questions of this product: what its Composer coordinate means, and whether a product of bodies can itself be a body.

10.2 THE FIRST VIEW: THE COUPLING TERM AS TWO MASSES

View 10.1 (The Coupling Term as Two Masses)**.** The Coupling term $(m_a q_b + m_b q_a)\,\mathcal{G}_{\bar{\phi}}$ has the unit kg^2, and it acts like the product of two masses in Newton's law: either of two masses $m_x m_y$, where x and y could be new particles, or of the total mass of the two bodies and one more mass, $(m_a + m_b)\,m_z$, where z could be a new particle.

The charge part $q\mathcal{G}_{\bar{\phi}}$ of a body is measured in kilograms, and for one elementary charge it is the Stoney mass $m_S \approx 1.85921 \times 10^{-9}$ kg (Definition 7.2). The second mass of the view gets a name.

Definition 10.2 (The Pair Mass)**.** For two bodies with $m_a + m_b \neq 0$, the **pair mass** is

$$m_z = |\mathcal{G}_{\bar{\phi}}|\,\frac{m_a q_b + m_b q_a}{m_a + m_b} = \frac{m_a\,(q_b \mathcal{G}_{\bar{\phi}}) + m_b\,(q_a \mathcal{G}_{\bar{\phi}})}{m_a + m_b},$$

the mean of the two charge parts, each weighted by the mass of the other body. Here, as on the identity card, $q\mathcal{G}_{\bar{\phi}}$ stands for the signed length $q\,|\mathcal{G}_{\bar{\phi}}|$ of the charge part.

Proposition 10.3 (The Coupling Term as Two Newton Pairs)**.** *Measured by its signed length, the Coupling term of two bodies is*

$$(m_a q_b + m_b q_a)\,|\mathcal{G}_{\bar{\phi}}| = m_a\,(q_b \mathcal{G}_{\bar{\phi}}) + m_b\,(q_a \mathcal{G}_{\bar{\phi}}) = (m_a + m_b)\,m_z,$$

a sum of two products of a mass and a charge part, or one product of the total mass and the pair mass. Read like a force, multiplied by G/d^2, it is

$$\frac{G\,m_a\,(q_b \mathcal{G}_{\bar{\phi}})}{d^2} + \frac{G\,m_b\,(q_a \mathcal{G}_{\bar{\phi}})}{d^2} = \frac{G\,(m_a + m_b)\,m_z}{d^2},$$

two terms of the form of Newton's law, each between the mass of one body and the charge part of the other, or one between the total mass and the pair mass. Its ratio to the gravity term is $\eta_a + \eta_b$, the sum of the two charge-to-mass ratios $\eta = q\mathcal{G}_{\bar{\phi}}/m$ of Remark 6.15.

Proof. The first equality multiplies out the bracket, and the second is Definition 10.2 multiplied by $m_a + m_b$. The force is the same line multiplied by G/d^2. Dividing the Coupling term by $m_a m_b$ gives $q_b \mathcal{G}_{\bar{\phi}}/m_b + q_a \mathcal{G}_{\bar{\phi}}/m_a = \eta_b + \eta_a$. □

For two protons $m_z = m_S$, and for two electrons $m_z = -m_S$; for a proton and an electron $m_z \approx -0.99891\,m_S$, and for a neutron and a proton $m_z \approx 0.50034\,m_S$.

THE STATUS OF THE FIRST VIEW.

HOLDS AS MATHEMATICS. The unit is right, both ways of writing the term are exact, and the formula names the two masses: the mass of one body and the charge part of the other (Proposition 10.3).

NOT UNIQUE. Any other split into two masses would need a reason from outside the formula.

A SIGN. The pair mass has the sign of the charges, so it behaves more like a charge written in kilograms than like the mass of a new particle.

EXCLUDED AS A FORCE ALONG THE LINE. A neutral body of mass m in an electric field E would feel the force $m\,E\,\sqrt{G/K}$, as if it carried the charge $\sqrt{G/K}\,m$: $9.0 \times 10^{-19}\,e$ for a neutron, whose charge is measured as $(-0.2 \pm 0.8) \times 10^{-21}\,e$ (Baumann et al., 1988; Bressi et al., 2011; Particle Data Group et al., 2024). The whole term misses by about eleven hundred standard errors, and any share of it acting along the line must be below about two parts in a thousand (Section 6.5.3).

OPEN. Read as a mass, the pair mass points to a definite place in the frame (Section 10.3).

10.3 THE DIAGONAL OF THE FRAME

For two bodies with the same charge, the pair mass is their charge part. A body with $m = |q\mathcal{G}_{\bar\phi}|$ lies on a diagonal of the Cosmic Mass Frame: the ray $\alpha = \bar\phi/2 \approx 39.93°$ for a positive charge, and $\alpha \approx 309.93°$ for a negative one (Section 7.3, reading 3).

Proposition 10.4 (Three Zones for Two Equal Bodies)**.** *Two equal bodies of positive mass m and charge q, the distance d apart, pull each other when $|q\mathcal{G}_{\bar\phi}| < m$, push each other when $|q\mathcal{G}_{\bar\phi}| > m$, and feel no force when $|q\mathcal{G}_{\bar\phi}| = m$. On the cards, where bodies of positive mass have charge phase angles between $-100.14°$ and $79.86°$: they pull when α lies between $-50.07°$ and $39.93°$, push when it lies between $39.93°$ and $79.86°$ or between $-100.14°$ and $-50.07°$, and balance on the two diagonals.*

Proof. By Definition 6.17 the force is $(Gm^2 - Kq^2)/d^2 = G\,(m^2 - (q\mathcal{G}_{\bar\phi})^2)/d^2$, since $K = G\,|\mathcal{G}_{\bar\phi}|^2$. On the cards, a body of positive mass has $\sin(\bar\phi - \alpha) > 0$ (Remark 6.11), so its angle lies between $\bar\phi - 180° \approx -100.14°$ and $\bar\phi$, and twice its angle between $2\bar\phi - 360°$ and $2\bar\phi$. In that range, Proposition 6.18 with $\alpha_a = \alpha_b = \alpha$ gives a pull exactly when 2α lies between $\bar\phi - 180°$ and $\bar\phi$, that is, when α lies between $\bar\phi/2 - 90° \approx -50.07°$ and $\bar\phi/2 \approx 39.93°$, a push in the rest of the range, and no force on the two diagonals. □

The diagonals cut the frame into zones (Figure 10.1): the *gravity side* near the mass axis, with the atoms, the neutron, planets and stars, and the *charge side* near the charge axes, with every charged particle, ion and nucleus, the electron 2.0×10^{21} times beyond the diagonal. On a diagonal two equal bodies can rest at any distance from each other (Remark 7.3).

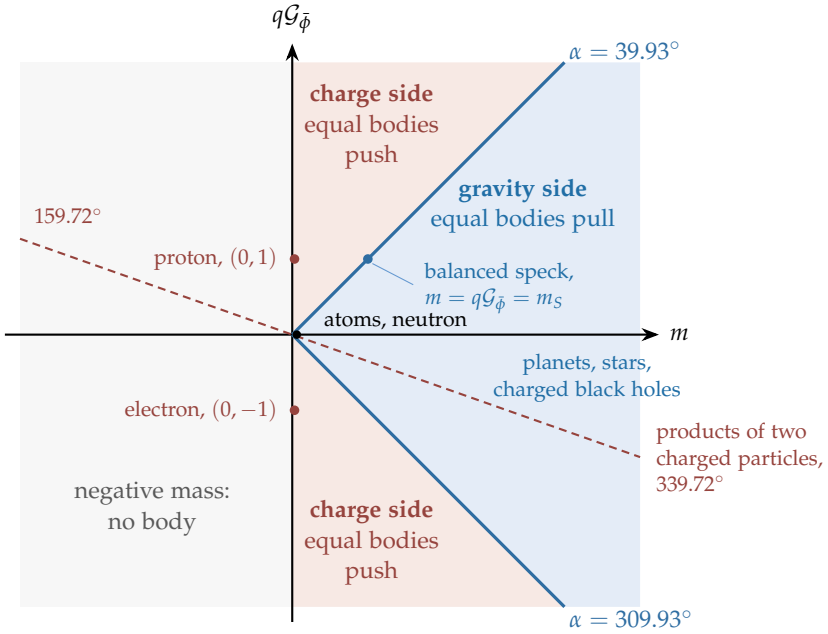


Figure 10.1: The Cosmic Mass Frame in three zones, with the mass m across and the charge part $q\mathcal{G}_{\bar{\phi}}$ up, in Stoney masses and not to scale beyond the marked points (Proposition 10.4). The two diagonals $|q\mathcal{G}_{\bar{\phi}}| = m$ (solid) are the lines of balance for two equal bodies, and the lines of extremal black holes in general relativity (Remark 10.5). The dashed line holds the products of two charged particles (Proposition 10.8).

The same line has a place in general relativity.

Remark 10.5 (The Diagonal in Einstein's Theory and Beyond).

1. *The edge of the black holes.* In 1916 Hans Reissner, and in 1918 Gunnar Nordström, found the field of a charged body in Einstein's theory (Reissner, 1916; Nordström, 1918). A body of mass m and charge q can be a black hole only if $Gm^2 \geq Kq^2$, that is, if $|q\mathcal{G}_{\bar{\phi}}| \leq m$: on the gravity side of the frame or on one of its diagonals. On a diagonal the black hole is called *extremal*, and for one elementary charge its mass is exactly the Stoney mass m_S.

2. *Balance at any distance.* In 1947 Sudhansu Datta Majumdar and Achilles Papapetrou found that bodies with $|q\mathcal{G}_{\bar{\phi}}| = m$, all charged with the same sign, can rest in Einstein's theory at any distances from one another, any number of them (Majumdar, 1947; Papapetrou, 1947). In 1972 James Hartle and Stephen Hawking showed that these solutions describe several extremal black holes held in balance (Hartle and Hawking, 1972). Remark 7.3 finds the same balance in the laws of Newton and Coulomb.

3. *Gravity as the weakest force.* In a paper of 2006, published in 2007, Nima Arkani-Hamed, Luboš Motl, Alberto Nicolis and Cumrun Vafa conjectured that any consistent quantum theory of gravity together with a force of charges must contain a particle whose charge part is at least as large as its mass, $|q\mathcal{G}_{\bar\phi}| \geq m$: a particle on the charge side of the diagonal, or on it (Arkani-Hamed et al., 2007). The electron lies there, by a factor of 2.0×10^{21}.

These results belong to general relativity and to the search for a quantum theory of gravity, not to this book. They fall on the diagonal because all of them compare Gm^2 with Kq^2. On the upper diagonal one elementary charge goes with 1.86 micrograms, about the mass of a grain of fine sand. Squeezed into a black hole, such a speck would be an extremal one, with a horizon of radius $Gm_S/c^2 \approx 1.38 \times 10^{-36}$ m, 0.0854 of the Planck length, where Einstein's theory alone cannot be trusted. The first view thus lands on a definite line, though by construction; whether the line means more than the condition $Gm^2 = Kq^2$, which the laws of Newton and Coulomb share with Einstein's, is open.

10.4 THE SECOND VIEW: A PARTICLE AS A PRODUCT

The identity card gives $m = r\cos(\alpha,\phi)$ and $q\mathcal{G}_{\bar\phi} = r\sin(\alpha,\phi)$, with $\alpha+\phi = \bar\phi$ (Remarks 6.11 and 6.13). Equations (A.64) and (A.65) of Appendix A.9 write each general function as a product of scaled copies of itself; for the card they read

$$m = r\prod_{k=0}^{2^n-1}\cos\Big(\frac{\alpha}{2^n},\frac{\phi}{2^n}+\pi\frac{k}{2^n}\Big), \qquad q\mathcal{G}_{\bar\phi} = r\prod_{k=0}^{2^n-1}\sin\Big(\frac{\alpha}{2^n}+\pi\frac{k}{2^n},\frac{\phi}{2^n}\Big).$$

View 10.6 (A Particle as a Product of Particles)**.** Since the mass of a body is a product of many general cosines, and its charge part a product of many general sines, at other angles, a particle could be the result of many other particles.

The products hold for any number of factors, not only for powers of two.

Proposition 10.7 (The Products for Every Number of Factors)**.** *For every integer $N \geq 1$, and for all real x and y with $x+y \notin \pi\mathbb{Z}$ (the domains of Appendix A.2),*

$$\sin(x,y) = \prod_{k=0}^{N-1}\sin\Big(\frac{x+\pi k}{N},\frac{y}{N}\Big),$$
$$\cos(x,y) = \prod_{k=0}^{N-1}\cos\Big(\frac{x}{N},\frac{y+\pi k}{N}\Big),$$

and every factor lies in the domain of its function. For $N = 2^n$ these are equations (A.64) and (A.65).

Proof. The classical product of sines,

$$\prod_{k=0}^{N-1} \sin\Big(\theta + \frac{\pi k}{N}\Big) = \frac{\sin(N\theta)}{2^{N-1}},$$

follows from $\prod_{k=0}^{N-1}(1 - t\,\omega^k) = 1 - t^N$, with $\omega = e^{2\pi i/N}$ and $t = e^{2i\theta}$. Apply it with $N\theta = x$, with $N\theta = y$ and with $N\theta = x + y$, and divide:

$$\frac{\sin x}{\sin(x+y)} = \prod_{k=0}^{N-1} \frac{\sin\big((x+\pi k)/N\big)}{\sin\big((x+y+\pi k)/N\big)},$$
$$\frac{\sin y}{\sin(x+y)} = \prod_{k=0}^{N-1} \frac{\sin\big((y+\pi k)/N\big)}{\sin\big((x+y+\pi k)/N\big)}.$$

By the definitions of Appendix A.1, the left sides are $\sin(x, y)$ and $\cos(x, y)$, and the factors are the general sine at the angles $\big((x + \pi k)/N, y/N\big)$ and the general cosine at $\big(x/N, (y + \pi k)/N\big)$, whose two angles add up to $(x + y + \pi k)/N$ in both. No denominator vanishes, because their product is $\sin(x + y)/2^{N-1} \neq 0$, so every factor lies in the domain of Appendix A.2. When $0 < y < \pi$, the second angles y/N and $(y + \pi k)/N$, with $k \leq N - 1$, also lie between 0 and π, as in Appendix A.1. □

WHAT THE FACTORS ARE. In the products for the card, the two angles of the factor numbered k add up to $(\alpha + \phi + \pi k)/N = (79.86^\circ + 180^\circ \cdot k)/N$, not to 79.86°. A factor is a pure number from a triangle of another shape, not the card of a body, and all the kilograms stay in r. For the proton one factor carries the whole smallness of the mass: two factors give its mass part as $6.90 \times 10^{-19} \times 1.304$, and three as $6.59 \times 10^{-19} \times 0.868 \times 1.574$. The proton does have constituents, three quarks, but they combine by addition: its charge is the sum of theirs, and most of its mass is the energy that binds them (Particle Data Group et al., 2024).

PRODUCTS INSIDE THE FRAME. With whole cards, the product multiplies the magnitudes and adds the charge phase angles, $\mu_a\mu_b = r_a r_b\, e^{(\alpha_a+\alpha_b)\mathcal{J}_{\bar\phi}}$ (Proposition 6.18 and Theorem 4.46). Divided by a positive reference mass, which the view would have to name, it lands on the ray of the sum of the two angles.

Proposition 10.8 (Where Two Charged Particles Land). *Let two bodies have charge parts much larger than their masses, $|q\mathcal{G}_{\bar\phi}| \gg m$, as every charged particle, ion and nucleus has. Then, up to terms of the order of $m/|q\mathcal{G}_{\bar\phi}|$, their product lies on one line through the origin, on which the charge part is $-2\cos\bar\phi \approx -0.35210$ times the mass part: on the ray $2\bar\phi \approx 159.72^\circ$, with a negative mass part, when the two charges have the same sign, and on the ray $2\bar\phi + 180^\circ \approx 339.72^\circ$, with a positive mass part and a negative charge part, when they have opposite signs. If instead one body is neutral, the product is a positive multiple of the other and keeps its ray.*

Proof. In $\mu_a\mu_b = q_aq_b\,\mathcal{G}_{\bar\phi}^2 + (m_aq_b + q_am_b)\,\mathcal{G}_{\bar\phi} + m_am_b$ the first term is larger than the others by the factor $|q\mathcal{G}_{\bar\phi}|/m$. By the Rule of the Square, $q_aq_b\,\mathcal{G}_{\bar\phi}^2$ has the real part $-|\mathcal{G}_{\bar\phi}|^2q_aq_b$ and the Composer coordinate $2\,|\mathcal{G}_{\bar\phi}|\cos\bar\phi\;q_aq_b$, so its charge part, the coordinate times $|\mathcal{G}_{\bar\phi}|$, is $-2\cos\bar\phi$ times its real part. The real part is negative when $q_aq_b > 0$ and positive when $q_aq_b < 0$. The angles follow from Proposition 6.18: the sum of two angles near $\bar\phi$, or near $\bar\phi + 180°$, is near $2\bar\phi$, and the sum of one of each is near $2\bar\phi + 180°$. If $q_a = 0$, then $\mu_a\mu_b = m_a\,\mu_b$, a positive multiple of μ_b. □

Of this line, dashed in Figure 10.1, the ray 159.72° lies in the half of negative mass and the ray 339.72° in the empty middle of the frame (Section 7.6, reading 7): no atom, ion or nucleus lies on either.

Remark 10.9 (Products Do Not Keep Charge). Read as bodies, products do not keep charge. A proton and an electron have together no charge, but their product has a charge part -0.35210 times its mass part (Proposition 10.8). A neutron and a proton have together the charge e, but their product, divided by a reference mass m_0, is $(m_n/m_0)\,\mu_p$, with the charge $e\,m_n/m_0$, equal to e only if $m_0 = m_n$. Every composition in this book keeps charge exactly: in the fusions of Chapter 8 the charge parts add (Proposition 8.2), and in Chapter 9 the Composer coordinate of the conserved source, the charge, is kept exactly (Theorem 9.8). The composition of bodies adds their cosmic masses, up to a real number, the mass of the energy that leaves or binds them (Proposition 8.2). The product has another job: its real part is the force between them.

THE STATUS OF THE SECOND VIEW.

HOLDS AS MATHEMATICS. The products are exact for every number of factors (Proposition 10.7).

THE FACTORS ARE NUMBERS. They are not the cards of bodies: their two angles do not add up to 79.86°, and they carry no kilograms.

EXCLUDED AS A LAW OF COMPOSITION. Read as bodies, products do not keep charge (Remark 10.9), and products of charged particles land where no particle lives (Proposition 10.8).

OPEN. Whether a measured quantity, such as a relation among the masses of particles that their sum does not explain, follows the products. None is known.

10.5 THREE MORE READINGS

10.5.1 *The Charge Share*

For charged particles the Composer coordinate is mostly the charge share (Table 10.1). Read along the line, the share is excluded by hy-

drogen (Section 6.5.3). There the charges are opposite, so it would push back with 0.352 of Coulomb's pull, and since the energy that holds the electron grows with the square of the pull (Bohr, 1913), it would be $0.648^2 \times 13.598 \approx 5.71$ eV instead of the measured 13.598 eV (Sansonetti and Martin, 2005).

10.5.2 *Flipping Every Charge*

Proposition 10.10 (Flipping Every Charge)**.** *If the charge of every body changes sign, $q \to -q$, and every mass is kept, then the real part of every product and its charge share are unchanged, and its Coupling term changes sign. In particular, the Coupling term of a particle and its antiparticle, $m\,(-q) + q\,m$, is zero.*

Proof. The real part $m_a m_b - |\mathcal{G}_{\bar\phi}|^2 q_a q_b$ and the charge share $2\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi \;\; q_a q_b$ hold the charges in pairs, and a pair of signs that both flip is kept. The Coupling term $m_a q_b + q_a m_b$ holds each charge once. □

So, set beside the same neutral mass, a particle and its antiparticle would differ only in the Coupling term: if it were a force, an electron and a positron near the Earth would be pushed in opposite directions. Antihydrogen falls as matter does, $(0.75 \pm 0.13 \pm 0.16)\,g$ (Anderson et al., 2023), but it is neutral and tests nothing here. The fall of a charged antiparticle has not been measured.

10.5.3 *A Body in a Field*

For a body in a cosmic field $\boldsymbol{\mathcal{E}} = \mathbf{g} + \mathbf{e}\,\mathcal{G}_{\bar\phi}$, Chapter 9 reads the force as the real part of $\mu\,\boldsymbol{\mathcal{E}}$ (Law 9.3), with the electric field $\mathbf{E} = -|\mathcal{G}_{\bar\phi}|^2\,\mathbf{e}$. The Composer coordinate of the same product is, by Proposition 4.18 and the Rule of the Square,

$$m\,\mathbf{e} + q\,\mathbf{g} + 2\,|\mathcal{G}_{\bar\phi}| \cos\bar\phi\; q\,\mathbf{e}\;:$$

mass with the charge part of the field, charge with gravity, and the charge share: Faraday's question again (Figure 6.1). Read as forces,

- a charge q in the gravity g would feel the force $|q\mathcal{G}_{\bar\phi}|\,g$, as if it sat in an electric field of $|\mathcal{G}_{\bar\phi}|\,g \approx 1.14 \times 10^{11}$ V/m: 1 nC would feel 114 N;
- a neutral mass m in an electric field E would feel the force $m\,E\sqrt{G/K}$, as if it carried the charge $\sqrt{G/K}\,m$ (Section 10.2).

Remark 10.11 (Hidden Inside Metal). A force that acts on each charge in proportion to its charge, whatever its mass, acts on charges exactly as an electric field does, and conductors and plasmas cancel such a field inside them, as they cancel any electric field: their free electrons

move until the force on them vanishes. Schiff and Barnhill argued in 1966 that, for the weight of the electrons themselves, a field of $m_e g/e \approx 5.6 \times 10^{-11}$ V/m forms in and near a metal and holds them up (Schiff and Barnhill, 1966); the size of that field in a real metal was soon disputed (Dessler et al., 1968). In 1967 Fred Witteborn and William Fairbank found that free electrons falling inside a copper tube feel almost no net force (Witteborn and Fairbank, 1967). The half of the Coupling term that acts on a charge in gravity would be hidden in the same way, inside any closed box of metal. The half that acts on a mass in an electric field cannot be hidden: a neutral body has no free charges to move. And the two halves are one term seen from its two bodies: the piece $m_a q_b$ acts on body a as a mass in the field of b, and on body b, equally and oppositely, as a charge in the gravity of a, with one strength. The limit that the neutron sets therefore holds for both halves.

The decisive test is therefore the neutron in an electric field (Section 6.5.3). It limits both halves, including the half that an electron beam would feel, so the conclusion of Section 2.4 stands.

10.6 WHAT MEASUREMENT ALREADY SAYS

Table 10.2 sets each reading beside the measurements that already limit it.

Measurement has excluded every reading of the Composer coordinate as a force along the line, at the size that the product gives it. It has said nothing about the coordinate as anything else, a mass, a property of a pair, a line of the frame, or a quantity that is not a force at all, as Weyl's phase was not a length. There the readings stay open.

10.7 QUESTIONS FOR VOLUME 2

Volume 2 can try to answer each question below in the form that Chapter 2 asks of a new prediction (Section 2.4): a number with its error, an experiment able to see it, agreement with every measurement that already limits it, and a result that would refute it, stated in advance.

1. *What reads the Composer coordinate?* A reading other than a force along the line, such as an energy, a phase or a property of pairs, must give a number that the measurements of Table 10.2 allow.

2. *Is there one pair mass?* Does the law itself select one way of writing the Coupling term as two masses, and does any body carry the pair mass? A new particle would need a mass, a charge, and a way to make it or see it.

3. *What is the diagonal?* Does the cosmic mass describe the extremal black hole, or the many bodies of Majumdar and Papapetrou, beyond the laws of Newton and Coulomb? The question needs the

READING	WHAT IT WOULD DO	MEASUREMENT	STATUS
Coupling term along the line (Section 10.2)	a neutron acts as if charged, $9.0 \times 10^{-19}\,e$	neutron charge $(-0.2 \pm 0.8) \times 10^{-21}\,e$	excluded; any share below about 0.002
charge share along the line (Section 10.5.1)	hydrogen holds its electron with 5.71 eV	13.598 eV	excluded
a charge in gravity (Section 10.5.3)	every charge as if in 1.14×10^{11} V/m	electrons in a copper tube	hidden by metal; its strength is the one the neutron limits
Coupling term inside the atom (Table 10.1)	9.0×10^{-19} of Coulomb's pull in hydrogen	atomic spectra	too small to see; no test
particle against antiparticle (Section 10.5.2)	opposite pushes near a neutral mass	antihydrogen falls, $(0.75 \pm 0.13 \pm 0.16)\,g$	neutral, so no test; charged antiparticles not measured
products as bodies (Section 10.4)	charge not kept	every fusion of Chapter 8	excluded as composition
the diagonal (Section 10.3)	the line of balance and of extremal black holes	none needed: the condition $Gm^2 = Kq^2$ of known laws	holds as mathematics; no new prediction

Table 10.2: What measurement already says about the readings of this chapter. The neutron charge is from Baumann et al. (1988), Bressi et al. (2011) and Particle Data Group et al. (2024), the energy of hydrogen from Sansonetti and Martin (2005), the electrons in copper from Witteborn and Fairbank (1967), and the fall of antihydrogen from Anderson et al. (2023); the diagonal is Proposition 10.4 and Remark 10.5.

strong fields where this volume gives way to general relativity (Chapter 9).

4. *Does anything follow the product?* Is there a measured relation among the masses or charges of particles that their sum does not explain? Since every composition keeps charge, it could hold only among quantities that compositions do not add.

5. *Does a charged antiparticle fall as its particle does?* Only the Coupling term would tell them apart near a neutral mass, within the limit that the neutron sets. The experiment is hard (Remark 10.11), and it has not been done.

6. *Does the angle belong to nature or to the units?* The angle of the Composer axis comes from one number, 1.16043, the size of the Composer Unit once the factor 10^{10} is given to its direction (Sec-

tion 6.3.2); in other units, $\sqrt{K/G}$ is written with another number. Is 79.86° a property of nature, or of the units in which K and G are written? No force in this volume can say, because the real part of a product does not contain the angle (Section 6.6.3); a measurement that reads the Composer coordinate, if one exists, would be the first that could.

Chapter 11 draws the frame in full, and the Coda looks back over the whole road.

What we have so far

- The product of two cosmic masses has two parts: the force, which is measured, and the Composer coordinate, which holds the Coupling term and the charge share and is not read.
- The first view reads the Coupling term as two masses. That is exact, and for two equal bodies the second mass lies on the diagonal of the frame, the line of extremal black holes in general relativity. As a force along the line, it is excluded by the neutron.
- The second view reads a particle as a product of particles. The products are exact for any number of factors, but the factors are numbers, not bodies, and products do not keep charge.
- Every reading of the Composer coordinate as a force along the line is excluded; every reading of it as something else is open, and those are the questions of Volume 2.

11

WHAT THE FRAME ALLOWS: ANTIMATTER, NEGATIVE MASS AND BLACK HOLES

Ordinary matter uses only three directions of the Cosmic Mass Frame. This chapter reads the whole of it, one step at a time: the mirror images of the particles, which are antimatter and lie on the same three directions; the half of negative mass, which nature has left empty so far; and the side where gravity wins over charge, where the black holes are. Each answer says what has been measured and what is only allowed.

PASADENA, AUGUST 1932

On 2 August 1932, at the California Institute of Technology, Carl Anderson photographed in a cloud chamber the track of a particle from the cosmic rays. A magnetic field bent the track, and a plate of lead across the chamber showed which way the particle had come: its charge was positive, yet its track ran far longer than a proton's could, so its mass was far smaller. Anderson called it the positive electron, and in the same paper named it the positron (Anderson, 1933). Paul Dirac had described such a particle the year before. His equation of 1928 for the electron has solutions of negative energy (Dirac, 1928), and in 1931 he wrote that a missing one would be "a new kind of particle, unknown to experimental physics, having the same mass and opposite charge to an electron" (Dirac, 1931). In the language of this book, the positron is the electron with its charge part turned around.

Chapter 7 found that ordinary matter uses only three directions of the frame: the mass axis and the two ends of the charge axis (Section 7.6). This chapter reads the whole frame in seven steps, with the tools of the earlier chapters. For black holes it quotes, and does not derive, Einstein's theory: the horizons of a charged body, and the rule that a settled black hole is described by its mass, charge and spin.

11.1 STEP 1: THE FRAME, ONCE MORE

A body of mass m and charge q has the cosmic mass $\mu = m + q\mathcal{G}_{\bar{\phi}}$ (Definition 6.5) and the identity card (r, α, ϕ), with $\alpha + \phi = 79.86^\circ$, $m = r\cos(\alpha, \phi)$ and $q\mathcal{G}_{\bar{\phi}} = r\sin(\alpha, \phi)$ (Remarks 6.13 and 6.11). As on the card, $q\mathcal{G}_{\bar{\phi}}$ is the signed length of the charge part in kilograms: one Stoney mass $m_S \approx 1.85921 \times 10^{-9}$ kg for each elementary charge (Definition 7.2). The Cosmic Mass Frame draws m across and $q\mathcal{G}_{\bar{\phi}}$ up (Definition 7.5). Four facts about it are all that this chapter needs.

1. *The rings lean.* Every ring is an ellipse whose short axis lies on the diagonal $q\mathcal{G}_{\bar{\phi}} = m$ and whose long axis lies on $q\mathcal{G}_{\bar{\phi}} = -m$ (Corollary 7.4). A point and its mirror image in the mass axis therefore lie, in general, on two different rings.
2. *Ordinary matter uses three directions.* Neutral bodies lie on the mass axis, positive charges near the ray 79.86° and negative charges near 259.86°, each leaning towards positive mass by the lean δ of Remark 7.6 (Section 7.6).
3. *The diagonals balance.* On the diagonals $|q\mathcal{G}_{\bar{\phi}}| = m$, two equal bodies neither pull nor push; nearer the mass axis they pull, and nearer the charge axes they push (Proposition 10.4).
4. *One half is empty.* No known body has $m < 0$ (Figure 7.7).

11.2 STEP 2: THE MIRROR OF A PARTICLE

Every particle has an antiparticle, with the same mass and the opposite charge: the positron for the electron, the antiproton for the proton. The masses of the antiproton and the proton agree to better than one part in 10^9, and their charge-to-mass ratios to 16 parts in 10^{12} (Particle Data Group et al., 2024; Borchert et al., 2022); antihydrogen, the atom of an antiproton and a positron, falls as ordinary matter does, within the errors of the experiment (Anderson et al., 2023).

Definition 11.1 (The Antiparticle on the Frame)**.** The antiparticle of a body with the cosmic mass $\mu = m + q\mathcal{G}_{\bar{\phi}}$ has the cosmic mass

$$\tilde{\mu} = m - q\mathcal{G}_{\bar{\phi}},$$

with the same mass and the opposite charge part. It is the skew flip of Remark 4.39 applied to μ, not its conjugate (Definition 4.37).

The conjugate would not do: it keeps the length of μ but changes its real part to $m + 2q|\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}$ (Definition 4.37 and Theorem 4.38), for the electron a negative mass about 7×10^{20} times larger than its own. The skew flip keeps the mass, as measurement asks, and pays for it with a change of length (Remark 4.39).

Proposition 11.2 (The Antiparticle on the Frame)**.** *Let a body have the mass $m > 0$ and the charge part $q\mathcal{G}_{\bar{\phi}}$, and let $\tilde{\mu}$ be the cosmic mass of its antiparticle.*

1. *In the frame, $\tilde{\mu}$ is the mirror image of μ in the mass axis.*
2. *Its cosmic magnitude $\tilde{r}$ obeys $\tilde{r}^2 = r^2 - 4m\,(q\mathcal{G}_{\bar{\phi}})\cos\bar{\phi}$. The antiparticle of a positive charge lies on a smaller ring than its particle, and that of a negative charge on a larger one; when the charge part is much larger than the mass, the two rings differ by about $2\cos\bar{\phi}\; m \approx 0.35210\, m$.*

3. *Its ray leans from the other end of the charge axis, also towards positive mass, by the lean of Remark 7.6 written for the opposite sign of the charge. For every particle, ion and nucleus the two leans agree to many digits.*

4. *A neutral body is its own mirror: if $q = 0$, then $\tilde{\mu} = \mu$.*

Proof. (1) The frame draws m across and $q\mathcal{G}_{\bar{\phi}}$ up (Definition 7.5); $\tilde{\mu}$ keeps the first and reverses the second. (2) Remark 4.39 with $a = m$ and $b = q$ gives $|m - q\mathcal{G}_{\bar{\phi}}|^2 = |\mu|^2 - 4mq\,|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}$. When $|q\mathcal{G}_{\bar{\phi}}|$ is much larger than m, $r \approx |q\mathcal{G}_{\bar{\phi}}| + \cos\bar{\phi}\ m$ for a positive charge and $r \approx |q\mathcal{G}_{\bar{\phi}}| - \cos\bar{\phi}\ m$ for a negative one, and the flip exchanges the two. (3) Remark 7.6 gives the lean of a body with charge number z from the size $|\eta| = |z|\,m_S/m$ of its charge-to-mass ratio, which the flip keeps: $\arctan\big(\sin\bar{\phi}/(|\eta| + \cos\bar{\phi})\big)$ for $z > 0$ and $\arctan\big(\sin\bar{\phi}/(|\eta| - \cos\bar{\phi})\big)$ for $z < 0$, both towards positive mass. Their ratio differs from 1 by about $2\cos\bar{\phi}/|\eta|$, less than 10^{-15} for every body of ordinary matter. (4) If $q = 0$, then $m - q\mathcal{G}_{\bar{\phi}} = m + q\mathcal{G}_{\bar{\phi}}$. □

So the positron sits beside the proton at the upper end of the charge axis, and the antiproton beside the electron at the lower end (Figure 11.2). The proton's ring is larger than the antiproton's by $0.35210\,m_p$, about three parts in 10^{19} of the ring. Every measured quantity of a particle and its antiparticle follows from their masses and charges, so the difference of rings is, so far, a feature of the leaning frame, not of anything measured.

11.3 STEP 3: A PARTICLE AND ITS MIRROR

The product of a particle and its antiparticle gives the force between them, and their sum gives the body they make together.

Proposition 11.3 (A Particle and Its Antiparticle)**.** *Let $\tilde{\mu} = m - q\mathcal{G}_{\bar{\phi}}$ be the antiparticle of $\mu = m + q\mathcal{G}_{\bar{\phi}}$.*

1. *Their product has the real part $m^2 + (q\mathcal{G}_{\bar{\phi}})^2$ and the Composer coordinate $-2|\mathcal{G}_{\bar{\phi}}|\cos\bar{\phi}\ q^2$: the Coupling term is zero, and only the charge share of Section 10.1 is left.*

2. *Two of them, the distance d apart, pull each other with the force*

$$F = \frac{Gm^2 + Kq^2}{d^2} :$$

gravity and the electric force pull together, each at full strength.

3. *Their sum is the real number $\mu + \tilde{\mu} = 2m$: the charge parts cancel, and the pair, taken as one body, lies on the mass axis, on the ring $2m$.*

Proof. (1) Proposition 6.16 with $m_a = m_b = m$, $q_a = q$ and $q_b = -q$ gives the real part $m^2 + |\mathcal{G}_{\bar{\phi}}|^2 q^2$ and the Composer coordinate

$-mq + qm - 2|\mathcal{G}_{\bar{\phi}}| \cos\bar{\phi}\ q^2$. (2) Definition 6.17, with $G\,|\mathcal{G}_{\bar{\phi}}|^2 = K$. (3) General Numbers add part by part (Definition 4.17). □

Annihilation. A particle and its antiparticle can disappear together into light: an electron and a positron at rest usually become two gamma rays of 511 keV each, the light that the scanners of positron emission tomography detect. In the language of Proposition 8.2, the loss of cosmic mass is the whole sum, $2m$, and it is real, so light, which carries no charge, can take all of it. A real sum is needed but is not enough: a single neutron, or an atom of hydrogen, also has a real cosmic mass, and neither can vanish into light. A particle and its antiparticle can, because every other number that nature keeps, such as the baryon number, cancels between them as well (Section 11.4). All of the pair can therefore leave as energy,

$$E = 2mc^2.$$

For an electron and a positron that is 1.022 MeV, while a fusion releases only a small real part of the sum (Proposition 8.2). The reverse also happens: a gamma ray of at least 1.022 MeV passing near a nucleus can turn into an electron and a positron, a real number splitting into two equal and opposite charge parts, so that the charge is kept.

11.4 STEP 4: WHAT TWO NUMBERS CANNOT TELL

Statement 4 of Proposition 11.2 has a consequence that the frame cannot escape. The antineutron has the mass of the neutron and no charge, so it lies at the same point of the frame; so do an atom of hydrogen and an atom of antihydrogen. Yet a neutron and an antineutron that meet annihilate. They differ in numbers that the cosmic mass does not hold, such as the baryon number, $+1$ for the neutron and -1 for the antineutron (Particle Data Group et al., 2024), and, by the symmetry that turns a particle into its antiparticle, the sign of the magnetic moment.

Remark 11.4 (The Limit of Two Numbers). The cosmic mass holds two numbers. For a charged body they tell a particle from its own antiparticle, through the sign of the charge part, though not, on their own, whether a body is matter or antimatter: the positron sits with the protons. For a neutral body they do not tell even a particle from its antiparticle. A cosmic mass that told every particle from its antiparticle would need at least one more number, such as the baryon number, and, in the language of this book, one more axis. Whether such an axis belongs in the General Numbers is a question for Volume 2.

11.5 STEP 5: THE HALF OF NEGATIVE MASS

The half $m < 0$ is open in the mathematics, and antimatter, whose mass is positive, does not live there. The real reading cannot say whether a body of negative mass exists, but it can say what one would do: its inertia, the real part of its cosmic mass (Law 9.3), would be negative, so it would move against the force that acts on it.

Proposition 11.5 (Negative Mass in the Real Reading)**.** *Let two neutral bodies with the masses m_a and m_b start at rest, the distance d apart, and move by the law of motion of Chapter 9. At the first instant:*

1. *if $m_a > 0$ and $m_b > 0$, each accelerates towards the other;*
2. *if $m_a < 0$ and $m_b < 0$, each accelerates away from the other;*
3. *if $m_a > 0 > m_b$, both accelerate in the same direction, from the negative mass towards the positive one: the positive mass runs away, and the negative mass chases it. If $m_b = -m_a$, the two accelerations are equal at every instant: the distance stays d, and the pair speeds up without end, while its total momentum and its total kinetic energy stay zero.*

Proof. By Definition 6.17, the force on body a along the direction from a to b is $F = Gm_am_b/d^2$, and by Law 9.3 its acceleration along that direction is $F/m_a = Gm_b/d^2$. In the same way, body b accelerates towards a at Gm_a/d^2. (1) Both accelerations are positive: each body moves towards the other. (2) Both are negative: each moves away. (3) Body a moves towards b at $Gm_b/d^2 < 0$, that is, away from b; body b moves towards a at $Gm_a/d^2 > 0$. Both move in the direction from b to a, with the sizes $G|m_b|/d^2$ and Gm_a/d^2, which are equal when $m_b = -m_a$. The two bodies start at rest with equal accelerations, so they keep one velocity v: the distance stays d, the forces keep their sizes, and the accelerations stay equal. Then $m_av + m_bv = 0$ and $\frac{1}{2}m_av^2 + \frac{1}{2}m_bv^2 = 0$. □

Hermann Bondi found the same behaviour in Einstein's theory in 1957 (Bondi, 1957). Nothing like it has been seen, and antimatter, which was once a candidate, falls down (Anderson et al., 2023). This half of the frame is room, not a prediction.

11.6 STEP 6: BLACK HOLES ON THE FRAME

A black hole is a region of space from which not even light can escape; its edge is called the horizon. Astronomers now observe them: in 2015 LIGO recorded the gravitational waves of two black holes merging into one (Abbott et al., 2016), and in 2019 the Event Horizon Telescope imaged the shadow of the black hole at the centre of the galaxy M87 (Event Horizon Telescope Collaboration, 2019). Chapter 10 recalled

from Einstein's theory where a charged body can be a black hole: on the gravity side of the frame, $|q\mathcal{G}_{\bar{\phi}}| < m$, or on its diagonals (Remark 10.5). The theory also gives the horizons, and the cosmic mass can write them. In this step r is the distance from the centre of the body, not a cosmic magnitude.

Proposition 11.6 (Horizons Written with the Cosmic Mass)**.** *In Einstein's theory, let a spherical body of mass $m > 0$ and charge q, with no spin, collapse. If $\Re(\mu^2) \geq 0$, it can become a black hole, with horizons at the radii*

$$r_{\pm} = \frac{G}{c^2}\Big(m \pm \sqrt{\Re(\mu^2)}\Big), \qquad \Re(\mu^2) = m^2 - (q\mathcal{G}_{\bar{\phi}})^2$$

(if $q = 0$, then $r_- = 0$ and only r_+ is a horizon); if $\Re(\mu^2) < 0$, its field has no horizon at any radius. A body of negative mass has no horizon. Since $G\,\Re(\mu^2)/d^2$ is the force between two copies of the body the distance d apart (Definition 6.17), a charged body of positive mass can be a black hole exactly when two copies of it would not push each other apart.

Proof. Outside the body its field is the one found by Reissner and Nordström (Reissner, 1916; Nordström, 1918). Its horizons lie where the time coefficient of that field vanishes (Misner et al., 1973),

$$1 - \frac{2Gm}{c^2 r} + \frac{GKq^2}{c^4 r^2} = 0, \qquad \text{that is,} \qquad r^2 - \frac{2Gm}{c^2}\,r + \frac{GKq^2}{c^4} = 0.$$

The roots are $r = (G/c^2)\big(m \pm \sqrt{m^2 - (K/G)\,q^2}\big)$. Since $K/G = |\mathcal{G}_{\bar{\phi}}|^2$ (Section 6.3.1), the quantity under the root is $m^2 - |\mathcal{G}_{\bar{\phi}}|^2 q^2$, the real part of μ^2 (Proposition 6.16 with $\mu_a = \mu_b = \mu$). Real roots exist exactly when it is not negative. If $q = 0$, the smaller root is the centre, $r = 0$, which is not a horizon. If $m < 0$, two real roots would have the sum $2Gm/c^2 < 0$ and the product $GKq^2/c^4 \geq 0$, so neither would be positive. The last sentence is Definition 6.17 for two equal bodies. □

This is the formula of Reissner and Nordström, written with the cosmic mass; the law of this book does not reach it, because horizons belong to the strong fields where that law gives way to Einstein's theory (Chapter 9). What the cosmic mass adds is a reading: the number that decides whether a body can be a black hole, $\Re(\mu^2)$, also decides whether two copies of it pull or push (Figure 11.1). For the mass of the Sun, $Gm/c^2 \approx 1.477$ km: with no charge the horizon is at 2.95 km, and with a charge part equal to the mass, about 1.7×10^{20} C, far more than any star could keep, the two horizons meet at 1.48 km.

Remark 11.7 (What the Cosmic Mass Leaves Out).

1. *Spin.* A black hole that has settled is described by three numbers: its mass, its charge and its spin (Misner et al., 1973). The cosmic mass holds two of them. Real black holes spin, and spin moves the horizons; a full card for a black hole would need a third number, as neutral antimatter did in Step 4.

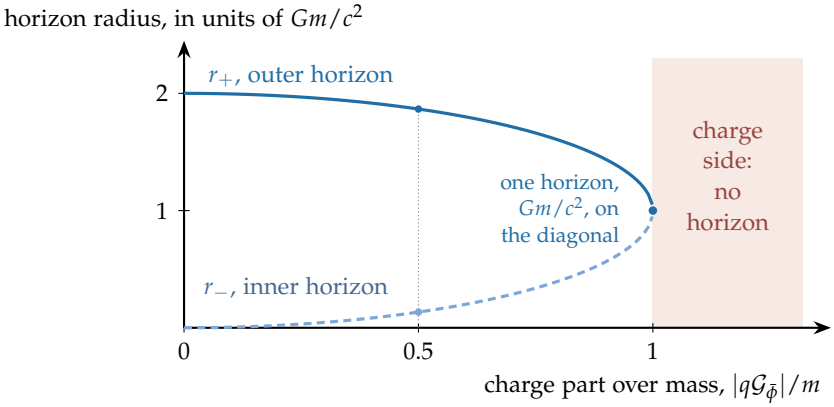


Figure 11.1: The two horizons of a charged body of positive mass (Proposition 11.6), in units of Gm/c^2, as its charge part grows. With no charge there is one horizon at $2Gm/c^2$; as the charge part grows the two horizons approach each other, and on the diagonal, $|q\mathcal{G}_{\bar{\phi}}| = m$, they meet at Gm/c^2. Beyond it, on the charge side, there is no horizon. The two dots mark a charge part of half the mass.

2. *Charge.* Real black holes are expected to carry almost no charge, because the gas around them holds charges of both signs and the kind that is pulled in cancels the charge. On the frame they lie on the mass axis, far out: the black hole of M87 on the ring of about 7×10^{48} Stoney masses. Charged and extremal black holes are allowed, but none has been observed.

3. *Matter and antimatter.* A black hole made of antimatter is, once it has settled, the same as one made of matter with the same mass, charge and spin. On the frame the two are one point, as neutral matter and antimatter were in Step 4.

4. *The diagonals.* On the diagonals sit the extremal black holes, which, when all their charges have the same sign, can rest beside one another at any distances (Remark 10.5).

11.7 STEP 7: THE WHOLE FRAME

Figure 11.2 and Table 11.1 put the six steps together. Every place of the frame has a status: measured, where bodies have been observed; allowed, where the mathematics and Einstein's theory leave room but nothing has been seen; and empty so far.

The frame, then, holds more than the bodies of Chapter 7: antimatter on the same three lines as matter, charged grains between the axes, and black holes far out on the mass axis. Nature does not use everything the frame allows, and four questions stay open for Volume 2.

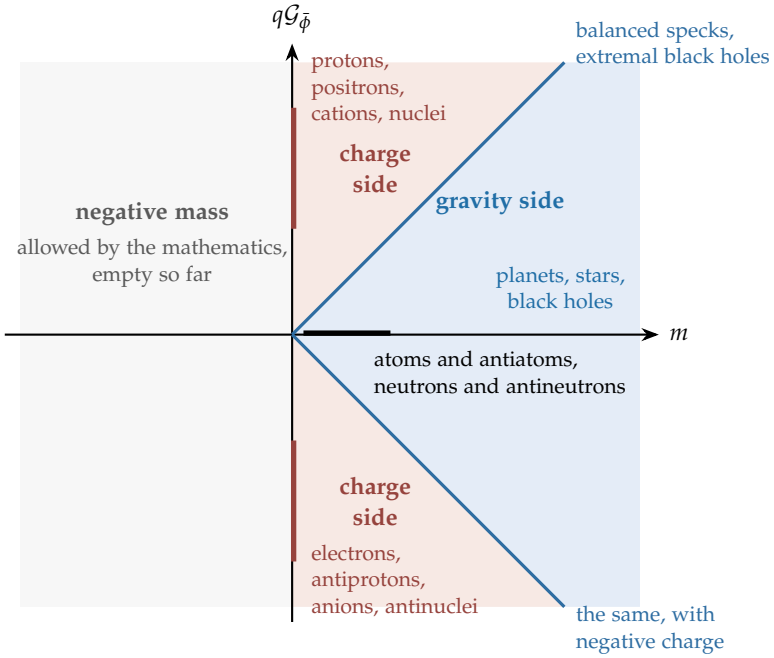


Figure 11.2: The whole frame, with the mass m across and the charge part $q\mathcal{G}_{\tilde{\phi}}$ up, not to scale. Charged particles and their antiparticles lie on the two ends of the charge axis, mirror images in the mass axis; neutral matter and neutral antimatter share the points of the mass axis; planets, stars and black holes lie on the gravity side, far out; balanced specks and extremal black holes lie on the diagonals; the half of negative mass is empty so far.

1. *A third number.* Neutral matter and neutral antimatter share their points, and a black hole's spin has no place on the card. Does the cosmic mass need a third number, and with it one more axis?

2. *The ring of an antiparticle.* A charged particle and its antiparticle lie on rings that differ by about 0.35 of their mass. Does anything that can be measured read the ring?

3. *The diagonals.* Extremal black holes are allowed, and none has been seen; a balanced speck is an ordinary grain. Does the diagonal mean more than the condition $Gm^2 = Kq^2$ (question 3 of Section 10.7)?

4. *The half of negative mass.* The mathematics allows it, and the real reading says what a body there would do. Is there one?

PLACE ON THE FRAME	WHAT LIES THERE	STATUS
upper end of the charge axis	protons, positive ions, nuclei; positrons and other positive antiparticles	measured
lower end of the charge axis	electrons, negative ions; antiprotons and antinuclei	measured
mass axis	atoms, neutrons; antiatoms and antineutrons, at the same points	measured; matter and antimatter not told apart
gravity side, far out	planets, stars, black holes	measured; black holes expected to be nearly neutral
diagonals	balanced specks, extremal black holes	specks: ordinary grains, one elementary charge for every 1.86 micrograms, nothing measured singles them out; extremal black holes: allowed, not observed
charge side, between the diagonals and the charge axes	charged grains and specks of dust, whose charge part is larger than their mass	measured
half of negative mass	nothing known	allowed by the mathematics; empty so far

Table 11.1: What the frame allows. The measurements are those of Sections 11.2 to 11.6 and, for charged grains, Millikan's oil drops (Millikan, 1913), recalled in Chapter 7.

WHAT WE HAVE SO FAR

- An antiparticle is the skew flip of its particle, $m - q\mathcal{G}_{\bar{\phi}}$: its mirror image in the mass axis.
- A particle and its antiparticle pull each other with the force $(Gm^2 + Kq^2)/d^2$; their sum, $2m$, is real and their other labels cancel, so all of it can leave as energy.
- Two numbers cannot tell neutral matter from neutral antimatter.
- A body of negative mass would move against every force on it; none has been seen.
- In Einstein's theory, a charged body of positive mass can be a black hole exactly when two copies of it would not push each other apart, and its horizons are

$$r_{\pm} = \frac{G}{c^2}\left(m \pm \sqrt{\Re(\mu^2)}\right).$$

CODA: THE NEXT RUNG

Where the new numbers stand on the long ladder of numbers, what they still need, and the question this book hands to its readers.

Pick up the pebble again, and let it go. Rub the balloon on your sleeve, and hold it near your hair. At the start of this book these were two pulls, described by two laws, and every particle carried two numbers that the two laws read separately. Now they can be written with one number for each particle and one law for each pair, and, once the bodies move, with one set of fields for both. In the new language, the pebble's fall belongs to the first term of the Cosmic Interaction Law, mass with mass, and your rising hair to the second, charge with charge. Between them the law holds a third term, mass with charge. Read by its real part, as Chapter 6 reads every force we measure, that term adds nothing to either pull.

THE LADDER, ONE RUNG HIGHER

Chapter 1 set out the ladder of numbers as far as the complex numbers (Table 1.1). The ladder did not stop there. The table below continues it with the numbers that came after $\mathbb{C}$, and with the rung proposed in this book. Each row records a question that the numbers of the time could not answer, and what the new numbers later allowed physics to say (Kline, 1972; Baez, 2002). A map of the whole road, from the first notched bones to the present day, is given in Appendix B.

NEW NUMBERS	THE QUESTION	WHAT FOLLOWED
negative numbers	What is $3 - 5$?	direction, deficit, opposite charges
irrational numbers	How long is the diagonal of a unit square?	continuous magnitude, the real line
complex numbers $\mathbb{C}$	Which number squares to -1?	rotation, phase, alternating currents, quantum amplitudes
quaternions $\mathbb{H}$, 1843	Can the arithmetic of the plane be carried into space?	the motion of rigid bodies, spin one half
Grassmann numbers, 1844	Can a product change sign when its order changes, $ab = -ba$?	the particles of matter in quantum theory; supersymmetry, not yet observed
Clifford algebras, 1878	Can the algebras of Hamilton and Grassmann be joined into one?	the Dirac equation of the electron, antimatter
octonions $\mathbb{O}$, 1843 to 1845	Can the quaternions be extended once more?	speculative: the symmetries of particle physics
General Numbers $\mathbb{G}$, this book	Must the second axis of the number plane stand at a right angle?	proposed: one number for mass and charge

Two of these rungs appeared earlier in this book. Hamilton's quaternions of 1843, cut into the stone of a Dublin bridge, opened Chapter 4 (Hamilton, 1844). Grassmann's numbers, whose products change sign when their order changes, return below (Grassmann, 1844). The Clifford algebras (Clifford, 1878) had to wait fifty years for Paul Dirac, who needed a square root of a differential operator, was led to matrices that anticommute, and found the equation that gave the electron its spin and predicted antimatter (Dirac, 1928).

WHAT EVERY RUNG NEEDED

Look down the ladder and a pattern appears. It is worth stating plainly, because it tells us what the General Numbers still need. Every rung needed four things.

1. *A question* that the old numbers could not answer, sharp enough that no amount of ingenuity inside the old system could dissolve it.

2. *A definition* that contradicted nothing, and that kept the old numbers inside the new ones as a special case.

3. *Consequences* that nobody had put in on purpose: the surprises that made the new numbers worth having.

4. In the end, *a fact about the world* that could not be said without them.

The General Numbers, as this book presents them, offer a candidate for each of the first three. The question is the one Chapter 1 asked: can a single number carry both the mass and the charge of a particle? Here honesty requires a remark. A complex number, such as $m + iq\sqrt{K/G}$, can also hold a mass and a charge in one number, and each relative plane of the General Number Plane is, algebraically, the complex plane drawn in a leaning frame. What the General Numbers add is the frame itself: the angle between the two axes, and the length of the second unit. Whether nature needs that freedom, as it turned out to need the number i, is part of what remains to be shown. The definition is Axiom 4.1, which contains the complex numbers as the relative plane $\bar{\phi} = \pi/2$ (Theorem 4.5). The consequences include the spiral of powers, the triangle of every general number, the new identity (r, α, ϕ) of a particle, and the Rule of the Square, whose minus sign pushes like charges apart and, in Chapter 9, gives the energy of the charge fields a plus sign and the energy of gravity a minus sign.

The fourth step is not in any book. For the complex numbers it took nearly three and a half centuries, from Cardano's puzzle of 1545 to Steinmetz's alternating currents of 1893, and longer still to quantum mechanics, where complex numbers turned out to be not merely convenient but necessary. For the General Numbers, the fourth step will be written, if it is written at all, by an experiment.

EARLIER DREAMS OF ONE NUMBER

The wish to hold mass and charge in a single mathematical object is not new, and it is only fair to place this book beside the attempts that came before it.

In 1921 Theodor Kaluza wrote Einstein's theory of gravity in five dimensions instead of four, and found that the extra parts of the geometry behave exactly like the electromagnetic field (Kaluza, 1921). In 1926 Oskar Klein added that the fifth dimension could be curled up into a tiny circle. Motion around that circle then comes in whole steps, and those steps behave like electric charge (Klein, 1926). In the theory of Kaluza and Klein, charge became motion in a direction we cannot see, and mass and charge became two parts of one geometric object. The theory ran into physical difficulties and did not become the physics of our world, although its central idea still lives in modern research.

In 1939 Eugene Wigner showed that the mass of a particle is a label: it tells how the particle behaves under the symmetries of space

and time (Wigner, 1939). Electric charge, it became clear, labels a symmetry of a different kind, an internal one, attached to each point of space without being part of its geometry. In 1967 Sidney Coleman and Jeffrey Mandula proved that, under very general assumptions, these two kinds of symmetry cannot be mixed in any straightforward way (Coleman and Mandula, 1967). But every good impossibility theorem is also a map of its own loophole. In 1975 Rudolf Haag, Jan Łopuszański and Martin Sohnius showed exactly where this one lies: the mixing becomes possible, and only then, if some quantities multiply with a change of sign, $ab = -ba$, as Grassmann's numbers of 1844 do (Haag et al., 1975). That loophole is supersymmetry, which physicists had begun to explore a few years earlier. So far, it has not been seen in any experiment.

Two lessons follow. The first is that the question of this book is a serious one: some of the best physicists of the twentieth century asked it too. The second is that every serious attempt to answer it has needed a new mathematical structure: an extra dimension of space, or a new kind of number. The General Numbers are a different and far more elementary proposal. They add no dimension to space and no new symmetry to physics. They change only the frame of the number plane, by letting its second axis lean. Whether so small a change can carry so large a load is exactly what remains to be seen.

THE OPEN QUESTION

Faraday asked with his falling coil whether there is a link between mass and charge beyond the separate laws of Newton and Coulomb. The Cosmic Mass Theory gives an answer, though not the one he looked for. Its law has a third term, the *Coupling Force*, which pairs the mass of each body with the charge of the other. Read by its real part, as Chapter 6 reads every measured force, that term adds nothing along the line between two bodies, and Faraday's needle, which stayed at zero, is not asked to move. The link is the law itself: one number for each body, one product for gravity and electricity, and, once the bodies move, one set of fields for both (Chapter 9).

If this book leaves one question in the reader's hands, it is what that link means in the world, and it is the question that Chapter 7 leaves open at its end: the part of a cosmic mass that lies across the mass axis, and the Composer coordinate of a product, which holds the Coupling term together with a share of the term of two charges, far larger for charged particles (Section 6.5.2), but is not read as a force. The Composer axis itself is held at 79.86°, the angle at which the plane shows why like charges push apart (Section 6.6.3). Volume 2 of this work takes up the Composer coordinate, beginning with the Coupling term (Section 2.4). Chapter 10 gathers the readings proposed for it, with what each would require and what measurement already says.

Chapter 11 draws the whole frame: antimatter, on the same lines as matter; the half of negative mass, empty so far; and the black holes, far out on the mass axis.

Whether it stands for something in the world is not for the author to decide, nor for the reader. It is for the world. The way forward is the way every rung of the ladder was made secure: a sharp prediction, and a measurement able to confirm it or to refute it. The first chapter of this book promised to state every step clearly, so that you could follow it and judge it for yourself. If you have come this far, you are in a position to do exactly that, and perhaps to design the experiment.

THE CHARACTERS OF THE BOOK

Galileo wrote that the book of nature is written in the language of mathematics, and that its characters are triangles, circles and other geometric figures. He did not say right triangles. For four centuries physics has read that book with an alphabet built on the right angle: the right triangle behind the sine and the cosine, the perpendicular axes of the complex plane. This book has tried to add one letter to that alphabet: a triangle that leans, and a number plane that leans with it.

Again and again, as the ladder shows, somebody has written down an object that seemed to have no right to exist, and found that it worked. There is no reason to think it has happened for the last time.

APPENDIX

A

THE GENERAL SINUS: A FORMULARY

This appendix collects the main formulas of the General Sinus, in the order of the paper *General Sinus Principles* (Derraz, 2026a). Chapter 3 explains where they come from. Here they are simply listed, with a sentence of guidance where it helps. As in the paper, the calculus formulas at the end use the letters (x, y) in place of (α, ϕ).

A.1 DEFINITIONS

For a studied angle α and a projection angle ϕ, the general trigonometric functions are the ratios of the sides of the oblique triangle $OZ\bar{Z}$ (Definition 3.3). By the law of sines (Theorem 3.4), for $\alpha \in \mathbb{R}$ and $\phi \in (0, \pi)$ with $\alpha + \phi \notin \pi\mathbb{Z}$,

$$\sin(\alpha, \phi) = \frac{\sin(\alpha)}{\sin(\alpha + \phi)}, \tag{A.1}$$

$$\cos(\alpha, \phi) = \frac{\sin(\phi)}{\sin(\alpha + \phi)}, \tag{A.2}$$

$$\tan(\alpha, \phi) = \frac{\sin(\alpha)}{\sin(\phi)}. \tag{A.3}$$

The reciprocal functions are

$$\csc(\alpha, \phi) = \frac{\sin(\alpha + \phi)}{\sin(\alpha)}, \tag{A.4}$$

$$\sec(\alpha, \phi) = \frac{\sin(\alpha + \phi)}{\sin(\phi)}, \tag{A.5}$$

$$\cot(\alpha, \phi) = \frac{\sin(\phi)}{\sin(\alpha)}, \tag{A.6}$$

and, as in classical trigonometry,

$$\frac{\sin(\alpha, \phi)}{\cos(\alpha, \phi)} = \frac{\sin(\alpha)}{\sin(\phi)} = \tan(\alpha, \phi). \tag{A.7}$$

When $\alpha + \phi = \pi/2$ the triangle is a right triangle, and the general functions become the classical ones (Corollary 3.5).

A.2 DOMAINS AND PERIODICITY

The six functions are defined on the following domains, and take real values there:

$$\sin(\alpha,\phi): \quad \alpha\in\mathbb{R},\ \phi\in\mathbb{R},\ \alpha+\phi\notin\pi\mathbb{Z}, \tag{A.8}$$
$$\cos(\alpha,\phi): \quad \alpha\in\mathbb{R},\ \phi\in\mathbb{R},\ \alpha+\phi\notin\pi\mathbb{Z}, \tag{A.9}$$
$$\tan(\alpha,\phi): \quad \alpha\in\mathbb{R},\ \phi\in\mathbb{R},\ \phi\notin\pi\mathbb{Z}, \tag{A.10}$$
$$\csc(\alpha,\phi): \quad \alpha\in\mathbb{R},\ \phi\in\mathbb{R},\ \alpha\notin\pi\mathbb{Z}, \tag{A.11}$$
$$\sec(\alpha,\phi): \quad \alpha\in\mathbb{R},\ \phi\in\mathbb{R},\ \phi\notin\pi\mathbb{Z}, \tag{A.12}$$
$$\cot(\alpha,\phi): \quad \alpha\in\mathbb{R},\ \phi\in\mathbb{R},\ \alpha\notin\pi\mathbb{Z}. \tag{A.13}$$

For any $(\alpha,\phi)\in\mathbb{R}^2$ there are unique $\alpha'\in[0,2\pi)$ and $\phi'\in[0,2\pi)$ with $\alpha=\alpha'+2k\pi$ and $\phi=\phi'+2m\pi$ for some $k,m\in\mathbb{Z}$, and

$$\sin(\alpha,\phi)=\sin(\alpha',\phi'), \tag{A.14}$$
$$\cos(\alpha,\phi)=\cos(\alpha',\phi'), \tag{A.15}$$
$$\tan(\alpha,\phi)=\tan(\alpha',\phi'). \tag{A.16}$$

A.3 RECIPROCITY AND RECIPROCAL IDENTITIES

Exchanging the two angles exchanges the sine and the cosine:

$$\sin(\alpha,\phi)=\cos(\phi,\alpha), \qquad \cos(\alpha,\phi)=\sin(\phi,\alpha), \tag{A.17}$$
$$\tan(\alpha,\phi)=\cot(\phi,\alpha), \qquad \csc(\alpha,\phi)=\sec(\phi,\alpha). \tag{A.18}$$

The reciprocal functions are themselves general functions with shifted arguments:

$$\csc(\alpha,\phi)=\sin(\alpha+\phi,-\phi), \tag{A.19}$$
$$\sec(\alpha,\phi)=\tan(\alpha+\phi,\phi), \tag{A.20}$$
$$\cot(\alpha,\phi)=\cos(\alpha-\phi,\phi). \tag{A.21}$$

A.4 SPECIAL CASES

When $\alpha+\phi=\pi/2$, the general functions coincide with the classical ones:

$$\sin(\alpha,\phi)=\sin(\alpha), \qquad \csc(\alpha,\phi)=\csc(\alpha), \tag{A.22}$$
$$\cos(\alpha,\phi)=\cos(\alpha), \qquad \sec(\alpha,\phi)=\sec(\alpha), \tag{A.23}$$
$$\tan(\alpha,\phi)=\tan(\alpha), \qquad \cot(\alpha,\phi)=\cot(\alpha). \tag{A.24}$$

When $\phi=\pi/2$, the right angle moves to the other end of the opposite side:

$$\sin(\alpha,\pi/2)=\tan(\alpha), \qquad \csc(\alpha,\pi/2)=\cot(\alpha), \tag{A.25}$$
$$\cos(\alpha,\pi/2)=\sec(\alpha), \qquad \sec(\alpha,\pi/2)=\cos(\alpha), \tag{A.26}$$
$$\tan(\alpha,\pi/2)=\sin(\alpha), \qquad \cot(\alpha,\pi/2)=\csc(\alpha). \tag{A.27}$$

When $\alpha = \phi = \theta$, the triangle is isosceles:

$$\sin(\theta,\theta) = \cos(\theta,\theta) = \tfrac{1}{2}\sec(\theta), \qquad \tan(\theta,\theta) = 1, \tag{A.28}$$
$$\csc(\theta,\theta) = \sec(\theta,\theta) = 2\cos(\theta), \qquad \cot(\theta,\theta) = 1. \tag{A.29}$$

For $\theta = \pi/3$ all six functions equal 1; for $\theta = \pi/5$, $\sin(\theta,\theta) = \varphi^{-1}$ and $\csc(\theta,\theta) = \varphi$, where $\varphi = (\sqrt{5}+1)/2$ is the golden ratio.

A.5 SYMMETRY AND π-ADDITION IDENTITIES

For $(i,i',j,j') \in \{-1,1\}^4$,

$$\sin(i\alpha, j\phi) = \sin(i'\alpha, j'\phi) \quad \text{whenever } ij = i'j', \tag{A.30}$$
$$\cos(i\alpha, j\phi) = \cos(i'\alpha, j'\phi) \quad \text{whenever } ij = i'j', \tag{A.31}$$
$$\tan(i\alpha, j\phi) = ij\tan(\alpha,\phi). \tag{A.32}$$

For $(p,q) \in \mathbb{Z}^2$,

$$\sin(\alpha + p\pi, \phi + q\pi) = (-1)^q \sin(\alpha,\phi), \tag{A.33}$$
$$\cos(\alpha + p\pi, \phi + q\pi) = (-1)^p \cos(\alpha,\phi), \tag{A.34}$$
$$\tan(\alpha + p\pi, \phi + q\pi) = (-1)^{p+q} \tan(\alpha,\phi). \tag{A.35}$$

The sign of the general sine depends only on the shift of the projection angle, and the sign of the general cosine only on the shift of the studied angle.

A.6 ADDITION FORMULAS

For the studied angle, for all α, β, ϕ in the respective domains,

$$\sin(\alpha+\beta,\phi) = \sin(\alpha,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi)\cos(\alpha,\phi+\beta), \tag{A.36}$$
$$\sin(\alpha+\beta,\phi) = \sin(\alpha,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi+\alpha), \tag{A.37}$$
$$\cos(\alpha+\beta,\phi) = \cos(\alpha,\phi)\cos(\beta,\phi) + \cos(\alpha,\phi)\sin(\beta,\phi)\sin(\alpha,\phi+\beta), \tag{A.38}$$
$$\cos(\alpha+\beta,\phi) = \cos(\alpha,\phi)\cos(\beta,\phi) + \sin(\alpha,\phi)\sin(\beta,\phi)\cos(\alpha+\beta,\phi). \tag{A.39}$$

For the projection angle,

$$\sin(\alpha,\phi+\theta) = \sin(\alpha,\phi)\sin(\alpha+\phi,\theta), \tag{A.40}$$
$$\cos(\alpha,\phi+\theta) = \cos(\alpha,\phi) + \sin(\alpha,\phi)\cos(\alpha+\phi,\theta). \tag{A.41}$$

Two direct consequences are

$$\sec(\alpha+\beta,\phi) = \sec(\alpha,\phi)\sec(\beta,\phi) - \tan(\alpha,\phi)\tan(\beta,\phi), \tag{A.42}$$
$$\cos(\alpha+\beta,\phi) = \frac{\cos(\alpha,\phi)\cos(\beta,\phi)}{1-\sin(\alpha,\phi)\sin(\beta,\phi)}. \tag{A.43}$$

A.7 DOUBLE-ANGLE AND HALF-ANGLE IDENTITIES

Doubling the studied angle gives

$$\sin(2\alpha,\phi) = \sin(\alpha,\phi)\big(\cos(2\alpha,\phi)+1\big), \tag{A.44}$$
$$\cos(2\alpha,\phi) = \cos(\alpha,\phi)^2 + \sin(\alpha,\phi)^2\cos(2\alpha,\phi). \tag{A.45}$$

The second identity contains $\cos(2\alpha,\phi)$ on both sides; solving it gives

$$\cos(2\alpha,\phi) = \frac{\cos(\alpha,\phi)^2}{1-\sin(\alpha,\phi)^2}, \tag{A.46}$$

in close analogy with the classical identity $\cos(2\alpha) = (1-\tan^2\alpha)/(1+\tan^2\alpha)$. Further consequences are

$$\sec(2\alpha,\phi) = \sec(\alpha,\phi)^2 - \tan(\alpha,\phi)^2, \tag{A.47}$$
$$\sin(2\alpha,\phi) - \sin(\alpha,\phi) = \sin(\alpha,\phi)\cos(2\alpha,\phi). \tag{A.48}$$

Halving the studied angle gives, in the same way,

$$\cos(\alpha,\phi) = \cos\left(\tfrac{\alpha}{2},\phi\right)^2 + \sin\left(\tfrac{\alpha}{2},\phi\right)^2\cos(\alpha,\phi), \tag{A.49}$$
$$\cos(\alpha,\phi) = \frac{\cos\left(\frac{\alpha}{2},\phi\right)^2}{1-\sin\left(\frac{\alpha}{2},\phi\right)^2}, \tag{A.50}$$
$$\sec(\alpha,\phi) = \sec\left(\tfrac{\alpha}{2},\phi\right)^2 - \tan\left(\tfrac{\alpha}{2},\phi\right)^2. \tag{A.51}$$

A.8 PARTIAL DERIVATIVES

In the formulas below, f'_x and f'_y are the ordinary partial derivatives of a general function $f(x,y)$ with respect to its first and second variable:

$$\sin(x,y)'_x = \frac{\sin(y)}{\sin(x+y)^2} = \frac{\cos(x,y)}{\sin(x+y)}, \tag{A.52}$$
$$\sin(x,y)'_y = -\frac{\sin(x)\cos(x+y)}{\sin(x+y)^2} = -\sin(x,y)\cot(x+y), \tag{A.53}$$
$$\cos(x,y)'_x = -\frac{\sin(y)\cos(x+y)}{\sin(x+y)^2} = -\cos(x,y)\cot(x+y), \tag{A.54}$$
$$\cos(x,y)'_y = \frac{\sin(x)}{\sin(x+y)^2} = \frac{\sin(x,y)}{\sin(x+y)}, \tag{A.55}$$
$$\tan(x,y)'_x = \frac{\cos(x)}{\sin(y)} = \tan\left(\tfrac{\pi}{2}\pm x,y\right), \tag{A.56}$$
$$\tan(x,y)'_y = -\frac{\sin(x)\cos(y)}{\sin(y)^2} = -\tan(x,y)\cot(y). \tag{A.57}$$

They satisfy the following relations:

$$\cos(x,y) - \sin(x+y)\ \sin(x,y)'_x = 0, \tag{A.58}$$
$$\sin(x,y) + \tan(x+y)\ \sin(x,y)'_y = 0, \tag{A.59}$$
$$\cos(x,y) + \tan(x+y)\ \cos(x,y)'_x = 0, \tag{A.60}$$
$$\sin(x,y) - \sin(x+y)\ \cos(x,y)'_y = 0, \tag{A.61}$$
$$\tan\left(\tfrac{\pi}{2}+x,y\right) - \tan(x,y)'_x = 0, \tag{A.62}$$
$$\tan(x,y) + \tan(y)\ \tan(x,y)'_y = 0. \tag{A.63}$$

A.9 THE GENERAL SINUS TRANSFORM

The general functions can be written as products of scaled copies of themselves. For all $n \in \mathbb{N}$ and (x,y) in the respective domains,

$$\sin(x,y) = \prod_{k=0}^{2^n-1} \sin\left(\frac{x}{2^n} + \pi\frac{k}{2^n}, \frac{y}{2^n}\right), \tag{A.64}$$
$$\cos(x,y) = \prod_{k=0}^{2^n-1} \cos\left(\frac{x}{2^n}, \frac{y}{2^n} + \pi\frac{k}{2^n}\right), \tag{A.65}$$
$$\tan(x,y) = \prod_{k=0}^{2^n-1} \tan\left(\frac{x}{2^n} + \pi\frac{k}{2^n}, \frac{y}{2^n} + \pi\frac{k}{2^n}\right). \tag{A.66}$$

For $n = 1$ the first product is the exact factorisation

$$\sin(x,y) = \sin\left(\frac{x}{2}, \frac{y}{2}\right) \sin\left(\frac{x}{2} + \frac{\pi}{2}, \frac{y}{2}\right), \tag{A.67}$$

which expresses a general sine as the product of two general sines at half the arguments.

As n grows, every argument $z/2^n$ becomes small, and each factor can be replaced by a ratio of two polynomials of the second degree. For $n \in \mathbb{N}$ and $k \in \mathbb{N}_0$, define

$$a_n = \frac{\pi}{2^n}, \qquad b_n = \frac{\pi^2}{2^{2n+1}}, \tag{A.68}$$

with $u_{n,0} = 0$, $v_{n,0} = 1$ and

$$u_{n,k+1} = a_n v_{n,k} + (1 - b_n)\, u_{n,k}, \tag{A.69}$$
$$v_{n,k+1} = (1 - b_n)\, v_{n,k} - a_n u_{n,k}. \tag{A.70}$$

These sequences approximate $u_{n,k} \approx \sin(\pi k/2^n)$ and $v_{n,k} \approx \cos(\pi k/2^n)$, more closely as n grows. The *General Sinus Trans-*

form then approximates the three functions by products of ratios of polynomials:

$$\sin(x,y) \approx \prod_{k=0}^{2^n-1} \frac{u_{n,k}x^2 - 2^{n+1}v_{n,k}x - 2^{2n+1}u_{n,k}}{u_{n,k}(x+y)^2 - 2^{n+1}v_{n,k}(x+y) - 2^{2n+1}u_{n,k}}, \tag{A.71}$$

$$\cos(x,y) \approx \prod_{k=0}^{2^n-1} \frac{u_{n,k}y^2 - 2^{n+1}v_{n,k}y - 2^{2n+1}u_{n,k}}{u_{n,k}(x+y)^2 - 2^{n+1}v_{n,k}(x+y) - 2^{2n+1}u_{n,k}}, \tag{A.72}$$

$$\tan(x,y) \approx \prod_{k=0}^{2^n-1} \frac{u_{n,k}x^2 - 2^{n+1}v_{n,k}x - 2^{2n+1}u_{n,k}}{u_{n,k}y^2 - 2^{n+1}v_{n,k}y - 2^{2n+1}u_{n,k}}. \tag{A.73}$$

The paper reports that $n \geq 20$ gives at least five correct decimal places. That accuracy holds when the factors use exact values of $\sin(\pi k/2^n)$ and $\cos(\pi k/2^n)$; with the recurrences above, the error is larger, especially when $x + y$ comes close to π. The transform plays for the general functions the part that Taylor series play for the classical ones, except that it multiplies instead of adding.

STILL TO COME

[Author: the tangent addition and double-angle formulas, two further identities for the general tangent, the second-order partial derivatives, the total derivatives under the convention that the projection angle varies clockwise, and the n-th derivatives and primitives of the paper will be added here once the checks listed in the action plan are settled.]

B

THE LONG ROAD

The ideas of this book stand at the end of a very long road. This appendix is a map of that road. It follows four threads through fifty-two *stations*, places and moments where an idea changed, and thirteen *meditations*, pauses to ask what the change meant. Its guiding thought is this:

> *In each of these histories a question was asked that the available concepts could not answer, and the age concluded that the question was defective. In each of them somebody eventually concluded instead that the concepts were defective, stipulated a new object, endured the ridicule, and handed the next generation a world with more in it.*

The tables below list the stations (numbered 1 to 52) and the meditations (numbered I to XIII) in order. Several of them appear as scenes in the chapters of this book.

PART I NUMBER

How counting became arithmetic, and how each new kind of number, from fractions to the complex numbers, was forced on its inventors by a question the old numbers could not answer (Chapter 1).

	TITLE	PLACE AND TIME
1	The Notch	southern Africa, roughly forty thousand years ago
I	The Limit and the Unlimited	Miletus and Croton, sixth and fifth centuries BCE
2	The Word	everywhere, before writing
3	The Clay	Sumer and Babylon, about 3300 to 1600 BCE
4	The Rope and the Fraction	Egypt, about 1850 BCE
5	The Crisis	Greece, fifth century BCE
6	The Debt and the Void	China, about 200 BCE; India, 628 CE
7	The House of Wisdom	Baghdad, ninth century
8	The Long Ride West	Béjaïa to Pisa, 1202
9	The Duel	Bologna and Milan, 1515 to 1545
10	The Wild Thought	Bologna, 1572
II	The Freedom to Stipulate	from Euclid's definitions to the imaginary unit
11	Two Centuries of Embarrassment	Europe, 1572 to 1777
III	The Amphibian	Europe, 1700 to 1830
12	The Picture	Copenhagen 1797, Geneva 1806, Göttingen 1831
13	The Legitimation	Dublin, 1837
14	The End of the Road	and why it ends there

PART II ANGLE

How the shadow of a stick became trigonometry: the right angle of the builders' rope, the tables of chords of the Greek astronomers, the Indian half-chord that became our sine through a chain of translations, and the moments when an angle became a number and a direction became a quantity (Chapter 3).

	TITLE	PLACE AND TIME
15	The Shadow	Egypt and Babylon, before 1500 BCE
16	The Rope and the Right Angle	the Near East, second millennium BCE
17	The Triangle Becomes an Argument	Ionia and Alexandria, sixth to third century BCE
18	The Chord	Rhodes and Alexandria, 150 BCE to 150 CE
IV	Is an Angle a Quality, a Quantity, or a Relation?	Alexandria and Athens, second to fifth century CE
19	The Half-Chord	India, fifth to fifteenth century
20	A Word Loses Its Way	from Sanskrit to Latin
21	The Discipline Comes of Age	Baghdad, Isfahan and Marāgha, ninth to thirteenth century
22	The Angle Becomes a Number	Europe, 1637 to 1748
23	The Hinge	the sentence this part was written for
24	The Direction Becomes a Quantity	Copenhagen 1797, Dublin 1843, New Haven 1901
25	The Payoff	Paris 1822, Chicago 1893

PART III MATTER AND FORCE

How the stuff of the world became a quantity: from the atoms of the Greek and Indian philosophers and the balance of Archimedes, through Galileo's falling bodies and Newton's quantity of matter, to the two inverse square laws and the units that finally let both laws be written as equations (Chapters 1 and 6).

	TITLE	PLACE AND TIME
26	The Cut	Abdera and India, fifth century BCE
V	The Cut	Elea, Abdera and Athens, fifth and fourth centuries BCE
27	The Rejection	Athens, fourth century BCE
VI	Abstraction	Athens, fourth century BCE
28	The Balance	Syracuse, third century BCE
29	The Push That Stays	Alexandria, Baghdad and Paris, sixth to fourteenth century
30	The Atom Returns	Europe, 1417 to 1661
31	Fall	Padua and Arcetri, 1590 to 1638
32	Quantity of Matter	Cambridge, 1687
33	The First Inverse Square	London and Cambridge, 1679 to 1687
VII	Necessity or Habit	Baghdad, Córdoba and Edinburgh, 1095 to 1748
34	Weighing the Earth	London, 1798
35	The Other Force	from amber to the Leyden jar, 600 BCE to 1750 CE
36	The Second Inverse Square	Leeds, London and Paris, 1767 to 1785
37	The Units	Paris 1799, Göttingen 1832, Chicago 1893, Paris 1889 and 1948
VIII	Force as a Fiction, Mass as a Ratio	Dublin 1721, Prague 1883
38	The Atom Becomes a Number	Manchester 1808, Paris 1913, Chicago 1913
IX	From Substance to Function	Berlin, 1910
39	The Definitions Change Places	Versailles, 20 May 2019

PART IV THE LANGUAGE

How mathematics came to be seen as the language of nature, from the Pythagoreans to Wigner, together with the doubts and the explanations that followed every such claim (Chapter 5).

	TITLE	PLACE AND TIME
40	All Is Number	Croton, fifth century BCE
41	God Ever Geometrizes	Athens, fourth century BCE
42	The Door and the Key	Oxford and Paris, thirteenth century
43	The Book of Nature	Rome, 1623
44	Geometry Shining in the Mind of God	Graz, Prague and Linz, 1596 to 1619
45	I Frame No Hypotheses	Cambridge, 1687 and 1713
46	As Much Science as There Is Mathematics	Königsberg 1786, Paris 1822
47	Translate It Out of Hieroglyphics	London and Aberdeen, 1857 to 1865
48	Wiser Than Their Discoverers	Karlsruhe, 1887
49	Express It in Numbers	London, 1883
X	Kant, Helmholtz, and the Enlargement of Perception	Königsberg 1781, Heidelberg 1870
50	Certain, or About Reality	Berlin, 1921
51	Beauty in the Equations	Cambridge, 1928 to 1963
XI	Labels on Representations	Princeton, 1939
XII	Why They Have Not Merged	1919 to 1975
52	Unreasonable Effectiveness	New York, 1960
XIII	The Ladder	what each new number bought

C

COMPLEX NUMBERS FROM SCRATCH

This appendix is a short course on the complex numbers, for readers who would like to see the rules of Chapter 1 in one place. It follows the teaching approach of the lecture notes of Mokhithi and Shock (2020), released under the Creative Commons Attribution 4.0 license; the text here is a new and shorter presentation. For a beautiful, picture-first treatment, see Needham (1997).

C.1 A NUMBER WHOSE SQUARE IS MINUS ONE

Every number on the number line gives a result of zero or more when it is multiplied by itself. So the equation $x^2 = -1$ has no solution on the line. Instead of giving up, we *define* a new number.

Definition C.1 (The imaginary unit)**.** The imaginary unit i is a number whose square is -1:

$$i^2 = -1.$$

Its opposite, $-i$, has the same property: $(-i)^2 = -1$. So i and $-i$ are the two numbers which, when squared, give -1.

A *complex number* is a number of the form

$$z = a + bi, \qquad a, b \in \mathbb{R}.$$

The number a is the *real part* of z, written $\operatorname{Re} z$, and b is its *imaginary part*, written $\operatorname{Im} z$. Two complex numbers are equal when their real parts are equal and their imaginary parts are equal. The real numbers are the complex numbers with $b = 0$, so nothing old is lost.

C.2 ARITHMETIC

Complex numbers are added and multiplied with the ordinary rules of algebra, remembering that $i^2 = -1$:

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)\, i,\\ (a + bi)(c + di) &= ac + adi + bci + bd\, i^2\\ &= (ac - bd) + (ad + bc)\, i.\end{aligned}$$

For example, $(2 + i)^2 = 4 + 4i + i^2 = 3 + 4i$, and $(2 + i)^3 = (3 + 4i)(2 + i) = 6 + 3i + 8i + 4i^2 = 2 + 11i$. This is the calculation with which Bombelli tamed his cubic equation in 1572 (Chapter 1).

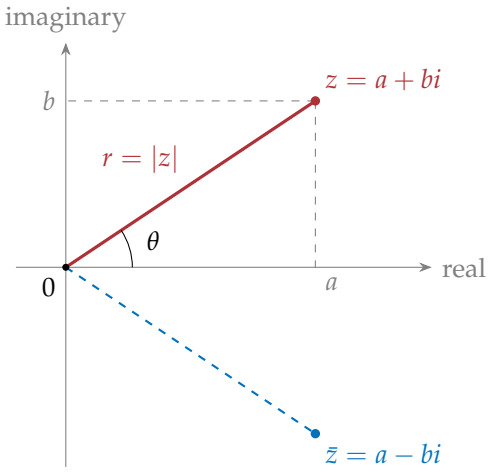


Figure C.1: A complex number $z = a + bi$ in the plane, with its modulus $r = |z|$, its argument θ, and its conjugate $\bar{z}$, the mirror image of z in the real axis.

The *conjugate* of $z = a + bi$ is $\bar{z} = a - bi$. Multiplying a number by its conjugate always gives a real number that is zero or more:

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2.$$

This is the key to division. To divide by $c + di$, multiply the top and the bottom by its conjugate $c - di$:

$$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{(ac + bd) + (bc - ad)\,i}{c^2 + d^2}.$$

C.3 THE PLANE

A complex number $a + bi$ is a point of the plane: a steps along the real axis and b steps along the imaginary axis (Figure C.1). Two numbers describe the point in another way.

- The *modulus* $|z| = \sqrt{a^2 + b^2}$ is the distance from 0 to z, by the theorem of Pythagoras. Note that $|z|^2 = z\bar{z}$.
- The *argument* θ is the angle from the positive real axis to the segment from 0 to z, measured counterclockwise. It satisfies $\tan\theta = b/a$, and the signs of a and b tell in which quarter of the plane the angle lies.

With $r = |z|$, the number is written in *polar form*:

$$z = r(\cos\theta + i\sin\theta).$$

The conjugate $\bar{z}$ is the mirror image of z in the real axis: same modulus, opposite argument.

C.4 MULTIPLYING MEANS TURNING

Take two numbers in polar form, $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. Multiplying them out and using the addition formulas of trigonometry gives

$$z_1 z_2 = r_1 r_2\big(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big).$$

This is the rule that Wessel found in 1797: *to multiply two complex numbers, multiply their lengths and add their angles*. In particular, $i = \cos(\pi/2) + i\sin(\pi/2)$ has length one and angle $\pi/2$, so multiplying by i turns any number by a quarter turn, and multiplying by i twice turns it by a half turn, which is multiplication by -1. That is the whole meaning of $i^2 = -1$.

C.5 EULER'S FORMULA AND DE MOIVRE'S FORMULA

Euler discovered that the exponential function and the circle are the same story:

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Every complex number can therefore be written as $z = re^{i\theta}$, and the multiplication rule becomes the ordinary rule for exponents: $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}$. Raising a number of length one to the power n turns it n times, which is *de Moivre's formula*:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta).$$

With $\theta = \pi$, Euler's formula gives $e^{i\pi} = -1$, often written $e^{i\pi} + 1 = 0$: one short line that joins five of the most important numbers of mathematics.

C.6 ROOTS OF UNITY

The equation $z^n = 1$ has exactly n complex solutions:

$$z_k = e^{2\pi i k/n} = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}, \qquad k = 0, 1, \ldots, n-1.$$

They are the corners of a regular polygon with n sides, drawn inside the circle of radius one with one corner at 1. For $n = 4$ they are 1, i, -1 and $-i$: the cycle of powers of i from Chapter 1. In Chapter 4, the powers of the Composer Unit at $\bar{\phi} = \pi/3$ turn by the same steps as the sixth roots of unity, but grow at every step (Figure 4.7).

C.7 EXERCISES

1. Compute $(1+i)^2$ and $(1+i)^4$.

2. Find the modulus and the argument of $-1 + i\sqrt{3}$.
3. Compute $\dfrac{3+4i}{1+2i}$.
4. Using polar form, explain why $i^3 = -i$ and $i^4 = 1$.

Answers

1. $(1+i)^2 = 1 + 2i + i^2 = 2i$, and $(1+i)^4 = (2i)^2 = -4$.
2. The modulus is $\sqrt{1+3} = 2$. Since $\cos\theta = -1/2$ and $\sin\theta = \sqrt{3}/2$, the argument is $\theta = 2\pi/3$.
3. $\dfrac{(3+4i)(1-2i)}{(1+2i)(1-2i)} = \dfrac{3-6i+4i-8i^2}{5} = \dfrac{11-2i}{5}$.
4. The number i has length one and angle $\pi/2$. Its third power has angle $3\pi/2$, which points straight down: $-i$. Its fourth power has angle 2π, a full turn: 1.

D

NOTATION, UNITS AND CONSTANTS

This appendix gathers the symbols of the book in one place, with the place where each is introduced, followed by the units and the physical constants that the theory uses.

D.1 SYMBOLS

The symbols of the mathematics come first and those of the physics after them; Table 6.1 in Chapter 6 pairs the words of the two.

SYMBOL	MEANING	SEE
The General Sinus		
α	studied angle	Def. 3.3
ϕ	projection angle	Def. 3.3
$\sin(\alpha,\phi)$, $\cos(\alpha,\phi)$, $\tan(\alpha,\phi)$	general sine, cosine and tangent	Def. 3.3, Thm. 3.4
$\csc(\alpha,\phi)$, $\sec(\alpha,\phi)$, $\cot(\alpha,\phi)$	their reciprocals	Appendix A.1
$\bar{\phi} = \alpha + \phi$	angle of the Com-axis of the composed circle	Def. 3.6
The General Numbers		
G	General Number Plane; also the set of all General Numbers	Axiom 4.1, Def. 4.14
$G_{\bar{\phi}}$	relative plane, or subgeneral plane, at the angle $\bar{\phi}$	Def. 4.3
$\bar{\phi}$	angle of the Composer axis, which the Axiom calls a projection angle	Axiom 4.1
$\mathcal{G}_{\bar{\phi}}$	Composer Unit	Axiom 4.1
$\mathcal{A}_{\bar{\phi}}$	amplitude, the length of the Composer Unit	Axiom 4.1, Def. 4.2
$\vec{u}_{\bar{\phi}}$	direction of the Composer Unit	Axiom 4.1
$\vec{x}$, $\vec{y}$	perpendicular unit arrows of the page	Axiom 4.1
$z = a + b\mathcal{G}_{\bar{\phi}}$	General Number	Axiom 4.1
$\Re(z) = a$	real part of z	Def. 4.11
$\mathrm{Com}(z) = b\mathcal{G}_{\bar{\phi}}$	Composer part of z	Def. 4.11
b	Composer coordinate of z, the number of steps of the Composer Unit	Def. 4.11

SYMBOL	MEANING	SEE
$\mathrm{Sh}(z)$	shadow of z on the real axis	Def. 4.19
M	magnitude of a General Number	Def. 4.8, Thm. 4.9
O, P, Q	corners of the triangle of a General Number: the origin, the end of the real part, and the tip z	Section 4.4.2, Rem. 4.30
η	ratio of the two parts, $\mathcal{A}_{\bar{\phi}}\, b/a$, which Theorem 4.10 writes $\mathcal{G}_{\bar{\phi}}\, b/a$	Thm. 4.10
k	a whole number, $k \in \mathbb{Z}$	Thm. 4.10
$\bar{\phi}_1$, $\bar{\phi}_2$	arcsine root and mirror root of a length between 1 and $\sqrt{2}$	Thm. 4.25
$\pi/4 < \bar{\phi} < 3\pi/4$	backward band, where the square of the Composer Unit points backward	Thm. 4.28
δ	angle between a General Number and the Composer axis	Prop. 4.32
(a, B), $B = b\,\mathcal{A}_{\bar{\phi}}$	chart of the two parts, ruled by rings and rays	Thm. 4.34, Def. 4.35
The Cosmic Mass		
$P(A)$	particle signature, the column (m_a, q_a)	Def. 6.1
m, q	mass in kilograms and electric charge in coulombs	Def. 6.1
$\vec{e}_r$	unit direction vector from particle A to particle B	Rem. 6.2
r	distance between two particles; also the cosmic mass magnitude	Defs. 6.3, 6.4; Rem. 6.13
$\vec{F}_N$, $\vec{F}_C$	the forces of Newton and Coulomb	Defs. 6.3, 6.4
μ	cosmic mass	Def. 6.5
$q\mathcal{G}_{\bar{\phi}}$	charge part of the cosmic mass, in kilograms	Def. 6.5
$\lvert\mathcal{G}_{\bar{\phi}}\rvert = \sqrt{K/G}$	size of the Composer Unit, in kilograms per coulomb	Section 6.3.1
$\vec{F}_G$	cosmic interaction force	Thm. 6.6
α, ϕ	charge phase angle and mass phase angle	Rem. 6.13
(r, α, ϕ)	identity card of a particle	Rem. 6.13
η	charge-to-mass ratio q/m	Rem. 6.15
$\Re(\mu_a \mu_b)$	real part of the product of two cosmic masses, read as the force along the line	Prop. 6.16, Def. 6.17
The Cosmic Mass Standard Model		
z	charge number of a body, $q = z\,e$	Section 7.1.1

SYMBOL	MEANING	SEE
m_S	Stoney mass, the charge part of one elementary charge	Def. 7.2
m_P	Planck mass	Rem. 7.3
α_{fs}	fine-structure constant	Rem. 7.3
Z, A	atomic number; atomic mass in daltons	Section 7.5
B_e	binding energy of the electrons of an atom	Section 7.5
δ	lean of a charged body from the charge axis	Rem. 7.6
The Closure Law		
N, $A = z + N$	number of neutrons of a nucleus with z protons, and its mass number	Section 8.1
$\mu(z, N)$, $m(z, N)$	cosmic mass and mass of that nucleus	Def. 8.1
$\mathcal{L}_A$, $\mathcal{L}_B$	the two closure loops of six neighbouring nuclei	Def. 8.3
w, $\tilde{m}$	cusp constant, 22.75 MeV, and the mass less its cusp term	Section 8.4
$\sigma(A)$	error of the closure estimate	Law 8.7
$\Delta\mu$, $Q = \Delta\mu\, c^2$	mass lost in a fusion, and the energy it releases	Prop. 8.2, Law 8.7
The Unified Cosmic Law		
$\boldsymbol{\mathcal{E}} = \mathbf{g} + \mathbf{e}\,\mathcal{G}_{\bar{\phi}}$	cosmic field; its real coordinate $\mathbf{g}$ is the field of gravity	Def. 9.2
$\boldsymbol{\mathcal{B}} = \mathbf{B}_g + \mathbf{b}\,\mathcal{G}_{\bar{\phi}}$	second cosmic field; $\mathbf{B}_g$ is the magnetic part of gravity	Law 9.3, Section 9.3
ρ_μ, $\mathbf{J}_\mu$	density and current of cosmic mass	Section 9.2
$\mathbf{E}$, $\mathbf{B}$	electric and magnetic fields, $-\lvert\mathcal{G}_{\bar{\phi}}\rvert^2\,\mathbf{e}$ and $-\lvert\mathcal{G}_{\bar{\phi}}\rvert^2\,\mathbf{b}$	Thm. 9.5
$\mathbf{v}$	velocity of a body	Law 9.3
u, $\mathbf{S}$	density and flow of energy	Thm. 9.9
Φ, $\boldsymbol{\mathcal{A}}$	potentials of the two cosmic fields	Def. 9.12
S	action	Def. 9.12

Five letters do double duty, as in the chemistry and nuclear physics that Chapters 7 and 8 borrow. The letter r is the distance between two particles in the force laws, and the cosmic mass magnitude of a single particle in Remarks 6.11 to 6.14. The letter η is the ratio of the two parts of a General Number in Theorem 4.10, and the charge-to-mass ratio of a particle in Remark 6.15. The letter z is a General Number in Chapter 4, and the charge number of a body in Chapter 7. The letter

A is the atomic mass of a body, in daltons, in Chapter 7, and the mass number $z + N$ of a nucleus in Chapter 8. The letter Q is a corner of the triangle of a General Number in Chapter 4, and the energy released by a fusion in Chapter 8. The context always makes clear which is meant. The angle δ is the same in both of its places: the lean of Remark 7.6 is the angle of Proposition 4.32 for a cosmic mass.

D.2 UNITS

The book uses the units of the International System (Bureau International des Poids et Mesures, 2019).

KILOGRAM, KG. The unit of mass. Since 20 May 2019 it is defined by fixing the numerical value of Planck's constant.

COULOMB, C. The unit of electric charge, the charge carried by a current of one ampere in one second. Since 20 May 2019 it is defined by fixing the numerical value of the elementary charge e, the size of the charge of the electron.

METRE, M, AND NEWTON, N. The units of length and of force; one newton is one kilogram metre per second squared.

KILOGRAM PER COULOMB, KG/C. The unit of the Composer Unit $\mathcal{G}_{\tilde{\phi}}$ (Definition 6.5), which turns a charge into a mass so that both parts of a cosmic mass are measured in kilograms.

DALTON, U. The unified atomic mass unit, one twelfth of the mass of an atom of carbon-12. Chapters 7 and 8 and Appendix E give the masses of atoms, ions and nuclei in daltons.

ELECTRONVOLT, EV. The energy that one elementary charge gains across one volt, $1.602\,176\,634 \times 10^{-19}$ joule; one MeV is a million electronvolts. Chapter 8 gives the energies of nuclei and of fusions in MeV, and Chapter 9 the energy of hydrogen in eV.

TESLA, T, AND VOLT PER METRE, V/M. The units of the magnetic field and of the electric field, which Chapter 9 uses for bodies in motion.

In the displays of Section 6.3 and of Remark 6.11, the kilogram is written Kg. Those displays are kept exactly as they were first written.

D.3 CONSTANTS

CONSTANT	SYMBOL	VALUE
gravitational constant	G	$6.67430 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$
Coulomb constant	$K = 1/(4\pi\varepsilon_0)$	$8.98755 \times 10^{9}\ \mathrm{N\,m^2\,C^{-2}}$
speed of light (exact)	c	$299\,792\,458\ \mathrm{m\,s^{-1}}$
magnetic constant	μ_0	$1.25664 \times 10^{-6}\ \mathrm{N\,A^{-2}}$
elementary charge (exact)	e	$1.602\,176\,634 \times 10^{-19}\ \mathrm{C}$
reduced Planck constant	$\hbar$	$1.054\,571\,817 \times 10^{-34}\ \mathrm{J\,s}$
fine-structure constant	α_{fs}	$1/137.036$
dalton	u	$1.660\,539\,07 \times 10^{-27}\ \mathrm{kg}$
electron mass	m_e	$9.10938 \times 10^{-31}\ \mathrm{kg} = 0.000548580\ \mathrm{u}$
proton mass	m_p	$1.67262 \times 10^{-27}\ \mathrm{kg}$
neutron mass	m_n	$1.67493 \times 10^{-27}\ \mathrm{kg}$
Derived in Chapter 6		
size of the Composer Unit	$\sqrt{K/G}$	$1.16043 \times 10^{10}\ \mathrm{kg\,C^{-1}}$
angle of the Composer axis	$\bar{\phi}$	$\approx 79.86°$ (Remark 6.10, Section 6.6.3)
Derived in Chapter 7, at $\bar{\phi} \approx 79.86°$		
Stoney mass	$m_S = e\sqrt{K/G}$	$1.85921 \times 10^{-9}\ \mathrm{kg}$ (Definition 7.2)
Planck mass	$m_P = \sqrt{\hbar c/G}$	$2.17643 \times 10^{-8}\ \mathrm{kg} \approx 11.706\, m_S$
lean of a charged body	δ	$\approx 5.0374 \times 10^{-17\circ} \times m/\lvert z\rvert$, m in u (Remark 7.6)

The values of G and K are those used in Chapter 6. The other values are rounded from the recommended values of the Committee on Data of the International Science Council (Tiesinga et al., 2021); the speed of light and the elementary charge are exact by definition (Bureau International des Poids et Mesures, 2019). The remarks of Chapter 6 use the arcsine root; from Section 6.6 on, the book uses the mirror root.

E

IDENTITY CARDS OF MATTER

This appendix lists the identity cards (r, α, ϕ) at the angle $\bar{\phi} \approx 79.86°$, of the 306 bodies of Chapter 7: the electron, the proton and the neutron; then, in the order of the periodic table, each of the 118 atoms, followed by its common ions and by its bare nucleus. The masses are those of Section 7.5: atoms from the standard atomic weights, ions as $A - z\,m_e$, and nuclei as $A - Z\,m_e + B_e/c^2$. Z is the atomic number and z the charge number. For a neutral body $r = m$, $\alpha = 0$ and $\phi = 79.86°$. For a charged body r is $|z|\,m_S$ to many digits, and the angles are written with the lean δ of Remark 7.6, which is far too small to be written as a plain decimal. The workbook *Identity-Cards-data.xlsx* holds the same numbers as live formulas.

	BODY	Z	z	m (u)	m (kg)	r (kg)	α (degree)	ϕ (degree)
1	electron		-1	0.0005485799091	9.10938×10^{-31}	1.85921×10^{-9}	$259.86 + 2.76342\times10^{-20}$	$-180 - 2.76342\times10^{-20}$
2	proton		$+1$	1.007276467	1.67262×10^{-27}	1.85921×10^{-9}	$79.86 - 5.07406\times10^{-17}$	5.07406×10^{-17}
3	neutron		0	1.008664916	1.67493×10^{-27}	1.67493×10^{-27}	0	79.86
4	H	1	0	1.008	1.67382×10^{-27}	1.67382×10^{-27}	0	79.86
5	H^+	1	$+1$	1.00745142	1.67291×10^{-27}	1.85921×10^{-9}	$79.86 - 5.07494\times10^{-17}$	5.07494×10^{-17}
6	H^-	1	-1	1.00854858	1.67473×10^{-27}	1.85921×10^{-9}	$259.86 + 5.08046\times10^{-17}$	$-180 - 5.08046\times10^{-17}$
7	H nucleus	1	$+1$	1.007451436	1.67291×10^{-27}	1.85921×10^{-9}	$79.86 - 5.07494\times10^{-17}$	5.07494×10^{-17}
8	He	2	0	4.002602	6.64648×10^{-27}	6.64648×10^{-27}	0	79.86
9	He nucleus	2	$+2$	4.001504921	6.64466×10^{-27}	3.71842×10^{-9}	$79.86 - 1.00786\times10^{-16}$	1.00786×10^{-16}
10	Li	3	0	6.94	1.15241×10^{-26}	1.15241×10^{-26}	0	79.86
11	Li^+	3	$+1$	6.93945142	1.15232×10^{-26}	1.85921×10^{-9}	$79.86 - 3.49568\times10^{-16}$	3.49568×10^{-16}
12	Li nucleus	3	$+3$	6.938354474	1.15214×10^{-26}	5.57763×10^{-9}	$79.86 - 1.16504\times10^{-16}$	1.16504×10^{-16}
13	Be	4	0	9.0121831	1.49651×10^{-26}	1.49651×10^{-26}	0	79.86
14	Be^{2+}	4	$+2$	9.01108594	1.49633×10^{-26}	3.71842×10^{-9}	$79.86 - 2.26962\times10^{-16}$	2.26962×10^{-16}
15	Be nucleus	4	$+4$	9.009989206	1.49614×10^{-26}	7.43684×10^{-9}	$79.86 - 1.13467\times10^{-16}$	1.13467×10^{-16}
16	B	5	0	10.81	1.79504×10^{-26}	1.79504×10^{-26}	0	79.86
17	B nucleus	5	$+5$	10.80725783	1.79459×10^{-26}	9.29605×10^{-9}	$79.86 - 1.08881\times10^{-16}$	1.08881×10^{-16}
18	C	6	0	12.011	1.99447×10^{-26}	1.99447×10^{-26}	0	79.86
19	C nucleus	6	$+6$	12.00770964	1.99393×10^{-26}	1.11553×10^{-8}	$79.86 - 1.00813\times10^{-16}$	1.00813×10^{-16}
20	N	7	0	14.007	2.32592×10^{-26}	2.32592×10^{-26}	0	79.86
21	N^{3-}	7	-3	14.00864574	2.32619×10^{-26}	5.57763×10^{-9}	$259.86 + 2.35224\times10^{-16}$	$-180 - 2.35224\times10^{-16}$
22	N nucleus	7	$+7$	14.00316156	2.32528×10^{-26}	1.30145×10^{-8}	$79.86 - 1.00771\times10^{-16}$	1.00771×10^{-16}
23	O	8	0	15.999	2.65670×10^{-26}	2.65670×10^{-26}	0	79.86
24	O^{2-}	8	-2	16.00009716	2.65688×10^{-26}	3.71842×10^{-9}	$259.86 + 4.02995\times10^{-16}$	$-180 - 4.02995\times10^{-16}$
25	O nucleus	8	$+8$	15.99461359	2.65597×10^{-26}	1.48737×10^{-8}	$79.86 - 1.00714\times10^{-16}$	1.00714×10^{-16}
26	F	9	0	18.99840316	3.15476×10^{-26}	3.15476×10^{-26}	0	79.86
27	F^-	9	-1	18.99895174	3.15485×10^{-26}	1.85921×10^{-9}	$259.86 + 9.57053\times10^{-16}$	$-180 - 9.57053\times10^{-16}$
28	F nucleus	9	$+9$	18.9934689	3.15394×10^{-26}	1.67329×10^{-8}	$79.86 - 1.06309\times10^{-16}$	1.06309×10^{-16}
29	Ne	10	0	20.1797	3.35092×10^{-26}	3.35092×10^{-26}	0	79.86
30	Ne nucleus	10	$+10$	20.17421801	3.35001×10^{-26}	1.85921×10^{-8}	$79.86 - 1.01626\times10^{-16}$	1.01626×10^{-16}
31	Na	11	0	22.98976928	3.81754×10^{-26}	3.81754×10^{-26}	0	79.86
32	Na^+	11	$+1$	22.9892207	3.81745×10^{-26}	1.85921×10^{-9}	$79.86 - 1.15806\times10^{-15}$	1.15806×10^{-15}
33	Na nucleus	11	$+11$	22.98373968	3.81654×10^{-26}	2.04513×10^{-8}	$79.86 - 1.05253\times10^{-16}$	1.05253×10^{-16}
34	Mg	12	0	24.305	4.03594×10^{-26}	4.03594×10^{-26}	0	79.86
35	Mg^{2+}	12	$+2$	24.30390284	4.03576×10^{-26}	3.71842×10^{-9}	$79.86 - 6.12142\times10^{-16}$	6.12142×10^{-16}
36	Mg nucleus	12	$+12$	24.29842292	4.03485×10^{-26}	2.23105×10^{-8}	$79.86 - 1.02001\times10^{-16}$	1.02001×10^{-16}

	BODY	Z	z	m (u)	m (kg)	r (kg)	α (degree)	ϕ (degree)
37	Al	13	0	26.9815384	4.48039×10^{-26}	4.48039×10^{-26}	0	79.86
38	Al^{3+}	13	+3	26.97989266	4.48012×10^{-26}	5.57763×10^{-9}	$79.86-4.53028\times10^{-16}$	4.53028×10^{-16}
39	Al nucleus	13	+13	26.97441399	4.47921×10^{-26}	2.41697×10^{-8}	$79.86-1.04524\times10^{-16}$	1.04524×10^{-16}
40	Si	14	0	28.085	4.66362×10^{-26}	4.66362×10^{-26}	0	79.86
41	Si nucleus	14	+14	28.07732839	4.66235×10^{-26}	2.60289×10^{-8}	$79.86-1.01026\times10^{-16}$	1.01026×10^{-16}
42	P	15	0	30.973762	5.14331×10^{-26}	5.14331×10^{-26}	0	79.86
43	P^{3-}	15	−3	30.97540774	5.14359×10^{-26}	5.57763×10^{-9}	$259.86+5.20118\times10^{-16}$	$-180-5.20118\times10^{-16}$
44	P nucleus	15	+15	30.96554333	5.14195×10^{-26}	2.78881×10^{-8}	$79.86-1.03991\times10^{-16}$	1.03991×10^{-16}
45	S	16	0	32.06	5.32369×10^{-26}	5.32369×10^{-26}	0	79.86
46	S^{2-}	16	−2	32.06109716	5.32387×10^{-26}	3.71842×10^{-9}	$259.86+8.07523\times10^{-16}$	$-180-8.07523\times10^{-16}$
47	S nucleus	16	+16	32.05123443	5.32223×10^{-26}	2.97473×10^{-8}	$79.86-1.00909\times10^{-16}$	1.00909×10^{-16}
48	Cl	17	0	35.45	5.88661×10^{-26}	5.88661×10^{-26}	0	79.86
49	Cl^{-}	17	−1	35.45054858	5.88670×10^{-26}	1.85921×10^{-9}	$259.86+1.78579\times10^{-15}$	$-180-1.78579\times10^{-15}$
50	Cl nucleus	17	+17	35.44068767	5.88506×10^{-26}	3.16066×10^{-8}	$79.86-1.05017\times10^{-16}$	1.05017×10^{-16}
51	Ar	18	0	39.95	6.63385×10^{-26}	6.63385×10^{-26}	0	79.86
52	Ar nucleus	18	+18	39.94014107	6.63222×10^{-26}	3.34658×10^{-8}	$79.86-1.11775\times10^{-16}$	1.11775×10^{-16}
53	K	19	0	39.0983	6.49243×10^{-26}	6.49243×10^{-26}	0	79.86
54	K^{+}	19	+1	39.09775142	6.49233×10^{-26}	1.85921×10^{-9}	$79.86-1.96951\times10^{-15}$	1.96951×10^{-15}
55	K nucleus	19	+19	39.08789464	6.49070×10^{-26}	3.53250×10^{-8}	$79.86-1.03632\times10^{-16}$	1.03632×10^{-16}
56	Ca	20	0	40.078	6.65511×10^{-26}	6.65511×10^{-26}	0	79.86
57	Ca^{2+}	20	+2	40.07690284	6.65493×10^{-26}	3.71842×10^{-9}	$79.86-1.00942\times10^{-15}$	1.00942×10^{-15}
58	Ca nucleus	20	+20	40.06704836	6.65329×10^{-26}	3.71842×10^{-8}	$79.86-1.00917\times10^{-16}$	1.00917×10^{-16}
59	Sc	21	0	44.955907	7.46510×10^{-26}	7.46510×10^{-26}	0	79.86
60	Sc^{3+}	21	+3	44.95426126	7.46483×10^{-26}	5.57763×10^{-9}	$79.86-7.54842\times10^{-16}$	7.54842×10^{-16}
61	Sc nucleus	21	+21	44.94440925	7.46319×10^{-26}	3.90434×10^{-8}	$79.86-1.07811\times10^{-16}$	1.07811×10^{-16}
62	Ti	22	0	47.867	7.94850×10^{-26}	7.94850×10^{-26}	0	79.86
63	Ti nucleus	22	+22	47.85495631	7.94650×10^{-26}	4.09026×10^{-8}	$79.86-1.09575\times10^{-16}$	1.09575×10^{-16}
64	V	23	0	50.9415	8.45904×10^{-26}	8.45904×10^{-26}	0	79.86
65	V nucleus	23	+23	50.92891055	8.45694×10^{-26}	4.27618×10^{-8}	$79.86-1.11543\times10^{-16}$	1.11543×10^{-16}
66	Cr	24	0	51.9961	8.63416×10^{-26}	8.63416×10^{-26}	0	79.86
67	Cr^{2+}	24	+2	51.99500284	8.63397×10^{-26}	3.71842×10^{-9}	$79.86-1.30960\times10^{-15}$	1.30960×10^{-15}
68	Cr^{3+}	24	+3	51.99445426	8.63388×10^{-26}	5.57763×10^{-9}	$79.86-8.73056\times10^{-16}$	8.73056×10^{-16}
69	Cr nucleus	24	+24	51.98296496	8.63197×10^{-26}	4.46210×10^{-8}	$79.86-1.09108\times10^{-16}$	1.09108×10^{-16}
70	Mn	25	0	54.938043	9.12268×10^{-26}	9.12268×10^{-26}	0	79.86
71	Mn^{2+}	25	+2	54.93694584	9.12249×10^{-26}	3.71842×10^{-9}	$79.86-1.38370\times10^{-15}$	1.38370×10^{-15}
72	Mn^{3+}	25	+3	54.93639726	9.12240×10^{-26}	5.57763×10^{-9}	$79.86-9.22455\times10^{-16}$	9.22455×10^{-16}
73	Mn^{4+}	25	+4	54.93584868	9.12231×10^{-26}	7.43684×10^{-9}	$79.86-6.91835\times10^{-16}$	6.91835×10^{-16}
74	Mn nucleus	25	+25	54.92436255	9.12040×10^{-26}	4.64802×10^{-8}	$79.86-1.10670\times10^{-16}$	1.10670×10^{-16}
75	Fe	26	0	55.845	9.27328×10^{-26}	9.27328×10^{-26}	0	79.86
76	Fe^{2+}	26	+2	55.84390284	9.27310×10^{-26}	3.71842×10^{-9}	$79.86-1.40654\times10^{-15}$	1.40654×10^{-15}
77	Fe^{3+}	26	+3	55.84335426	9.27301×10^{-26}	5.57763×10^{-9}	$79.86-9.37685\times10^{-16}$	9.37685×10^{-16}
78	Fe nucleus	26	+26	55.83077432	9.27092×10^{-26}	4.83394×10^{-8}	$79.86-1.08170\times10^{-16}$	1.08170×10^{-16}
79	Co	27	0	58.933194	9.78609×10^{-26}	9.78609×10^{-26}	0	79.86
80	Co^{2+}	27	+2	58.93209684	9.78590×10^{-26}	3.71842×10^{-9}	$79.86-1.48432\times10^{-15}$	1.48432×10^{-15}
81	Co^{3+}	27	+3	58.93154826	9.78581×10^{-26}	5.57763×10^{-9}	$79.86-9.89539\times10^{-16}$	9.89539×10^{-16}
82	Co nucleus	27	+27	58.91842328	9.78363×10^{-26}	5.01986×10^{-8}	$79.86-1.09924\times10^{-16}$	1.09924×10^{-16}
83	Ni	28	0	58.6934	9.74627×10^{-26}	9.74627×10^{-26}	0	79.86
84	Ni nucleus	28	+28	58.67808442	9.74373×10^{-26}	5.20579×10^{-8}	$79.86-1.05566\times10^{-16}$	1.05566×10^{-16}
85	Cu	29	0	63.546	1.05521×10^{-25}	1.05521×10^{-25}	0	79.86
86	Cu^{+}	29	+1	63.54545142	1.05520×10^{-25}	1.85921×10^{-9}	$79.86-3.20104\times10^{-15}$	3.20104×10^{-15}
87	Cu^{2+}	29	+2	63.54490284	1.05519×10^{-25}	3.71842×10^{-9}	$79.86-1.60051\times10^{-15}$	1.60051×10^{-15}
88	Cu nucleus	29	+29	63.53013976	1.05494×10^{-25}	5.39171×10^{-8}	$79.86-1.10354\times10^{-16}$	1.10354×10^{-16}
89	Zn	30	0	65.38	1.08566×10^{-25}	1.08566×10^{-25}	0	79.86
90	Zn^{2+}	30	+2	65.37890284	1.08564×10^{-25}	3.71842×10^{-9}	$79.86-1.64670\times10^{-15}$	1.64670×10^{-15}
91	Zn nucleus	30	+30	65.3635953	1.08539×10^{-25}	5.57763×10^{-8}	$79.86-1.09754\times10^{-16}$	1.09754×10^{-16}
92	Ga	31	0	69.723	1.15778×10^{-25}	1.15778×10^{-25}	0	79.86
93	Ga^{3+}	31	+3	69.72135426	1.15775×10^{-25}	5.57763×10^{-9}	$79.86-1.17071\times10^{-15}$	1.17071×10^{-15}
94	Ga nucleus	31	+31	69.70605103	1.15750×10^{-25}	5.76355×10^{-8}	$79.86-1.13270\times10^{-16}$	1.13270×10^{-16}
95	Ge	32	0	72.63	1.20605×10^{-25}	1.20605×10^{-25}	0	79.86
96	Ge nucleus	32	+32	72.61250696	1.20576×10^{-25}	5.94947×10^{-8}	$79.86-1.14306\times10^{-16}$	1.14306×10^{-16}

	BODY	Z	z	m (u)	m (kg)	r (kg)	α (degree)	ϕ (degree)
97	As	33	0	74.921595	1.24410×10^{-25}	1.24410×10^{-25}	0	79.86
98	As^{3-}	33	−3	74.92324074	1.24413×10^{-25}	5.57763×10^{-9}	$259.86+1.25806\times10^{-15}$	$-180-1.25806\times10^{-15}$
99	As nucleus	33	+33	74.90355809	1.24380×10^{-25}	6.13539×10^{-8}	$79.86-1.14339\times10^{-16}$	1.14339×10^{-16}
100	Se	34	0	78.971	1.31134×10^{-25}	1.31134×10^{-25}	0	79.86
101	Se^{2-}	34	−2	78.97209716	1.31136×10^{-25}	3.71842×10^{-9}	$259.86+1.98907\times10^{-15}$	$-180-1.98907\times10^{-15}$
102	Se nucleus	34	+34	78.95241943	1.31104×10^{-25}	6.32131×10^{-8}	$79.86-1.16975\times10^{-16}$	1.16975×10^{-16}
103	Br	35	0	79.904	1.32684×10^{-25}	1.32684×10^{-25}	0	79.86
104	Br^{-}	35	−1	79.90454858	1.32685×10^{-25}	1.85921×10^{-9}	$259.86+4.02511\times10^{-15}$	$-180-4.02511\times10^{-15}$
105	Br nucleus	35	+35	79.88487598	1.32652×10^{-25}	6.50723×10^{-8}	$79.86-1.14975\times10^{-16}$	1.14975×10^{-16}
106	Kr	36	0	83.798	1.39150×10^{-25}	1.39150×10^{-25}	0	79.86
107	Kr nucleus	36	+36	83.77833274	1.39117×10^{-25}	6.69315×10^{-8}	$79.86-1.17229\times10^{-16}$	1.17229×10^{-16}
108	Rb	37	0	85.4678	1.41923×10^{-25}	1.41923×10^{-25}	0	79.86
109	Rb^{+}	37	+1	85.46725142	1.41922×10^{-25}	1.85921×10^{-9}	$79.86-4.30533\times10^{-15}$	4.30533×10^{-15}
110	Rb nucleus	37	+37	85.44758972	1.41889×10^{-25}	6.87907×10^{-8}	$79.86-1.16333\times10^{-16}$	1.16333×10^{-16}
111	Sr	38	0	87.62	1.45496×10^{-25}	1.45496×10^{-25}	0	79.86
112	Sr^{2+}	38	+2	87.61890284	1.45495×10^{-25}	3.71842×10^{-9}	$79.86-2.20686\times10^{-15}$	2.20686×10^{-15}
113	Sr nucleus	38	+38	87.59924691	1.45462×10^{-25}	7.06499×10^{-8}	$79.86-1.16124\times10^{-16}$	1.16124×10^{-16}
114	Y	39	0	88.905838	1.47632×10^{-25}	1.47632×10^{-25}	0	79.86
115	Y^{3+}	39	+3	88.90419226	1.47629×10^{-25}	5.57763×10^{-9}	$79.86-1.49282\times10^{-15}$	1.49282×10^{-15}
116	Y nucleus	39	+39	88.88454232	1.47596×10^{-25}	7.25092×10^{-8}	$79.86-1.14807\times10^{-16}$	1.14807×10^{-16}
117	Zr	40	0	91.222	1.51478×10^{-25}	1.51478×10^{-25}	0	79.86
118	Zr nucleus	40	+40	91.20016196	1.51441×10^{-25}	7.43684×10^{-8}	$79.86-1.14853\times10^{-16}$	1.14853×10^{-16}
119	Nb	41	0	92.90637	1.54275×10^{-25}	1.54275×10^{-25}	0	79.86
120	Nb nucleus	41	+41	92.88398982	1.54237×10^{-25}	7.62276×10^{-8}	$79.86-1.14120\times10^{-16}$	1.14120×10^{-16}
121	Mo	42	0	95.95	1.59329×10^{-25}	1.59329×10^{-25}	0	79.86
122	Mo nucleus	42	+42	95.92707791	1.59291×10^{-25}	7.80868×10^{-8}	$79.86-1.15053\times10^{-16}$	1.15053×10^{-16}
123	Tc	43	0	97.0	1.61072×10^{-25}	1.61072×10^{-25}	0	79.86
124	Tc nucleus	43	+43	96.97653624	1.61033×10^{-25}	7.99460×10^{-8}	$79.86-1.13607\times10^{-16}$	1.13607×10^{-16}
125	Ru	44	0	101.07	1.67831×10^{-25}	1.67831×10^{-25}	0	79.86
126	Ru nucleus	44	+44	101.0459948	1.67791×10^{-25}	8.18052×10^{-8}	$79.86-1.15684\times10^{-16}$	1.15684×10^{-16}
127	Rh	45	0	102.90549	1.70879×10^{-25}	1.70879×10^{-25}	0	79.86
128	Rh nucleus	45	+45	102.8809436	1.70838×10^{-25}	8.36644×10^{-8}	$79.86-1.15167\times10^{-16}$	1.15167×10^{-16}
129	Pd	46	0	106.42	1.76715×10^{-25}	1.76715×10^{-25}	0	79.86
130	Pd nucleus	46	+46	106.3949126	1.76673×10^{-25}	8.55236×10^{-8}	$79.86-1.16512\times10^{-16}$	1.16512×10^{-16}
131	Ag	47	0	107.8682	1.79119×10^{-25}	1.79119×10^{-25}	0	79.86
132	Ag^{+}	47	+1	107.8676514	1.79118×10^{-25}	1.85921×10^{-9}	$79.86-5.43373\times10^{-15}$	5.43373×10^{-15}
133	Ag nucleus	47	+47	107.8425719	1.79077×10^{-25}	8.73828×10^{-8}	$79.86-1.15584\times10^{-16}$	1.15584×10^{-16}
134	Cd	48	0	112.414	1.86668×10^{-25}	1.86668×10^{-25}	0	79.86
135	Cd^{2+}	48	+2	112.4129028	1.86666×10^{-25}	3.71842×10^{-9}	$79.86-2.83134\times10^{-15}$	2.83134×10^{-15}
136	Cd nucleus	48	+48	112.3878314	1.86624×10^{-25}	8.92420×10^{-8}	$79.86-1.17946\times10^{-16}$	1.17946×10^{-16}
137	In	49	0	114.818	1.90660×10^{-25}	1.90660×10^{-25}	0	79.86
138	In^{3+}	49	+3	114.8163543	1.90657×10^{-25}	5.57763×10^{-9}	$79.86-1.92792\times10^{-15}$	1.92792×10^{-15}
139	In nucleus	49	+49	114.7912912	1.90615×10^{-25}	9.11012×10^{-8}	$79.86-1.18010\times10^{-16}$	1.18010×10^{-16}
140	Sn	50	0	118.71	1.97123×10^{-25}	1.97123×10^{-25}	0	79.86
141	Sn^{2+}	50	+2	118.7089028	1.97121×10^{-25}	3.71842×10^{-9}	$79.86-2.98992\times10^{-15}$	2.98992×10^{-15}
142	Sn^{4+}	50	+4	118.7078057	1.97119×10^{-25}	7.43684×10^{-9}	$79.86-1.49495\times10^{-15}$	1.49495×10^{-15}
143	Sn nucleus	50	+50	118.6827512	1.97077×10^{-25}	9.29605×10^{-8}	$79.86-1.19571\times10^{-16}$	1.19571×10^{-16}
144	Sb	51	0	121.76	2.02187×10^{-25}	2.02187×10^{-25}	0	79.86
145	Sb nucleus	51	+51	121.7322115	2.02141×10^{-25}	9.48197×10^{-8}	$79.86-1.20238\times10^{-16}$	1.20238×10^{-16}
146	Te	52	0	127.6	2.11885×10^{-25}	2.11885×10^{-25}	0	79.86
147	Te^{2-}	52	−2	127.6010972	2.11887×10^{-25}	3.71842×10^{-9}	$259.86+3.21389\times10^{-15}$	$-180-3.21389\times10^{-15}$
148	Te nucleus	52	+52	127.5716721	2.11838×10^{-25}	9.66789×10^{-8}	$79.86-1.23583\times10^{-16}$	1.23583×10^{-16}
149	I	53	0	126.90447	2.10730×10^{-25}	2.10730×10^{-25}	0	79.86
150	I^{-}	53	−1	126.9050186	2.10731×10^{-25}	1.85921×10^{-9}	$259.86+6.39271\times10^{-15}$	$-180-6.39271\times10^{-15}$
151	I nucleus	53	+53	126.8756029	2.10682×10^{-25}	9.85381×10^{-8}	$79.86-1.20589\times10^{-16}$	1.20589×10^{-16}
152	Xe	54	0	131.293	2.18017×10^{-25}	2.18017×10^{-25}	0	79.86
153	Xe nucleus	54	+54	131.2635939	2.17968×10^{-25}	1.00397×10^{-7}	$79.86-1.22450\times10^{-16}$	1.22450×10^{-16}
154	Cs	55	0	132.905452	2.20695×10^{-25}	2.20695×10^{-25}	0	79.86
155	Cs^{+}	55	+1	132.9049034	2.20694×10^{-25}	1.85921×10^{-9}	$79.86-6.69495\times10^{-15}$	6.69495×10^{-15}
156	Cs nucleus	55	+55	132.8755072	2.20645×10^{-25}	1.02257×10^{-7}	$79.86-1.21699\times10^{-16}$	1.21699×10^{-16}

	BODY	Z	z	m (u)	m (kg)	r (kg)	α (degree)	ϕ (degree)
157	Ba	56	0	137.327	2.28037×10^{-25}	2.28037×10^{-25}	0	79.86
158	Ba^{2+}	56	+2	137.3259028	2.28035×10^{-25}	3.71842×10^{-9}	$79.86 - 3.45883\times10^{-15}$	3.45883×10^{-15}
159	Ba nucleus	56	+56	137.2965169	2.27986×10^{-25}	1.04116×10^{-7}	$79.86 - 1.23503\times10^{-16}$	1.23503×10^{-16}
160	La	57	0	138.90547	2.30658×10^{-25}	2.30658×10^{-25}	0	79.86
161	La^{3+}	57	+3	138.9038243	2.30655×10^{-25}	5.57763×10^{-9}	$79.86 - 2.33238\times10^{-15}$	2.33238×10^{-15}
162	La nucleus	57	+57	138.8744488	2.30606×10^{-25}	1.05975×10^{-7}	$79.86 - 1.22731\times10^{-16}$	1.22731×10^{-16}
163	Ce	58	0	140.116	2.32668×10^{-25}	2.32668×10^{-25}	0	79.86
164	Ce^{3+}	58	+3	140.1143543	2.32665×10^{-25}	5.57763×10^{-9}	$79.86 - 2.35271\times10^{-15}$	2.35271×10^{-15}
165	Ce nucleus	58	+58	140.084441	2.32616×10^{-25}	1.07834×10^{-7}	$79.86 - 1.21666\times10^{-16}$	1.21666×10^{-16}
166	Pr	59	0	140.90766	2.33983×10^{-25}	2.33983×10^{-25}	0	79.86
167	Pr^{3+}	59	+3	140.9060143	2.33980×10^{-25}	5.57763×10^{-9}	$79.86 - 2.36600\times10^{-15}$	2.36600×10^{-15}
168	Pr nucleus	59	+59	140.8755634	2.33929×10^{-25}	1.09693×10^{-7}	$79.86 - 1.20279\times10^{-16}$	1.20279×10^{-16}
169	Nd	60	0	144.242	2.39519×10^{-25}	2.39519×10^{-25}	0	79.86
170	Nd^{3+}	60	+3	144.2403543	2.39517×10^{-25}	5.57763×10^{-9}	$79.86 - 2.42199\times10^{-15}$	2.42199×10^{-15}
171	Nd nucleus	60	+60	144.2093661	2.39465×10^{-25}	1.11553×10^{-7}	$79.86 - 1.21073\times10^{-16}$	1.21073×10^{-16}
172	Pm	61	0	145.0	2.40778×10^{-25}	2.40778×10^{-25}	0	79.86
173	Pm^{3+}	61	+3	144.9983543	2.40775×10^{-25}	5.57763×10^{-9}	$79.86 - 2.43472\times10^{-15}$	2.43472×10^{-15}
174	Pm nucleus	61	+61	144.9668292	2.40723×10^{-25}	1.13412×10^{-7}	$79.86 - 1.19714\times10^{-16}$	1.19714×10^{-16}
175	Sm	62	0	150.36	2.49679×10^{-25}	2.49679×10^{-25}	0	79.86
176	Sm^{3+}	62	+3	150.3583543	2.49676×10^{-25}	5.57763×10^{-9}	$79.86 - 2.52472\times10^{-15}$	2.52472×10^{-15}
177	Sm nucleus	62	+62	150.3262925	2.49623×10^{-25}	1.15271×10^{-7}	$79.86 - 1.22138\times10^{-16}$	1.22138×10^{-16}
178	Eu	63	0	151.964	2.52342×10^{-25}	2.52342×10^{-25}	0	79.86
179	Eu^{3+}	63	+3	151.9623543	2.52339×10^{-25}	5.57763×10^{-9}	$79.86 - 2.55165\times10^{-15}$	2.55165×10^{-15}
180	Eu nucleus	63	+63	151.9297561	2.52285×10^{-25}	1.17130×10^{-7}	$79.86 - 1.21481\times10^{-16}$	1.21481×10^{-16}
181	Gd	64	0	157.249	2.61118×10^{-25}	2.61118×10^{-25}	0	79.86
182	Gd^{3+}	64	+3	157.2473543	2.61115×10^{-25}	5.57763×10^{-9}	$79.86 - 2.64039\times10^{-15}$	2.64039×10^{-15}
183	Gd nucleus	64	+64	157.21422	2.61060×10^{-25}	1.18989×10^{-7}	$79.86 - 1.23742\times10^{-16}$	1.23742×10^{-16}
184	Tb	65	0	158.925354	2.63902×10^{-25}	2.63902×10^{-25}	0	79.86
185	Tb^{3+}	65	+3	158.9237083	2.63899×10^{-25}	5.57763×10^{-9}	$79.86 - 2.66854\times10^{-15}$	2.66854×10^{-15}
186	Tb nucleus	65	+65	158.8900382	2.63843×10^{-25}	1.20849×10^{-7}	$79.86 - 1.23137\times10^{-16}$	1.23137×10^{-16}
187	Dy	66	0	162.5	2.69838×10^{-25}	2.69838×10^{-25}	0	79.86
188	Dy^{3+}	66	+3	162.4983543	2.69835×10^{-25}	5.57763×10^{-9}	$79.86 - 2.72856\times10^{-15}$	2.72856×10^{-15}
189	Dy nucleus	66	+66	162.4641488	2.69778×10^{-25}	1.22708×10^{-7}	$79.86 - 1.24000\times10^{-16}$	1.24000×10^{-16}
190	Ho	67	0	164.930329	2.73873×10^{-25}	2.73873×10^{-25}	0	79.86
191	Ho^{3+}	67	+3	164.9286833	2.73871×10^{-25}	5.57763×10^{-9}	$79.86 - 2.76937\times10^{-15}$	2.76937×10^{-15}
192	Ho nucleus	67	+67	164.8939426	2.73813×10^{-25}	1.24567×10^{-7}	$79.86 - 1.23976\times10^{-16}$	1.23976×10^{-16}
193	Er	68	0	167.259	2.77740×10^{-25}	2.77740×10^{-25}	0	79.86
194	Er^{3+}	68	+3	167.2573543	2.77737×10^{-25}	5.57763×10^{-9}	$79.86 - 2.80847\times10^{-15}$	2.80847×10^{-15}
195	Er nucleus	68	+68	167.2220788	2.77679×10^{-25}	1.26426×10^{-7}	$79.86 - 1.23877\times10^{-16}$	1.23877×10^{-16}
196	Tm	69	0	168.934219	2.80522×10^{-25}	2.80522×10^{-25}	0	79.86
197	Tm^{3+}	69	+3	168.9325733	2.80519×10^{-25}	5.57763×10^{-9}	$79.86 - 2.83660\times10^{-15}$	2.83660×10^{-15}
198	Tm nucleus	69	+69	168.8967632	2.80460×10^{-25}	1.28285×10^{-7}	$79.86 - 1.23304\times10^{-16}$	1.23304×10^{-16}
199	Yb	70	0	173.045	2.87348×10^{-25}	2.87348×10^{-25}	0	79.86
200	Yb^{3+}	70	+3	173.0433543	2.87345×10^{-25}	5.57763×10^{-9}	$79.86 - 2.90563\times10^{-15}$	2.90563×10^{-15}
201	Yb nucleus	70	+70	173.00701	2.87285×10^{-25}	1.30145×10^{-7}	$79.86 - 1.24501\times10^{-16}$	1.24501×10^{-16}
202	Lu	71	0	174.96669	2.90539×10^{-25}	2.90539×10^{-25}	0	79.86
203	Lu^{3+}	71	+3	174.9650443	2.90536×10^{-25}	5.57763×10^{-9}	$79.86 - 2.93790\times10^{-15}$	2.93790×10^{-15}
204	Lu nucleus	71	+71	174.9281662	2.90475×10^{-25}	1.32004×10^{-7}	$79.86 - 1.24110\times10^{-16}$	1.24110×10^{-16}
205	Hf	72	0	178.486	2.96383×10^{-25}	2.96383×10^{-25}	0	79.86
206	Hf nucleus	72	+72	178.4469426	2.96318×10^{-25}	1.33863×10^{-7}	$79.86 - 1.24848\times10^{-16}$	1.24848×10^{-16}
207	Ta	73	0	180.94788	3.00471×10^{-25}	3.00471×10^{-25}	0	79.86
208	Ta nucleus	73	+73	180.9082894	3.00405×10^{-25}	1.35722×10^{-7}	$79.86 - 1.24837\times10^{-16}$	1.24837×10^{-16}
209	W	74	0	183.84	3.05274×10^{-25}	3.05274×10^{-25}	0	79.86
210	W nucleus	74	+74	183.7998766	3.05207×10^{-25}	1.37581×10^{-7}	$79.86 - 1.25118\times10^{-16}$	1.25118×10^{-16}
211	Re	75	0	186.207	3.09204×10^{-25}	3.09204×10^{-25}	0	79.86
212	Re nucleus	75	+75	186.1663441	3.09136×10^{-25}	1.39441×10^{-7}	$79.86 - 1.25039\times10^{-16}$	1.25039×10^{-16}
213	Os	76	0	190.23	3.15884×10^{-25}	3.15884×10^{-25}	0	79.86
214	Os nucleus	76	+76	190.1888119	3.15816×10^{-25}	1.41300×10^{-7}	$79.86 - 1.26060\times10^{-16}$	1.26060×10^{-16}
215	Ir	77	0	192.217	3.19184×10^{-25}	3.19184×10^{-25}	0	79.86
216	Ir nucleus	77	+77	192.1752801	3.19115×10^{-25}	1.43159×10^{-7}	$79.86 - 1.25723\times10^{-16}$	1.25723×10^{-16}

	BODY	Z	z	m (u)	m (kg)	r (kg)	α (degree)	ϕ (degree)
217	Pt	78	0	195.084	3.23945×10^{-25}	3.23945×10^{-25}	0	79.86
218	Pt nucleus	78	+78	195.0417487	3.23874×10^{-25}	1.45018×10^{-7}	$79.86 - 1.25962\times10^{-16}$	1.25962×10^{-16}
219	Au	79	0	196.96657	3.27071×10^{-25}	3.27071×10^{-25}	0	79.86
220	Au nucleus	79	+79	196.9237876	3.27000×10^{-25}	1.46878×10^{-7}	$79.86 - 1.25568\times10^{-16}$	1.25568×10^{-16}
221	Hg	80	0	200.592	3.33091×10^{-25}	3.33091×10^{-25}	0	79.86
222	Hg^{2+}	80	+2	200.5909028	3.33089×10^{-25}	3.71842×10^{-9}	$79.86 - 5.05228\times10^{-15}$	5.05228×10^{-15}
223	Hg nucleus	80	+80	200.5486869	3.33019×10^{-25}	1.48737×10^{-7}	$79.86 - 1.26281\times10^{-16}$	1.26281×10^{-16}
224	Tl	81	0	204.38	3.39381×10^{-25}	3.39381×10^{-25}	0	79.86
225	Tl^{+}	81	+1	204.3794514	3.39380×10^{-25}	1.85921×10^{-9}	$79.86 - 1.02954\times10^{-14}$	1.02954×10^{-14}
226	Tl^{3+}	81	+3	204.3783543	3.39378×10^{-25}	5.57763×10^{-9}	$79.86 - 3.43179\times10^{-15}$	3.43179×10^{-15}
227	Tl nucleus	81	+81	204.3361565	3.39308×10^{-25}	1.50596×10^{-7}	$79.86 - 1.27077\times10^{-16}$	1.27077×10^{-16}
228	Pb	82	0	207.2	3.44064×10^{-25}	3.44064×10^{-25}	0	79.86
229	Pb^{2+}	82	+2	207.1989028	3.44062×10^{-25}	3.71842×10^{-9}	$79.86 - 5.21872\times10^{-15}$	5.21872×10^{-15}
230	Pb^{4+}	82	+4	207.1978057	3.44060×10^{-25}	7.43684×10^{-9}	$79.86 - 2.60935\times10^{-15}$	2.60935×10^{-15}
231	Pb nucleus	82	+82	207.1556266	3.43990×10^{-25}	1.52455×10^{-7}	$79.86 - 1.27259\times10^{-16}$	1.27259×10^{-16}
232	Bi	83	0	208.9804	3.47020×10^{-25}	3.47020×10^{-25}	0	79.86
233	Bi nucleus	83	+83	208.935497	3.46946×10^{-25}	1.54314×10^{-7}	$79.86 - 1.26806\times10^{-16}$	1.26806×10^{-16}
234	Po	84	0	209.0	3.47053×10^{-25}	3.47053×10^{-25}	0	79.86
235	Po nucleus	84	+84	208.9545679	3.46977×10^{-25}	1.56174×10^{-7}	$79.86 - 1.25308\times10^{-16}$	1.25308×10^{-16}
236	At	85	0	210.0	3.48713×10^{-25}	3.48713×10^{-25}	0	79.86
237	At nucleus	85	+85	209.9540391	3.48637×10^{-25}	1.58033×10^{-7}	$79.86 - 1.24426\times10^{-16}$	1.24426×10^{-16}
238	Rn	86	0	222.0	3.68640×10^{-25}	3.68640×10^{-25}	0	79.86
239	Rn nucleus	86	+86	221.9535108	3.68562×10^{-25}	1.59892×10^{-7}	$79.86 - 1.30008\times10^{-16}$	1.30008×10^{-16}
240	Fr	87	0	223.0	3.70300×10^{-25}	3.70300×10^{-25}	0	79.86
241	Fr^{+}	87	+1	222.9994514	3.70299×10^{-25}	1.85921×10^{-9}	$79.86 - 1.12334\times10^{-14}$	1.12334×10^{-14}
242	Fr nucleus	87	+87	222.9529828	3.70222×10^{-25}	1.61751×10^{-7}	$79.86 - 1.29092\times10^{-16}$	1.29092×10^{-16}
243	Ra	88	0	226.0	3.75282×10^{-25}	3.75282×10^{-25}	0	79.86
244	Ra^{2+}	88	+2	225.9989028	3.75280×10^{-25}	3.71842×10^{-9}	$79.86 - 5.69224\times10^{-15}$	5.69224×10^{-15}
245	Ra nucleus	88	+88	225.9524553	3.75203×10^{-25}	1.63610×10^{-7}	$79.86 - 1.29342\times10^{-16}$	1.29342×10^{-16}
246	Ac	89	0	227.0	3.76942×10^{-25}	3.76942×10^{-25}	0	79.86
247	Ac^{3+}	89	+3	226.9983543	3.76940×10^{-25}	5.57763×10^{-9}	$79.86 - 3.81161\times10^{-15}$	3.81161×10^{-15}
248	Ac nucleus	89	+89	226.9519282	3.76863×10^{-25}	1.65470×10^{-7}	$79.86 - 1.28455\times10^{-16}$	1.28455×10^{-16}
249	Th	90	0	232.0377	3.85308×10^{-25}	3.85308×10^{-25}	0	79.86
250	Th nucleus	90	+90	231.9891015	3.85227×10^{-25}	1.67329×10^{-7}	$79.86 - 1.29847\times10^{-16}$	1.29847×10^{-16}
251	Pa	91	0	231.03588	3.83644×10^{-25}	3.83644×10^{-25}	0	79.86
252	Pa nucleus	91	+91	230.9867552	3.83563×10^{-25}	1.69188×10^{-7}	$79.86 - 1.27865\times10^{-16}$	1.27865×10^{-16}
253	U	92	0	238.02891	3.95256×10^{-25}	3.95256×10^{-25}	0	79.86
254	U nucleus	92	+92	237.9792594	3.95174×10^{-25}	1.71047×10^{-7}	$79.86 - 1.30304\times10^{-16}$	1.30304×10^{-16}
255	Np	93	0	237.0	3.93548×10^{-25}	3.93548×10^{-25}	0	79.86
256	Np nucleus	93	+93	236.949824	3.93464×10^{-25}	1.72906×10^{-7}	$79.86 - 1.28345\times10^{-16}$	1.28345×10^{-16}
257	Pu	94	0	244.0	4.05172×10^{-25}	4.05172×10^{-25}	0	79.86
258	Pu nucleus	94	+94	243.9492991	4.05087×10^{-25}	1.74766×10^{-7}	$79.86 - 1.30731\times10^{-16}$	1.30731×10^{-16}
259	Am	95	0	243.0	4.03511×10^{-25}	4.03511×10^{-25}	0	79.86
260	Am nucleus	95	+95	242.9487747	4.03426×10^{-25}	1.76625×10^{-7}	$79.86 - 1.28824\times10^{-16}$	1.28824×10^{-16}
261	Cm	96	0	247.0	4.10153×10^{-25}	4.10153×10^{-25}	0	79.86
262	Cm nucleus	96	+96	246.9482507	4.10067×10^{-25}	1.78484×10^{-7}	$79.86 - 1.29581\times10^{-16}$	1.29581×10^{-16}
263	Bk	97	0	247.0	4.10153×10^{-25}	4.10153×10^{-25}	0	79.86
264	Bk nucleus	97	+97	246.9477272	4.10066×10^{-25}	1.80343×10^{-7}	$79.86 - 1.28245\times10^{-16}$	1.28245×10^{-16}
265	Cf	98	0	251.0	4.16795×10^{-25}	4.16795×10^{-25}	0	79.86
266	Cf nucleus	98	+98	250.9472042	4.16708×10^{-25}	1.82202×10^{-7}	$79.86 - 1.28992\times10^{-16}$	1.28992×10^{-16}
267	Es	99	0	252.0	4.18456×10^{-25}	4.18456×10^{-25}	0	79.86
268	Es nucleus	99	+99	251.9466817	4.18367×10^{-25}	1.84062×10^{-7}	$79.86 - 1.28198\times10^{-16}$	1.28198×10^{-16}
269	Fm	100	0	257.0	4.26759×10^{-25}	4.26759×10^{-25}	0	79.86
270	Fm nucleus	100	+100	256.9461596	4.26669×10^{-25}	1.85921×10^{-7}	$79.86 - 1.29434\times10^{-16}$	1.29434×10^{-16}
271	Md	101	0	258.0	4.28419×10^{-25}	4.28419×10^{-25}	0	79.86
272	Md nucleus	101	+101	257.9456381	4.28329×10^{-25}	1.87780×10^{-7}	$79.86 - 1.28651\times10^{-16}$	1.28651×10^{-16}
273	No	102	0	259.0	4.30080×10^{-25}	4.30080×10^{-25}	0	79.86
274	No nucleus	102	+102	258.9451171	4.29988×10^{-25}	1.89639×10^{-7}	$79.86 - 1.27883\times10^{-16}$	1.27883×10^{-16}
275	Lr	103	0	266.0	4.41703×10^{-25}	4.41703×10^{-25}	0	79.86
276	Lr nucleus	103	+103	265.9445966	4.41611×10^{-25}	1.91499×10^{-7}	$79.86 - 1.30065\times10^{-16}$	1.30065×10^{-16}

	BODY	Z	z	m (u)	m (kg)	r (kg)	α (degree)	ϕ (degree)
277	Rf	104	0	267.0	4.43364×10^{-25}	4.43364×10^{-25}	0	79.86
278	Rf nucleus	104	+104	266.9440766	4.43271×10^{-25}	1.93358×10^{-7}	$79.86-1.29298\times10^{-16}$	1.29298×10^{-16}
279	Db	105	0	268.0	4.45024×10^{-25}	4.45024×10^{-25}	0	79.86
280	Db nucleus	105	+105	267.9435572	4.44931×10^{-25}	1.95217×10^{-7}	$79.86-1.28547\times10^{-16}$	1.28547×10^{-16}
281	Sg	106	0	267.0	4.43364×10^{-25}	4.43364×10^{-25}	0	79.86
282	Sg nucleus	106	+106	266.9430383	4.43269×10^{-25}	1.97076×10^{-7}	$79.86-1.26858\times10^{-16}$	1.26858×10^{-16}
283	Bh	107	0	270.0	4.48346×10^{-25}	4.48346×10^{-25}	0	79.86
284	Bh nucleus	107	+107	269.94252	4.48250×10^{-25}	1.98935×10^{-7}	$79.86-1.27085\times10^{-16}$	1.27085×10^{-16}
285	Hs	108	0	271.0	4.50006×10^{-25}	4.50006×10^{-25}	0	79.86
286	Hs nucleus	108	+108	270.9420022	4.49910×10^{-25}	2.00795×10^{-7}	$79.86-1.26374\times10^{-16}$	1.26374×10^{-16}
287	Mt	109	0	278.0	4.61630×10^{-25}	4.61630×10^{-25}	0	79.86
288	Mt nucleus	109	+109	277.941485	4.61533×10^{-25}	2.02654×10^{-7}	$79.86-1.28450\times10^{-16}$	1.28450×10^{-16}
289	Ds	110	0	281.0	4.66611×10^{-25}	4.66611×10^{-25}	0	79.86
290	Ds nucleus	110	+110	280.9409684	4.66513×10^{-25}	2.04513×10^{-7}	$79.86-1.28656\times10^{-16}$	1.28656×10^{-16}
291	Rg	111	0	282.0	4.68272×10^{-25}	4.68272×10^{-25}	0	79.86
292	Rg nucleus	111	+111	281.9404524	4.68173×10^{-25}	2.06372×10^{-7}	$79.86-1.27950\times10^{-16}$	1.27950×10^{-16}
293	Cn	112	0	285.0	4.73254×10^{-25}	4.73254×10^{-25}	0	79.86
294	Cn nucleus	112	+112	284.9399369	4.73154×10^{-25}	2.08231×10^{-7}	$79.86-1.28157\times10^{-16}$	1.28157×10^{-16}
295	Nh	113	0	286.0	4.74914×10^{-25}	4.74914×10^{-25}	0	79.86
296	Nh nucleus	113	+113	285.9394221	4.74814×10^{-25}	2.10091×10^{-7}	$79.86-1.27468\times10^{-16}$	1.27468×10^{-16}
297	Fl	114	0	289.0	4.79896×10^{-25}	4.79896×10^{-25}	0	79.86
298	Fl nucleus	114	+114	288.9389079	4.79794×10^{-25}	2.11950×10^{-7}	$79.86-1.27676\times10^{-16}$	1.27676×10^{-16}
299	Mc	115	0	290.0	4.81556×10^{-25}	4.81556×10^{-25}	0	79.86
300	Mc nucleus	115	+115	289.9383944	4.81454×10^{-25}	2.13809×10^{-7}	$79.86-1.27003\times10^{-16}$	1.27003×10^{-16}
301	Lv	116	0	293.0	4.86538×10^{-25}	4.86538×10^{-25}	0	79.86
302	Lv nucleus	116	+116	292.9378814	4.86435×10^{-25}	2.15668×10^{-7}	$79.86-1.27211\times10^{-16}$	1.27211×10^{-16}
303	Ts	117	0	294.0	4.88198×10^{-25}	4.88198×10^{-25}	0	79.86
304	Ts nucleus	117	+117	293.9373691	4.88094×10^{-25}	2.17527×10^{-7}	$79.86-1.26554\times10^{-16}$	1.26554×10^{-16}
305	Og	118	0	294.0	4.88198×10^{-25}	4.88198×10^{-25}	0	79.86
306	Og nucleus	118	+118	293.9368575	4.88094×10^{-25}	2.19387×10^{-7}	$79.86-1.25481\times10^{-16}$	1.25481×10^{-16}

F

THE ALGEBRA OF GENERAL NUMBERS: A FORMULARY

This appendix lists the laws of Section 4.9, each beside its complex case, in the order of Appendix C. The letters follow that appendix, so that each line can be read against it. The same laws are collected in the formulary of Derraz (2026b, Appendix A).

- $z = a + b\mathcal{G}_{\bar\phi}$ and $w = c + d\mathcal{G}_{\bar\phi}$ are General Numbers of one true relative plane $\mathbb{G}_{\bar\phi}$ (Remark 4.13).
- $|\mathcal{G}_{\bar\phi}| = |\cos\bar\phi + \sin\bar\phi|$ is the length of the Composer Unit, $\vec{u}_{\bar\phi} = \mathcal{G}_{\bar\phi}/|\mathcal{G}_{\bar\phi}|$ is its direction, and $\mathcal{J}_{\bar\phi}$ is the Turning Unit (Definition 4.42).
- θ is the angle of a number, the studied angle of Definition 4.8, and $\bar\phi - \theta$ is its projection angle. For short, $C_\theta = \cos(\theta, \bar\phi - \theta)$ and $S_\theta = \sin(\theta, \bar\phi - \theta)$.

The right-hand column, the complex case, gives the same law at $\bar\phi = \pi/2$. There $|\mathcal{G}_{\bar\phi}| = 1$, the three numbers $\mathcal{G}_{\bar\phi}$, $\vec{u}_{\bar\phi}$ and $\mathcal{J}_{\bar\phi}$ are all i, and C_θ and S_θ are $\cos\theta$ and $\sin\theta$ (Corollary 3.5). Otherwise the two columns differ only in names, which the right-hand column writes as Appendix C does: Re and Im for the parts, and r for $|z|$ in the polar forms. A few laws are not stated in Section 4.9; a line under each table says where they come from.

F.1 THE TWO UNITS

IN A RELATIVE PLANE	THE COMPLEX CASE
the Rule of the Square (Proposition 4.22)	
$\mathcal{G}_{\bar\phi}^2 = 2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi\ \mathcal{G}_{\bar\phi} - \lvert\mathcal{G}_{\bar\phi}\rvert^2$, $\vec{u}_{\bar\phi}^{\,2} = 2\cos\bar\phi\ \vec{u}_{\bar\phi} - 1$	$i^2 = -1$
the Composer Unit and its conjugate (Theorem 4.38)	
$\mathcal{G}_{\bar\phi} + \overline{\mathcal{G}_{\bar\phi}} = 2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi$, $\mathcal{G}_{\bar\phi}\,\overline{\mathcal{G}_{\bar\phi}} = \lvert\mathcal{G}_{\bar\phi}\rvert^2$	$i + (-i) = 0$, $i\,(-i) = 1$
the Turning Unit (Theorem 4.43)	
$\mathcal{J}_{\bar\phi} = \dfrac{\vec{u}_{\bar\phi} - \cos\bar\phi}{\sin\bar\phi} = -\cot\bar\phi + \dfrac{\mathcal{G}_{\bar\phi}}{\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi}$, $\mathcal{J}_{\bar\phi}^2 = -1$	$i^2 = -1$

IN A RELATIVE PLANE	THE COMPLEX CASE
the square roots of -1	
$\mathcal{J}_{\bar\phi}$ and $-\mathcal{J}_{\bar\phi}$	i and $-i$
shadow and height (Proposition 4.44)	
$\mathrm{Sh}(z) = a + b\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi$, $\quad \mathrm{Ht}(z) = b\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi$	a and b, the real and imaginary parts
the two frames	
$z = a + b\mathcal{G}_{\bar\phi} = \mathrm{Sh}(z) + \mathrm{Ht}(z)\,\mathcal{J}_{\bar\phi}$	$z = a + bi$
back to the leaning frame	
$a = \mathrm{Sh}(z) - \mathrm{Ht}(z)\cot\bar\phi$, $\quad b = \dfrac{\mathrm{Ht}(z)}{\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi}$	a and b
from the complex plane	
$x + iy = (x - y\cot\bar\phi) + \dfrac{y}{\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi}\,\mathcal{G}_{\bar\phi}$	$x + iy$
to another true relative plane $\mathbb{G}_{\bar\psi}$	
$z = a' + b'\mathcal{G}_{\bar\psi}$ with $b' = \dfrac{\mathrm{Ht}(z)}{\lvert\mathcal{G}_{\bar\psi}\rvert\sin\bar\psi}$, $a' = \mathrm{Sh}(z) - \mathrm{Ht}(z)\cot\bar\psi$	at $\bar\psi = \pi/2$: $a' = \mathrm{Sh}(z)$, $\quad b' = \mathrm{Ht}(z)$

The shadow and the height are read on the page, so they do not depend on the frame: a number keeps them when it is written in the complex plane or in another relative plane, and the last two lines follow from Proposition 4.44.

F.2 ARITHMETIC

IN A RELATIVE PLANE	THE COMPLEX CASE
sum	
$(a + b\mathcal{G}_{\bar\phi}) + (c + d\mathcal{G}_{\bar\phi}) = (a + c) + (b + d)\,\mathcal{G}_{\bar\phi}$	$(a + c) + (b + d)\,i$
product (Proposition 4.23)	
$(a + b\mathcal{G}_{\bar\phi})(c + d\mathcal{G}_{\bar\phi})$ $= (ac - \lvert\mathcal{G}_{\bar\phi}\rvert^2 bd) + (ad + bc + 2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi\; bd)\,\mathcal{G}_{\bar\phi}$	$(ac - bd) + (ad + bc)\,i$
the rules of arithmetic (Theorem 4.36)	
order, grouping, distribution; 0, 1, opposites and inverses: a field	the same
conjugate (Definition 4.37)	
$\bar z = (a + 2b\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi) - b\mathcal{G}_{\bar\phi}$, the mirror image in the real axis	$\bar z = a - bi$

IN A RELATIVE PLANE	THE COMPLEX CASE
the parts of the conjugate (Definitions 4.11 and 4.37)	
$\Re(\bar{z}) = a + 2b\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi}$, $\operatorname{Com}(\bar{z}) = -b\mathcal{G}_{\bar{\phi}} = -\operatorname{Com}(z)$	$\operatorname{Re}\bar{z} = \operatorname{Re} z$, $\operatorname{Im}\bar{z} = -\operatorname{Im} z$
a number times its conjugate	
$z\bar{z} = a^2 + 2ab\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi} + b^2\lvert\mathcal{G}_{\bar{\phi}}\rvert^2 = \lvert z\rvert^2$	$z\bar{z} = a^2 + b^2$
a number plus or minus its conjugate	
$z + \bar{z} = 2\operatorname{Sh}(z)$, $\quad z - \bar{z} = 2\operatorname{Ht}(z)\,\mathcal{J}_{\bar{\phi}}$	$z + \bar{z} = 2a$, $z - \bar{z} = 2bi$
the conjugate of a sum and of a product	
$\overline{z+w} = \bar{z} + \bar{w}$, $\quad \overline{zw} = \bar{z}\,\bar{w}$, $\quad \bar{\bar{z}} = z$	the same
inverse (Theorem 4.41)	
$\dfrac{1}{z} = \dfrac{(a + 2b\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi}) - b\mathcal{G}_{\bar{\phi}}}{a^2 + 2ab\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi} + b^2\lvert\mathcal{G}_{\bar{\phi}}\rvert^2}$, $\dfrac{1}{\mathcal{G}_{\bar{\phi}}} = \dfrac{2\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi} - \mathcal{G}_{\bar{\phi}}}{\lvert\mathcal{G}_{\bar{\phi}}\rvert^2}$	$\dfrac{1}{z} = \dfrac{a - bi}{a^2 + b^2}$, $\quad \dfrac{1}{i} = -i$
quotient (Theorem 4.41)	
$\dfrac{a + b\mathcal{G}_{\bar{\phi}}}{c + d\mathcal{G}_{\bar{\phi}}} =$ $\dfrac{(ac + 2ad\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi} + bd\lvert\mathcal{G}_{\bar{\phi}}\rvert^2) + (bc - ad)\,\mathcal{G}_{\bar{\phi}}}{c^2 + 2cd\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi} + d^2\lvert\mathcal{G}_{\bar{\phi}}\rvert^2}$	$\dfrac{(ac + bd) + (bc - ad)\,i}{c^2 + d^2}$
the skew flip, which is not the conjugate (Remark 4.39)	
$(a + b\mathcal{G}_{\bar{\phi}})(a - b\mathcal{G}_{\bar{\phi}}) =$ $a^2 + b^2\lvert\mathcal{G}_{\bar{\phi}}\rvert^2 - 2b^2\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi}\;\mathcal{G}_{\bar{\phi}}$	$(a + bi)(a - bi) =$ $a^2 + b^2$

F.3 THE PLANE

IN A RELATIVE PLANE	THE COMPLEX CASE
magnitude (Theorem 4.9)	
$\lvert z\rvert = \sqrt{a^2 + 2ab\lvert\mathcal{G}_{\bar{\phi}}\rvert \cos\bar{\phi} + b^2\lvert\mathcal{G}_{\bar{\phi}}\rvert^2} =$ $\sqrt{\operatorname{Sh}(z)^2 + \operatorname{Ht}(z)^2}$	$\lvert z\rvert = \sqrt{a^2 + b^2}$
the magnitude of a product (Theorem 4.40)	
$\lvert zw\rvert = \lvert z\rvert\,\lvert w\rvert$	the same

IN A RELATIVE PLANE	THE COMPLEX CASE
the two-squares identity, with $zw = p + q\mathcal{G}_{\bar\phi}$	
$(a^2 + 2ab\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi + b^2\lvert\mathcal{G}_{\bar\phi}\rvert^2) \times (c^2 + 2cd\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi + d^2\lvert\mathcal{G}_{\bar\phi}\rvert^2) = p^2 + 2pq\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi + q^2\lvert\mathcal{G}_{\bar\phi}\rvert^2$	$(a^2+b^2)(c^2+d^2) = (ac-bd)^2 + (ad+bc)^2$
angle (Theorem 4.10)	
$\tan\theta = \dfrac{\mathrm{Ht}(z)}{\mathrm{Sh}(z)} = \dfrac{b\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi}{a + b\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi}$	$\tan\theta = \dfrac{b}{a}$
polar form (Theorem 4.45)	
$z = \lvert z\rvert\left[\cos(\theta,\bar\phi-\theta) + \sin(\theta,\bar\phi-\theta)\,\vec{u}_{\bar\phi}\right] = \lvert z\rvert\,(C_\theta + S_\theta\,\vec{u}_{\bar\phi})$	$z = r(\cos\theta + i\sin\theta)$
the parts in polar form	
$a = \lvert z\rvert\,C_\theta, \quad b\,\lvert\mathcal{G}_{\bar\phi}\rvert = \lvert z\rvert\,S_\theta$	$a = r\cos\theta, \quad b = r\sin\theta$
the general Pythagoras identity	
$C_\theta^2 + 2\cos\bar\phi\,C_\theta S_\theta + S_\theta^2 = 1$	$\cos^2\theta + \sin^2\theta = 1$
the opposite angle	
$C_{-\theta} = C_\theta + 2\cos\bar\phi\,S_\theta, \quad S_{-\theta} = -S_\theta$	$\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$
the conjugate in polar form	
$\bar z = \lvert z\rvert\,(C_{-\theta} + S_{-\theta}\,\vec{u}_{\bar\phi})$: the same magnitude, the opposite angle	$\bar z = r(\cos\theta - i\sin\theta)$
the mirror in the bisector of the two axes	
$z \mapsto \vec{u}_{\bar\phi}\,\bar z$ sends $x + y\,\vec{u}_{\bar\phi}$ to $y + x\,\vec{u}_{\bar\phi}$	$z \mapsto i\bar z$ sends $a + bi$ to $b + ai$
reciprocity (Appendix A)	
$C_{\bar\phi-\theta} = S_\theta$ and $S_{\bar\phi-\theta} = C_\theta$: the unit number at the angle $\bar\phi - \theta$ has the two parts of the one at θ, swapped	$\cos(\frac{\pi}{2} - \theta) = \sin\theta$, $\sin(\frac{\pi}{2} - \theta) = \cos\theta$

The angle is the one of Theorem 4.10, whose ratio $\eta = \lvert\mathcal{G}_{\bar\phi}\rvert\,b/a$ gives $\eta\sin\bar\phi/(1+\eta\cos\bar\phi) = \mathrm{Ht}(z)/\mathrm{Sh}(z)$. The opposite angle follows from $\sin(\bar\phi+\theta) = \sin(\bar\phi-\theta) + 2\cos\bar\phi\sin\theta$. The mirror in the line at the angle $\bar\phi/2$, which halves the angle between the two axes, keeps the length of a number and turns its angle θ into $\bar\phi - \theta$; reciprocity, $\sin(\alpha,\phi) = \cos(\phi,\alpha)$, says the same in the General Sinus.

F.4 MULTIPLYING MEANS TURNING

IN A RELATIVE PLANE	THE COMPLEX CASE
product in polar form (Theorem 4.46)	
$\lvert z\rvert\,(C_{\theta_1} + S_{\theta_1}\vec{u}_{\bar\phi}) \cdot \lvert w\rvert\,(C_{\theta_2} + S_{\theta_2}\vec{u}_{\bar\phi}) =$ $\lvert z\rvert\,\lvert w\rvert\,(C_{\theta_1+\theta_2} + S_{\theta_1+\theta_2}\vec{u}_{\bar\phi})$	$r_1 r_2\big(\cos(\theta_1+\theta_2) +$ $i\sin(\theta_1+\theta_2)\big)$
quotient in polar form	
$\dfrac{z}{w} = \dfrac{\lvert z\rvert}{\lvert w\rvert}\,(C_{\theta_1-\theta_2} + S_{\theta_1-\theta_2}\vec{u}_{\bar\phi})$	$\dfrac{r_1}{r_2}\big(\cos(\theta_1-\theta_2) +$ $i\sin(\theta_1-\theta_2)\big)$
addition along the Composer axis, cosine	
$C_{\alpha+\beta} = C_\alpha C_\beta - S_\alpha S_\beta$	$\cos(\alpha+\beta) =$ $\cos\alpha\cos\beta - \sin\alpha\sin\beta$
addition along the Composer axis, sine	
$S_{\alpha+\beta} = C_\alpha S_\beta + S_\alpha C_\beta + 2\cos\bar\phi\; S_\alpha S_\beta$	$\sin(\alpha+\beta) =$ $\sin\alpha\cos\beta + \cos\alpha\sin\beta$
subtraction, cosine	
$C_{\alpha-\beta} = C_\alpha C_\beta + S_\alpha S_\beta + 2\cos\bar\phi\; C_\alpha S_\beta$	$\cos(\alpha-\beta) =$ $\cos\alpha\cos\beta + \sin\alpha\sin\beta$
subtraction, sine	
$S_{\alpha-\beta} = S_\alpha C_\beta - C_\alpha S_\beta$	$\sin(\alpha-\beta) =$ $\sin\alpha\cos\beta - \cos\alpha\sin\beta$
double angle	
$C_{2\alpha} = C_\alpha^2 - S_\alpha^2, \quad S_{2\alpha} = 2S_\alpha\,(C_\alpha + \cos\bar\phi\; S_\alpha)$	$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha,$ $\sin 2\alpha = 2\sin\alpha\cos\alpha$
a turn by $\bar\phi$, the product by $\vec{u}_{\bar\phi}$	
$C_{\theta+\bar\phi} = -S_\theta, \quad S_{\theta+\bar\phi} = C_\theta + 2\cos\bar\phi\; S_\theta$	$\cos(\theta+\frac{\pi}{2}) = -\sin\theta,$ $\sin(\theta+\frac{\pi}{2}) = \cos\theta$
a quarter turn, the product by $\mathcal{J}_{\bar\phi}$	
$C_{\theta+\pi/2} = -C_\theta\cot\bar\phi - \dfrac{S_\theta}{\sin\bar\phi},$ $S_{\theta+\pi/2} = \dfrac{C_\theta}{\sin\bar\phi} + S_\theta\cot\bar\phi$	$\cos(\theta+\frac{\pi}{2}) = -\sin\theta,$ $\sin(\theta+\frac{\pi}{2}) = \cos\theta$
a half turn, the product by -1	
$C_{\theta+\pi} = -C_\theta, \quad S_{\theta+\pi} = -S_\theta$	$\cos(\theta+\pi) = -\cos\theta,$ $\sin(\theta+\pi) = -\sin\theta$

The subtraction formulas are the addition formulas with the opposite angle of Appendix F.3, and the double angle is $\beta = \alpha$. A turn by $\bar\phi$ is the product by $\vec{u}_{\bar\phi} = e^{\bar\phi\mathcal{J}_{\bar\phi}}$, a quarter turn the product by $\mathcal{J}_{\bar\phi}$, and

a half turn the product by $e^{\pi\mathcal{J}_{\bar\phi}} = -1$; the last is also the π-addition identity of Appendix A. In the complex plane the turn by $\bar\phi$ and the quarter turn are one and the same.

F.5 EULER'S FORMULA AND DE MOIVRE'S FORMULA

IN A RELATIVE PLANE	THE COMPLEX CASE
Euler's formula (Theorem 4.48)	
$e^{\theta\mathcal{J}_{\bar\phi}} = \cos(\theta, \bar\phi - \theta) + \sin(\theta, \bar\phi - \theta)\,\vec{u}_{\bar\phi} = C_\theta + S_\theta\,\vec{u}_{\bar\phi}$	$e^{i\theta} = \cos\theta + i\sin\theta$
exponential form	
$z = \lvert z\rvert\, e^{\theta\mathcal{J}_{\bar\phi}}, \quad \bar z = \lvert z\rvert\, e^{-\theta\mathcal{J}_{\bar\phi}}$	$z = re^{i\theta}, \quad \bar z = re^{-i\theta}$
the two units as exponentials	
$\mathcal{G}_{\bar\phi} = \lvert\mathcal{G}_{\bar\phi}\rvert\, e^{\bar\phi\mathcal{J}_{\bar\phi}}, \quad \mathcal{J}_{\bar\phi} = e^{(\pi/2)\mathcal{J}_{\bar\phi}}$	$i = e^{i\pi/2}$
Euler's identity	
$e^{\pi\mathcal{J}_{\bar\phi}} + 1 = 0$	$e^{i\pi} + 1 = 0$
the rate of turning (Remark 4.49)	
$\dfrac{d}{d\theta}\, e^{\theta\mathcal{J}_{\bar\phi}} = \mathcal{J}_{\bar\phi}\, e^{\theta\mathcal{J}_{\bar\phi}}$	$\dfrac{d}{d\theta}\, e^{i\theta} = i\, e^{i\theta}$
de Moivre's formula (Theorem 4.50)	
$(C_\theta + S_\theta\,\vec{u}_{\bar\phi})^n = C_{n\theta} + S_{n\theta}\,\vec{u}_{\bar\phi}$	$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$
the powers of the Composer Unit (Corollary 4.51)	
$\mathcal{G}_{\bar\phi}^n = \lvert\mathcal{G}_{\bar\phi}\rvert^n C_{n\bar\phi} + \lvert\mathcal{G}_{\bar\phi}\rvert^{n-1} S_{n\bar\phi}\,\mathcal{G}_{\bar\phi}$ $= -\lvert\mathcal{G}_{\bar\phi}\rvert^n \dfrac{\sin(n-1)\bar\phi}{\sin\bar\phi} + \lvert\mathcal{G}_{\bar\phi}\rvert^{n-1} \dfrac{\sin n\bar\phi}{\sin\bar\phi}\,\mathcal{G}_{\bar\phi}$	$i^n = \cos\dfrac{n\pi}{2} + i\sin\dfrac{n\pi}{2}$
the step rule	
$\mathcal{G}_{\bar\phi}^{n+1} = 2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi\;\mathcal{G}_{\bar\phi}^n - \lvert\mathcal{G}_{\bar\phi}\rvert^2\,\mathcal{G}_{\bar\phi}^{n-1}$	$i^{n+1} = -i^{n-1}$
the numbers $\sin n\bar\phi / \sin\bar\phi$*, for* $n = 1, 2, 3, 4$	
$1, \quad 2\cos\bar\phi, \quad 4\cos^2\bar\phi - 1, \quad 8\cos^3\bar\phi - 4\cos\bar\phi$	$1, \quad 0, \quad -1, \quad 0$
the exponential of a General Number	
$e^z = e^{\mathrm{Sh}(z)}\left(C_{\mathrm{Ht}(z)} + S_{\mathrm{Ht}(z)}\,\vec{u}_{\bar\phi}\right)$	$e^{a+bi} = e^a(\cos b + i\sin b)$
the logarithm, for $-\pi < \theta \le \pi$	
$\ln z = \ln\lvert z\rvert + \theta\mathcal{J}_{\bar\phi} =$ $(\ln\lvert z\rvert - \theta\cot\bar\phi) + \dfrac{\theta}{\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi}\,\mathcal{G}_{\bar\phi}$	$\ln z = \ln r + i\theta$

The numbers $\sin n\bar{\phi}/\sin\bar{\phi}$ are polynomials in $\cos\bar{\phi}$, the Chebyshev polynomials $U_{n-1}(\cos\bar{\phi})$ of the second kind; the step rule makes each one from the two before it. The exponential of a General Number follows from $z = \mathrm{Sh}(z) + \mathrm{Ht}(z)\mathcal{J}_{\bar{\phi}}$ and the rule $e^{w_1+w_2} = e^{w_1}e^{w_2}$, and the logarithm from $z = |z|\, e^{\theta\mathcal{J}_{\bar{\phi}}}$.

F.6 ROOTS

IN A RELATIVE PLANE	THE COMPLEX CASE
the n-th roots of z (Theorem 4.52)	
$w_k = \lvert z\rvert^{1/n}\,(C_{\theta_k} + S_{\theta_k}\,\vec{u}_{\bar{\phi}})$, $\quad \theta_k = \dfrac{\theta + 2\pi k}{n}$, $k = 0, \ldots, n-1$	$w_k = \lvert z\rvert^{1/n}\, e^{i(\theta+2\pi k)/n}$
the roots of unity	
$C_{2\pi k/n} + S_{2\pi k/n}\,\vec{u}_{\bar{\phi}}$	$e^{2\pi i k/n}$
the square roots of the Composer Unit (Corollary 4.53)	
$\pm\dfrac{\sqrt{\lvert\mathcal{G}_{\bar{\phi}}\rvert}}{2\cos(\bar{\phi}/2)}\,(1 + \vec{u}_{\bar{\phi}})$	$\pm\dfrac{1+i}{\sqrt{2}}$
one step along each axis, squared	
$(1 + \vec{u}_{\bar{\phi}})^2 = 4\cos^2(\bar{\phi}/2)\;\vec{u}_{\bar{\phi}}$	$(1+i)^2 = 2i$
the quadratic equation $v^2 + Pv + Q = 0$	
$v = \dfrac{-P \pm \sqrt{P^2 - 4Q}}{2}$, with the square roots of the first line	the same

The fourth line follows from $\vec{u}_{\bar{\phi}}^2 = 2\cos\bar{\phi}\;\vec{u}_{\bar{\phi}} - 1$ and $1 + \cos\bar{\phi} = 2\cos^2(\bar{\phi}/2)$. The fifth holds in every field in which each number has its two square roots, and the first line gives them.

F.7 THE MATRIX OF A GENERAL NUMBER

IN A RELATIVE PLANE	THE COMPLEX CASE
the matrix of z	
$\begin{pmatrix} a & -\lvert\mathcal{G}_{\bar\phi}\rvert^2 b \\ b & a + 2b\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi \end{pmatrix}$	$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$
products	
the matrix of zw is the product of the matrices of z and w	the same
determinant and trace	
$\det = \lvert z\rvert^2$, $\quad$ trace $= 2\,\mathrm{Sh}(z)$	$\det = a^2 + b^2$, $\quad$ trace $= 2a$
the matrix of the Composer Unit	
$\begin{pmatrix} 0 & -\lvert\mathcal{G}_{\bar\phi}\rvert^2 \\ 1 & 2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi \end{pmatrix}$, which obeys the Rule of the Square: $M^2 = 2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi\, M - \lvert\mathcal{G}_{\bar\phi}\rvert^2$	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, with $M^2 = -1$

Multiplying by z sends 1 to $a + b\mathcal{G}_{\bar\phi}$ and $\mathcal{G}_{\bar\phi}$ to $-\lvert\mathcal{G}_{\bar\phi}\rvert^2 b + (a + 2b\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi)\,\mathcal{G}_{\bar\phi}$, by Proposition 4.23; these two pairs of parts are the columns of the matrix.

F.8 THE PARTS BESIDE THE SHADOW AND THE HEIGHT

A number has two pairs of coordinates: its parts, $\Re(z) = a$ and $\mathrm{Com}(z) = b\mathcal{G}_{\bar\phi}$ (Definition 4.11), counted in the leaning frame, and its shadow and height, counted in the upright frame (Proposition 4.44). In each row the law for the parts comes first, and under it the same law for the shadow and the height; beside them stands the complex case, where the two pairs are one. Each row cites the statements that prove it, most of them in Section 4.9.7.

IN A RELATIVE PLANE	THE COMPLEX CASE
the two pairs (Definitions 4.11 and 4.19, Proposition 4.44)	
$\Re(z) = a$, $\quad \mathrm{Com}(z) = b\mathcal{G}_{\bar\phi}$; $\mathrm{Sh}(z) = a + b\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi$, $\quad \mathrm{Ht}(z) = b\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi$	$\mathrm{Re}\,z = a$, $\mathrm{Com}(z) = bi$; the two pairs are one

IN A RELATIVE PLANE	THE COMPLEX CASE
from one pair to the other (Proposition 4.54)	
$\Re(z) = \mathrm{Sh}(z) - \mathrm{Ht}(z)\cot\bar{\phi}$, $\mathrm{Com}(z) = \mathrm{Ht}(z)\,\mathcal{G}_{\bar{\phi}}/(\lvert\mathcal{G}_{\bar{\phi}}\rvert\sin\bar{\phi})$; $\mathrm{Sh}(z) = \Re(z) + \mathrm{Sh}\big(\mathrm{Com}(z)\big)$, $\mathrm{Ht}(z) = \mathrm{Ht}\big(\mathrm{Com}(z)\big)$	the shadow is the real part, and $\mathrm{Com}(z) = i\,\mathrm{Im}\,z$
sum and real multiple (Proposition 4.54)	
$\Re(z+w) = \Re(z) + \Re(w)$ and $\Re(kz) = k\,\Re(z)$ for every real k, and the same for Com, for Sh and for Ht	the same for Re and for Im
magnitude (Theorem 4.9, Proposition 4.44)	
$\lvert z\rvert^2 = \Re(z)^2 + 2\cos\bar{\phi}\;\Re(z)\,b\lvert\mathcal{G}_{\bar{\phi}}\rvert + b^2\lvert\mathcal{G}_{\bar{\phi}}\rvert^2$; $\lvert z\rvert^2 = \mathrm{Sh}(z)^2 + \mathrm{Ht}(z)^2$	$\lvert z\rvert^2 = a^2 + b^2$
from a number and its conjugate (Proposition 4.54)	
$\Re(z) = (\mathcal{G}_{\bar{\phi}}\bar{z} - \overline{\mathcal{G}_{\bar{\phi}}}z)/(\mathcal{G}_{\bar{\phi}} - \overline{\mathcal{G}_{\bar{\phi}}})$, $\mathrm{Com}(z) = \mathcal{G}_{\bar{\phi}}\,(z-\bar{z})/(\mathcal{G}_{\bar{\phi}} - \overline{\mathcal{G}_{\bar{\phi}}})$, with $\mathcal{G}_{\bar{\phi}} - \overline{\mathcal{G}_{\bar{\phi}}} = 2\lvert\mathcal{G}_{\bar{\phi}}\rvert\sin\bar{\phi}\;\mathcal{J}_{\bar{\phi}}$; $\mathrm{Sh}(z) = (z+\bar{z})/2$, $\quad\mathrm{Ht}(z)\,\mathcal{J}_{\bar{\phi}} = (z-\bar{z})/2$	$\mathrm{Re}\,z = (z+\bar{z})/2$, $i\,\mathrm{Im}\,z = (z-\bar{z})/2$
real numbers, pure Composer numbers and multiples of $\mathcal{J}_{\bar{\phi}}$ (Proposition 4.54)	
$\mathrm{Com}(z) = 0$ exactly when $\bar{z} = z$, $\Re(z) = 0$ exactly when $\mathcal{G}_{\bar{\phi}}\bar{z} = \overline{\mathcal{G}_{\bar{\phi}}}z$; $\mathrm{Ht}(z) = 0$ exactly when $\bar{z} = z$, $\mathrm{Sh}(z) = 0$ exactly when $\bar{z} = -z$	real numbers: $\bar{z} = z$, imaginary numbers: $\bar{z} = -z$
product (Theorem 4.55)	
$\Re(zw) = \Re(z)\Re(w) - \lvert\mathcal{G}_{\bar{\phi}}\rvert^2 bd$, $\mathrm{Com}(zw) = \mathrm{Sh}(z)\,\mathrm{Com}(w) + \mathrm{Sh}(w)\,\mathrm{Com}(z)$; $\mathrm{Sh}(zw) = \mathrm{Sh}(z)\,\mathrm{Sh}(w) - \mathrm{Ht}(z)\,\mathrm{Ht}(w)$, $\mathrm{Ht}(zw) = \mathrm{Sh}(z)\,\mathrm{Ht}(w) + \mathrm{Sh}(w)\,\mathrm{Ht}(z)$	$\mathrm{Re}(zw) = ac - bd$, $\mathrm{Im}(zw) = ad + bc$
square (Theorem 4.55)	
$\Re(z^2) = \Re(z)^2 - \lvert\mathcal{G}_{\bar{\phi}}\rvert^2 b^2$, $\mathrm{Com}(z^2) = 2\,\mathrm{Sh}(z)\,\mathrm{Com}(z)$; $\mathrm{Sh}(z^2) = \mathrm{Sh}(z)^2 - \mathrm{Ht}(z)^2$, $\mathrm{Ht}(z^2) = 2\,\mathrm{Sh}(z)\,\mathrm{Ht}(z)$	$\mathrm{Re}(z^2) = a^2 - b^2$, $\mathrm{Im}(z^2) = 2ab$
conjugate (Proposition 4.56)	
$\Re(\bar{z}) = 2\,\mathrm{Sh}(z) - \Re(z)$, $\quad\mathrm{Com}(\bar{z}) = -\,\mathrm{Com}(z)$, so $\mathrm{Sh}(z) = \big[\Re(z) + \Re(\bar{z})\big]/2$; $\mathrm{Sh}(\bar{z}) = \mathrm{Sh}(z)$, $\quad\mathrm{Ht}(\bar{z}) = -\,\mathrm{Ht}(z)$	$\mathrm{Re}\,\bar{z} = a$, $\mathrm{Im}\,\bar{z} = -b$

IN A RELATIVE PLANE	THE COMPLEX CASE
inverse (Proposition 4.56)	
$\Re(1/z) = \left[2\,\mathrm{Sh}(z) - \Re(z)\right]/\lvert z\rvert^2$, $\mathrm{Com}(1/z) = -\,\mathrm{Com}(z)/\lvert z\rvert^2$; $\mathrm{Sh}(1/z) = \mathrm{Sh}(z)/\lvert z\rvert^2$, $\mathrm{Ht}(1/z) = -\,\mathrm{Ht}(z)/\lvert z\rvert^2$	the two parts of $\dfrac{a - bi}{a^2 + b^2}$
quotient, with the top $z\bar{w}$ (Proposition 4.56)	
$\Re(z/w) = \Re(z\bar{w})/\lvert w\rvert^2$ with $\Re(z\bar{w}) = \Re(z)\Re(\bar{w}) + \lvert\mathcal{G}_{\bar{\phi}}\rvert^2 bd$, $\mathrm{Com}(z/w) = \mathrm{Com}(z\bar{w})/\lvert w\rvert^2$ with $\mathrm{Com}(z\bar{w}) = \mathrm{Sh}(w)\,\mathrm{Com}(z) - \mathrm{Sh}(z)\,\mathrm{Com}(w)$; $\mathrm{Sh}(z/w) = \mathrm{Sh}(z\bar{w})/\lvert w\rvert^2$ with $\mathrm{Sh}(z\bar{w}) = \mathrm{Sh}(z)\,\mathrm{Sh}(w) + \mathrm{Ht}(z)\,\mathrm{Ht}(w)$, $\mathrm{Ht}(z/w) = \mathrm{Ht}(z\bar{w})/\lvert w\rvert^2$ with $\mathrm{Ht}(z\bar{w}) = \mathrm{Sh}(w)\,\mathrm{Ht}(z) - \mathrm{Sh}(z)\,\mathrm{Ht}(w)$	the two parts of $\dfrac{(ac + bd) + (bc - ad)\,i}{c^2 + d^2}$
a quarter turn, the product by $\mathcal{J}_{\bar{\phi}}$ (Corollary 4.57)	
$\Re(\mathcal{J}_{\bar{\phi}} z) = -\,\mathrm{Ht}(z) - \mathrm{Sh}(z)\cot\bar{\phi}$, $\mathrm{Com}(\mathcal{J}_{\bar{\phi}} z) = \mathrm{Sh}(z)\,\mathcal{G}_{\bar{\phi}}/(\lvert\mathcal{G}_{\bar{\phi}}\rvert \sin\bar{\phi})$; $\mathrm{Sh}(\mathcal{J}_{\bar{\phi}} z) = -\,\mathrm{Ht}(z)$, $\mathrm{Ht}(\mathcal{J}_{\bar{\phi}} z) = \mathrm{Sh}(z)$	$\mathrm{Re}(iz) = -b$, $\mathrm{Im}(iz) = a$
the product by the Composer Unit (Corollary 4.57)	
$\Re(\mathcal{G}_{\bar{\phi}} z) = -\lvert\mathcal{G}_{\bar{\phi}}\rvert^2 b$, $\mathrm{Com}(\mathcal{G}_{\bar{\phi}} z) = \Re(\bar{z})\,\mathcal{G}_{\bar{\phi}}$; $\mathrm{Sh}(\mathcal{G}_{\bar{\phi}} z) = \lvert\mathcal{G}_{\bar{\phi}}\rvert\left(\mathrm{Sh}(z)\cos\bar{\phi} - \mathrm{Ht}(z)\sin\bar{\phi}\right)$, $\mathrm{Ht}(\mathcal{G}_{\bar{\phi}} z) = \lvert\mathcal{G}_{\bar{\phi}}\rvert\left(\mathrm{Sh}(z)\sin\bar{\phi} + \mathrm{Ht}(z)\cos\bar{\phi}\right)$	the product by i, as in the line above
the powers of the Composer Unit (Theorem 4.20, Corollaries 4.51 and 4.59)	
$\Re(\mathcal{G}_{\bar{\phi}}^n) = -\lvert\mathcal{G}_{\bar{\phi}}\rvert^n \sin(n-1)\bar{\phi}/\sin\bar{\phi}$, $\mathrm{Com}(\mathcal{G}_{\bar{\phi}}^n) = \lvert\mathcal{G}_{\bar{\phi}}\rvert^{n-1}(\sin n\bar{\phi}/\sin\bar{\phi})\,\mathcal{G}_{\bar{\phi}}$; $\mathrm{Sh}(\mathcal{G}_{\bar{\phi}}^n) = \lvert\mathcal{G}_{\bar{\phi}}\rvert^n \cos n\bar{\phi}$, $\mathrm{Ht}(\mathcal{G}_{\bar{\phi}}^n) = \lvert\mathcal{G}_{\bar{\phi}}\rvert^n \sin n\bar{\phi}$	$\mathrm{Re}(i^n) = \cos(n\pi/2)$, $\mathrm{Im}(i^n) = \sin(n\pi/2)$
powers, and for $n = 1$ the polar form (Corollary 4.59)	
$\Re(z^n) = \lvert z\rvert^n C_{n\theta}$, $\mathrm{Com}(z^n) = \lvert z\rvert^n S_{n\theta}\,\vec{u}_{\bar{\phi}}$; $\mathrm{Sh}(z^n) = \lvert z\rvert^n \cos n\theta$, $\mathrm{Ht}(z^n) = \lvert z\rvert^n \sin n\theta$	$\mathrm{Re}(z^n) = r^n \cos n\theta$, $\mathrm{Im}(z^n) = r^n \sin n\theta$
exponential (Corollary 4.59)	
$\Re(e^z) = e^{\mathrm{Sh}(z)}\,C_{\mathrm{Ht}(z)}$, $\mathrm{Com}(e^z) = e^{\mathrm{Sh}(z)}\,S_{\mathrm{Ht}(z)}\,\vec{u}_{\bar{\phi}}$; $\mathrm{Sh}(e^z) = e^{\mathrm{Sh}(z)}\cos\mathrm{Ht}(z)$, $\mathrm{Ht}(e^z) = e^{\mathrm{Sh}(z)}\sin\mathrm{Ht}(z)$	$\mathrm{Re}(e^z) = e^a \cos b$, $\mathrm{Im}(e^z) = e^a \sin b$
signed area and dot product (Proposition 4.58)	
$\mathrm{Com}(\bar{z}w) = (ad - bc)\,\mathcal{G}_{\bar{\phi}}$, $\Re(\bar{z}w) = \Re(\bar{z})\Re(w) + \lvert\mathcal{G}_{\bar{\phi}}\rvert^2 bd$; $\mathrm{Ht}(\bar{z}w) = \mathrm{Sh}(z)\,\mathrm{Ht}(w) - \mathrm{Ht}(z)\,\mathrm{Sh}(w)$, which is $(ad - bc)\,\lvert\mathcal{G}_{\bar{\phi}}\rvert \sin\bar{\phi}$, $\mathrm{Sh}(\bar{z}w) = \mathrm{Sh}(z)\,\mathrm{Sh}(w) + \mathrm{Ht}(z)\,\mathrm{Ht}(w)$	$\mathrm{Im}(\bar{z}w) = ad - bc$, $\mathrm{Re}(\bar{z}w) = ac + bd$

IN A RELATIVE PLANE	THE COMPLEX CASE
size (Remark 4.60)	
$\lvert\Re(z)\rvert \le \lvert z\rvert/\sin\bar\phi$, $\lvert\mathrm{Com}(z)\rvert \le \lvert z\rvert/\sin\bar\phi$;	$\lvert\mathrm{Re}\, z\rvert \le \lvert z\rvert$,
$\lvert\mathrm{Sh}(z)\rvert \le \lvert z\rvert$, $\lvert\mathrm{Ht}(z)\rvert \le \lvert z\rvert$	$\lvert\mathrm{Im}\, z\rvert \le \lvert z\rvert$

From the third row on, the laws for the shadow and the height are the laws of the complex numbers, read in the upright frame, word for word; the first two rows pass from one frame to the other. The laws for the parts are the same arithmetic counted in the leaning frame, where the lean shows. Written out in the coordinates a, b, c and d, two of them carry neither $\cos\bar\phi$ nor $\sin\bar\phi$, and nor do their special cases: the real part of zw, which is $ac - |\mathcal{G}_{\bar\phi}|^2 bd$, and the Composer part of $\bar z w$, which is $(ad - bc)\,\mathcal{G}_{\bar\phi}$. The angle enters the first only through the length $|\mathcal{G}_{\bar\phi}|$. In the row of real numbers, the lines for the shadow and the height follow from $z + \bar z = 2\,\mathrm{Sh}(z)$ and $z - \bar z = 2\,\mathrm{Ht}(z)\,\mathcal{J}_{\bar\phi}$.

The parts, the shadow and the height of six numbers that recur:

	REAL PART	COMPOSER PART	SHADOW	HEIGHT
1	1	0	1	0
$\mathcal{G}_{\bar\phi}$	0	$\mathcal{G}_{\bar\phi}$	$\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi$	$\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi$
$\overline{\mathcal{G}_{\bar\phi}}$	$2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi$	$-\mathcal{G}_{\bar\phi}$	$\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi$	$-\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi$
$\mathcal{G}_{\bar\phi}^2$	$-\lvert\mathcal{G}_{\bar\phi}\rvert^2$	$2\lvert\mathcal{G}_{\bar\phi}\rvert\cos\bar\phi\,\mathcal{G}_{\bar\phi}$	$\lvert\mathcal{G}_{\bar\phi}\rvert^2\cos 2\bar\phi$	$\lvert\mathcal{G}_{\bar\phi}\rvert^2\sin 2\bar\phi$
$1/\mathcal{G}_{\bar\phi}$	$2\cos\bar\phi/\lvert\mathcal{G}_{\bar\phi}\rvert$	$-\mathcal{G}_{\bar\phi}/\lvert\mathcal{G}_{\bar\phi}\rvert^2$	$\cos\bar\phi/\lvert\mathcal{G}_{\bar\phi}\rvert$	$-\sin\bar\phi/\lvert\mathcal{G}_{\bar\phi}\rvert$
$\mathcal{J}_{\bar\phi}$	$-\cot\bar\phi$	$\mathcal{G}_{\bar\phi}/(\lvert\mathcal{G}_{\bar\phi}\rvert\sin\bar\phi)$	0	1

At $\bar\phi = \pi/2$ the six numbers are 1, i, $-i$, -1, $-i$ and i. The first two columns then give the real part and i times the imaginary part, and the last two the real part and the imaginary part.

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COLOPHON

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